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1999

SPECIAL FEATURES

Tax system

Public sector reforms

JAPAN

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1998-1999

JAPAN

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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Table of contents

Assessment and recommendations 9

I. Recent trends and prospects 29

Shrinking private demand in 1998 29
Effects of policy measures 42
Signs of a pick-up in private activity in 1999 44
Restructuring in the non-financial business sector 46
The outlook for the period to 2001 59

II. Monetary, exchange rate and financial-sector policy and developments 63

Policy easings 63
The recent debate 68
Other supportive actions by the authorities 73
The impact of the measures 88

III. Fiscal policy and medium-term public sector issues 95

Expansionary fiscal policy and its consequences 95
Public sector reforms 117

IV. The tax system: a need for comprehensive reform 135

Considerations influencing tax policy 135
Main features of the tax system 143
Scope for strengthening the system: increasing revenues, efficiency and equity 168
Conclusion and recommendations 187

V. Structural reforms to revitalise the economy 191

Financial sector reform: ongoing liberalisation to enhance competition 191
Corporate reform: enhancing flexibility and accountability 200
Corporate pension reform: filling the hole 202
Labour market policies: cyclical reactions and structural reforms 203
Deregulation: further progress toward more market-determined outcomes 208
Consumer and competition policy: rebalancing the rules 212
Agricultural policy: revising the framework 213
Environmental protection: putting some of the pieces in place 217
Commercial and external relations: coping with persistent tensions 220

Notes 227

Bibliography 254

Annexes

I. Institutional framework of local allocation tax (LAT) 262
II. Statutory features of the tax system and key elements of recent reforms 263

III. Taxation and private savings 283
IV. Chronology of main economic events 287

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Boxes

1. The international retreat of Japan's banks 83
2. Unsubsidised investments and local governments 103
3. Bankruptcies in the third sector 113
4. The pension system in Japan and its recent reforms 131
5. Economic impact of major tax hikes 138
6. Short-term economic impact of tax reform 141
7. Comparison of tax to GDP ratios 146
8. Labour supply elasticities 151
9. Reservation price of landholders and the tax system 160
10. Inefficiency of tax reliefs 172
11. Earmarking of VAT revenues for social security 175
12. Imputation versus a classical system 180
13. Increasing the valuation for property taxation 182
14. Blue and white returns 183
15. Italy's new local corporate tax 186
16. Options for reform 188

Annexes

A1. Tax treatment of private pensions 284

Tables

1. Business investment by industry and firm size 33
2. Severity of unemployment 35
3. Consumption growth by household type 37
4. External trade 38
5. Recent foreign direct investment flows 40
6. Recent monthly indicators 45
7. Employment adjustment measures in historical perspective 49
8. Short-term outlook 61
9. The evolution of the Bank of Japan's balance sheet, FY 1996-98 69
10. Capital losses in the 1990s 77
11. Bank failures requiring public funds to cover capital shortfalls, 1998-99 78
12. Second round of public capital injections into the banking system, March 1999 80
13. International lending position of Japanese banks by geographic zone 84
14. Revenue and expense ratios of major banks 85
15. The recent evolution of problem loans 87
16. Loans and discounts of domestically licensed banks by interest rate 91
17. Loans and discounts classified by borrower 92
18. Fiscal measures in the November 1998 package 96
19. Fiscal packages in the 1990s 97
20. Central government budget 99
21. Fiscal Investment and Loan Programme (FILP) 100

22. Basic market liquidity measures in five large OECD countries 102
23. Local public finance programmes (LPFP) and local budgets 103
24. General government deficits 105
25. Balance of social security funds by account 109
26. Detailed balance sheet for general government and public institutions, March 1998 111
27. Needed fiscal correction by 2010: an illustration 114
28. The number of municipal governments 129
29. Dependency ratios 136
30. Tax increasing episodes in OECD countries 139
31. Model simulations of tax reform 142
32. Average implicit tax rates on continued work due to old-age pension systems 152
33. Marginal effective tax wedges in manufacturing in major OECD countries 154
34. Dimensions of capital taxation in selected OECD countries 156
35. Tax treatment of pensions in major OECD countries 157
36. Tax effects on landholders' reservation prices 161
37. Changes in capital gains tax rates for individuals 163
38. Redistribution through taxes and transfers 165
39. Revenue losses due to special tax measures, FY 1999 168
40. Effectiveness of value added taxes 176
41. Relative disposable income by age of the household head 179
42. Blue Return diffusion ratios 184
43. Local tax revenues 185
44. Recommendations for further structural reform 192
45. Some structural labour market indicators 204
46. Support to agricultural producers 215
47. Agriculture-related indicators 215
48. Tariff and non-tariff barriers to trade in the Quad countries 221

Annexes

A1. Key statutory features of the Japanese tax system, 1999 264
A2. Personal income taxation in major OECD countries 275
A3. Corporate taxation in major OECD countries 276
A4. Main features of VAT systems in major OECD countries 277
A5. Estimated effect on national tax revenue of tax changes in FY 1998 and 1999 280

Figures

1. Overall output and productivity performance, 1985-2000 30
2. Industrial production and inventories 31
3. Contributions to real GDP growth 32
4. Unemployment by age 34
5. Wage and salary developments 36
6. Import penetration and relative import prices 39
7. Inflation indicators 41
8. Public works and its leading indicator 42
9. The effect of credit guarantees on the number of bankruptcies 43
10. Housing starts and the Housing Loan Corporation's (HLC) loans 44
11. Firms' judgements as to their employment levels 48
12. Firms' judgements on their production capacity 51

13. Capacity utilisation in manufacturing 52
14. The deviation of the capital – potential output ratio from its trend 52
15. Capital-output ratios: an international comparison 53
16. The relationship between investment share and real growth 54
17. Corporate sector financial indicators 56
18. Corporate restructuring announcements 57
19. Interest rates and monetary conditions 65
20. Call money market: transactions and outstandings 67
21. Interest differentials and the effective exchange rate of the yen 71
22. Monetary aggregates 73
23. The public sector's share in total assets 75
24. Loans and discounts of financial institutions 76
25. Reduced credit risk premia in the banking system 82
26. Bank lending margins 90
27. Yield spreads of corporate over government bonds 92
28. Public works by local governments 105
29. Net lending: an international comparison 106
30. Intra-governmental transfers 108
31. Various definitions of Japan's government debt 110
32. Central government reorganisation 118
33. Local and central government shares in receipts and expenditures 124
34. Inter-governmental financial transfers to local governments 125
35. Regional distribution of costs and benefits of government activities 126
36. Effects of the local allocation tax 127
37. Total tax revenues 136
38. Highest all-in tax rates for top income earners 144
39. Tax mix in major OECD countries 145
40. Average effective tax rates 148
41. Average tax wedge for a production worker 149
42. Marginal tax wedges on labour income 150
43. Effective property tax rates 162
44. Overall and marginal progressivity in major OECD countries 166
45. Actual tax payments by income group 167
46. Comprehensive versus actual tax rates by income group 170
47. VAT payments based on household surveys 177

BASIC STATISTICS OF JAPAN

THE LAND

Area (1 000 sq. km), 1995	377.8
Cultivated agricultural land (1 000 sq. km), 1995	51.3
Forest (1 000 sq. km), 1994	251.4
Densely inhabited districts[1] (1 000 sq. km), 1995	12.3

Major cities, October 1996 estimate (10 000 inhabitants):	
Tokyo (23 wards)	796
Yokohama	332
Osaka	260
Nagoya	215
Sapporo	177
Kobe	142
Kyoto	146

THE PEOPLE

Population, October 1998 estimate (1 000)	126 490
Number of persons per sq. km in 1998	335
Percentage of population living in densely inhabited districts in 1995[1]	64.7
Net annual rate of population increase (1987-98)	0.3

Labour force in per cent of total population, October 1998	53.9
Percentage distribution of employed persons, 1998:	
Agriculture and forestry	4.9
Manufacturing	21.2
Service	25.9
Other	48.0

PRODUCTION

Gross domestic product in 1998 (billion yen)	495 211
Growth of real GDP, 1998	–2.8
Gross fixed investment in 1998 (per cent of GDP)	26.2

Growth of real gross fixed investment, 1998	–8.8
Net domestic product of agriculture, forestry and fishery, at market prices, in 1997 (billion yen)	7 076
Growth of industrial production, 1998	–7.1

THE GOVERNMENT

Public consumption in 1998 (in per cent of GDP)	10.1
Current public revenue in 1997 (in per cent of GDP)	31.7
Government employees in per cent of total employment, 1998	8.3
Outstanding long-term national bonds in per cent of GDP (FY 1998)	61.5

	House of Representatives	House of Councillors
Composition of Parliament, September 1998:		
Liberal Democratic Party	264	103
Democratic Party	93	55
Peace and Reform (*Komei*)	47	24
Liberal Party	40	12
Communist Party	26	23
Others	30	35
Total	500	252
Last elections	Nov. 1995	July 1998

FOREIGN TRADE AND PAYMENTS

(1998, million yen)

Commodity exports (fob)	488 665
Commodity imports (fob)	328 820
Services	–64 546
Investment income	74 011
Current balance	157 846
Exports of goods and services in per cent of GDP	11.2
Imports of goods and services in per cent of GDP	9.2

	Exports	Imports
Percentage distribution:		
OECD countries	60.0	52.5
of which: North America	33.6	27.2
Far East	30.7	32.8
Other	9.3	14.7
Total	100.0	100.0
Crude material and fuels (SITC 2, 3, 4)	1.1	23.2
Semi-manufactured goods (5, 6)	17.9	18.0
Machinery and transport equipment (7)	69.0	27.6
Other (0, 1, 8, 9)	12.0	31.2
Total	100.0	100.0

THE CURRENCY

Monetary unit: Yen

Currency units per US$ average of daily figures:	
Year 1998	130.9
October 1999	106.0

1. Areas whose population density exceeds 5 000 persons per sq. km.

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Japan by the Economic and Development Review Committee on 11 October 1999.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 10 November 1999.

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The previous Survey of Japan was issued in November 1998.

Assessment and recommendations

The economic recovery probably got underway earlier in the year

It now looks likely that the recent downturn in the Japanese economy has ended and that some modest yet fragile recovery is underway. This survey reviews the evidence for this improvement and examines the risks and uncertainties still evident and the short-term policy requirements to ensure a durable recovery (Chapter I). The recovery has been prompted by policy measures in the monetary and financial sphere (examined in Chapter II) as well as by significant fiscal stimulus (reviewed in Chapter III). However, the massive build-up in public-sector debt will require significant fiscal consolidation over the medium-term, necessitating both spending cuts and revenue increases. Achieving the latter in the most efficient way will entail a major overhaul of the tax system (an issue taken up in Chapter IV). More generally, the authorities should pursue a broad range of structural reforms (as reviewed in Chapter V) in order to revitalise the economy in the medium term.

The slump resulted from earlier foreign, financial and fiscal shocks

The worst of the 1997-98 downturn apparently ended at the close of 1998, even if the turn-around in the situation passed generally unnoticed until late in the spring of 1999. This was indeed a welcome change for an economy that was mired in a slump throughout 1998 as a result of a combination of earlier negative shocks – foreign, financial and fiscal. The cumulative loss in output by year-end of around 5¼ per cent since the peak seven quarters earlier was the most severe since the 1970s. Businesses sought ways of cutting spending to maintain cash flow in response to plunging sales. Other than outright dismissals, which are not as rare as they used to be, this inevitably led to reduced overtime, lower regular wages and diminished bonuses, all of which cut into disposable incomes and household expenditure.

Householders' anxieties about their jobs mounted, as they observed increasing numbers of workers involuntarily dismissed and the inexorable rise of the unemployment rate, which surpassed its US counterpart for the first time in post-war history. This amplified the weakness in households' expenditure, as did the uncertainties surrounding the future path of their health care costs and the value of their pensions and tax liabilities in view of the increasingly severe public finance situation. Even the external sector provided little buffering of the downturn in the domestic economy: exports fell despite a small rise in overall trading partner imports.

Macroeconomic stimulus and banking system measures have played decisive roles

Indicators through 1999 to date now suggest a modest pick-up even though there remain reasonable doubts about its sustainability. However, it is not unusual for economic data to provide conflicting evidence early in a recovery. Labour market indicators in particular are still flashing red – the unemployment rate has risen to nearly 5 per cent and may well stay around that level for some time yet, but the unemployment rate always lags behind output in the business cycle. There can be no doubt that the upturn owes much to earlier policy easings. Unsurpassed amounts of budgetary stimulus applied over the past year and a half, combined with the world's first attempt to run a zero-interest-rate policy, an effective stabilisation of the banking system, at least for now, and lavish amounts of new public loans and credit guarantees finally proved sufficient to convince both consumers and investors that an implosion would be avoided and that a turn-around was at hand. With a return of confidence, financial markets rebounded, and private spending immediately picked up. Once the resulting sales cleared away the last vestiges of the inventory overhang, moderate growth in industrial production was set to resume as from the summer.

Output apparently increased modestly in 1999

Modest growth may have settled in during the second half of 1999. It looks like households continued to increase their spending, cutting their saving rates, as they have remained fairly confident that management of the economy is being handled prudently, both for the short and longer terms. And while business may not have jettisoned earlier plans to cut

back on investment this year, the size of that retrenchment will surely be revised down, reflecting the new more optimistic mood observable in each of the political, financial and economic spheres. Housing construction should remain active in view of the strength in starts thus far this year. Exports too could pick up somewhat, despite the strength in the yen, in view of the robust recovery that appears to be underway in the economies of Japan's Asian neighbours, signs of a pickup in Europe and continued strong demand for imports in the United States. Overall, real GDP growth on both a calendar and a fiscal year basis may be of the order of 1½ per cent in 1999, a figure that does not excite, but one which should be compared with earlier predictions. Last year's Survey, for example, projected growth for 1999 would be barely positive, and forecasts made earlier this year were almost universally for yet another year of falling output.

Corporate sector restructuring may limit the prospective strength of the recovery

It is the outlook for the year 2000 and beyond that is more difficult to ascertain because of the unusual difficulty of assessing the strength of contractionary and expansionary factors. As to the former, the major influence is from corporate-sector restructuring. The need to engage in such measures is not new: it is one of the legacies of the 1980s asset price bubble. But it is only fairly recently – when finally confronted with the unsustainability of their imbalances in personnel, fixed capital and financing – that firms have begun to take it seriously. They have been adjusting their labour inputs downwards, but no more than in past recessions of this magnitude and largely by the same means (mainly by shrinking the workweek), although recourse to redundancies has been on the rise. Nonetheless, if prevalent estimates of the size of the imbalance (upwards of 2¼ million workers or 3½ per cent of employment) are reliable, downward pressure on total labour compensation will be more severe than in previous recovery periods, delaying the pickup in employment and restraining gains in real compensation per head. Similarly, low capacity utilisation in manufacturing, an investment share of GDP more appropriate for an economy whose trend growth is 4 per cent per year than 1, and unusually rapid declines in capital productivity point to the existence of excessive amounts of capital

in a wide variety of sectors. Again the estimates are worrisome: they are centred on a range from 40 to 60 trillion yen (roughly 3 to 5 per cent of the capital stock), once cyclical effects are removed.

Poor investment returns require a sizeable reduction in labour's share of income

Because this unnecessary capital was largely financed by external funds, it is no surprise that the corporate sector's balance sheet is loaded with excessive amounts of debt and that financial returns have been inadequate to remunerate shareholders properly. While deleveraging has been underway for several decades already, it has been insufficient to offset the long-term slowdown in sales growth, especially for non-manufacturing firms. Since firms have allowed both personnel and sales and administrative costs to rise unremittingly in relation to sales, their profitability has dropped to extremely low levels. In many cases, that has resulted in bankruptcy, the frequency and average size of which soared in the recession. Surviving companies have begun to respond to financial market pressures by cutting back on investment more savagely than at any time in the post-war era, by engaging in rationalisation of their business through mergers, plant closures and asset sales and by promising to boost returns on assets and equity in the coming years towards western norms. To achieve those profit goals will require a combination of increased sales and enhanced margins well beyond what can be expected from cyclical recovery. This could be facilitated by ongoing corporate governance and pension management changes that would concentrate managers' minds on profitability, as will the expected move to consolidated accounting and taxation and the planned easing of rules applying to corporate divestiture. Bankruptcy reform could also help free up resources currently tied to their present uses. In macroeconomic terms labour's share of income and output may have to fall by around 3 percentage points beyond cyclical changes. That will be challenging, but, since the adjustment phase is likely to take at least three to five years, not so formidable as to abort the recovery, assuming the policy environment is sufficiently supportive.

The economy will benefit from a more supportive environment for dynamic small businesses, pent-up consumer demand and stronger export markets

Fortunately, there are some expansionary factors as well. First, it must be recognised that most of the restructuring will be done by existing large firms who employ less than 20 per cent of the work force. A similar process of shedding workers by large established firms has been underway in the United States for nearly 20 years, and that has not prevented economic growth from occurring, even if it did slow it down when it initially got underway. The basis for economic expansion in the future must be the ability of the economy to spawn new firms and to generate growth at existing smaller firms in more dynamic sectors. Many of the necessary conditions for such a transformation are gradually falling into place, as was discussed in more detail in the previous Survey. In addition, the government has just made a number of changes that will ease corporate restructuring, promote technological innovation and encourage the private sector to take more business venture risks by, for example, facilitating mergers and acquisitions and employee buyouts, handing back intellectual property rights to private researchers sponsored by the government, and providing loans and credit insurance to start-ups and venture firms. This should be useful in stimulating economic activity, even if it proves insufficient to jump start the small-business sector without stepped-up regulatory reform. Second, following such a severe and lengthy recession, there may be a substantial backlog of pent-up household demands that, along with the healthy state of household finances, could sustain consumer spending for some time to come. Third, there is some chance that foreign market conditions will be more favourable than in the past few years: for the first time in a decade all the major regions of the world look set to be growing at or above potential.

Moderate growth could continue into next year, assuming current policy support continues

Assuming the current fiscal and monetary policy settings remain unchanged over the next year or so, the economy is judged to be able to withstand the adjustment burdens resulting from restructuring without falling back into recession. There is little doubt, however, that the recovery will not match that of 1994-96 in terms of its robustness. At that point it was still widely believed that the economy could grow its way out of most of its problems, and therefore adjustment was postponed. A moderate increase in real

GDP is about all that can be expected in the environment projected for next year: something similar to this year's figure of 1½ per cent would be most likely. But there exist important risks of an even less satisfactory outcome, based, for example, on a faster pace of restructuring or on a relapse in confidence, possibly resulting from an excessive rise in the value of the yen or renewed financial instability. This could entail a further sharp cutback in business investment plans and a dangerous setback to the income-generating process. But the scenario in the centre of a wide range of possible outcomes involves growth continuing to be led by household spending, with both private consumption and residential investment retaining some of the momentum built up in recent quarters, despite the lack of much growth in disposable income. A rebound in export growth may also help sustain demand, as may a deliberate attempt to restock by manufacturing firms. Some time next year, despite a heavy rate of scrapping, especially by large firms in the traditional sectors, greater risk-taking in various forms could take hold, so long as the easy monetary policy persists and the balance sheets of financial institutions continue to improve, allowing an incipient rebound in capital formation. This could boost business investment in more dynamic sectors and overall. The projected reduction in public investment, especially in 2001, is unlikely to have more than a limited restraining effect. In such a scenario the output gap – the aggregate amount of capacity underutilisation – would shrink very slightly, even though the unemployment rate would at best stabilise at a high level (due to ongoing labour shedding). Accordingly, moderate downward pressures on prices would persist, and labour's share would fall a little more, after a sharp drop in 1999. Assuming a further supplementary budget the general government deficit could be stable in 2000 in both actual and cyclically-adjusted terms before falling modestly in 2001 but would remain large by any standard of comparison. Finally, despite an oil-related deterioration in the terms of trade over the next year (under the assumption of an unchanging nominal exchange rate), the current account surplus is projected to edge up to 3 per cent of GDP once again mainly because of faster export growth to Asian markets.

The zero interest-rate policy should be maintained

With such a scenario in view the current monetary policy settings still seem broadly appropriate. The concerns that have confronted the monetary authorities should be only slightly alleviated over the projection horizon. Prices may still be edging down, and the yen may be well supported by renewed confidence and persistent capital inflows. Projected rises in long-term rates reflect expectations that the zero-interest-rate policy would eventually no longer be necessary, and the resulting steepening of the yield curve would be entirely consistent with normal cyclical developments. But the debate about whether sufficient stimulus is being provided in what had appeared to be a "liquidity trap" has given way to redoubled efforts by the Bank of Japan to ensure its monetary easing fully permeates the financial markets. Financial market stability has largely returned, as reflected in the resurgence in the prices of equities and of the yen, as well as the disappearance of the infamous "Japan premium" on loans to Japanese banks, even if their lending has continued to shrink slightly. The major remaining questions are how long to maintain the current zero-interest-rate policy and whether anything would be gained by adopting an explicit inflation target. The former is a function of what happens to budgetary policy and to the banking system. But if and when recovery proceeds and the credit mechanism gradually returns to normal – next year in the central scenario – the Bank of Japan might do best to no longer try so hard to hold rates further out on the yield curve near zero, even if overnight rates remain there. Setting an inflation target could potentially help in attaining price stability by enhancing the Bank's communications with outsiders once the zero interest rate constraint can be abandoned, but this is not likely to be possible at least for the time being.

Public lending and credit guarantees should be scaled back as private financial intermediaries regain strength

Monetary policy has been ably assisted in supporting real activity by increased lending by public financial institutions at modest interest rates and by an expanded system of credit guarantees. Under normal conditions neither of these measures would be called for, but in the context of a dysfunctional banking system in the midst of a severe downturn, they were justified. However, private-sector risk should be assumed by the public sector only in

extraordinary circumstances because of the longer-term moral hazard implications. Thus, with the banking system recapitalised and a more suitable regulatory system put in place, once private financial intermediaries regain strength these programmes would best be progressively scaled back in order not to have further permanent increases in the public sector's share of lending and in its role in the financial system more generally.

Laudable progress has been made in overcoming some of the problems in the banking system, but its future profitability is not assured

The last year has marked a watershed in the government's handling of the evolving financial system crisis. Two large banks have been effectively nationalised, most others have received a further and larger dose of public capital in order to resolve their non-performing loans problem, regulatory powers have been transferred to a new agency that has quickly gained credibility, and provisioning standards have been tightened. No longer will "window dressing" or even fraudulent operations be able to conceal financial losses with impunity, even though that lesson has been a costly one for the Japanese taxpayer. The two failed long-term credit banks alone had over 6 trillion yen ($55 billion) in capital deficits, much more than had been known as late as the moment of their closure. In total, the final cost to the government will probably surpass the 7 trillion yen authorised in the 1998 legislation. The result is that the government is expected to have to return to the Diet to seek authorisation for more funds to finish the job of cleaning up the banks' balance sheets. But in recapitalising the solvent large banks the opportunity seemed to have been missed of implementing more of the changes that will surely be required in order to put the industry on a footing that will be sustainable in the global marketplace. While foreign operations are being downsized, director numbers slashed and operating expenses reduced, incentive compensation for executives was not introduced, and a reduction of capacity through mergers was not insisted on. Fortunately, a significant realignment now seems more likely, and this should allow the new bigger banks to undertake the key investments in technology that have been in too short supply heretofore. But cutting operating expenses cannot play much of a role, given their already low levels. Rather, unless more revenues can be generated from existing assets

through greater securitisation, for example, and unless credit risk is henceforth assessed and priced properly (and there are as yet no signs of improvement), the banks will fail to meet their profit commitments, and the government's stake will have to remain of indefinite duration. Furthermore, while the government seems to have finally dealt with some of the banking sector's problems, there remain some bankruptcy risks among life insurance companies, which are now attracting increased regulatory attention. The gap between the actual rates earned and the rates these firms promised their policy-holders ("negative spread") has caused their financial conditions to deteriorate. In short, the financial sector has been stabilised, but there is still a long way to go before it is fully transformed.

Fiscal stimulus should not be withdrawn prematurely

The more controversial question is what should happen to budgetary settings. The government seems intent on keeping its foot all the way to the floor so as to ensure that the economy does not stall in a repeat of what happened in 1997. While the present circumstances are not those of that earlier period, the current recovery is still tenuous, with a fine balance between positive and negative forces. Implementing a programme of fiscal consolidation now might well tip that balance unfavourably. Indeed, the limitations on available policy tools with which the authorities could confront any renewed weakness in private spending and the persistent risks of deflation are reasons to seek the added assurance of sufficient public demand. Thus, a carefully designed supplementary budget is justified in order to signal the priority placed on maintaining the recovery. In that regard, heightened attention should be paid to the variable quality of public spending, and greater focus should be put on areas where returns are likely to be highest.

But consolidation should begin as soon as a clear recovery is underway

But longer-term considerations cannot be ignored. Already in last year's Survey the Committee warned that the nation was not far from the point when counter-cyclical fiscal policy would no longer be available because of the deterioration of the public finances and that aiming for a deficit on central and local government account of 3 per cent of GDP in FY 2005 was not sufficiently ambitious. Since that time fully 38 per cent of the national budget has had to be bond

financed. Local government finances have worsened further, even abstracting from the notorious "third sector" companies, over which they have joint responsibility and about which there is shockingly little information. And the underlying situation of government institutions, albeit opaque, has undoubtedly deteriorated to the point where their indebtedness recently amounted to some two-thirds of GDP. Thus, overall gross public sector debt (leaving out public financial institutions) reached 129 per cent of GDP in March 1998, even excluding contingent liabilities in the pension, banking and credit guarantee systems. For the narrowly defined general government sector debt is set to reach 114 per cent of GDP in gross terms (and 45 per cent in net terms) next year, and Japan is set to become the OECD country with the highest gross debt-to-GDP ratio by 2001. Although there are substantial measured government-held assets, their worth is difficult to assess, and the trend in net indebtedness is just as worrisome.

Over the next decade that consolidation will have to be massive and must include both spending cuts and revenue increases

To make the situation sustainable (as determined by a stabilisation of the government debt ratio) over a horizon as long as 10 years, even taking the recently announced pension reform on board, will require a massive consolidation. As stressed in the previous Survey, as much of that as possible should be carved out of spending. It seems as though public investment could yield much in the way of savings. But slowing the rate of increase in health care expenditure and shrinking the size of the civil service are other avenues that will surely also have to be included. Yet even after these moves, as well as assumed increases in social security contributions, there will probably remain a large portion to be made up by other revenue increases. To accommodate the need to increase the tax-to-GDP ratio substantially will require major changes to the tax system. The challenge – daunting, to be sure – will be to increase revenues sufficiently while at the same time avoiding raising economic distortions or compromising distributional objectives unduly.

The tax system is reasonably well designed to meet historical revenue needs

Up to this point, the tax system appears to have worked reasonably well overall. Relatively low marginal income tax rates for the vast majority of economic agents, in combination with moderate tax elasticities of the bases, indicate that the overall distortion is probably modest compared with that of most other OECD countries. In particular, it appears that neither the labour force potential, including human capital formation, nor aggregate savings and investment decisions are unduly hampered by taxation. This feature has recently been strengthened further by the reduction of the top marginal tax rate on labour income from 65 to 50 per cent (including local taxes), bringing Japan broadly into line with the other major countries. The tax-exempt threshold is among the highest in OECD, and since the brackets are also wide, around 95 per cent of all personal income earners pay less than 20 per cent of their income in taxes and social security contributions. The effective tax rate on labour is thus at a level that is comparable to those of the United States and the United Kingdom at the bottom end of the OECD range. Previously fairly high effective corporate rates have recently been reduced by 9 percentage points and are now in line with, or only slightly above, those of many other OECD countries. As well, corporations will be assisted in their restructuring by improved loss carry-forward provisions and the planned introduction of consolidated taxation. The value-added tax (VAT) applies the lowest standard rates among those OECD countries with VAT and is broadly based in terms of the goods and services it covers. It has also been a success in terms of its effectiveness in raising revenue and in contributing to balancing revenues from different bases.

But it does suffer from narrow tax bases, non-neutralities and the lack of a taxpayer identification number system

The Japanese tax system is, however, not problem-free. The need for its improvement will increase with the prospect of having to raise tax revenues substantially in relation to GDP. Tax bases are narrow in some parts of the system, and significant non-neutralities exist in other parts. The allowances and tax credits given in the personal tax system dilute the base and hence revenues, contribute to horizontal as well as vertical inequity and distort incentives for labour market participation for some groups, in particular dependent spouses. The potential for base broadening,

though more limited, also exists for social security contributions, where bonuses are only partially included, and in the VAT system, where the tax-exempt threshold for small retailers is much higher than elsewhere in the OECD. Non-neutralities are present in the incentives for corporate financing – where the tax advantages to debt relative to retained earnings and new equity are even larger than in other countries – and in the taxation of land and property. The inheritance tax distorts market prices for land, especially agricultural land, largely to the benefit of a small number of urban farmers, but the tax on holding property as well as the capital gains taxes and transaction taxes also contribute to inefficient land use. The tax treatment of personal savings, in particular pension savings, appears to be relatively lax compared with other major countries, especially considering the generational imbalances in Japan. On the administrative side, Japan stands out as one of the few OECD countries not applying a taxpayer identification number system.

Higher tax revenues should be raised through a broad package of measures focused on widening the personal tax base, increasing consumption taxes and introducing an extensive information system

A key conclusion which follows from the analysis of the Japanese tax system presented in Chapter IV is that a significant part of the required revenue increase should come from broadening the personal income tax base, reversing the tendency in recent years. Although precise estimates are not readily available, it is not unrealistic to assume that the revenue shortfall from these tax preferences could amount to as much as 10 per cent of GDP. Eliminating only half of them would thus contribute significantly to closing the financing gap. Even though exemptions can serve some economically and socially legitimate goals, these can generally be attained more efficiently through targeted transfers, and an alternative way of increasing revenues by raising rates would have large harmful effects on the economy as distortions tend to rise disproportionately with rates. Another important recommendation is to exploit more fully the potential for taxing consumption through an increase in the VAT rate and, to a lesser extent, a widening of the base. This tax has several attractive features in terms of horizontal equity and neutrality with respect to decisions to save and invest. Both an increased consumption tax and reduced personal allowances would spread the burden of

population aging and of servicing the existing government debt more evenly across generations. Successful implementation of a meaningful tax reform requires that a broad range of measures be applied all at once so that the burden of higher taxes is widely shared and the perceived sense of unfairness minimised. The introduction of a sufficiently extensive information system, including taxpayer identification numbers, could improve the accuracy of data on individuals' income status – thereby raising the efficiency and effectiveness of administering the tax and transfers systems. It could also contribute to the alleviation of the widespread sense of unfairness, notably between wage earners and the self-employed. Even with improved transparency and enforcement, however, some degree of horizontal inequity is likely to persist between capital and labour incomes in an increasingly globalised environment, which puts a limit on taxing mobile resources at rates significantly above those prevailing elsewhere.

Various public sector reforms are intended to enhance efficiency by reducing the government's role, but further changes are required to ensure success

The need for increasing tax revenues will be lessened by stepped-up efforts to reform the public sector. Enhanced transparency is acknowledged to be necessary, and to this end access to information legislation has been passed and a system of public comment procedures on regulatory changes is being instituted. This, along with increased internal assessment and evaluation of policies and actions, should serve to increase the quality of the government's economic management. The now-decided reorganisation both of the ministries and of various other institutions may also augment efficiency, but only if real public-sector cost savings result. More broadly, the government's reforms are predicated on a desire to reduce its role in the economy and to allow a greater role for both markets and lower levels of government, even though progress here has been limited and public ownership in the economy has continued to trend up to surprisingly high levels. Some privatisation has taken place, but no clear strategy has been worked out, and thus it has appeared haphazard and revenue-driven. For example, there appears to be little justification for continued government ownership of NTT. The Private Finance Initiative scheme, as appropriately modified by the Diet to reduce moral hazard, may lead to fewer poor-quality public

works projects. Similarly, the forthcoming reform of the Fiscal Investment and Loan Programme is intended to move to a more market-based allocation of funds among competing public institutions. The additional financial discipline should boost the average quality of their outlays. In addition, the central government will have to accept a smaller role if decentralisation is to lead to decision-making at the most appropriate level, as in the "subsidiarity" principle espoused earlier in the European Union context. In any case, the current system of inter-governmental relations seems to be excessively paternalistic, which results in serious incentive problems in local government financing.

The draft pension reform seems an appropriate response to future demographic constraints, but health care reform looks likely to be unsatisfactory

The government's social security policies are also under review, largely because they are not sustainable as the population ages. The recent draft law to reform the public pension system represents a reasonable compromise designed to ensure long-term actuarial balance. By gradually raising the retirement age as from 2013, making a small cut in benefits and indexing to prices rather than wages, the plan allows a delay in the schedule of contribution increases and a moderate reduction in future peak rates from what they otherwise would have been. In addition, consideration is being given to increasing the funding from general revenues from one-third to one-half (at a cost of around ½ per cent of GDP). But progress in designing the intended overhaul of the health care system has been disappointing. Few important decisions have been taken, even though the intention was to implement the reform next April. The government has not yet persuaded doctors of the desirability of changes, despite evidence of unnecessary diagnostic testing and excessive consumption of drugs. On the other hand, next spring's introduction of long-term care insurance should move the elderly who presently occupy acute-care beds into more appropriate structures at significant savings.

Some useful initiatives have been taken to deal with problems in corporate pensions

Corporate pensions also have major under-funding problems, estimated by some to be as much as 80 trillion yen (around $700 billion or 14 per cent of GDP). The government has decided that funding shortfalls will have to be reported as from next year and then eliminated over 15 years. Companies will be allowed to contribute cross-held shares to their pension plans – with potentially enormous corporate governance implications – but the call on profits will still be substantial. Another vastly important proposal that seems to have government support is to introduce a system of defined-contribution pension plans along the lines of the US 401(k) scheme. If the annual ceiling on tax-assisted contributions is set as high as is being mooted – nearly $4 000 – the fiscal implications could be significant, although the benefit in terms of pension portability and thus deepening the mid-career job market would be immense.

The structural reform process includes a myriad of other changes, and speedy implementation would be beneficial

The authorities are engaged in planning and implementing a plethora of other structural reforms as well; some of the major ones are briefly reviewed below. Most are intended to revitalise the economy and enhance medium-term growth prospects, which have deteriorated because of well-known demographic changes. In view of the lengthy period required to reach consensus it would no doubt be best if agreed reforms were carried out as quickly as possible, so that the synergies among them can begin to work without delay. But in no case should reform be postponed by reopening the debate once decisions have been taken, nor should interest groups be accorded a right of veto. In the current circumstances it would be wise to put priority on those reforms that promise new entry in dynamic sectors, attenuate workers' income uncertainty and facilitate the mobility of resources to further the economic restructuring process.

Financial market reform is on schedule, and the benefits are starting to be visible

Liberalisation of the financial markets forms the backbone of the structural reform efforts. Implementation of the Financial System Reform Law is proceeding on time, and, although many of the changes remain to be implemented between now and FY 2001, the effects are now easily seen. The transformation is comprehensive, and that is entirely

welcome. The only part of the overhaul that still does not seem set on a sound footing is customer insurance: the securities industry has two rival funds that should eventually agree to merge to gain full risk-pooling benefits; the life insurance scheme is almost certainly underfunded; and the schedule for the normalisation of bank deposit protection is under attack, despite plenty of advance notice that full compensation would be available only during the intensive reform period. One general strategy that would be advisable in this delicate area is to move towards risk-based premia in order to reduce moral hazard problems.

Major changes in regulation are underway, which could yield substantial benefits

Regulatory reform is another domain on which official attention has long been focussed because of its potential to both reduce costs and prices towards world levels and stimulate domestic demand, as indicated in the recent OECD *Review of Regulatory Reform* in Japan. The most important advance of late was the strategic decision to move towards a regulatory system based on retroactive supervision rather than on prior intervention. If fully and consistently implemented, this will be a far-reaching and beneficial change. Another significant move will be to modify public utilities' price-setting methods: price flexibility will be introduced, subject only to a cap, and information disclosure requirements will be introduced. Such a system has already been announced in the telecommunications sector (except for basic local services). In that industry competition will also grow as a result of several changes that will effectively boost foreign and new domestic firms' ability to challenge incumbents, such as a reduction in interconnection charges, even though the competitive effects of the recent split of NTT remain uncertain because of the adoption of the holding company structure. Electric utilities have been benefiting from competitive tendering by independent power producers for several years, and now the largest users will be allowed to choose their suppliers as well. If these changes were supplemented by others put forward in the OECD *Review* – such as opening access to the transmission grid – the official objective of reducing prices to internationally comparable levels may be realisable. The various parts of the transportation sector have also been accorded increasing pricing freedom, and demand-supply adjustment regulation is on

the way out. Overall, it seems fair to say that within a few years Japan's regulatory landscape may well have been radically transformed.

Recent measures to ease the burden of unemployment are appropriate, but a greater focus on longer-term problems would be beneficial

The government has taken a number of measures that mainly respond to the recent deterioration of labour-market outcomes: the rises in the overall unemployment rate, the average duration of spells and the share of involuntary joblessness in the total. While there has been some increase in the generosity of the longstanding Employment Adjustment Subsidy programme, that scheme has often been accused of preventing adjustment, rather than assisting it. Over the past year the government has been subsidising hiring at recent start-ups, job-related education and training at professional schools, local government temporary jobs, and employment of older workers, especially in high-unemployment regions or in new and growing fields. This would seem a better way of spending funds to deal with cyclical problems than would merely paying out more unemployment benefits. But to the extent that many of those involved, especially the elderly, may well be without jobs for a long time, outcomes may not prove as satisfactory as is hoped. It is therefore entirely appropriate that the unemployment insurance system be reviewed so as to determine whether there should be changes in the benefit structure, duration and targeting by age group. All workers are likely to benefit from the recent opening-up of the private job-placement business, but the parallel liberalisation of occupational limitations in the temporary-job-placement business should not be held up unnecessarily.

Agriculture is one sector where there has been no decisive move to allow markets a greater role

In agriculture there has been clear reluctance to allow markets a much greater influence. The recently adopted Basic Law on Food, Agriculture and Rural Areas makes no mention of competition among its guiding principles, citing the primacy of food security and the multi-functionality of agriculture and rural development. The cost of this approach is overall producer support levels that are high and that have shown little tendency to decline in recent years, leaving taxpayers and especially consumers with heavy financial burdens. An explicit target for food self-sufficiency is being developed, one that will likely try to reverse the

longstanding trend decline. Such a policy would require producing more of what consumers want despite the relatively hostile environment. That would take either a great deal more intervention, with all its attendant costs to Japanese taxpayers, or a massive increase in efficiency, something that would necessarily entail substantial consolidation to reap available economies of scale. The latter strategy might be more easily adopted, given the concentration of elderly among the farming population, were it not for a variety of land and tax policies that impede new entry, rationalisation and shifting land to other uses.

Market-based instruments are also not seen as highly important in meeting environmental objectives, leading to doubts as to their attainability

The government has not seen fit to make much use of markets for meeting its environmental objectives either. The most important effort will be to meet the nation's commitment under the Kyoto protocol to cut greenhouse gas emissions by what amounts to over 13 per cent from 1997 levels through a reduction in energy demand growth – from changes in life style – and in emissions – from shifts in the energy mix mainly towards nuclear power and natural gas. The strategy to reducing emissions from mobile sources may include setting various taxes related to automobile usage based on emissions levels. But in the increasingly competitive marketplace it remains to be seen whether Japanese firms will respond sufficiently to moral suasion and other voluntary measures, without the adoption of a system of carbon-based fuel taxes, which would provide the necessary incentives for fuel substitution and conservation. Charging for disposing of household garbage by weight or by bag would also go some way to making consumers aware of the full costs of their buying patterns.

The recovery is still fragile, necessitating the continuation of policy support, but eventual fiscal consolidation and further structural reform are prerequisites for improved medium-term performance

The overall assessment then is certainly more favourable than the gloomy judgement issued in the previous Survey. The economy has probably turned the corner on the worst recession of the last quarter century. The banking system has been recapitalised and is now reorganising, and financial supervision has been improved, leaving financial markets more buoyant than at any time in recent years. Consumer and business sentiment have bounced back. Output growth looks set to be around potential rates for both this year and over the projection horizon, even though the recovery is still tenuous. Fiscal stimulus should therefore not be withdrawn until private demand-led growth is solidly in place, and, in the context of the zero-interest-rate policy, ample liquidity should be provided until deflationary concerns are dispelled. But the needed fiscal consolidation should be sought as soon as possible once the economy is on a clear recovery path. Progress has also been made in taking some of the key structural reform decisions and implementing others that will no doubt serve the country well in the future by bolstering the role of markets rather than government in allocating resources. There should be no let-up in this process: the ability of the nation to revitalise its economy in the new decade depends on it.

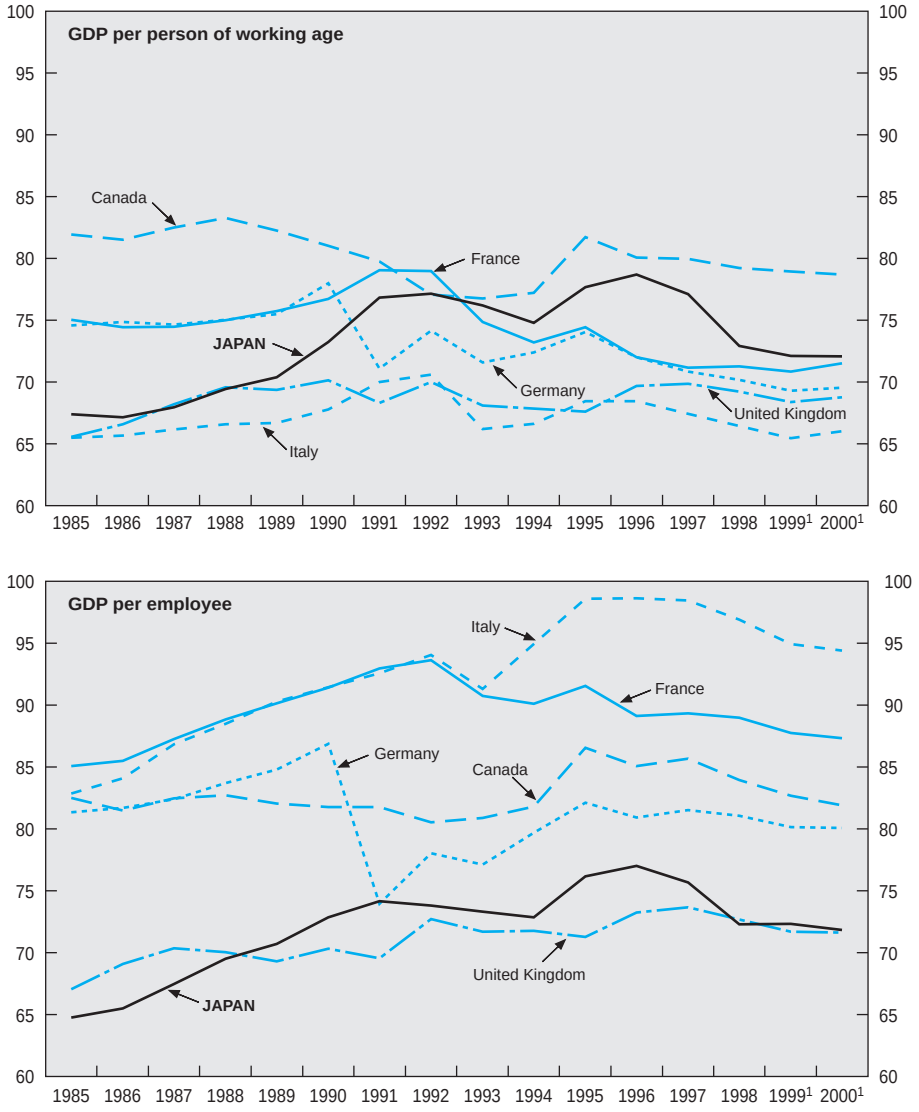
I. Recent trends and prospects

The Japanese economy experienced a severe recession in 1998, recording negative growth rates in five successive quarters to the fourth quarter of 1998, with real GDP reaching a level about 5¼ per cent lower than the previous peak in the first quarter of 1997. Corporate restructuring, which was originally triggered by the balance-sheet problems that resulted from asset-price deflation and a series of negative shocks in 1997 was a drag on the economy. Although the economy was boosted by the favourable effects of various stimulative policy measures as from the summer 1998, leading to a jump in the first quarter of 1999, there still seems some way to go before the economy is restored to a steady autonomous growth path. However, the widespread belief that Japan's performance throughout the 1990s has been dismal is exaggerated: it has suffered from a substantial demographic transition, which has lowered its potential output growth rate. But looking through the cycle and taking changes from 1990 to 1997 shows that real output both per person of working age and per employee grew slightly faster than in the United States (Figure 1), although this growth trend has reversed since 1998. It is only by the high standards that Japan itself set in previous decades that such performance could be seen as disappointing.

Shrinking private demand in 1998

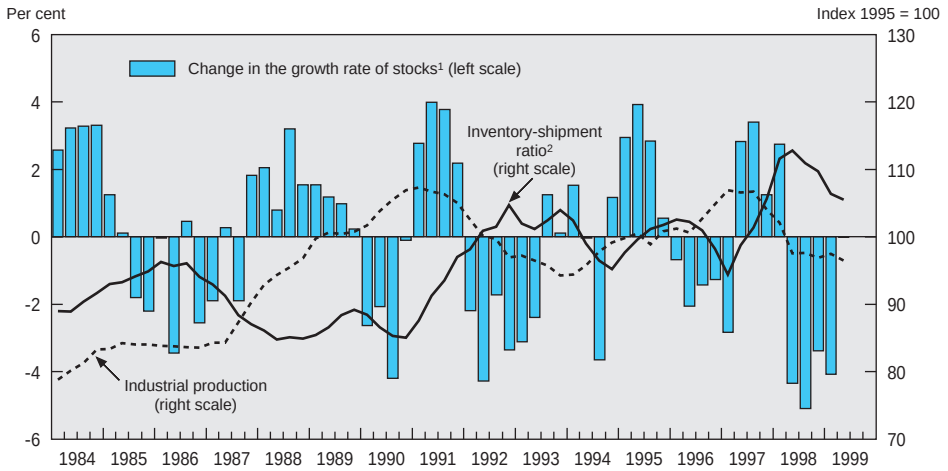
A series of negative shocks to private consumption and exports in 1997, including severe fiscal tightening, the Asian crisis and the banking crisis (see the previous Survey for details), led to inventory and production adjustment. In response to undesired increases in inventories, production adjustment was reinforced as from the second quarter of 1998, and inventories finally started to fall (Figure 2). Lower production levels decreased capacity utilisation, and corporate profits plummeted by more than 25 per cent. In addition to these cyclical factors, a rising capital-output ratio and a decline in output growth expectations[1] encouraged firms to undertake restructuring (see below). As a result, business investment was slashed, making the largest contribution to the negative growth in 1998, accounting for nearly half the 2.8 per cent decline (Figure 3). While a fall in

Figure 1. **Overall output and productivity performance, 1985-2000**
Purchasing power parity terms, USA = 100



1. Preliminary projections for OECD *Economic Outlook 66.*
Source: OECD.

Figure 2. **Industrial production and inventories**



1. Change in the year-on-year growth rate of stocks of finished products in manufacturing industry.
2. The inventory-shipment ratio covers finished products of manufacturing industry.

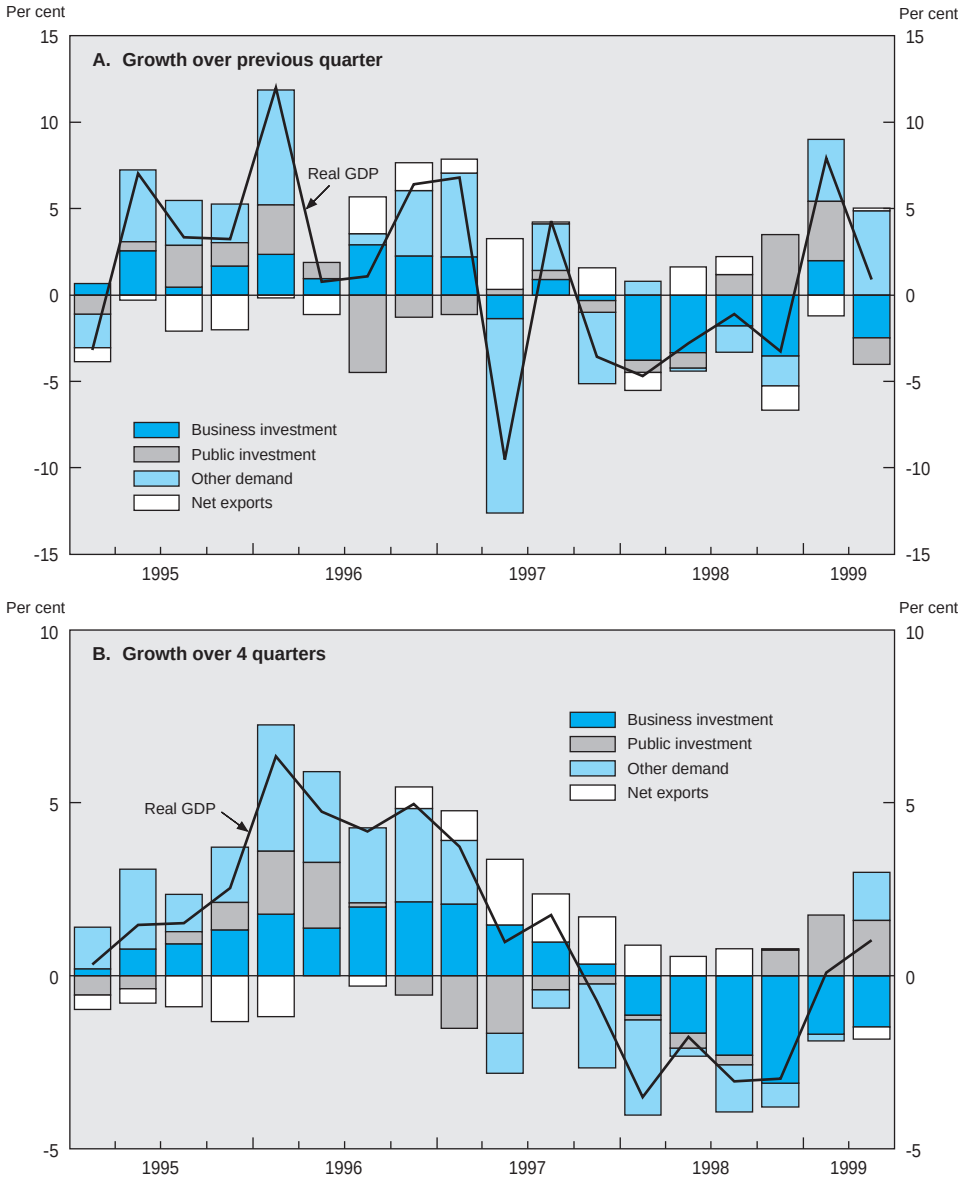
Source: Ministry of International Trade and Industry.

business investment was commonly observed across industries and firm sizes (Table 1), non-manufacturing sectors initiated the cuts, and manufacturing industry followed suit in the middle of the year. Small and medium-sized firms (SMEs), especially highly leveraged firms in the non-manufacturing sector, were also likely to be constrained by tighter credit availability.

Aggregate employment also dropped by 0.7 per cent in 1998, the first decline since 1975, with a large fall in the construction and manufacturing sectors. In addition to a steep decline in blue-collar workers, management jobs, which – according to the Ministry of Labour's survey – have long been judged as the job type that is in greatest excess supply, declined in both 1997 and 1998, by 5.8 and 1.8 per cent, respectively. These employment adjustments have increasingly taken the form of redundancies, driving up the overall unemployment rate: an increase in those unemployed due to involuntary separation accounted for more than 60 per cent of the increase in the rate in 1998.[2] The Japanese rate exceeded the US rate for the first time in post-war history in December 1998, when the two registered 4.4 and 4.3 per cent, respectively. Although male as well as female labour force participation played a buffering role for the increase in the

Figure 3. **Contributions to real GDP growth**

Seasonally adjusted annual rates



Source: OECD.

Table 1. **Business investment by industry and firm size**

Growth rate, per cent

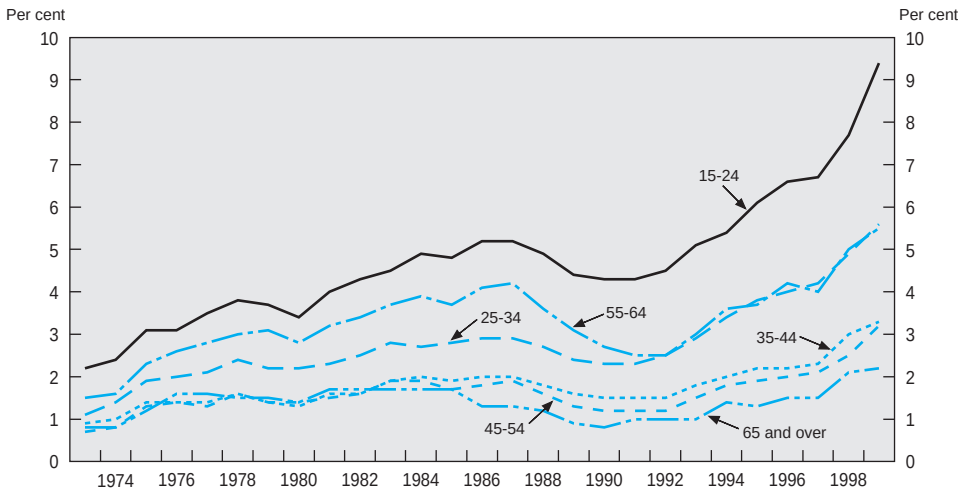
	1990	1991	1992	1993	1994	1995	1996	1997	1998
A. Business investment by industry (SNA base)									
All industry	11.1	6.4	–5.8	–10.4	–5.5	4.7	11.9	7.3	–11.7
of which:									
Construction	19.7	11.2	–1.7	–6.2	–4.0	–1.1	–11.3	–5.4	–16.9
Manufacturing	11.6	11.4	–15.6	–18.3	–9.2	12.2	11.9	11.1	–4.4
Wholesale and retail	10.3	–0.9	–6.8	–7.6	–11.6	1.0	3.3	1.1	–14.0
Finance and insurance	15.0	–2.8	–9.3	–9.4	–3.3	–9.7	22.1	12.6	–1.5
Real estate	21.0	–10.2	4.3	–6.0	–33.3	–6.3	16.2	–4.8	–9.7
Transport and telecommunication	17.9	1.3	4.3	–6.0	9.0	5.8	11.0	16.6	–19.1
Public utilities	–3.4	22.2	5.2	9.6	12.1	–4.7	10.3	–4.3	–8.4
Services	10.3	4.8	0.1	–13.5	–8.4	7.2	29.2	16.4	–21.4
B. Business investment by firm size (fiscal year)									
Manufacturing companies									
– Large	18.4	3.8	–20.4	–20.7	–9.9	10.6	7.3	5.7	–12.2
– Medium	27.4	9.3	–19.6	–24.0	–5.1	10.6	8.7	–0.8	–8.7
– Small	15.1	21.2	–21.2	–24.4	–6.0	–4.1	2.2	3.7	–12.1
Non-manufacturing companies									
– Large	18.4	14.1	–0.7	–9.0	–10.3	–2.1	4.1	–6.0	–1.0
– Medium	10.7	12.9	–4.5	–22.0	0.2	–7.5	–0.3	–6.2	–2.3
– Small	6.9	8.4	–13.6	–14.5	–2.9	–8.3	4.1	–7.9	–6.1

Source: Economic Planning Agency and Bank of Japan.

unemployment rate by easing by 0.4 and 0.3 percentage points, respectively, this effect seems to have been more limited than in the past.[3] As employment adjustment often takes the form of reducing new hiring and encouraging early retirement, youth[4] and (male) elderly unemployment shot up (Figure 4). The same overall developments are reflected in alternative unemployment rates: those for people seeking primary jobs and for heads of households, which probably reflect the social severity of the situation more accurately (Table 2).

The rising number of unemployed put strong downward pressures on labour compensation. Although regular wage payments rose very slightly, overtime payments plummeted in line with the shrinkage in production (Figure 5). Moreover, reduced profits had a considerable negative impact on bonus payments in 1998. As a result, total payments per worker registered a decline, –1.3 per cent, with the largest cuts in the construction and finance and insurance industries (3 and 3.2 per cent, respectively). With both employment and wage rates falling, labour compensation dropped for the first time in the post-war period by 0.7 per cent.[5] Despite the employment adjustment, the still larger plunge in output led to a record decline in productivity, –2.2 per cent, as well as an equal rise of unit labour costs, the largest increase since 1992. This led to a massive squeeze on profitability in the deflationary environment and provided a further incentive for firms to slash investment and engage in restructuring.

Figure 4. **Unemployment by age**



1. Unemployment rates for 1999 are through July.
Source: Management and Coordination Agency, *Labour Force Survey.*

Table 2. **Severity of unemployment**

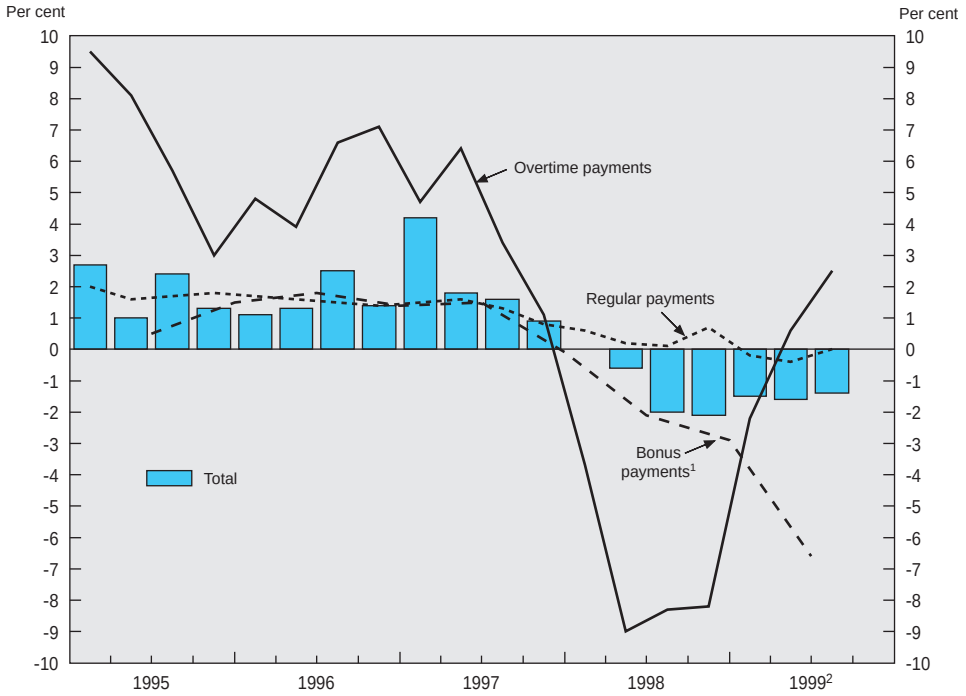
Per cent, not seasonally adjusted

	Overall rate	*Of which:*			Unemployment rates			
		Involuntary separations	Voluntary separations	New graduates	Those seeking primary jobs	Those seeking secondary jobs	Heads of households	Non-heads of households[1]
1990	2.1	0.5	0.8	0.09	1.9	3.3	1.5	2.5
1991	2.1	0.5	0.8	0.08	1.9	3.2	1.4	2.6
1992	2.2	0.5	0.9	0.09	2.0	3.5	1.4	2.7
1993	2.5	0.6	1.0	0.11	2.3	4.1	1.8	3.0
1994	2.9	2.1	1.2	0.14	2.7	4.4	2.0	3.5
1995	3.2	0.8	1.2	0.17	2.9	4.8	2.2	3.8
1996	3.4	0.9	1.3	0.19	3.1	4.9	2.4	4.0
1997	3.4	0.8	1.4	0.18	3.2	5.0	2.3	4.1
1998	4.1	1.3	1.5	0.22	3.9	5.6	2.9	4.9
1998 S1	4.0	1.2	1.4	0.25	3.7	5.7	2.9	4.7
1999 S1	4.8	1.6	1.6	0.30	4.6	6.8	3.4	5.8

1. Includes spouses and other members of households, as well as one-person households.

Source: Management and Coordination Agency and OECD.

Figure 5. **Wage and salary developments**
Per cent change over four quarters



1. Only summer (June to August) and winter (November to January) bonuses are plotted.
2. Data for third quarter 1999 are for July and August.

Source: Ministry of Labour, *Monthly Labour Survey.*

The weakness in household incomes that resulted from shrinking pay packets discouraged spending: private consumption and residential investment decreased by 1.1 per cent and 13.7 per cent in 1998, respectively. The Household Survey showed expenditures declined not only for the self-employed and low-income households (those most readily affected by the business cycle), but also those in the high-income class (Table 3). Spending on all product categories fell, with the exception of durable goods, which increased by more than 5 per cent. These expenditures were supported by the introduction of new products in the second half of the year (personal computers and compact cars) and the replacement cycle of certain household durable goods.[6] Consumers' caution was also confirmed by a decrease in employee households' propensity to consume, largely

Table 3. **Consumption growth by household type**

Constant price terms, per cent

	All households	Employee households						Non-employee households	
		Yearly income quintile groups[2]						Total	*of which:* Individual owners
		Total	I	II	III	IV	V		
1989-91[1]	1.0	1.1	1.0	0.8	1.9	1.2	0.7	1.2	0.8
1992-94[1]	–0.4	–0.4	0.5	–0.1	–0.6	–0.2	–0.8	–0.7	0.3
1995-97[1]	–0.5	0.0	–1.1	–0.5	0.2	0.6	0.2	–1.1	–1.1
1995	–1.1	–0.7	–1.1	0.0	1.2	–1.5	–1.5	–1.5	–1.2
1996	–0.1	0.6	–0.4	–0.9	–1.9	3.2	1.6	–0.9	–1.9
1997	–0.2	0.1	–1.7	–0.7	1.3	0.0	0.6	–0.8	–0.2
1998	–2.2	–1.8	–2.4	–1.4	–2.3	–3.3	–0.4	–2.8	–3.2

1. Average of three yearly growth rates.
2. Group I is the lowest income group and V is the highest.
Source: Management and Co-ordination Agency.

the result of justifiable anxiety over job security. This prudence might have been strengthened by concerns about the sustainability of the pensions and the social security system more generally.[7]

Housing investment could also have been adversely affected by a stock-adjustment mechanism: the per capita housing stock has continued to rise,[8] and the ratio of housing stock to disposable income was still at a historically high level at the end of 1997.[9] In addition, deteriorating income conditions contributed to the double-digit decline in starts of owner-occupied houses and condominiums. Weak income prospects and a lack of confidence also probably played a role, in view of the minimal response to the improvement in affordability[10] due to lower interest rates. Stagnation of starts for rental housing, most of which are small and for single-person households, might reflect future demographic prospects. Corporate restructuring may also have reduced employer demand for company-provided houses. Thus, the number of housing starts declined to an annual rate of 1.1 million units in the fourth quarter of 1998, about a third less than at the previous peak in the fourth quarter of 1996.

External developments

The only component of private spending which managed to contribute positively to real GDP growth in 1998 was net exports (0.6 percentage point) (Figure 2). However, that reflected the weakness in the domestic economy, rather than any source of strength, as a significant fall in import volumes overwhelmed a decline in real exports (Table 4). The weakness of export volumes was due to

Table 4. **External trade**

	1995	1996	1997	1998	Seasonally adjusted annual rates: 1998 1st half	1998 2nd half	1999 1st half
	Trillion yen						
A. Balance of payments							
Trade	12.3	9.1	12.3	16.0	16.5	15.5	15.1
Exports	40.3	43.6	49.5	48.9	50.1	47.6	45.4
Imports	27.9	34.5	37.2	32.9	33.6	32.1	30.3
Services	–5.4	–6.8	–6.5	–6.5	–6.4	–6.5	–6.1
Investment income	4.2	5.8	6.7	7.4	6.4	8.4	5.9
Transfers	–0.7	–1.0	–1.1	–1.1	–1.0	–1.3	–1.4
Current account	10.4	7.2	11.4	15.8	15.4	16.2	13.6
Per cent of GDP	2.2	1.4	2.3	3.2	3.1	3.3	2.7
	Percentage change from previous year						
B. Trade in goods (customs basis), 1995 = 100							
Volume							
Exports	3.2	1.2	11.8	–1.3	2.0	–4.3	–2.2
United States	–6.2	–4.0	13.1	5.4	7.9	3.2	1.4
EU	3.2	–1.0	18.2	13.4	18.4	8.8	–3.7
Asia	9.9	1.7	6.4	–15.3	–12.9	–17.4	1.4
Imports	11.7	5.6	1.7	–5.3	–5.0	–5.7	6.3
Manufactures	22.1	8.3	3.1	–5.9	–5.6	–6.3	8.0
Unit value							
Exports	–0.7	6.4	1.9	0.7	0.3	1.1	–6.8
Imports	0.5	14.0	6.0	–5.5	–5.3	–5.7	–14.6

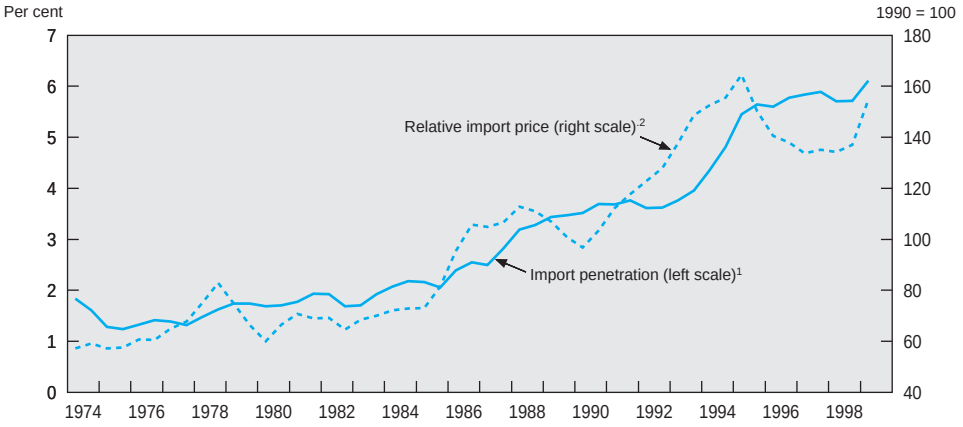
	1996	1997	1998	1998 Q1	1998 Q2	1998 Q3	1998 Q4	1999 Q1	1999 Q2
	Percentage change from previous year								
C. Effects of Asian crisis									
Export value									
World	7.7	13.9	–0.6	3.9	0.7	4.2	–10.2	–9.5	–8.3
Asia	8.9	8.6	–17.9	–11.8	–18.0	–16.7	–24.4	–11.6	–3.1
ASEAN4	10.6	4.4	–32.0	–30.1	–35.9	–31.2	–30.4	–11.8	2.5
Korea	9.0	–1.2	–36.4	–38.5	–38.8	–35.7	–32.5	12.0	23.6
Import value									
World	20.4	7.8	–10.5	–9.4	–10.5	–5.9	–16.2	–13.3	–5.0
Asia	23.0	6.6	–10.4	–8.9	–12.2	–6.6	–13.8	–9.9	3.2
ASEAN4	25.9	8.2	–14.7	–15.7	–16.7	–9.1	–16.9	–13.9	3.2
Korea	7.0	1.6	–10.6	–4.5	–13.4	–14.4	–9.2	–4.7	12.7

Source: Bank of Japan, *Balance of Payments Monthly*; Japan Tariff Association, *Summary Report of Trade of Japan*, and OECD.

plunging shipments to the crisis-afflicted countries in Asia, which offset persistently strong upward momentum in sales to the European Union. Nonetheless, export performance deteriorated, at it has steadily since the mid-1980s, as on average export markets grew slightly. A large appreciation of the yen against the Asian currencies still did not bring about a rise in import penetration (Figure 6), probably because of various supply-side problems in these countries' export sectors.[11] Thanks to lower oil and other commodity prices, import prices for goods declined by 5.5 per cent, thereby yielding a gain in the terms of trade. The lower prices of primary products also accounted for a rise in the share of manufactures in total import values to a record, 62.1 per cent, despite a decline in terms of volumes.

The simultaneous improvement in volumes and terms of trade resulted in a larger merchandise trade surplus and also pushed up the current account surplus to 3.2 per cent of GDP, the first time it has been over 3 per cent in the last five years. The current account also benefited from a widening in the investment income balance, despite a decline of payments from the Asian crisis-afflicted countries.[12] A noteworthy feature on the capital side of the balance of payments was that inward direct investment shot up, almost doubling to about

Figure 6. **Import penetration and relative import prices**



1. Import penetration is measured as the ratio of manufactured imports to total domestic demand.
2. The relative price of imports is measured by the ratio of the domestic demand deflator to the import price of manufactured products, 1990 = 100.

Source: OECD.

Table 5. **Recent foreign direct investment flows**

Notifications, billion yen, fiscal years beginning 1 April

	1993	1994	1995	1996	1997	1998
Outward investment	4 151	4 281	4 957	5 409	6 623	5 217
of which to:						
United States	1 694	1 802	2 185	2 479	2 549	1 321
Cayman Islands, Panama	256	201	226	231	449	709
Asia	767	1 008	1 192	1 308	1 495	836
United Kingdom	295	226	333	387	505	1 252
Netherlands	249	110	144	124	404	271
Other Europe	377	317	351	320	466	271
Oceania	228	151	272	101	253	283
of which:						
Manufacturing	1 277	1 443	1 824	2 282	2 373	1 569
Finance and insurance	726	687	527	876	1 469	2 096
Real estate	705	539	581	700	679	360
Services	411	718	1 035	456	795	263
Inward investment	359	433	370	771	678	1 340
of which from:						
United States	93	160	177	239	152	808
Latin America	17	14	14	66	59	34
Asia	54	27	25	137	74	21
United Kingdom	7	12	11	41	49	37
Netherlands	28	52	54	80	146	128
Other Europe	68	87	63	99	117	137
Foreign-owned companies in Japan	34	31	23	100	84	173
of which:						
Manufacturing	156	197	141	311	267	313
Finance and insurance	3	67	100	27	162	457
Real estate	9	3	2	27	48	42
Services	21	36	49	236	89	318

Source: Ministry of Finance.

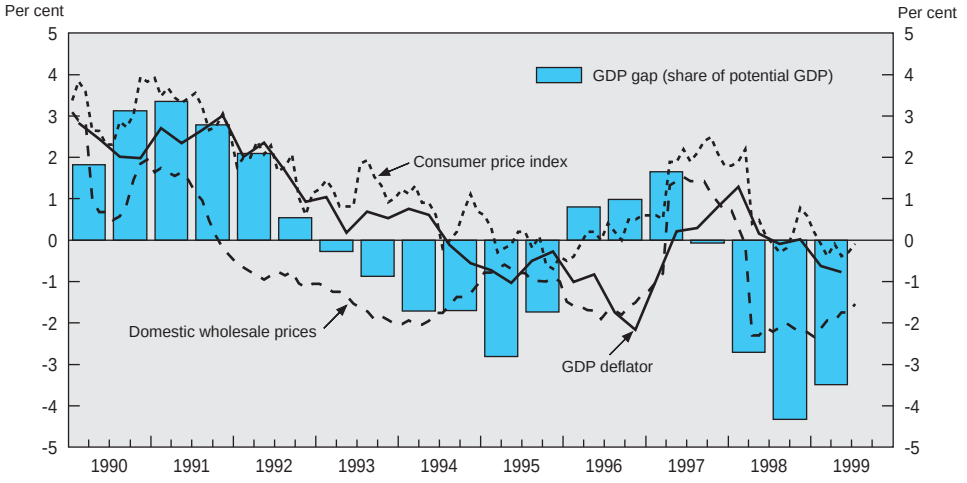
1.3 trillion yen in FY 1998 (about 0.3 per cent of GDP)[13] (Table 5). This was led by investment into the non-manufacturing sector: finance and insurance, and services accounted for about 80 per cent of the increase. On the other hand, the recession in both the domestic and Asian countries' economies led to a decline in outward direct investment of about 20 per cent in the same period after four straight years of increase. It should be noted, however, that direct investment outflows to ASEAN4 and Korea significantly increased in the first half of 1998 on a balance-of-payments basis, reflecting financial support to affiliated companies in these countries.[14]

Downward pressures on prices

The large output gap, which the OECD estimates reached 2½ per cent of potential GDP on average in 1998 (Figure 7), imparted substantial downward pressure on prices. Domestic wholesale prices declined by 2.1 per cent in FY 1998[15] due to slack demand conditions and lower import prices. On the other hand, the decline in the underlying consumer price index (CPI excluding fresh food prices)[16] was rather small, 0.2 per cent in the same period, although biases in its formulation imply rather greater price deflation probably occurred.[17] While commodity price components (excluding fresh food) in the CPI declined by 0.8 per cent,[18] services prices increased by 0.5 per cent, decelerating to stability in the fourth quarter of 1998, mainly owing to the decline in wages. According to the Economic Planning Agency's estimates, about a half percentage point out of the decline in CPI in 1998 was attributable to items subject to deregulation measures taken in the past, pointing to a limited effect of the slack in the economy on CPI inflation. The GDP deflator declined only slightly (by 0.1 per cent) in FY 1998. A 2.1 per cent terms-of-trade gain partially offset a 0.2 per cent decline in the domestic demand deflator. This decline was largely due to investment prices, which declined by 1.7 per cent, while falls in the private and public consumption deflators were small.

Figure 7. **Inflation indicators**

Percentage changes from previous year



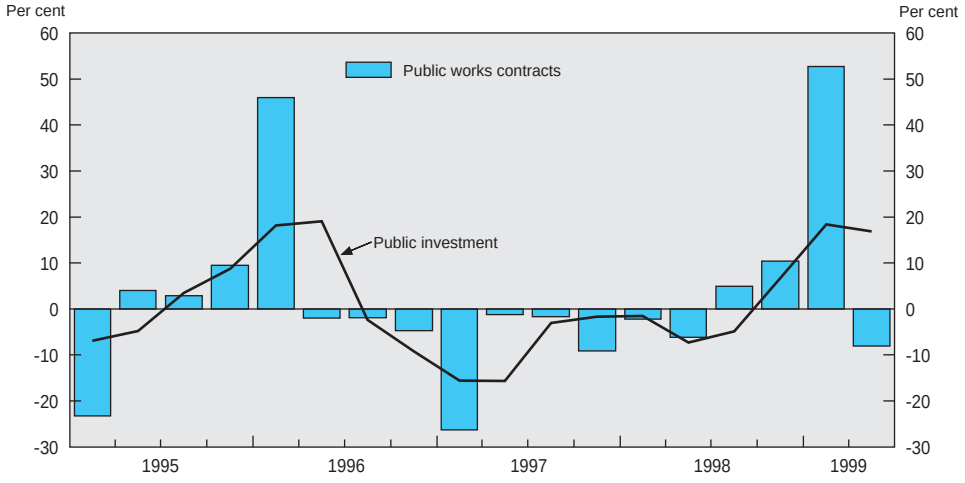
Source: OECD.

Effects of policy measures

In order to combat deflationary pressures, macroeconomic policies have become increasingly expansionary since April 1998 (see the previous Survey, and Chapters II and III below). Their effects started to be felt in the second half of 1998, and the economy bounced back in the first quarter of 1999, registering a robust growth rate of 8.1 per cent (s.a.a.r.), the first positive growth since summer 1997. The favourable effects of policy stimulus have been observable in the following developments. First, after declining for three consecutive quarters and a cumulative 21.3 per cent since the 1996 peak, public investment rebounded in the third quarter of 1998, reflecting the implementation of public works projects included in the April fiscal package (Figure 8). However, the increase in public investment was insufficient to compensate for the contraction in private economic activities in the second half of 1998: the 1.1 percentage point contribution of public demand was outweighed by the negative contribution of 3.2 percentage points (s.a.a.r.) by private demand in that half-year. The momentum of public investment was carried over to the first quarter of 1999, when public works orders surged as a result of the implementation of the second major fiscal package announced in November 1998 (see Chapter III). The increase in public investment accounted for nearly half the growth recorded in the first quarter.

Figure 8. **Public works and its leading indicator**

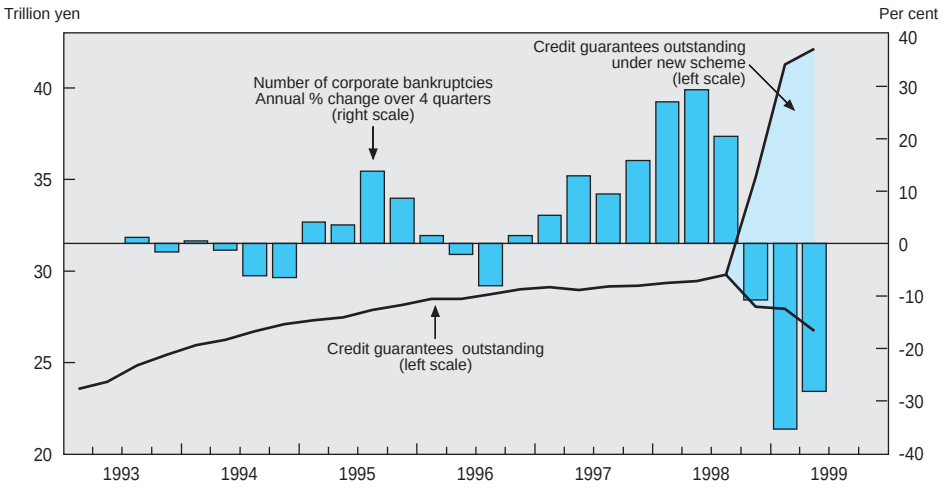
Nominal growth over four quarters



Source: Economic Planning Agency.

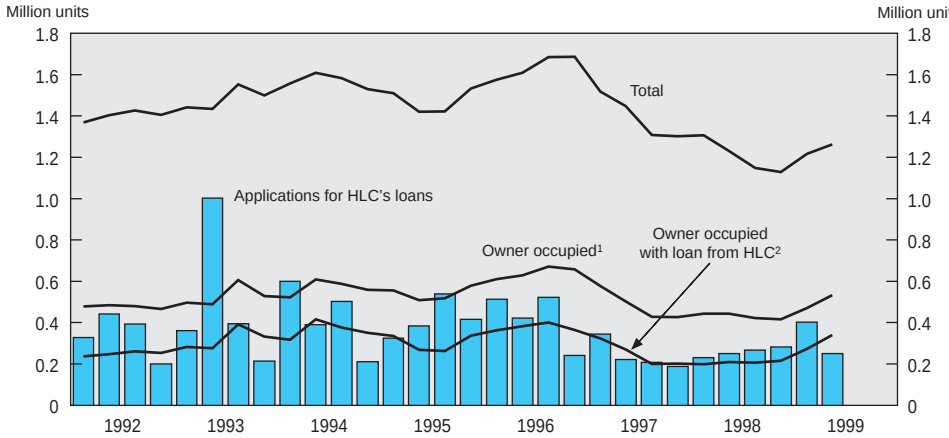
Second, anti-credit-crunch measures and a further easing in monetary policy have been effective in stabilising financial conditions (see Chapter II). In particular, the credit guarantee system for SMEs, which was greatly expanded in October 1998, reduced the number of bankruptcies substantially as from the fourth quarter (Figure 9), thereby contributing to an improvement of business and consumer confidence (see below). Third, tax credits were greatly expanded for borrowing to buy houses from 1999 to 2000.[19] Furthermore, by raising interest rates on loans provided by the Housing Loan Corporation (HLC)[20] in December 1998 by less[21] than the sharp rise in long-term interest rates in the market since autumn 1998, potential house buyers came to expect a future rise in loan rates (Figure 10). As a result, the number of applications for HLC's loans increased sharply. Almost all of the increase in housing starts in the first half of 1999, 18.5 per cent (s.a.a.r.), is attributable to an increase in owner-occupied houses financed by HLC loans. Tax credits also had a favourable effect on the stock of unsold condominiums,[22] thereby making developers more willing to undertake new construction.

Figure 9. **The effect of credit guarantees on the number of bankruptcies**



Source: Tokyo Shoko Research Ltd., Bank of Japan and Ministry of International Trade and Industry.

Figure 10. **Housing starts and the Housing Loan Corporation's (HLC) loans**
Annual rates



1. Seasonally adjusted.
2. Not seasonally adjusted.

Source: Ministry of Construction and Economic Planning Agency.

Signs of a pick-up in private activity in 1999

There are some signs of economic recovery, but as is typical at this stage of the business cycle, the evidence is conflicting. Various surveys show that consumer and especially business sentiment have recovered since the fourth quarter of 1998, supported by the reduced number of bankruptcies, easier monetary conditions and diminished concerns about the financial system. Indeed, output increased in two successive quarters, by 3.4 per cent (s.a.a.r.) in the first half of 1999. Private consumption recovered, led by durable goods sales[23] (Table 6), despite weak income conditions. While business investment declined, signs of recovery could be read from a pick-up in machinery orders and improved profit prospects (a 17.6 per cent increase is expected in FY 1999, according to the September Tankan survey). Recovery in Asian trading-partner economies allowed a progressive pickup in exports.[24] As a result of the pick-up in demand, major progress has been made in reducing manufacturers' inventories,[25] and manufacturers are now more willing to increase their production. However, it may take another quarter or so until companies intentionally seek to build up their inventories, even assuming demand continues to recover. Both long-term interest rates

Table 6. **Recent monthly indicators**

Year-on-year growth rates

	1998		1999		1999		
	Q3	Q4	Q1	Q2	June	July	August
Household consumption expenditure (all households, real)	–2.5	–0.2	–1.4	0.5	–0.1	1.4	0.1
Retail industry store sales	–4.1	–4.3	–4.3	–2.5	–2.1	–2.4	–1.4
Auto sales (including compact cars)	–3.8	–3.3	4.3	4.4[2]	3.9	–4.2	6.8
Household appliance sales	8.2	10.1	10.4	9.7	11.2	6.7	2.5
Housing starts (s.a.a.r. 10 000 units)	116	112	121	127	131	115	128
Machinery orders	–1.3	–2.7	0.7	–6.9	6.3	–5.4	2.7
Public works contracts	4.9	10.4	52.7	–8.1	–2.7	–0.9	–6.6
Exports volume (s.a.)[1]	–0.4	–2.9	1.4	0.6	7.4	–0.7	1.7
Imports volume (s.a.)[1]	1.5	0.5	7.7	–1.1	5.7	–3.7	9.2
Production (s.a.)[1]	0.1	–0.7	0.6	–1.0	3.2	–0.6	4.4
Inventories (s.a.)[1]	–2.2	–3.7	–2.3	–1.0	–0.3	–1.3	0.3
Inventory/shipments (s.a. 1995 = 100)	109.8	108.4	104.0	102.9	102.9	101.7	100.5
Job offers to job applicants ratio (s.a.)	0.49	0.47	0.49	0.47	0.46	0.46	0.46
Unemployment rate (s.a. %)	4.3	4.4	4.6	4.8	4.9	4.9	4.7
Overtime working hours	–8.7	–7.3	–5.7	–2.7	–2.1	–2.0	0.0
Number of employees	–0.7	–0.5	–1.1	–0.9	–1.3	–1.2	–0.3
Total wage payments	–2.0	–2.3	–1.5	–1.5	–3.3	–2.1	–0.4
Domestic wholesale prices	–2.1	–2.1	–2.1	–1.7[3]	–1.7	–1.5	–1.3
Consumer price index	–0.2	0.5	–0.1	–0.3	–0.3	–0.1	0.3
Money supply (M2 + CD)	3.7	4.0	3.6	4.1[4]	4.3	3.9	3.5
Number of bankruptcies	20.3	–10.6	–35.2	–28.0	–27.3	–21.2	–5.5

1. Change from previous period.

Source: Economic Planning Agency and Bank of Japan.

and the yen have tended to rise in recent months (see Chapter II), reflecting market expectations of economic recovery, even though the easy monetary policy stance has been maintained.

On the other hand, there are still several downward pressures on the economy:

– First, ongoing restructuring may continue to be a drag. In particular, the labour market has continued to deteriorate: both the job-offers-to-seekers ratio and unemployment rate remained at record levels until July, 0.46 and 4.9 per cent, respectively. The largest job losses thus far this year have been registered in companies with more than 500 employees, evidence that restructuring of major companies is

under way. Although overtime payments have started to recover, summer bonus payments are believed to have decreased significantly. As a result, total wage compensation declined by 1.1 per cent on a year-on-year basis in the first half and may continue to decline for a while yet, considering the possible size of labour hoarding (see below). Excess capacity in the form of equipment is also likely to restrain the upturn of business investment, even though recent start-ups and venture firms are beginning to boost their capital outlays.

– Second, there are also concerns about when the effects of the budget stimulus will peter out. Although residential investment surged by 11.1 per cent (s.a.a.r.) in the first half of the year, the number of applications to the Housing Loan Corporation's loans seems to suggest that the effects of tax credits and the expected rise in interest rates may be fading. Public investment, a major driving force of the growth in the past four quarters, reached its peak in the first quarter and decreased in the second quarter of the year, although remaining substantially higher than its level a year earlier.

A still large GDP gap has continued to exert downward pressure on prices. Most deflators declined in the first half, although the GDP deflator rose by 0.4 per cent, due to a terms-of-trade gain of 5¾ per cent. Deflationary pressures are less visible in domestic wholesale prices, reflecting progress in inventory adjustment (and the oil price hike). Consumer prices have remained perfectly stable, once adjusted for volatile fresh food prices.

In short, the current economic situation in Japan can be characterised as resting on a fragile balance of positive and negative forces. On the one hand, expansionary macroeconomic policies and measures to deal with the financial sector problems have succeeded in stopping the economy from shrinking further and in reducing the inventory stock to a level that is no longer considered to be excessive, and industrial production began to rise over the summer. In particular, there are signs that SMEs may be starting to restore the resiliency that was the usual feature of past recoveries.[26] On the other hand, the corporate sector as a whole has been responding to the need to improve profitability through restructuring efforts which are likely to continue to weigh on economic activity over the next several years largely though their labour market implications. It is necessary to understand these restructuring efforts more fully in order to assess the economic outlook.

Restructuring in the non-financial business sector

Confronted with severe financial pressures caused not only by the unforeseen depth of the recent recession but also by disappointed growth expectations

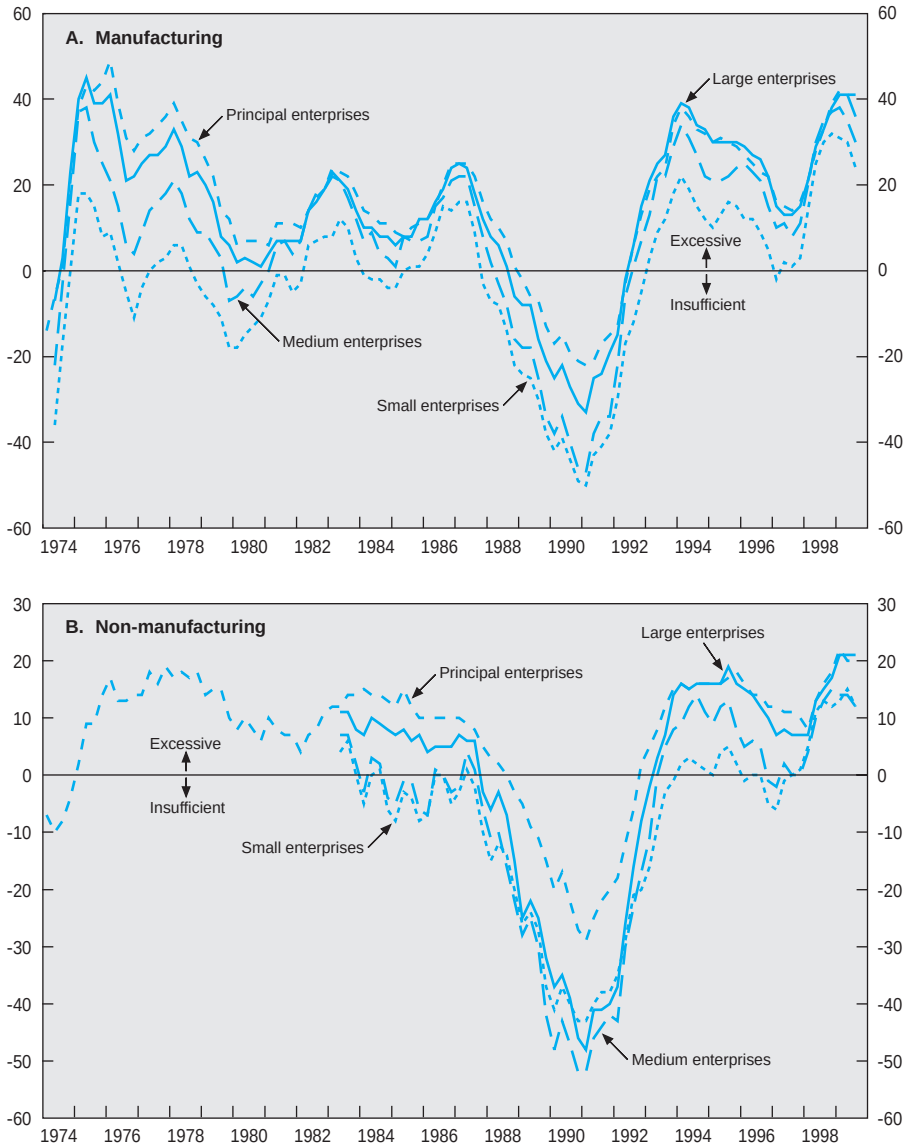
for much of the decade, the business sector has been in increasing need of restructuring. The confluence of imbalances in labour, fixed capital and financing has led to very poor profit outcomes and an increasing awareness of the unsustainable nature of the situation, even though the corporate governance system has not traditionally placed primary emphasis on shareholder returns.

The extent of excess labour

The symptoms of excess labour are multiple. First, firms themselves are regularly surveyed by the Bank of Japan as to their judgements on their employment levels (Figure 11). These judgements have moved steadily towards the "excessive" position over the course of the recent recession. That move has been especially marked in the manufacturing sector,[27] where both inputs and outputs are inherently more cyclically volatile. It has also been sharper for SMEs. However, it is noteworthy that in most cases firms are no more adamant about this problem of excess labour than they have been at similar times in previous recessions. Only SMEs would seem to be suffering from this problem to an unprecedented extent.[28] Regression analysis shows that, at least through most of the recent recession the reduction in labour input was not unusual relative to the pattern of real output; this is perhaps surprising, given the steady downward revisions that firms have made to their short- and medium-term output expectations. Indeed, until the past year, firms seem not to have adjusted even their wage levels in line with slowing productivity outcomes as much as in the past, thereby putting strong downward pressure on profit levels.

Consistent with this picture of little change, firms have not been very aggressive in implementing employment adjustment measures (Table 7). Only one in two manufacturing firms and one in four others have recently indicated they have taken such decisions. That is below the corresponding shares at the height of the last recession in 1993 and the earlier downturn following the first oil shock in 1975. Restricting overtime (downward adjustment of the workweek) remains the most prevalent form of labour adjustment, although the relative importance of internal transfers (secondments) has grown in the 1990s, possibly because of its eligibility for employment adjustment subsidies from the government (see Chapter V). Dismissals are still used sparingly – even if slightly more over time – in part because of legal barriers that boost the cost to very high levels (see Chapter V).[29] Nonetheless, private-sector analysts have stated that excess labour may be as much as 7.8 million workers (11.6 per cent of total employment).[30] The official government view, as revealed in the recent annual EPA White Paper (Economic Planning Agency, 1999) is that excess labour is probably on the order of 2¼ million workers, some 3½ per cent of total employment.[31] Managers, unskilled office workers and those in the construction trades are generally believed to be in greatest excess supply, while some occupations are still in short supply, information technology workers for example.

Figure 11. **Firms' judgements as to their employment levels**[1]



1. Diffusion index derived from Tankan survey responses: number of firms expressing the judgement that the number of employees is "excessive" minus those answering "insufficient".

Source: Bank of Japan.

Table 7. **Employment adjustment measures in historical perspective**

Per cent

	Manufacturing						Wholesale and retail trade, restaurants						Services					
	(1)	(2)	(3)	(4)	(5)	(6)	(1)	(2)	(3)	(4)	(5)	(6)	(1)	(2)	(3)	(4)	(5)	(6)
All firms taking measures	71	40	50	46	45	42	37	14	36	23	29	26	–	15	31	23	22	23
of which:																		
Restrictions on overtime	54	26	38	31	29	27	16	8	25	13	15	15	–	5	19	12	10	11
Increase of vacation time	14	4	9	7	6	6	–	1	6	2	3	2	–	–	4	4	3	2
Suspension of contract renewals and dismissal of temporary, seasonal and part-time workers	16	6	10	8	8	5	5	1	2	2	4	3	–	4	5	3	4	2
Reduction or suspension of recruitment of mid-career workers	50	12	24	14	13	12	24	4	14	9	8	6	–	5	13	7	7	5
Internal worker reallocation or transfer	23	20	29	28	26	25	6	6	13	8	13	10	–	3	14	8	8	11
Internal worker re-allocation	–	11	17	16	15	14	–	4	9	6	8	6	–	–	9	6	6	8
Transfer	–	9	12	12	11	11	–	2	4	2	5	4	–	–	5	2	1	3
Temporary suspension of business	20	3	7	7	7	6	0	–	0	0	0	0	–	0	1	1	1	1
Voluntary retirements and dismissals	5	3	2	4	6	5	1	0	0	3	4	3	–	1	2	2	3	3

1. 1975 oil crisis (Q2 for manufacturing and Q3 for wholesale and retail trade and restaurants).
2. 1986-87 yen appreciation (1986 Q4 for manufacturing, 1987 Q1 for wholesale and retail trade and restaurants, 1986 Q1 for services).
3. 1993 Q4 (Q3 for wholesale and retail trade and restaurants).
4. 1998 Q4.
5. 1999 Q1.
6. 1999 Q2.

Source: Ministry of Labour, *Survey on Labour Economy Trends.*

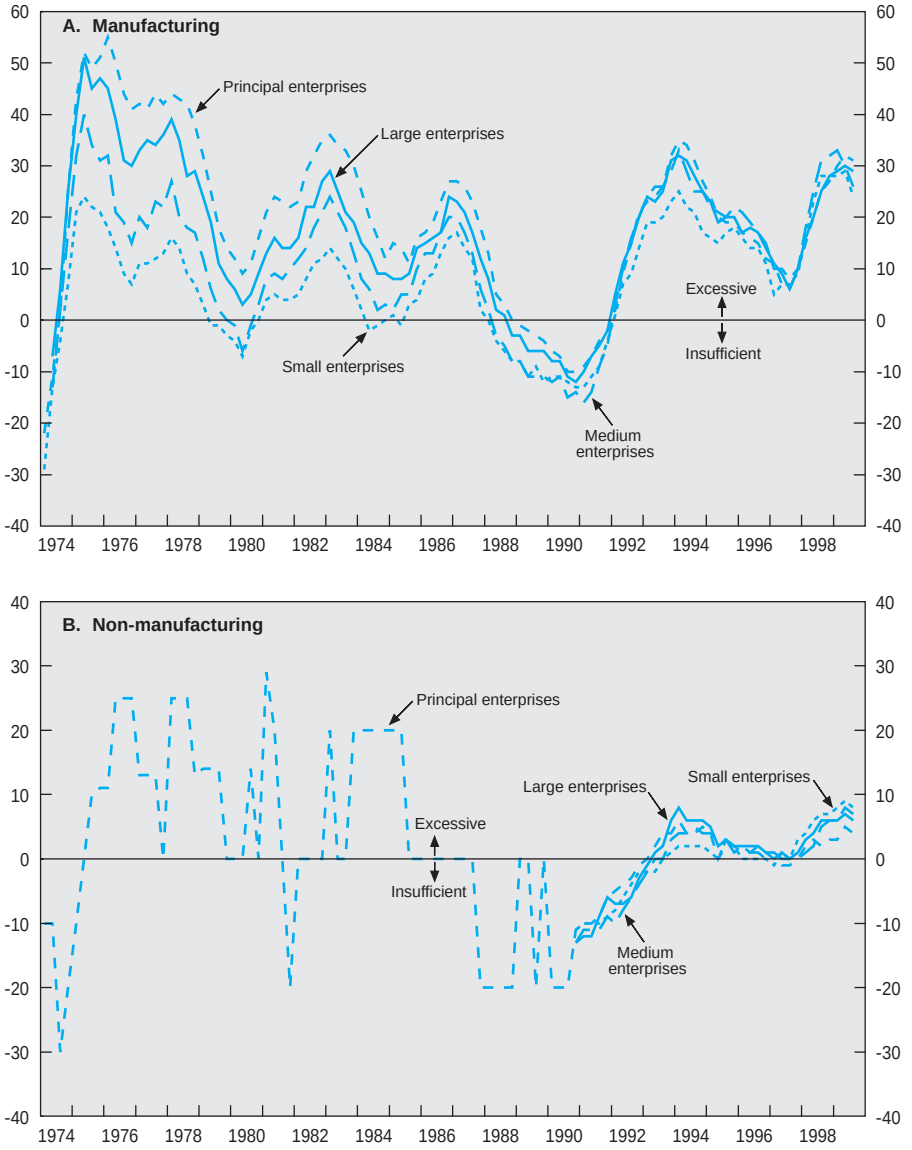
Nevertheless, there is at least anecdotal evidence that firms have begun to adjust their work forces over the past year through the increased use of early retirement as a supplement to attrition and across-the-board wage cuts. This may be one reason for the steep fall in employment and rise in involuntary unemployment as well as the corresponding surge in the number of job applicants citing "employers convenience" as the reason for their separation (93 000 or 29 per cent of all new applicants in August 1999, with a year-on-year growth rate of 10 per cent) in the Ministry of Labour's Report on Employment Service. Indirect evidence might also include the sharp reversal in manufacturing unit labour costs in 1999: by the second quarter they were running nearly 4 per cent below year-earlier figures. A booming business for outplacement firms might also support the claim: the number of such companies is said to be around 50, up from 12 just three or four years ago, despite the fee of 1 to 1½ million yen per case. Successful re-employment was managed for 21 800 people in FY 1997.

The problem of excess capital

The evidence that capital is in generalised excess supply is even stronger, although firms' own impressions seem to reflect nothing more than a cyclical problem (Figure 12). However, a quick look at the history of manufacturing capacity utilisation (Figure 13) confirms that at least in that sector the 1990s have been a period of unusually low capacity utilisation, with recent figures more than two standard deviations from the 30-year mean, and that there has been a negative trend in the series. But capacity figures do not exist for the wider economy. For the aggregate business sector the most commonly cited evidence favouring the excess capital proposition is the fact that capital productivity outcomes have been weaker in the 1990s than long-run trend developments; in other words, the capital-output ratio has diverged from its trend line (Figure 14). Even if the trend is calculated using potential rather than actual output (in order to abstract from cyclical shortfalls), the existence of a growing gap is unmistakable. The indicated divergence by the second half of 1998 is approximately 62 trillion yen – around 13 per cent of output and more than 5 per cent of the capital stock. That excess represents about ten months worth of business-sector investment. There are other estimates that are both higher and lower. For example, the EPA has recently opined that the excess is between 35 and 41 trillion yen,[32] but private estimates are higher, although they often include cyclical components.[33]

International comparisons allow a further avenue for testing the excess-capital hypothesis. Falling capital productivity – a rising capital-output ratio – is also observed in a number of other G7 countries, especially Canada, although none has experienced such a steep descent[34] (Figure 15, Panel A). Both the United States and, more emphatically, the United Kingdom have managed to

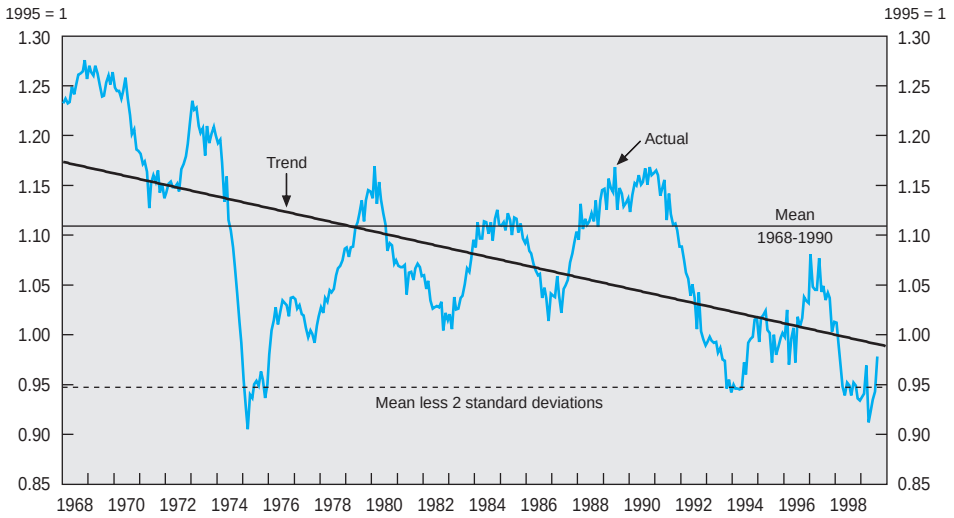
Figure 12. **Firms' judgements on their production capacity**[1]



1. Diffusion index derived from Tankan survey responses: number of firms expressing the judgement that their capacity is "excessive" minus those answering "insufficient".

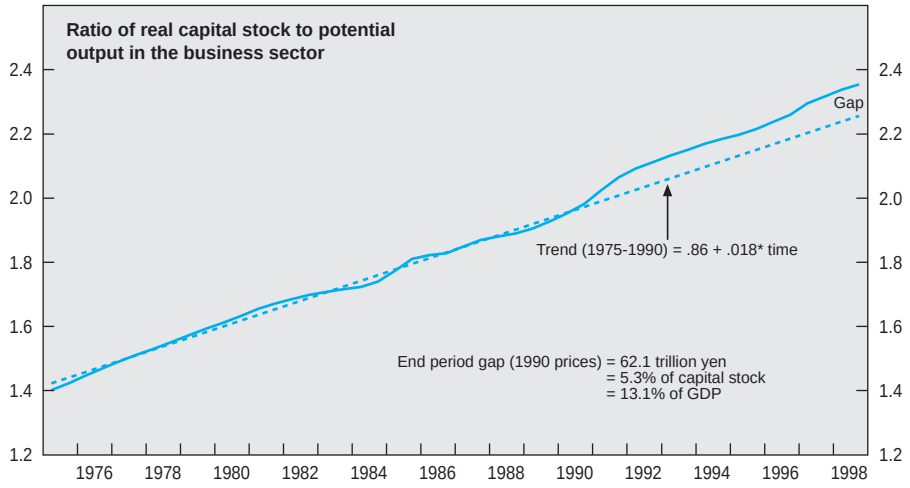
Source: Bank of Japan.

Figure 13. **Capacity utilisation in manufacturing**



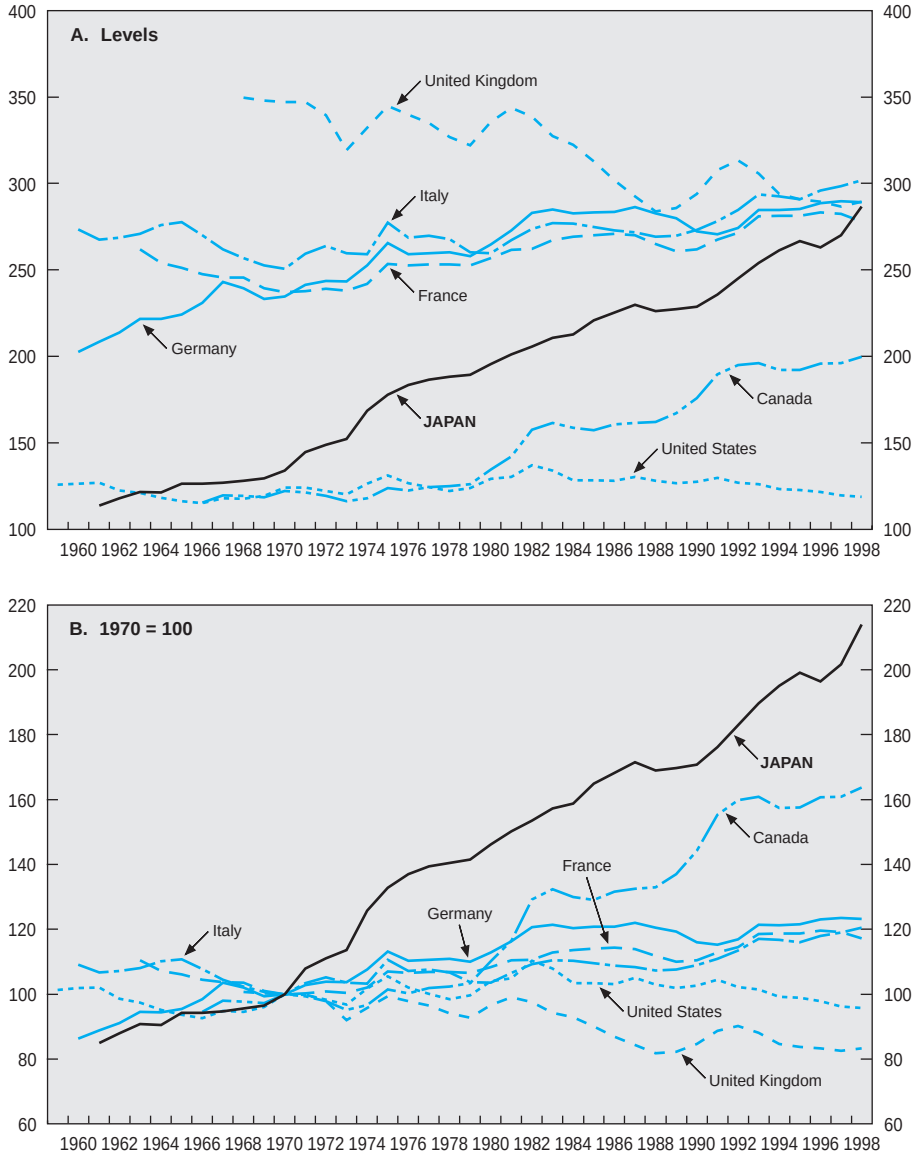
Source: OECD.

Figure 14. **The deviation of the capital – potential output ratio from its trend**



Source: OECD.

Figure 15. **Capital-output ratios: an international comparison**
Total business sector, constant prices



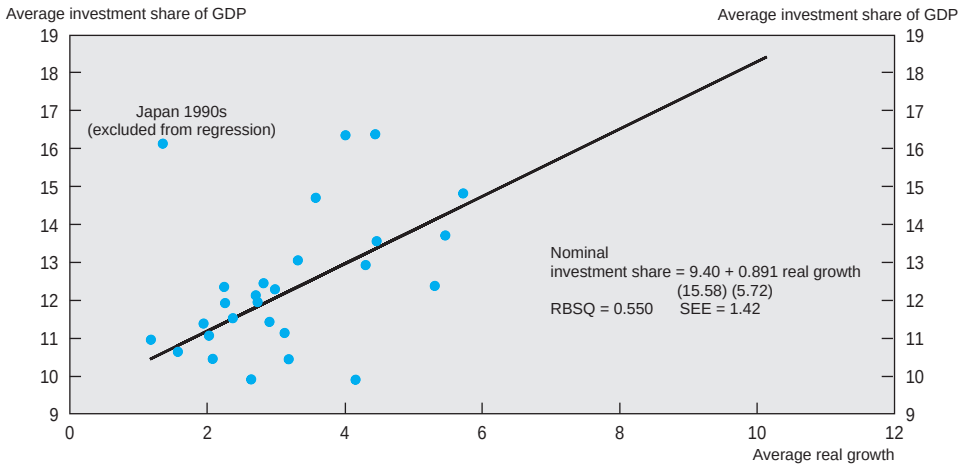
Source: OECD.

register gains in capital productivity since the early 1980s. However, the level of the ratio in Japan does not seem extraordinary by international standards, even though such comparisons may be sensitive to the choice of base years (Panel B).

Another perspective can be gained by looking at the share of business investment in GDP. Lower real growth rates should be associated with lower investment shares, and this relationship is indeed borne out in panel data using decade-long averages for the G7 countries (Figure 16): for every additional percentage point of growth, the investment share is around 0.9 percentage point higher. But Japan has become an outlier. The relationship would predict that with its growth record in the 1990s, an investment share of 10 to 11 per cent of GDP would have sufficed, whereas it has averaged 16 per cent, imperceptibly less than it did in the 1970s and 1980s when its real growth averaged 4 per cent per year. Some progress in reducing the share was observed in the recent recession: by 1998 it had fallen to almost 14 per cent, with a 13 per cent outcome likely in 1999.

Excess capacity estimates abound for individual sectors. Gas stations are a case in point. There are about 58 000 stations in Japan, but sectoral experts believe there is a need for only about 30 000, and the number has begun to fall steadily. Within manufacturing, steel is perhaps the best example: the gap between the recent production (93.5 million tonnes in 1998) and maximum

Figure 16. **The relationship between investment share and real growth**[1]



1. Observations are for each G7 country for each decade: 1960s, 1970s, 1980s and 1990s. The data for all countries are prior to the ongoing switchover in the System of National Accounts.

Source: OECD.

capacity under the current situation was around 20 per cent in 1998, estimated by one private research group administered by the Ministry of International Trade and Industry. Motor vehicles are in a similar state, with rated capacity said to be around 13 million units per year, while annual output is close to 10 million. Press reports also cite shipbuilding, department stores, metalworking and machine tools, construction, oil refining, cement, polyester and paper as suffering from too much capacity (Spindle, 1999).

Excesses on the balance sheet: too much long-term debt and not enough profit

The third excess is financial. Put simply, Japanese firms are too highly leveraged and earning inadequate returns on their assets, resulting in extremely low returns on equity. In a way this problem follows naturally from the surfeit of real-side excess capacity, as the unnecessary investments in the late 1980s and early 1990s were largely financed by external funds. Looking at the accounting data (to the extent that they can be relied on) from the Ministry of Finance reveals that the sector has been engaged in deleveraging since the mid-1970s (Figure 17), even if the debt-equity ratio remains high by international standards. A sharp decline in short-term debt has resulted in an increase in the equity share, but this healthy trend has been held back by the aforementioned long-term borrowing for investment purposes. Overall, if looked at in relation to sales, indebtedness soared in the 1980s and early 1990s, especially for non-manufacturing firms.[35] This is because, even though the economy was again, according to OECD estimates, in full employment in 1997, demand growth was insufficient to reverse more than a small part of the slide in the ratio of sales to assets that got underway at the end of the 1970s.

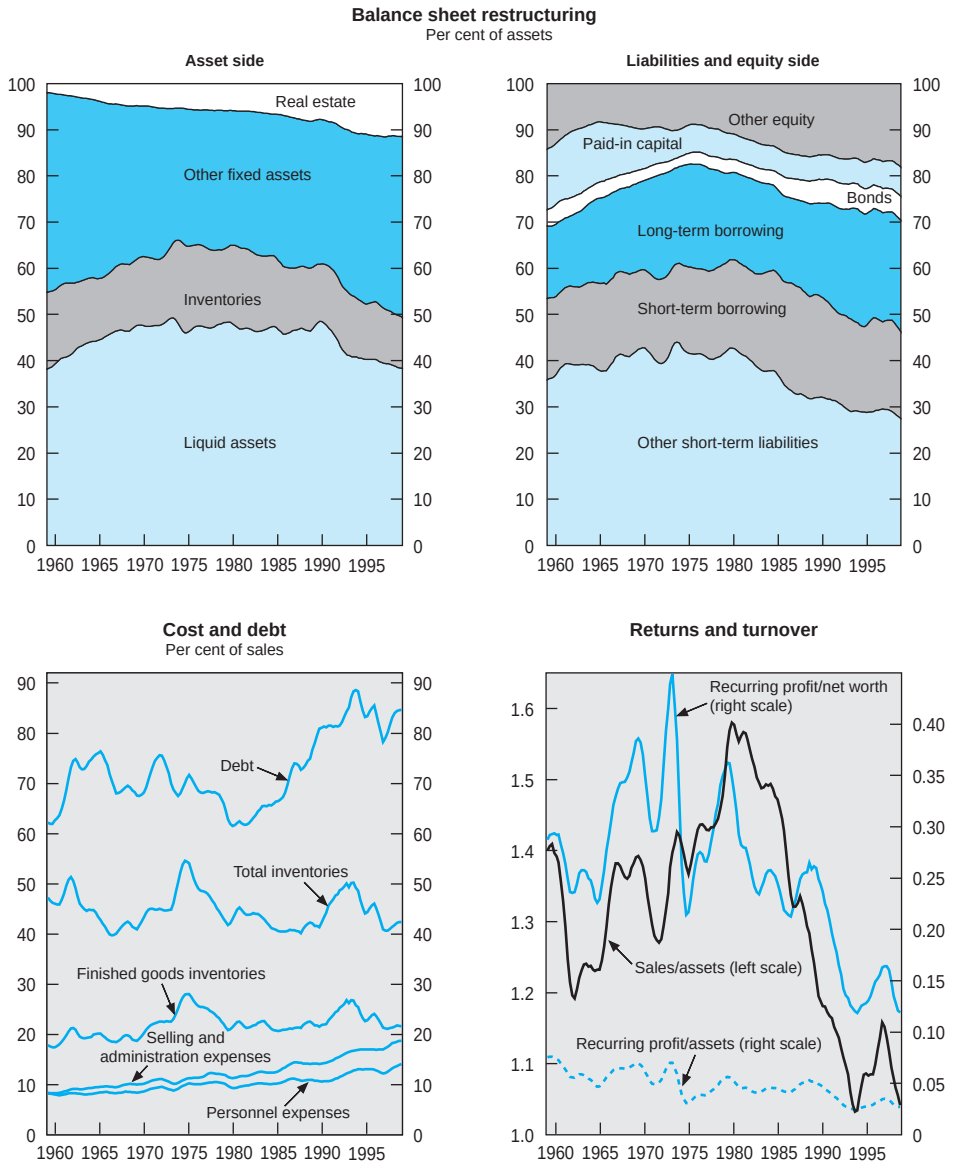
It would have been no surprise then if firms had already engaged in a round of severe cost reduction years ago, in response to the disappointing sales and profit outcomes. But this did not occur, in part for reasons of corporate governance (see below). Instead, both personnel, and sales and administrative expenses have continued their inexorable rise as a share of sales. The result has been a further dwindling in recurring profit, swamping the rise that occurred in the brief recovery from the post-bubble recession. Even at the 1996-97 cyclical peak, such profits were still less than 3 per cent of total assets, and they once again fell back to around 2 per cent of total assets in the recent recession. Indeed, the return on equity for all firms listed on the first section of the Tokyo Stock Exchange was only 1 per cent in FY 1998.

Efforts to alleviate the debt/capacity overhang and raise investment returns

Under the influence of the deflationary environment, recent and imminent changes in accounting rules[36] and pressure by foreigners, firms have clearly

Figure 17. **Corporate sector financial indicators**

Per cent, four-quarter moving averages, current prices



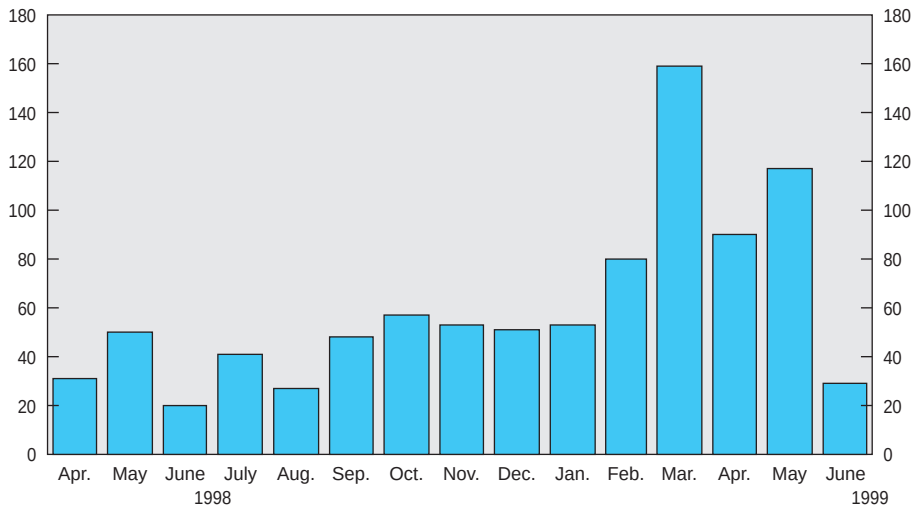
Source: Ministry of Finance.

taken the first steps to correct this unsustainable situation. The most obvious strategy they have used to try to reduce capacity is to cut back on investment as described above. The cutback has easily been the most severe of the post-war era, with declines of 16 per cent since the peak and more projected. This has been enough to remove about half the apparent gap in its share of GDP with what would have been predicted based on international experience, as described above, but it puts only a small dent in the capacity overhang estimated above.

Firms have also increasingly been making claims that they intend to redeploy or sell their assets[37] and restructure their businesses. The number of restructuring announcements has surged from an average of 42 per month in 1998 to 88 per month thus far this year (Figure 18). But there is a legitimate concern among observers as to whether many of the restructurings will be truly carried out or whether, similar to the previous fad to grip the markets – share buybacks[38] – they are being trotted out largely for the hoped-for favourable effect on the share price. This latter is unlikely to be the case,[39] as respondents to an EPA survey indicated that they are placing less emphasis on sales and total profits and more on returns on equity and efficiency of capital use. Other surveys show that although respondents expect sales to drop further in FY 1999, they

Figure 18. **Corporate restructuring announcements**

Number of companies



Source: Goldman Sachs.

believe a moderate recovery in profit is in view.[40] Thus, measures are being taken to cut costs. Plant closures are one still fairly rarely-used means of achieving such cuts. Prominent examples of such closures include department stores, producers of cars and semiconductors. But many restructuring announcements lack any targets for cost cutting by which they can be judged, and others that do set such objectives often set the bar very low[41] and rather distant – usually 3 to 5 years in the future.

Mergers are another form of restructuring that has gained sharply in importance over the recent past: in 1998 there were over 900 involving Japanese firms, more than double the 1993 figure, but still tiny relative to the 11 400 seen in the United States or even the 2 300 in the United Kingdom. The value of such activity rose from $20.6 billion to over $57 billion. The most prominent have been in the financial sector (see Chapter V). But other sectors in which important combinations have occurred are paper and oil-product distributors, drugs wholesalers and producers of textiles and steel pipe. Under government pressure (through heightened standards for eligibility for public works contract bidding), the besieged construction sector has also been implementing a variety of changes, all designed to stave off bankruptcy. Private buy-out funds are another small but growing phenomenon that makes rationalisation easier to undertake. Yet many mergers have apparently led to little change, and leading firms in many sectors remain too small or reliant on the domestic market to compete on the world market: pharmaceuticals and telecoms are two prominent examples.

The most drastic solution to the overhang is through bankruptcy itself. The number of corporate bankruptcies – as measured by the number of suspensions of dealings with banks[42] – surged from an average of fewer than 1 000 per month in most of the 1990s to around 1 200 per month through much of 1998. Eventually, however, the government's anti-credit crunch measures (see Chapter II) – and the additional credit guarantees in particular – led to a sharp decline (Figure 9 above). But the upward trend in the amount of liabilities has been smoother, with typical results moving from around 0.6 trillion yen per month in the first half of the decade to around 1.0 trillion yen just before the recession and near 1.5 trillion yen of late. The implication is that average bankruptcy size has risen over the past year or so, owing to the inclusion in the figures of such failures as those of the two major banks.

Overall, how big is the task of generating the kinds of returns seen in the past and that will be needed to satisfy global investors in the future?[43] As stated above, the return on assets in 1998 was just 2 per cent, not much more than half the average outcome in the 15-year period ending in 1991. To make up the gap of 1.8 percentage points would require an increase in recurring profits from 25.2 to 47.2 trillion yen or 22 trillion yen. Alternatively, assets could be reduced, but the cuts would have to be swingeing: around 578 trillion yen (47 per cent). Since the

total amount of equity capital is only around 278 trillion yen, even massive write-downs would not suffice. Thus, a combination of sales increases and cost reductions will have to play a major role in any rise in rates of return. Sales increases and improvement in profit margins should be available from cyclical recovery. During the last episode of recovery in the 1994-97 period, for example, recurring profits increased by some 40 per cent cumulatively. Cyclical considerations alone could thus reduce the "required" increase in profits by almost half. If the rest of the adjustment were imposed on sales and administrative expenses and personnel charges, then "required" cuts would be less than 3 per cent. Both an analysis based on the national income accounts and the result of the calculation of an equilibrium labour income share referred to above suggest a "required" reduction in the labour share of income and output of the order of 3 percentage points over and above cyclical changes. At the aggregate level a shift of such a magnitude will not come easily. On the other hand, it should not be considered an insurmountable task, all the more so since such a shift is likely to take at least three to five years. In any case, it will be facilitated by moves to boost the productivity of and returns to capital, so that the pie is enlarged without having to cut back on the remuneration of labour. This implies a shift in activity from traditional to dynamic sectors and from existing large firms to start-ups and venture firms. Shrinkage will require scrapping and redundancies, but with an improved framework being put in place there would appear to be ample opportunities for expansion by innovative firms in new areas.

The outlook for the period to 2001

The projections for the period to the year 2001 are based on the following assumptions:

– The central government is assumed to maintain its expansionary fiscal stance throughout the year 2000. An additional 5.5 trillion yen (1.1 per cent of GDP) of public investment (on a SNA basis) is assumed during FY 1999. However, the net increase in public investment would be smaller: 3.5 trillion yen (0.7 per cent of GDP) is assumed here because of an offsetting cut by local governments' own spending. The initial FY 2000 budget will be formulated following the guidelines established in July 1999; these foresee almost the same amount of public investment as in the initial FY 1999 budget. In addition, a freeze in contributions to the long-term care insurance scheme to be implemented in April 2000 is assumed to reduce revenues by 0.5 trillion yen (0.1 per cent of GDP). That freeze is assumed to be lifted by year-end. Otherwise the same fiscal stance is also assumed in FY 2001.

- As for social security reforms, the draft pension reform law, which was submitted to the Diet in July 1999, is assumed to pass the Diet and to be implemented without any significant revisions: contributions will be frozen and benefits will be reduced (see Chapter III for details). No major changes in the health care system are assumed, other than the long-term care insurance programme. No reduction in health care contributions is assumed in response to the start of the long-term care insurance programme, as has been proposed by the Ministry of Health and Welfare with respect to the government-run health care scheme.
- Easy monetary policy, anti-credit crunch and appropriate supervisory measures will be maintained. In particular, the enhancement of the credit guarantee system will be maintained. As a result, both severe credit crunch conditions and instability in the financial system are assumed to be avoided.
- The exchange rate is fixed at the levels of 106 yen to the dollar and the oil price is assumed to be $21.50 per barrel in the second half of 1999 and to remain unchanged in real terms thereafter.

The economic recovery that began at the turn of the year is expected to continue, mainly driven by an increase in household expenditures. Improved consumer confidence may cause private consumption to continue to recover modestly (Table 8), despite still meagre income growth: the saving rate is therefore expected to decline in 1999 and rebound modestly in 2000 and 2001. However, residential investment may stagnate for a while, before it picks up once again in the second semester of 2000 because of a forward shift in demand in response to the expiration of the tax credit at the end of that year. Overall, household expenditures may account for more than 1 percentage point of real GDP growth in both 2000 and 2001.

There seem likely to be two other sources of growth. First, export volumes may pick up, driven by a recovery of the other Asian economies, although the favourable effect may be partially offset by losses in market shares, attributable only to some extent to competitiveness effects resulting from the yen's appreciation. Because import volumes may also increase with the recovery of domestic demand, the contribution of net exports to growth may be quite limited in both 2000 and 2001, following a sizeable contribution in the current half-year. Second, after inventory adjustment ends in the second half of 1999, thanks to a recovery in consumers' demand and favourable external demand, companies may start to build up their stocks intentionally. As for negative forces, business investment may no longer be much of a drag on the economy after this year's further sharp fall. Indeed, it may start to recover over the course of 2000 (as profits and capacity utilisation rebound), even though its recovery may be modest due to persistent corporate restructuring effects. The decrease in public investment is expected to be limited in 2000, thanks to the additional fiscal stimulus in FY 1999, and is likely

Table 8. **Short-term outlook**[1]

	1998	1999	2000	2001	1999 1st half	1999 2nd half	2000 1st half	2000 2nd half	2001 1st half	2001 2nd half
Demand and output (volumes)										
Consumption										
Private	–1.1	1.7	1.6	1.8	3.2	2.0	1.7	1.8	1.7	1.7
Government	0.7	0.6	0.8	0.7	0.6	0.6	0.8	0.9	0.5	1.0
Gross fixed investment										
Public[2]	–0.3	13.9	–1.5	–7.8	26.6	–9.5	7.7	–10.2	–7.7	–5.5
Private residential	–13.7	3.1	4.8	0.9	11.1	9.1	–1.9	15.0	–5.0	0.0
Private non-residential	–11.3	–5.6	–0.6	4.0	–3.4	0.3	–3.0	3.3	4.0	4.7
Stockbuilding[3]	–0.1	0.1	0.2	0.1	0.2	0.2	0.2	0.1	0.0	0.0
Total domestic demand	–3.5	1.5	1.2	1.1	4.3	0.3	1.4	1.4	0.8	1.5
Exports of goods and services	–2.3	0.3	4.6	3.6	–2.2	7.6	4.3	2.5	3.7	4.5
Imports of goods and services	–7.5	1.1	3.3	3.6	5.6	0.9	4.5	3.2	3.6	4.0
Net exports[3]	0.6	–0.1	0.3	0.1	–0.9	0.9	0.1	0.0	0.1	0.2
GDP	–2.8	1.4	1.4	1.2	3.3	1.2	1.5	1.4	0.9	1.6
Inflation and capacity utilisation										
GDP deflator	0.4	–0.6	–0.5	–0.3	0.3	–1.3	–0.1	–0.4	–0.1	–0.5
Private consumption deflator	0.4	–0.3	–0.3	–0.3	–0.4	–0.2	–0.3	–0.3	–0.3	–0.3
Unemployment rate	4.1	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.7
Output gap	–3.5	–3.5	–3.4	–3.4	–3.5	–3.5	–3.4	–3.4	–3.5	–3.4
Memorandum items:										
Net government lending[4] (per cent of GDP)	–6.0	–7.6	–7.9	–7.2						
Gross debt[4]	97.3	105.4	114.1	122.1						
Net debt[4]	29.8	37.2	44.8	51.6						
Short-term interest rate	0.7	0.3	0.3	0.8	0.3	0.2	0.3	0.4	0.5	1.0
Long-term interest rate	1.5	1.8	2.2	2.5	1.7	1.9	2.1	2.3	2.5	2.6
Current account (per cent of GDP)	3.2	2.7	3.8	3.0	2.7	2.8	2.8	2.9	3.0	3.0

1. Assuming an exchange rate of 110 to the dollar.
2. Including public corporations.
3. Contribution to GDP growth.
4. Net lending excludes 5.3 percentage points of GDP for the assumption of debts of Japan National Railway Settlement Corporation and the National Forest Service, while gross and net debt figures include it.

Source: OECD.

to be more visible in 2001. On balance, real GDP is expected to rise by around 1½ per cent throughout the projection period. This recovery would be the weakest on record, but output would nonetheless grow by slightly more than potential rates – only 1¼ to 1½ per cent per year according to the latest OECD estimates.

The resulting edging-down of the output gap is likely to prevent deflationary pressures on prices from building. In addition, the higher oil prices may also push up import prices and then domestic prices with a lag. Thus, the decline

in the private consumption deflator may remain stable during the projection period, although fresh food prices are likely to fall to more normal levels in 1999. Despite the yen's appreciation, there may be an oil-related loss in the terms of trade in the second half of 1999 and the first half of 2000. As a result, the GDP deflator may edge down in both 1999 and 2000, following a slight rise in 1998, itself largely the result of terms-of-trade gains. The current-account surplus is expected to edge up to 3 per cent of GDP in the projection period from 2¾ per cent in the first half of 1999 because of the impact of the Asian recovery on export volumes.

A recovery in employment may not get underway until next year and would be extremely modest, given the weakness in projected output growth and the persistent labour hoarding. Wages are also unlikely to pick up before then, but there may be a recovery in overtime payments as production expands and a rebound in bonus payments following the turnaround in profits. Therefore, labour compensation may increase by ¾ per cent in 2000, after two years of declines. Increased job opportunities may induce a rise in labour force participation, thereby preventing any decline in the unemployment rate, which may remain around 4¾ per cent. On the other hand, progress in restructuring may be reflected in renewed labour productivity growth as well as in a decline in labour's share in GDP through the projection period.

The projection is subject to both upside and downside risks. First, the precise size and nature of the coming supplementary budget and the local governments' reactions to it could be larger or smaller than assumed here. Its impact could also be reduced by Ricardian reactions by consumers and interest- and exchange-rate crowding-out. Second, the yen's recent appreciation may constrain export growth, given the assumed exchange rate for business plans of 113½ yen per dollar, according to the September Tankan survey. Third, the projection crucially depends on the sustained recovery of consumer confidence. Whether that recovery is sufficient to support the projected increase in consumption and the resulting fall in the saving rate is unclear, and the projection is therefore vulnerable to any shocks to confidence. These might include concerns about the pensions system, government fiscal conditions, bankruptcies of life insurance companies or other instability in financial markets. It also relies on the ready availability of finance for creditworthy borrowers. Finally, even if consumers' sentiment remains robust in coming months, the speed of corporate restructuring efforts in the form of capacity downsizing cannot be accurately gauged. Thus, business investment could continue to shrink for longer than is here projected. On the other hand, there is a widely acknowledged shortfall in the amount of capital in some higher technology areas, such as those related to information technology, which could lead to increased investment once firms' confidence is re-established.

II. Monetary, exchange rate and financial-sector policy and developments

Until around the middle of 1999 the monetary authorities continued to struggle with the same questions they had had to face since the recession began in 1997: how best to provide support to the economy and the financial system and mitigate deflationary pressures without sowing the seeds of future problems when activity returns to normal levels. Once signs of recovery began to surface at that point, however, the nature of policy focus shifted slightly towards offsetting the possible negative impact on activity of upward pressures on the yen and long-term interest rates. Policymakers have succeeded in recapitalising the banking system, thereby fending off the risks of a systemic collapse under the weight of the long-standing non-performing loans problem. But on the monetary front, despite a great deal of debate both inside and outside the Bank of Japan, the monetary authorities have kept to a single approach: orthodox expansionary moves through interest-rate channels have been relied upon for their stimulative effects, and more radical alternatives have thus far been rejected. Nonetheless, the monetary authorities have also focused on ensuring that financing for creditworthy borrowers has been available in this period of dysfunction of the credit system, mainly by encouraging banks to provide more business loans. And the government has also boosted direct lending by public financial institutions and lending guarantees to smaller business borrowers.

Policy easings

On two occasions over the past year or so the Bank of Japan's Policy Board decided to implement clear reductions in its main policy interest rate: the overnight rate on uncollateralised loans. Prior to the first easing, its policy directive was to keep that rate on average slightly below the official discount rate, which to this day remains at the 0.5 per cent level it has occupied since 1995. But by September 1998 the economic situation had turned from weak to disturbing. Figures were published showing a third consecutive quarter of shrinkage in output combined with a further deterioration in both sentiment and prospects. This,

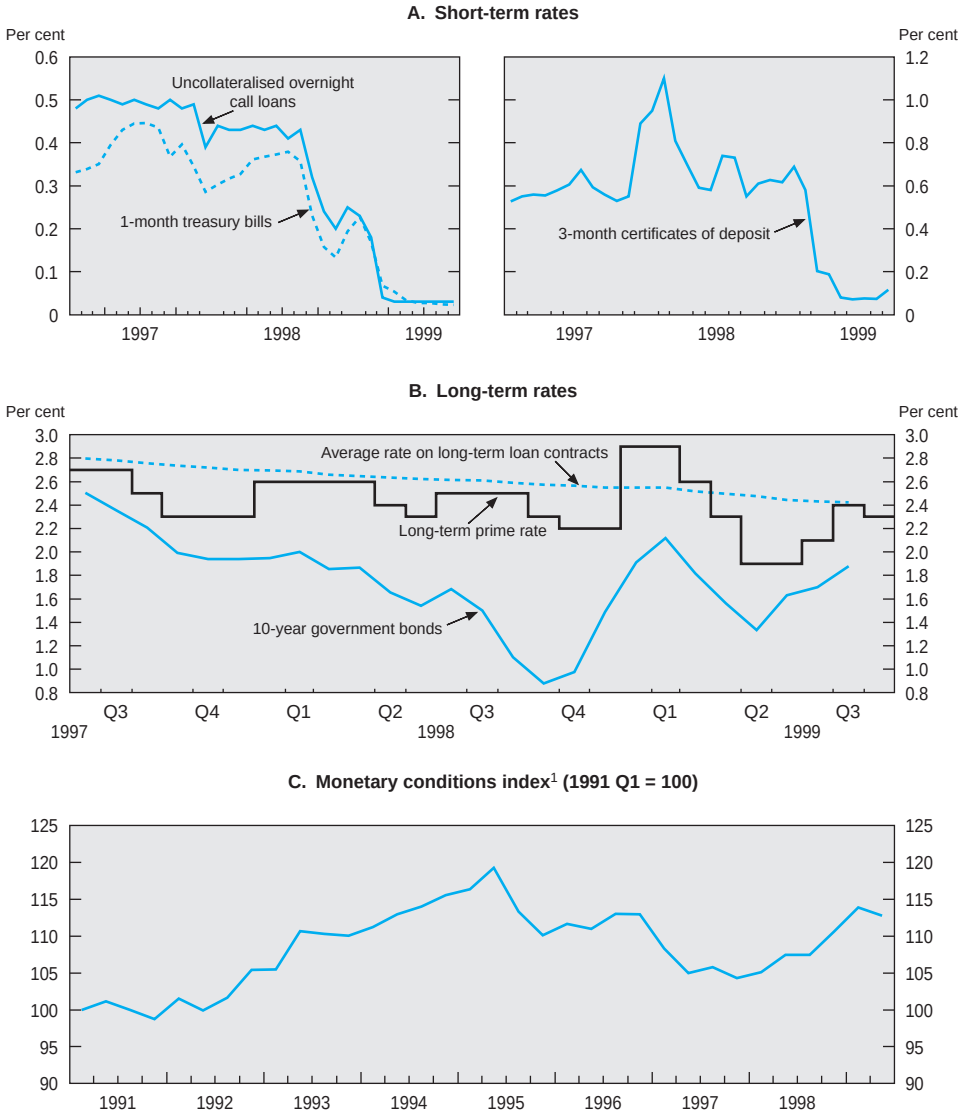
along with an unusual rise in risk aversion[44] and precautionary demand for liquidity resulting from the global financial market instability that had emanated from the summer's Russian default, and the heightened uncertainty pertaining to Japan's financial system were sufficient to convince the Board to cut the target for the call rate to 0.25 per cent. At the same time it announced that, regardless of the call rate target, more ample funds would be provided, if necessary, to maintain market stability.

In the event, the overnight rate fell to an average of around 0.2 per cent following the September decision, with other short-term market rates marking declines of a similar magnitude (Figure 19). The effect on intermediated rates was smaller, with three-month deposit rates merely returning to the 0.6 per cent level seen through most of 1997. Long-term rates eased a bit further but soon initiated a rising trend that would continue for four months (see below). Bond demand/supply considerations were the main factor pushing for higher rates in the autumn, with pressures for a flat yield curve emanating from expectations of persistently low short-term rates and from the risks of deflation.

The Bank followed up its September rate cut with a November package of measures in response to growing anxiety regarding business financing difficulties, both actual and prospective (see below), paying due attention to their potential impact on the soundness of the Bank's balance sheet.[45] First, it enlarged the scope for eligible commercial paper (CP) repurchase operations by extending maturity limits to one year from three months, a liberalisation expected at the time to raise the amount of eligible CP by around 2 trillion yen. Consequently, the amount of CP held by the Bank under repurchase operations increased from 3.9 trillion yen in October 1998 to more than 7 trillion in December and January.[46] Next, the Bank decided to establish a temporary lending facility to support business borrowing during the remainder of the fiscal year when "credit crunch" conditions were expected to prevail. Specifically, half of the increase in financial institutions' loans to business borrowers would be refinanced at 0.5 per cent so long as at least half the collateral took the form of corporate debt. This facility was responsible for a peak of around 1 trillion yen (0.2 per cent of GDP) in additional liquidity before it was terminated on 15 April, rather less than the hoped-for boost of up to 3 trillion yen predicted at the time and the ceiling of 4.1 trillion yen established at the outset. Finally, it decided to study the feasibility of setting up on a permanent basis a new lending scheme to financial institutions using corporate debt as collateral. This was later approved and implemented for the first time on 31 March 1999.

The second major interest-rate easing was approved at the Board's meeting on 12 February. That move was prompted by a sceptical assessment of the likelihood that a self-sustaining recovery would get underway in the second half of FY 1999, even though implementation of the government's 1998 stimulus packages was taking place as expected and credit conditions for private

Figure 19. **Interest rates and monetary conditions**



1. Defined as the level change in the three-month deposit rate deflated by the four-quarter percentage change in the GDP deflator (adjusted for the 1989 and 1997 tax hikes) plus one-quarter of the percentage change in the real effective exchange rate of the yen (unit labour cost definition) against 29 OECD and 12 non-OECD trading partners. The weight (one quarter) was derived from INTERLINK model simulation properties.

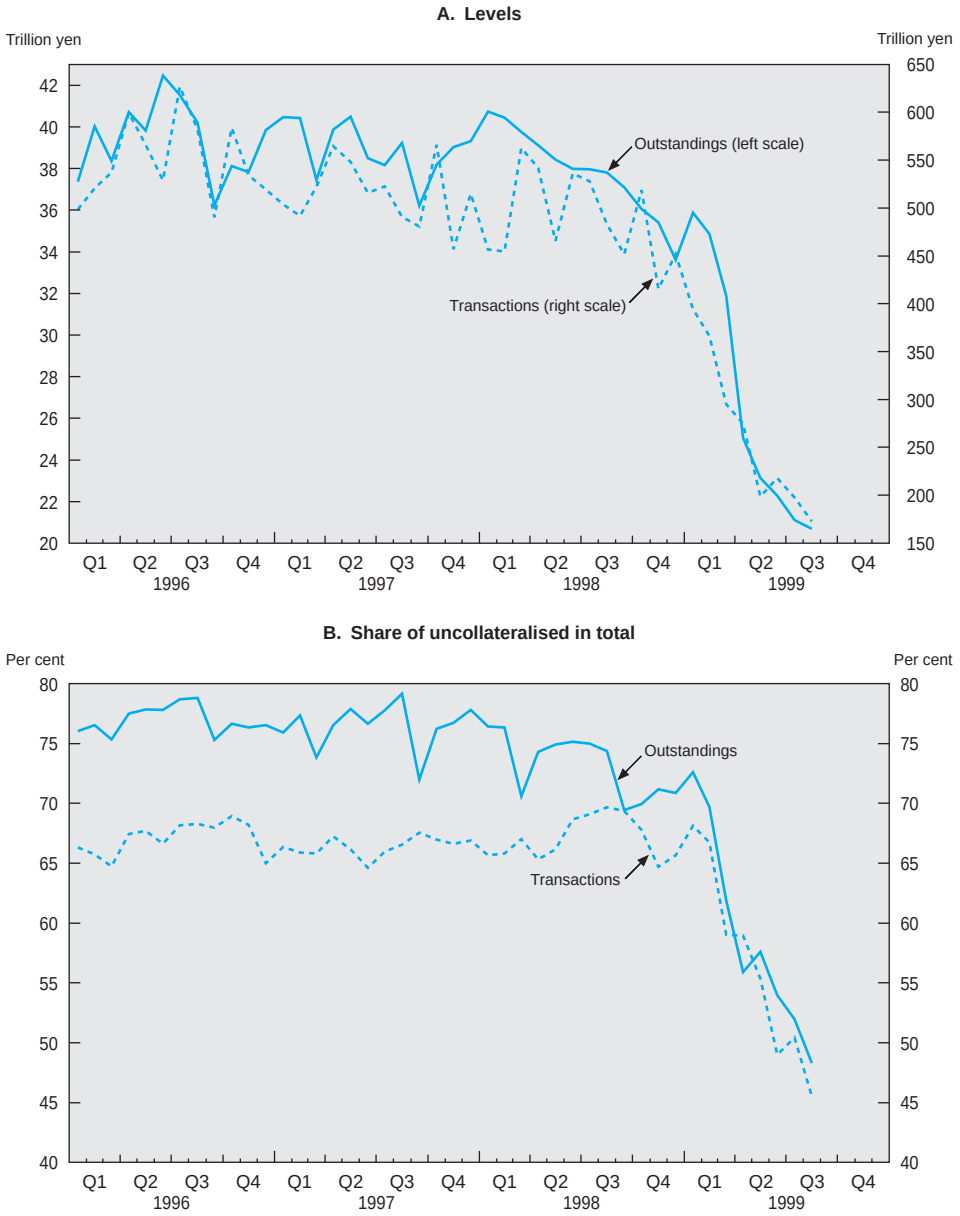
Source: Bank of Japan and OECD.

borrowers were likely to improve as bank recapitalisation was imminent. The main factor underlying this judgement was what had been happening in both bond and foreign exchange markets. Benchmark ten-year government bond yields had risen sharply since their trough of around 0.7 per cent in the early autumn to around 2.4 per cent in early February.[47] Assuming this reflected a rising risk premium,[48] rather than higher inflation expectations, then it would further depress activity. And despite the obvious displeasure of the authorities when the yen pierced the level of 110 to the dollar in January, the exchange rate remained about 24 per cent above its summer 1998 lows in effective terms, even though there was no clear sign of economic recovery. These two developments were viewed as boosting the downside risks to the economy in the near term, both in terms of activity and price movements, although at least one member of the Board persisted in the opinion that lower rates, if anything, were a contractionary influence because of the diminished interest receipts by creditors, especially households.

The upshot of the lengthy debate was to lower the target call rate first to 0.15 per cent and subsequently encourage it to move as low as possible – effectively a zero-interest-rate policy.[49] This was followed up by agreement in April that this policy would be maintained "until deflationary concerns are over". Liquidity surged and led to a jump in both monthly excess reserves from around 1 trillion yen (less than 50 billion yen per day) to more than 10 trillion yen (more than 300 billion yen per day) as from March and in current account deposits at the central bank by other financial institutions (mainly *tanshi* or money market dealers) from less than 2 trillion yen to around 20 trillion yen. Nonetheless, the impact on interest rates was clear. The overnight rate was allowed to fall to as low as 0.02 per cent, a rate which yielded effectively no return to lenders because of a commission of that amount due to the *tanshi* brokers for arranging the transactions. Similarly, market rates at longer terms also fell to record lows, such as around 0.05 for one-month treasury bills,[50] and intermediated rates also plunged to levels never before observed: around 0.2 per cent for three-month deposits and 1.9 per cent for long-term prime loans by May (down a full percentage point since February). Long-term bond rates turned around and reversed more than half their run-up in the ensuing three months. The stock market surged, largely on the back of heavy foreign purchases,[51] especially in March. But once talk of further budgetary stimulus began to spread in May and signs of economic revival emerged in June, a moderate uptrend in long-term rates resumed.

It was feared that in the uncharted waters of zero overnight rates the call money market would dry up. There has indeed been significant shrinkage – about 40 per cent year on year by the end of May (Figure 20, Panel A), mainly in the uncollateralised part of the market (Panel B) – due to short-term exit by some of the major lenders of such funds – trust banks, insurance companies and agricultural co-operative financial institutions – in favour of slightly better returns on ordinary saving deposits, but this has taken place against a backdrop of already

Figure 20. **Call money market: transactions and outstandings**



Source: Bank of Japan.

declining volumes transacted and outstanding on this market. The Bank's judgement is that this contraction has not hindered smooth settlement in the market to this point (Bank of Japan, 1999*a*).

The recent debate

Is there a need to provide more stimulus?

With the February decisions came the end of the line in terms of orthodox policy actions. But the debate has continued as to whether the Bank should go further and, if so, in what direction. Some observers have been concerned that, despite the reductions in short-term interest rates, the transmission mechanism did not seem to be functioning, given the prolonged slide in the money multiplier and in velocity (or, as some would prefer, the rise in Marshallian k).[52] Professor Krugman, in particular, argued that Japan was in a liquidity trap (Krugman, 1998*b*): even with zero nominal interest rates investors are indifferent between holding bonds and money, rendering further monetary easing ineffective. In order to provide further meaningful stimulus he argued that the Bank could follow either of two strategies. It could *extend its open market operations* to assets whose prices could still be driven up (longer-term domestic currency instruments, such as government bonds, and foreign currencies). Alternatively, it could try to persuade agents to spend rather than save by convincing them that prices will move up, not down, either by *setting an explicit inflation target*[53] or through what has become known in Japan as "quantitative easing", an attention-grabbing increase in base money of, say, at least 15 per cent.

Expanding the balance sheet of the central bank

The Bank refused to officially adopt any of these options, given the legal,[54] institutional and economic difficulties involved. It already regularly buys some 400 billion yen per month in government bonds outright. While this amount has been officially justified by pointing to the long-run trend increase in banknotes outstanding on the liabilities side of its balance sheet, going further, it is claimed, would impair confidence, boost risk premiums and might impart a moral hazard aspect to policy by loosening fiscal discipline. Yet, in truth, the increase in banknotes outstanding has not been that large in recent years, averaging only 3 trillion per year. The Bank's holdings of government securities have recently appeared to move well ahead of the level of outstanding banknotes (Table 9). But that figure includes the Bank's holdings of short-term government Financing Bills,[55] which were cut by more than 8 trillion yen in FY 1998 (a decline of 30 per cent) and are expected to gradually disappear, and excludes a long-term repurchase agreement with the Trust Fund Bureau of the Ministry of Finance and

Table 9. **The evolution of the Bank of Japan's balance sheet, FY 1996-98**

Trillion yen

	End of			
	FY 1996	FY 1997	FY 1998	September 1999
Assets				
Market operations	6.5	22.7	10.6	8.7
Loans	1.1	5.2	1.3	1.4
Bills purchased (inc. CP)	5.4	10.6	5.2	5.6
Cash collateral (repo transactions)	0	6.9	4.1	1.7
Government securities in custody (repo transactions)	0	6.1	3.9	1.6
Government securities	46.4	52.8	49.5	50.0
Financing bills	18.8	27.6	19.3	17.7[1]
Foreign exchange	3.0	3.4	3.9	3.8
Deposits with agencies	3.9	3.4	3.4	1.2
Loans to Deposit Insurance Corporation	0.5	1.8	6.7	2.4
Other	2.1	1.3	1.7	1.5
Total	62.4	91.5	79.7	69.2
Liabilities and capital accounts				
Market operations (bills sold)	5.7	20.3	10.0	0.0
Government securities (repo transactions)	0	6.1	3.9	1.6
Financial institution deposits	3.7	5.8	6.2	6.4
Banknotes	45.3	49.0	51.3	51.4
Government deposits	2.6	5.0	2.0	3.5
Allowances and accrued liabilities	2.5	3.1	2.7	2.7
Reserves and capital	2.0	2.1	2.1	2.2
Other	0.6	0.1	1.5	1.4
Total	62.4	91.5	79.7	69.2

1. August 1999.
Source: Bank of Japan.

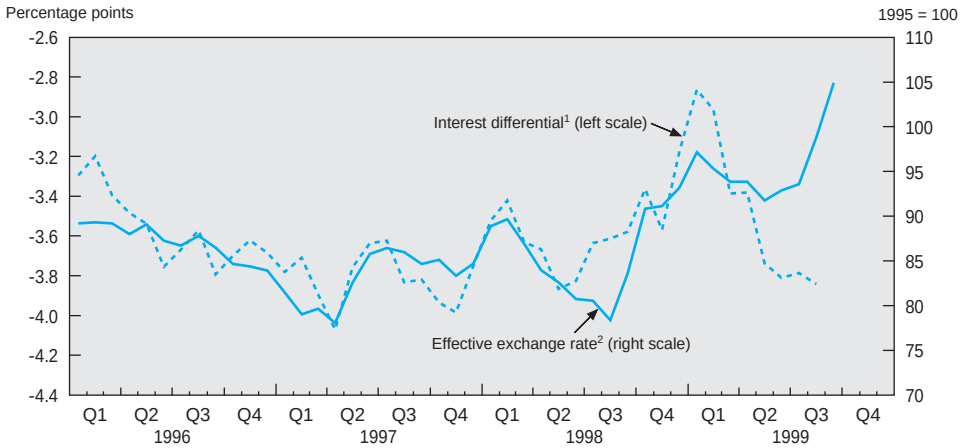
foreign central banks. With those adjustments the Bank's holdings of longer-term government securities (42 trillion yen) are well below the amount of banknotes outstanding (51 trillion yen). The Bank's balance sheet shrank by 12.9 per cent in FY 1998, reversing nearly half FY 1997's expansion. This reduction is mainly the result of reduced double-counted operations because of more stable financial markets: towards the end of 1998 such operations reflected the Bank's provision of longer-term liquidity in exchange for excess short-term funds, in order to offset upward pressure on rates on term instruments (resulting from commercial banks' dollar-funding difficulties) and prevent the overnight call rate from falling too much.[56] Since the beginning of 1999 the Bank's balance sheet has continued to contract, again largely reflecting stable developments in financial markets as a result of the financial reconstruction measures and earlier additional monetary easing.

Intervening in exchange markets

Buying foreign currency through unsterilised intervention is a theoretical option, but one that has rarely been debated among central banks,[57] mainly because money-market operations treat the various flows of funds (currency in circulation, Treasury funds and those resulting from interventions) as homogenous. In addition, it is always difficult to decide whether the level of a currency is inappropriate. Even the lower average rate over the past year or two has not been sufficient, however, to generate any export market share gains of late, as was shown in Chapter I, lending credibility to the argument voiced by some commentators that a lower value might be more consistent with cost and other competitiveness considerations. The long-term equilibrium value of the yen in dollar terms has recently been estimated to be anywhere from 100 to 163, depending on the approach taken.[58]

Over the past year or so the exchange rate has fluctuated over a wide range. After reaching a trough of around 147 to the dollar in the summer of 1998, the culmination of a 3½ year-long plunge that took the currency down by around 45 per cent against the dollar and 28 per cent on an effective basis, the yen rocketed back in the following months. It reached a peak of around 108 against the dollar in January, bringing about, reportedly, official intervention in favour of the dollar for the first time in several years. This change in trend was probably primarily attributable to the unwinding of the so-called "carry trade": investors who borrow in yen and sell the proceeds in favour of currencies, especially dollars, in which they can make a low-risk return far in excess of their borrowing costs. The strategy was very successful in the years when the yen was falling because substantial capital gains were also earned.[59] However, once most of the players in this market had to unwind many of their positions in the wake of the Russia crisis, the yen reversed course. The reversal was all the more violent, since it was accompanied by continued narrowing of interest-rate differentials with abroad[60] (Figure 21). Despite the February easing, it traded in a fairly narrow range centred on 120 to the dollar, somewhat stronger than the average rate in recent years, until the late spring. In part, this may be explained by foreign expectations of economic recovery in Japan and the associated profit opportunities. But a certain reticence to take exchange risk by Japanese institutional investors[61] and foreign hedge funds meant that capital outflows must have been of the same order of magnitude as the current account surplus. Nevertheless, once the markets were surprised by the publication of the first estimate of first-quarter growth in June, the authorities undertook a month-long series of largely unilateral exchange-market interventions (apparently mostly sterilised) to prevent the yen from appreciating before the economy had begun a self-sustaining recovery. Foreign currency reserves rose by some $37 billion, and the yen was pushed down from 118 to 122 to the dollar. The effort was apparently

Figure 21. **Interest differentials and the effective exchange rate of the yen**



1. Weighted average of short- and long-term differentials against the United States and Germany with equal weights on the terms and twice the weight for the United States as for Germany.
2. Against 40 trading partners.

Source: Bank of Japan and OECD.

interrupted in mid-July, and the yen rose by 10 per cent against the dollar and 8 per cent on an effective basis in the ensuing weeks as market confidence in the recovery built up. It resumed in September when the yen pushed through previous resistance levels, briefly reaching a three-year high of 103 to the dollar, despite widening interest differentials in favour of the dollar.

Setting an inflation target

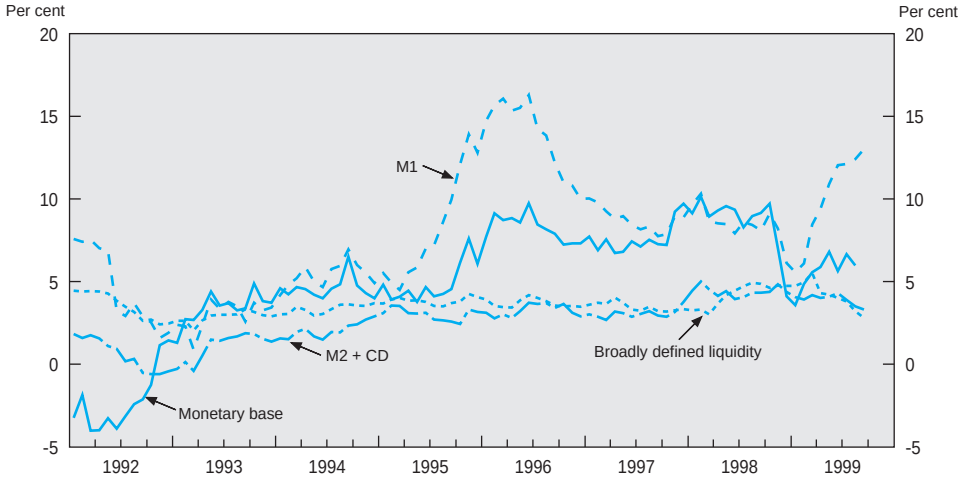
An explicit inflation target would be a valid option only if the Bank clearly had the wherewithal to attain that target; otherwise announcement of such a policy would have little credibility and would only undermine the Bank's reputation. The role of price expectations in determining the effectiveness of the low interest-rate regime is clearly crucial. Opposing deflation is at least as important as countering inflation. Accordingly, the Bank has recently committed itself to maintaining low interest rates until the economy returns to a steady growth path and the threat of deflation has passed (Bank of Japan, 1999*c*), a sort of open-ended, unspecified inflation target, but one with a clear floor of zero. If followed too literally, that might be too great a commitment, since monetary policy should normally aim to rein in some stimulus before a self-sustaining recovery is achieved and prices start to accelerate or some significant overshoot would likely

result, given the long lags with which it operates. But presumably the Bank will itself judge when this point has been reached, rather than merely relying on a single imperfect price measure such as the consumer price index, and Board members recognise the need to be forward-looking (Ueda, 1999). Indeed, there seems to be an active debate within the Bank and outside it (see, for example, Okina, 1999) as to whether an explicit inflation target should be adopted, either immediately or once it is clear that the deflationary risk has passed and that the banking system is again functioning normally, which, in the view of some participants, are some of the necessary conditions for the adoption of such a target to be credible. Inflation targeting would help to improve the communication between the Bank and outsiders.

Money supply targeting

The final subject of recent debate has been the advisability of "quantitative easing", *i.e.* moving more aggressively to boost some measure of money, most often specified as the monetary base.[62] In some quarters the Bank's post-February policy of providing liquidity to the money market has been likened to quantitative easing of a sort, but without an explicit target growth rate for any particular aggregate.[63] But the Bank's Governor denied in March that the February move represented anything other than another move to lower the overnight interest rate to as close to zero as possible, since interest rates and the stock of money are inextricably related, and the easing inevitably brings about a higher level of the monetary base and other monetary aggregates. The Bank's view is that, in the situation where the overnight interest rate is virtually zero and about three-quarters of excess funds provided by the Bank are ending up at *tanshi* (money market) brokers who re-deposit them at the Bank, any further quantitative easing would be ineffective, even if the aggregates were to increase to some extent. There has been a noticeable increase in both base money[64] and M1 growth since the beginning of 1999, with about a 2 percentage point acceleration in year-on-year growth in currency in circulation, a 10 percentage point surge in its reserves component and an increase of 15 percentage points in the growth of demand deposits. To some extent these movements have compensated for a somewhat unusual slowdown late last year. While there was a modest change in trend in 1998 in both broad money and liquidity increases (Figure 22), especially resulting from claims on the public sector, that was reversed earlier this year. In recent years the money multiplier has continued to slide, and in any case there has been little correlation between movements in any of the aggregates and output and spending. While the faster base money growth probably implies easier money market conditions, this may still reflect the move to a zero overnight rate. Such a rate does not yield a determinate rate of growth of any aggregate.

Figure 22. **Monetary aggregates**
Per cent change over 12 months



Source: Bank of Japan.

The latest measures

But in autumn 1999 the Bank did make a series of further expansionary changes. In order to ensure further permeation of its monetary easings it decided to: introduce outright purchases (and sales) of Treasury and Financing Bills; expand the range of government securities for repurchase operations to include two-year government notes; and take measures against potential year-end (Y2K) problems, such as accepting US Treasury securities as collateral. The first measure could potentially be significant, but no indication of the Bank's intentions as to the magnitude of such purchases has been provided.

Other supportive actions by the authorities

Expanded credit guarantees

The authorities also reacted to financing difficulties in the economy by a number of other means. First, in the autumn they authorised a hefty increase in the ceiling on credit guarantee obligations to SMEs that the 52 mixed private-local authority credit guarantee corporations were allowed to authorise in order to

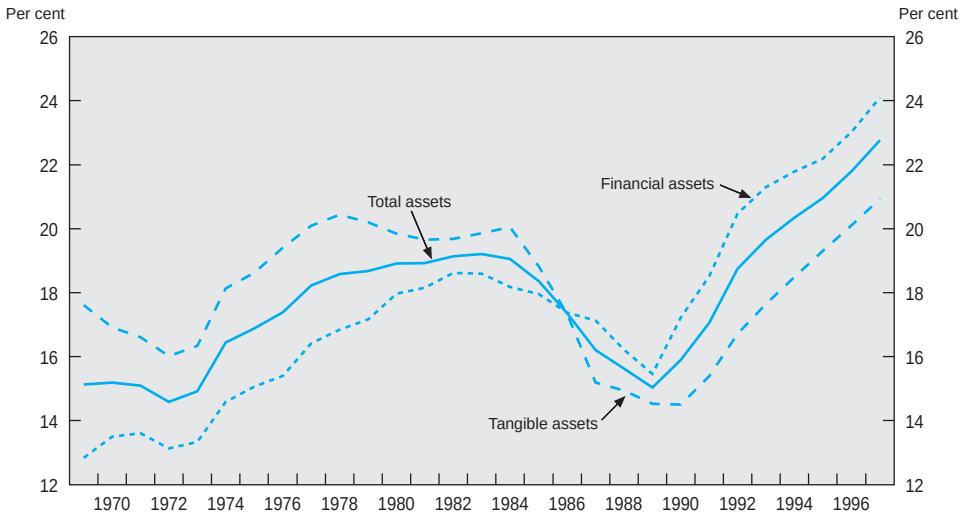
overcome the "credit crunch". New guarantees of up to 10 million yen became available without any collateral, 50 million with a third-party guarantor and up to 200 million yen with collateral. An overall limit of 30 trillion yen (6 per cent of GDP) had been in place (on nearly 4 million separate guarantees) for some years, with losses having surged to over 2¼ per cent at an annual rate to that point in 1998, up from less than half of 1 per cent in CY 1990 and around 1½ per cent as recently as 1997. The government decided to allocate a further 20 trillion yen to the programme in the autumn to be used by March 2000. Some 15.4 trillion yen of that had already been allocated to 832 000 borrowers by end-May, and there have been indications that an additional 10 trillion yen in funding for FY 2000 would be forthcoming because of the success achieved in slowing the rate of small-firm bankruptcies and to assist start-ups (see Chapter I).

Fears have been expressed that the guarantees were accorded indiscriminately to both deserving, high-potential firms and badly managed ventures alike. If true, the authorities may not have properly accounted for the budgetary implications of the increased guarantees, choosing to assume a default rate of only 10 per cent as well as a recovery rate of 50 per cent, implying a loss rate of 5 per cent (still more than double recent rates). But with claims that the only grounds for refusal of an application have been writing cheques on accounts that have insufficient funds, rigging financial statements or owing back taxes (Malcolm, 1999) and with 94 per cent of loans having been made without collateral, it would seem possible that the ½ trillion yen budgeted (200 billion yen to the credit guarantee corporations and 350 billion yen to the Small Business Credit Insurance Corporation) to cover eventual losses might well prove inadequate.[65] Furthermore, with both adjusted total and SME bank loans down several trillion yen since the guarantee increase took effect (only regional bank lending seems to have turned up since then), it seems fair to conclude that the biggest effect of the enrichment of the programme has been to transfer private risk to the public sector. At a time when the public sector – despite the objectives of the "Big Bang" (to encourage the development of financial markets and reduce state involvement) – is already assuming an ever-increasing direct ownership role in the economy (Figure 23), it seems questionable whether the public interest was well served by such limited conditionality. Finally, critics of the move have also alleged that the most significant impact of the expansion is to postpone needed restructuring (see Chapter I). Nonetheless, it must be admitted that the goal of short-run economic stabilisation has been well served.

Increased public lending and restrained interest rates thereon

Another action taken by the authorities to ease financial stress was to hold down housing loan rates at the Housing Loan Corporation. The rate on the first tranche of FY 1999 loans was set at 2.4 per cent, a rise of only 0.2 percentage

Figure 23. **The public sector's share in total assets**



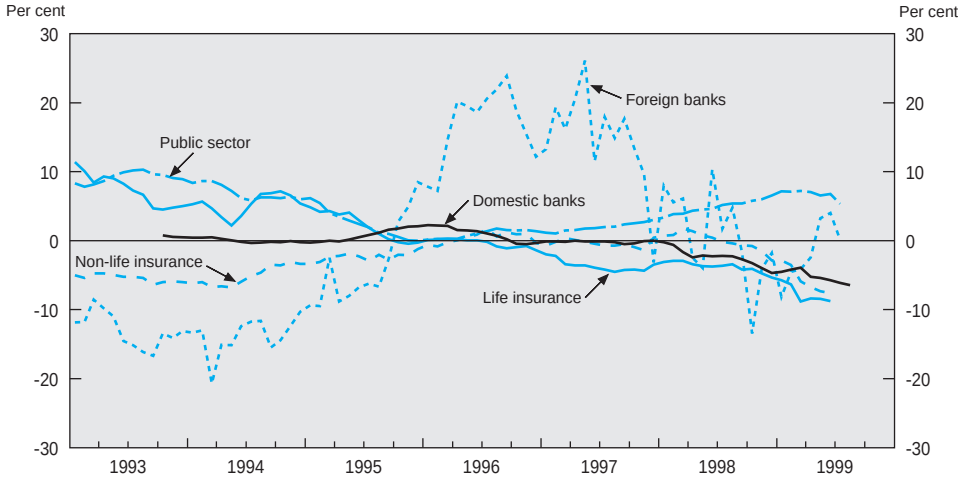
Source: Economic Planning Agency, *Annual Report on National Accounts* and OECD.

points, even though normal procedures would have yielded an increase of a further 0.45 points. Furthermore, they have continued to make more credit available through public financial institutions. In the three years to end-1998 the share of these institutions in total borrowing by domestic non-financial borrowers rose by nearly 3 percentage points to over 31 per cent.[66] This has offset to some extent the decrease in lending by domestic banks, insurance companies (especially life insurance) and, more recently, foreign banks (Figure 24). The various moves to directly ease corporate financing problems (credit guarantees, public lending and central bank commercial paper purchases) no doubt contributed to a reduction in bankruptcies and an avoidance of job losses in the short run.[67]

Dealing with problems in the banking system

But it is likely that the most significant measures were those taken to confront the balance-sheet weaknesses in the banking system. These involved aggressive closure of insolvent institutions, a tightening of standards and, above all, an injection of public capital in return for a variety of commitments, most importantly to resolve the non-performing loans (NPLs) problem. It had long

Figure 24. **Loans and discounts of financial institutions**
Annual per cent change



Source: Bank of Japan and OECD.

been acknowledged that the erosion of its capital following the bursting of the asset price bubble at the beginning of the 1990s (Table 10) had been the single most important difficulty confronting the financial system and jeopardising economic performance in the intervening years.[68] But it was only in 1998 that the authorities made a full frontal assault on the situation, by passing legislation that provided up to 60 trillion yen ($500 billion, 12 per cent of GDP) to clean up the sector[69] and by wresting virtually all of the regulatory and supervisory responsibilities from the Ministry of Finance and passing them to the Financial Supervisory Agency (FSA) and its parent body, the Financial Reconstruction Commission (FRC).[70] This latter body will cease to exist at the beginning of 2001, when the period of intensive reform is expected to be over, leaving the future Ministry of Finance (to be renamed, in Japanese at least, Ministry of the Treasury) with few powers in this area other than emergency planning (see Chapter V). The new institutions have demonstrated their willingness to act – for example, by forcibly taking over the ailing Nippon Credit Bank (see below) in late 1998 and by dealing firmly with the Credit Suisse Group in mid-1999 – and this has made a good impression on market participants.

Table 10. **Capital losses in the 1990s**

Changes in trillion yen, fiscal years

	1990	1991	1992	1993	1994	1995	1996	1997	Cumulative since 1989-90 peak
Total nation									
Gross assets	–37.7	–203.4	–446.7	–86.5	–21.8	–84.0	–113.1	–124.6	–993.2
of which:									
Equities	–306.7	–4.8	–178.1	4.1	48.4	–0.0	–24.5	–93.3	–554.9
Land	–	–192.1	–229.1	–79.6	–40.7	–49.2	–66.4	–49.3	–706.4
Net worth	–	–198.5	–247.8	–107.7	–83.8	–83.4	–75.5	–37.8	–834.5
Non-financial corporations									
Gross assets	–20.4	–69.0	–161.3	–30.4	6.7	–26.4	–50.1	–43.0	–393.9
of which:									
Equities	–106.9	–3.9	–79.4	–0.0	21.0	0.9	–11.6	–34.8	–214.7
Land	–	–62.6	–66.1	–24.7	–1.6	–9.8	–23.4	–16.9	–205.1
Net worth	–	–66.6	–2.0	–44.2	–48.3	–22.4	–42.6	17.4	–208.7
Financial institutions									
Gross assets	–95.8	–13.8	–64.1	4.6	6.2	0.8	–5.2	–27.9	–195.2
of which:									
Equities	–123.1	–2.9	–44.5	6.6	11.1	–1.0	–9.2	–33.3	–196.3
Land	–	–5.8	–6.7	–2.2	–1.9	–4.8	–3.5	–2.9	–27.8
Net worth	–37.7	–12.7	–29.1	–2.4	6.5	–0.6	18.3	–6.0	–63.7
General government									
Gross assets	–	–11.5	–19.4	–15.1	–14.4	–27.4	–21.7	–5.3	–114.8
of which:									
Equities	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Land	–	–11.1	–13.9	–7.9	–6.7	–7.1	–10.8	–6.6	–64.1
Net worth	–	–11.3	–19.4	–14.8	–15.0	–27.5	–23.3	–5.3	–116.6
Private non-profit institutions serving households									
Gross assets	–	–1.2	–2.3	–0.6	–0.2	–0.4	–1.0	1.3	–4.4
of which:									
Equities	–	–0.0	0.0	–0.0	–0.0	0.0	0.0	0.0	0.0
Land	–	–2.4	–2.9	–1.2	–0.7	–0.8	–1.9	–0.3	–10.2
Net worth	–	–1.2	–2.3	–0.6	–0.2	–0.4	–1.0	1.3	–4.4
Households (including private unincorporated non-financial enterprises)									
Gross assets	–	–107.9	–199.6	–45.0	–20.1	–30.6	–35.0	–49.7	–487.9
of which:									
Equities	–76.8	2.0	–54.2	–2.5	16.1	0.1	–3.7	–25.1	–144.1
Land	–	–110.2	–139.5	–43.6	–29.9	–26.6	–26.9	–22.6	–399.3
Net worth	–	–106.7	–195.1	–45.7	–26.7	–32.5	–26.9	–45.1	–478.7

Source: Economic Planning Agency, *Annual Report on National Accounts,* 1999 and OECD.

Closure of insolvent financial institutions

A number of banks and even a few other financial institutions have been closed because of effective bankruptcy over the past year.[71] In each case the ultimate condition of the balance sheet was much worse than the institution had reported prior to failure, placing various individuals and other institutions under suspicion of fraud in designing and abetting accounting/derivatives schemes to conceal the scale of financial losses through "window dressing" operations and making clear how weak Japan's auditing practices have been (Tett, 1999). The first was the Long-Term Credit Bank of Japan (LTCB), which was voluntarily taken over by public administrators in October 1998. Its capital deficit was initially estimated at 340 billion yen, but after closer inspection, the FSA raised its estimate to 2.65 trillion yen (Table 11) and ultimately in August 1999 to 3.5 trillion yen.

Table 11. **Bank failures requiring public funds to cover capital shortfalls, 1998-99**

Billion yen

Institution	Amount	Status
Major banks		
Nippon Credit Bank	3 047	Final
Long Term Credit Bank	2 654	Final
Regional banks		
Tokyo Sowa Bank	164	After FSA inspection (3/99)
Kokumin Bank	110	After FSA inspection (3/99)
Kofuku Bank	57	After FSA inspection (3/99)
Shinkin banks		
Ryugasaki Bank	3	Application (3/99)
Fudo Bank	2	After MOF inspection (3/98)
Kanda Bank	Unknown	–
Tamano Bank	Unknown	–
Credit unions		
Osaka Shogin	92	Application (3/98)
Nihon Shinpan	32	Application (3/99)
Tokyo Towa	4	Application (3/99)
Sobu	3	Application (3/99)
Adachi Sogo	3	Application (3/99)
Mie-ken	2	After prefectural inspection (3/98)
Heiwa	2	Application (3/99)
Kihoku	0.4	Application (3/98)
Taito	0.2	Application (3/99)
Kyodo	Unknown	–
Chitose	Unknown	–
Tochi Shogin	Unknown	–
Grand total	6 173	

Source: Nihon Keizai Shimbun.

Charges of fraud are expected to be filed against former executives. In December 1998 the Nippon Credit Bank was forcibly taken over in a similar procedure. It too was initially found to be short of only 94.4 billion yen in capital, but that estimate was subsequently likewise boosted to around 3 trillion yen, despite its having received 80 billion yen from the Bank of Japan and 210 billion yen from other financial institutions in the spring of 1997, as well as a further 60 billion yen in the first public capital injection in March 1998.

Credit unions (*shinkumi*), which will come under FSA authority only as from April 2000 but whose closure costs are nonetheless borne by the central government, have been failing at a rate of about one a month, most often with just a few billion yen in capital shortfalls. More bank failures have also been taking place in 1999, but to this point only among regional banks. Thus, taking into account the 1.5 trillion yen in taxpayer funds needed to deal with the aftermath of the 1997 bankruptcy of the Hokkaido Takushoku Bank,[72] the Midori Bank and other smaller institutions, it has become increasingly certain that the 7 trillion yen available for bank closures under the 1998 legislation will not suffice.[73] (As this Survey goes to press, the government has announced an additional 0.9 trillion yen.)

The small banks are likely to be liquidated, but the majors have been divested of some of their problem loans,[74] and attempts are being made to sell them back to the private sector. The target had been to find buyers within six months, but those deadlines have not been met. It appears that the Long Term Credit Bank of Japan will be sold to a consortium of 10 foreign financial institutions led by Ripplewood Holdings of the United States. But no buyer has as yet been found for the Nippon Credit Bank, nationalised in late 1998. And, as for the insurance sector, it was well over two years after the failure of Nissan Mutual Life that its successor company, Aoba Life insurance, finally decided to cede its shares to a French buyer. There is some chance that the slowness of the process might in some cases have led to moral hazard incentives to take unnecessary risks, causing additional budgetary costs.

The Ministry of Finance's Financial System Council has been considering how to handle bank failures once the "payoff" system comes into force in April 2001 (when the government will no longer take responsibility for all depositor losses incurred because of financial institution failures[75]). Ideas being mooted are either to transfer the bank's assets and liabilities to another institution or to extend existing mechanisms including bridge banks. In October the Council decided to endorse the idea of selling, with the help of public funds, rather than liquidating failed banks, even though it recommended retaining the current bridge bank scheme in the event purchasers do not emerge soon enough. However, whether deposit insurance is likely to be able to cover the costs of such inevitable failures without an overhaul remains an open question.

Recapitalising the solvent large banks

The bank recapitalisation bill (whose official title was the Financial Function Early Strengthening Law) of October 1998 for dealing with the problems in the banking system included 25 trillion yen in public funds for recapitalising solvent banks. The process got underway once the new FRC came into being in December. After applications were made, the FRC proceeded to ascertain the solvency and soundness of the capital bases of the individual banks (based on the 1998 intensive on-site inspections (*kensa* and *kosa*) undertaken by the FSA and the Bank of Japan[76]), as of end-September 1998 and then to negotiate the terms of the capital injections, taking into account each bank's risk management system. The final outcomes are given in Table 12. Fifteen large banks – eight city banks, five trust banks, a long-term credit bank and a large regional bank – were awarded public funds with amounts ranging from 150 billion to 1 trillion yen. The total injection was 7.5 trillion yen, more than four times as much as in the first round a year earlier when 21 banks received some 1.8 trillion yen. Most took the form of convertible preferred equity (which qualifies for Tier I status in meeting capital adequacy minima), and the rest was subordinated debt (Tier II capital). The price charged for the capital varied widely. For the preferred equity it ranged from a low of 0.35 up to 2.38 per cent, with a weighted average of 1.05 per cent, lower

Table 12. **Second round of public capital injections into the banking system, March 1999**

Billion yen

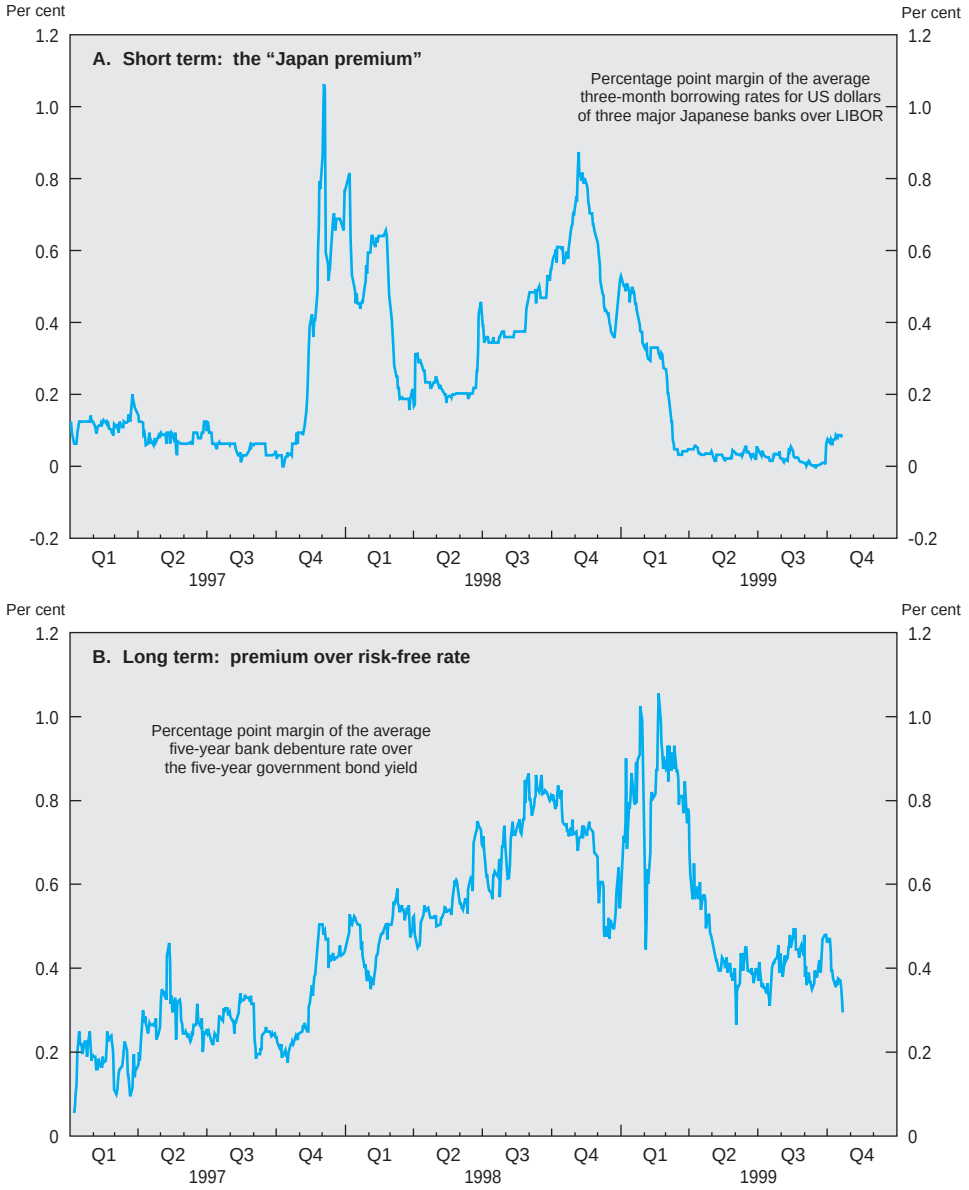
	Public funds		Other funds		Total
	Preferred stock	Subordinated debt	Equity	Other	
Industrial Bank of Japan	350	250	67	251	918
Daiichi-Kangyo Bank	700	200	–	–	900
Sakura Bank	800	–	86.2	258.8	1 145
Fuji Bank	800	200	217	–	1 217
Sumitomo Bank	501	–	–	340	841
Daiwa Bank	408	–	52.2	–	460.2
Sanwa Bank	600	100	–	180	880
Tokai Bank	600	–	100	–	700
Asahi Bank	400	100	144.8	–	644.8
Bank of Yokohama	100	100	–	–	200
Mitsui Trust	250.2	150	111	–	511.2
Mitsubishi Trust	200	100	–	–	300
Sumitomo Trust	100	100	90	83	373
Toyo Trust	200	–	100	–	300
Chuo Trust	150	–	37.5	34.2	221.7
Total	6 159.2	1 300	1 089.2	1 063.5	9 611.9

Source: Financial Reconstruction Commission.

than the 1.66 per cent average in the previous injection but greater than the government's own cost of funds. The government can convert its shares into common equity in nearly all cases, beginning as early as June 1999 for four of the weakest institutions, thereby ensuring that the recipient banks meet their commitments or risk losing control. For the subordinated debt the interest rate was, like last year, calculated as a premium over the six-month Euroyen rate. In general, initial spreads (most often for the first five years) were similar to those in the previous round for given banks, varying from 0.34 to 1.75 percentage points, while the yield spread to be paid thereafter is lower (consistent with the decline in long-term interest rates) for the trust and regional banks, lying between 1.25 and 2.54 percentage points. Recipient banks also planned to raise a further 2.2 trillion yen in additional capital from private sources, bringing their total recapitalisation to over 9.6 trillion yen ($80 billion). Including this injection, the government has provided nearly a quarter of these banks' total capital and around a third of their Tier I capital. Their BIS capital adequacy ratios vary from 9.5 to 15.1.

There were, of course, a variety of strings attached to the capital infusions. But thus far existing management has been allowed to remain in place in all cases (with only the threat that in the event "plans for restoring sound management" were not implemented, "managerial responsibility should be clarified"), raising moral hazard risks for the future. Some critics argued that the restructuring seems to have been designed to preserve surplus jobs, keep dicey borrowers afloat and minimise short-term financial losses to creditors and stockholders, rather than removing excess capacity, raising efficiency and creating strategic synergies. They noted that there were no tangible moves even to encourage the disposal of real estate collateral, the spin-off of unprofitable domestic operations, the introduction of incentive compensation, greater investment in technology and enhanced transparency of financial information.[77] Nonetheless, the authorities point out that much of the problem of non-disposal of real estate collateral is attributable to complex legal claims that have to be worked out. In any case, the market's assessment has been very favourable thus far, as the "Japan premium" has essentially disappeared and the interest-rate spread of bank debentures over government bonds has been cut in half compared with late 1998/early 1999 levels (Figure 25). Some of the conditions are worthy of mention. First, banks were encouraged to withdraw from unprofitable operations, especially those abroad. Five agreed to pull out of the foreign marketplace by March 2001 (as did another major bank that did not receive any public money), and the total number of foreign branches and subsidiaries for the remaining 10 is expected to fall from 384 in March 1998 to 230 in March 2003, a drop of 40 per cent. Much of this retrenchment was already underway and would probably have happened in any case, as Japan's share of international lending has fallen by an average of 2½ percentage points a year in the most recent three years (Box 1). Of more concern is

Figure 25. **Reduced credit risk premia in the banking system**



Source: Bloomberg, Bank of Japan and OECD.

Box 1. The international retreat of Japan's banks

Japanese banks continue to withdraw from global lending markets. The decline in their outstanding international lending during 1998, as reported by the Bank for International Settlements, was an enormous $170 billion, a drop of 19.1 per cent (Table 13). Around $70 billion of that fall represented claims on borrowers in the BIS reporting area and the rest on those outside the reporting area. Major retreats were observed in nearly all regions, but the largest declines were recorded for the United States, Hong Kong and Singapore.

Not surprisingly, Japan's banks have been losing market share, mainly to European banks. In 1998 the loss on claims within the BIS reporting area was 2.9 percentage points, bringing the cumulative fall to 7.2 points in three years. These declines are occurring nearly everywhere, but particularly in Asia: the cumulative decline in lending to Asia from end-1997 to mid-1999 was 35 per cent.

the fact that individual banks seem not to have developed a well-defined strategic role for their foreign operations, without which the downsizing represents mere cost cutting without a clear strategy to compete. Second, expense reduction, including the abolition of honorary posts for ex-presidents, was viewed favourably. The banks responded by promising to slash the number of directors by more than a third (from around 500 to 329, still a huge number for 15 institutions), cut back employment by nearly 20 000 (14 per cent) and hold down expenses other than mechanisation and personnel by over 5 per cent.

Realignments were another class of action that the authorities strongly supported – because of perceived economies of scale and excess capacity and because of the threats and opportunities provided by the "Big Bang" – and there have been several, most prominently the recent decision by Industrial Bank of Japan, Fuji Bank and Dai Ichi Kangyo Bank to effectively merge in autumn 2000.[78] Prior to that unexpected development, too many banks seemed to be pursuing non-distinctive strategies, based on universal banking models and a greater focus on lending to SMEs. In view of the challenge facing merging institutions to cut personnel costs in the context of the lifetime employment system heretofore in force in this industry and with already strong elements of cross-selling between sub-sectors, the case is weak for expecting fundamental improvement to result from any restructuring other than within-sector consolidation, which has the potential to at least gradually reduce capacity and exploit economies of scale (Atkinson *et al.*, 1999). Based on what is happening to the banking systems in other OECD countries,[79] these economies are sufficiently important as to question the need for any major banks greater in number than about four, compared with the

Table 13. **International lending position of Japanese banks by geographic zone**

	End-1997			End-1998		
	$ billion	Per cent of reporting bank lending	Per cent of Japanese bank international lending	$ billion	Per cent of reporting bank lending	Per cent of Japanese bank international lending
Total	895.1	n.a.	100.0	724.3	n.a.	100.0
Countries within BIS reporting area	515.3	n.a.	57.6	444.9	n.a.	61.4
of which:						
Europe	238.4	n.a.	26.6	218.2	n.a.	30.1
North America	276.9	n.a.	30.9	226.7	n.a.	31.3
United States	260.4	n.a.	29.1	214.0	n.a.	29.5
Countries within BIS non-reporting area	379.8	12.9	42.4	279.4	11.8	38.6
Developing countries	135.6	17.5	15.1	106.2	15.0	14.7
of which:						
Asia	114.7	30.1	12.8	85.9	28.8	11.9
of which:						
China	19.6	31.0	2.2	15.1	26.0	2.1
Indonesia	22.0	37.7	2.5	16.4	36.6	2.3
Korea	20.3	21.5	2.3	16.9	25.9	2.3
Malaysia	8.6	31.1	1.0	6.6	31.8	0.9
Thailand	33.2	56.4	3.7	22.4	55.1	3.1
Other	11.1	13.9	1.2	8.4	12.4	1.2
Latin America	14.7	5.2	1.6	14.5	5.0	2.0
Middle East	3.4	6.5	0.4	3.9	6.2	0.5
Africa	2.8	4.8	0.3	1.9	3.3	0.3
Eastern Europe	4.2	3.4	0.5	3.9	3.2	0.5
Developed countries	23.6	12.1	2.6	17.4	7.6	2.4
Offshore banking centres	216.4	29.6	24.2	151.9	24.0	21.0
of which:						
Cayman Islands	52.2	28.1	5.8	54.9	24.0	7.6
Hong Kong, China	76.3	36.0	8.5	38.7	29.4	5.3
Singapore	58.6	30.1	6.6	29.5	23.6	4.1
Panama	15.7	51.0	1.8	17.5	55.4	2.4

Source: Bank of Japan, Bank for International Settlements and OECD.

current 16 (Atkinson *et al.*, 1998). Such economies are said to be especially important in computer systems, an area where Japanese banks have been underspending for a long time relative to their global competitors: in FY 1996 that figure averaged 55 billion yen per top six city bank and 18.75 billion yen per top six trust bank, compared to an average of 110.9 billion yen for the top dozen US banks (Atkinson *et al.*, 1999). The implications may be serious, since in a deregulated environment IT spending is said to be the key to competitive strength.

The recipient banks also made a series of financial commitments over both the short and medium terms, plans which even the head of the FRC saw as "minimum targets" and "only the beginning". They agreed to expand domestic credit excluding impact loans (relending of foreign currency to domestic residents) over the current financial year by some 6.7 trillion yen (precisely 90 per cent of the capital injection), of which 3.0 trillion to SMEs.[80] They also promised to raise their operating profits through improving loan spreads so that the public capital can be repaid within an average of 6.8 years.[81] But the fundamental problem of Japanese banks remains one of insufficient revenues relative to their balance sheets (Table 14). Despite having the lowest ratio of operating expenses to recurring revenues among banks domiciled in the major seven OECD countries other than Italy,[82] those revenues are so meagre – around half of what the average is in the other major OECD countries – that net operating revenues are the lowest in relation to equity and not much more than half the rates achieved by banks from the three major English-speaking countries.

Table 14. **Revenue and expense ratios of major banks**

Percentages, fiscal years ending between 31 December 1997 and 30 June 1998

	Number of banks included	Recurring revenues	Operating expenses	Difference
		Equity	Equity	
United States	7	84.3	52.7	31.6
Japan	9	40.8	22.8	18.0
Germany	4	70.7	48.6	22.1
France	3	71.9	52.0	19.9
United Kingdom	7	79.4	44.9	34.5
Canada	5	91.6	58.5	33.1

Source: Atkinson *et al.* (1998) and OECD.

Finally, the recipient banks agreed to dispose of substantially all problem loans and to meet revised stricter FRC standards for the loan classification and provisioning of non-performing loans, now called "risk management loans".[83] With provisioning of around 9 trillion yen again in FY 1998, recipient banks are expecting to have to write off only about 1.1 trillion yen in the current year. The new standards were first announced in autumn 1998 and were applied to the major banks for the year ending March 1999 and for the others when they publish their first-half results for FY 1999. According to the new supervisory inspection manual, borrowers will be considered at risk of failure only if it is judged that they are unlikely to achieve their restructuring goals within a horizon of roughly five

years. Loans can even be considered sound if the borrower is incurring net losses but is expected to return to profits within a short period. Those in between are in the grey zone and are in need of monitoring or special monitoring. On average for the nine recipient city banks these new standards boosted problem loan estimates by 4.2 trillion yen (40.9 per cent), with increases ranging from 2 to 131 per cent for individual banks. Not surprisingly, the overall problem-loan situation deteriorated slightly over the course of FY 1998 (Table 15), as borrowers suffered during the severe recession. The current estimates for all recipient banks at 31 March 1999 are 17.7 trillion yen[84] (3.5 trillion yen in loans to failed borrowers, 10.6 trillion yen in those to borrowers in danger of failure and 3.6 trillion yen worth of lending to those in the grey zone), approximately double the amount of new capital they acquired. For some individual banks the latest estimates exceed 2 trillion yen. Mandatory provisioning requirements for the recipient banks are tougher than those currently applied in the United States (which is reasonable in light of relative macroeconomic conditions): 100 per cent for those in bankruptcy, around 70 per cent for those in danger of bankruptcy and without coverage by collateral or guarantees, and around 15 per cent for those in need of special attention. The major banks had previously provisioned only 52.1 per cent for those in danger of bankruptcy and 1.6 per cent for those in need of special attention, 10.4 and 1.2 per cent of their loan books, respectively.[85]

Developments in the rest of the banking and insurance sectors

Following the completion of the inspection of the major banks in 1998, the authorities have been inspecting the regional banks and insurance companies.[86] Most of the weak regional banks are believed to be among the second-tier institutions, which were formerly mutual savings banks. Thus far the FRC's position as to capital injections is that infusions will be considered favourably if the bank is indispensable to the local economy. This lack of generosity is justifiable in view of the substantial excess capacity in the industry. In any case in November 1998 only one in four regional banks felt in need of public capital. In June 1999 the FRC announced that it would target a capital adequacy ratio of 8 per cent following provision of public capital, rather than the legal cut-off of 4 per cent for banks without international operations. Then, in September 1999, it approved a capital infusion of 260 billion yen into three regional banks and one second-tier regional bank. In return, those institutions promised to cut staff by 12 to 22 per cent and shrink the number of branches by 12 to 17 per cent over a three- to four-year horizon. Repayment is slated for six to ten years in the future.

As for the life insurance industry, these mainly mutual companies' financial conditions continue to deteriorate due to the persistence of so-called "negative spread": a situation where the returns on assets fall short of those on

Table 15. **The recent evolution of problem loans**[1]

Trillion yen

	March 1998		March 1999	
	Level	% of total credit exposure	Level	% of total credit exposure
1. Risk management loans[2]				
All deposit-taking institutions	38.0 (34.8)	5.0 (4.7)	38.7	5.6
All banks	29.8 (26.7)	4.8 (4.5)	29.6	5.4
Major banks	22.0 (18.9)	5.2 (4.8)	20.3	5.7
Regional and regional II banks	7.8	4.0	9.4	4.9
Other co-operative-type institutions	8.2	5.6	9.0	6.3
2. Loan quality as determined by self-assessment				
All deposit-taking institutions	87.1 (81.4)	11.4 (10.9)	80.6	11.6
of which:				
Category II loans	80.3 (75.3)	10.5 (10.2)	76.6	11.0
Category III loans	6.9 (5.8)	0.9 (0.8)	3.9	0.6
Category IV loans	0.0 (0.0)	0.0 (0.0)	0.0	0.0
All banks	71.4 (65.7)	11.6 (11.2)	64.3	11.7
of which:				
Category II loans	65.5 (60.7)	10.6 (10.3)	61.0	11.1
Category III loans	6.1 (5.0)	1.0 (0.8)	3.0	0.6
Category IV loans	0.0 (0.0)	0.0 (0.0)	0.0	0.0
Major banks	50.1 (44.1)	11.9 (11.2)	41.6	11.6
of which:				
Category II loans	45.2 (40.2)	10.7 (10.2)	39.2	10.9
Category III loans	4.8 (3.8)	1.1 (1.0)	2.3	0.6
Category IV loans	0.0 (0.0)	0.0 (0.0)	0.0	0.0
Regional and regional II banks	21.3	10.9	22.7	11.7
of which:				
Category II loans	20.2	10.4	21.8	11.3
Category III loans	1.2	0.6	0.9	0.5
Category IV loans	0.0	0.0	0.0	0.0
Other co-operative-type institutions	15.7	10.7	16.3	11.4
of which:				
Category II loans	14.8	10.1	15.6	10.9
Category III loans	0.8	0.6	0.7	0.5
Category IV loans	0.0	0.0	0.0	0.0

1. Figures in parentheses and those for March 1999 exclude Long-Term Credit Bank and Nippon Credit Bank. These two institutions, nationalised in 1998, had about 6 to 7 trillion yen in risk management loans in March 1999.
2. Comprises loans to borrowers in legal bankruptcy, loans in arrears by 3 months or more and restructured loans.

Source: Financial Supervisory Agency.

liabilities.[87] While the major companies cut the guaranteed returns they offer on group pension funds from 2.5 to 1.5 per cent and on individual life policies with annual dividends from 2.75 to 1.75 per cent as from April 1999, the sector remains confronted by the problem of outstanding commitments to policyholders, made at a time when realised returns were much higher than current levels.[88] The FY 1998 results of the top eight firms showed a second straight year of declining revenues and new policies, although the number of policy cancellations edged down.[89] In most cases the companies' returns slid to below 2 per cent, although solvency margin ratios strangely improved. As from April 1999 it is this measure that the FSA has been using as an early warning system, parallel to Prompt Corrective Action in the banking sector. No existing firm has a solvency margin ratio[90] anywhere near as low as the 200 level above which no action is taken. However, some doubts have been raised as to whether this measure accurately reflects insurance companies' solvency. The recent failure of Toho Mutual Life, whose capital deficit is reportedly 200 billion yen, will severely deplete the industry's policyholders' protection fund, which aims to have only about 400 billion yen available (see below). If holders lose confidence in the adequacy of the safety net, a run could ensue. Another issue of short-term interest to the weaker companies is the need to raise more capital and the quasi-impossibility to do so without demutualisation. The Ministry of Finance plans to propose enabling legislation next year, and at least four of the 14 mutual life insurers have announced their intention to take advantage of that option once it is available. Finally, consolidation is also underway in the non-life sector where three large providers have just announced their intention effectively to merge by April 2002, thereby forming the nation's largest non-life insurer.

The impact of the measures

One of the hoped-for short-term effects of the recapitalisation was to ease the credit crunch. The logic of this expectation might be doubted, since better capitalised banks in an environment of full deposit insurance would be less prone to make risky loans as moral hazard considerations are weaker (Krugman, 1998*a*), at least if they are acting on behalf of their shareholders' interests. There are not many observations yet for the period since 31 March 1999, but there is no evidence at all of increased lending[91] (Figure 24 above): year-on-year changes in loans and discounts outstanding had averaged –4.1 per cent in the half-year prior to the infusion; since that time the shrinkage has been about 2 percentage points faster. However, after adjusting for special items such as loan securitisation, write-offs and valuation effects due to exchange rate changes, the rate of decline initially apparently eased, albeit modestly, but in recent months the decline has picked up once again, reaching the same 1.3 per cent in the first half of the current

year as was recorded in the second half of FY 1998. In any case, it is only regional banks that are raising their lending in absolute terms, whereas with one exception, the capital injection went to city, long-term credit and trust banks alone. If these latter institutions are boosting their lending to SMEs, as they promised to do in accepting the capital injection, then they would seem to have more than offset such increases with cutbacks to larger borrowers. More likely, they are not lifting lending to SMEs – given that they missed their end-of-year targets for such activity – possibly because of successful efforts by such firms to deleverage.

Another goal has been to completely remove the fully provisioned bad loans from banks' balance sheets. But banks have been reticent to make final disposal of such loans in large part because they hope that land prices will rebound, but also because their carrying costs are low and because many of them are inexperienced at assessing the fair market value of their bad loans.[92] Optimistic beliefs about land prices are perhaps supported by the actions of the public Resolution and Collection Corporation (RCC), the new specialist repository of bad loans[93] born in April out of the merger of the Resolution and Collection Bank and the Housing Loan Administration Corporation. Instead of packaging the real estate it has acquired and securitising it, its announced policy is to collect on loans, especially those to companies linked to *yakuza* (gangsters) who previously refused to pay. Its first-year target is to collect 710 billion yen worth of loans. In any case, the result is that the real estate market remains severely illiquid. However, securitisation overall is progressing. Early this year the first mortgage-backed securities, amounting to 50 billion yen, were sold by Sanwa Bank. Issuance of collateralised bond obligations began in December 1998 and had reached 300 billion yen by the end of March 1999. Moody's estimates that the overall asset-backed securities market almost doubled again in 1998 to some $12.5 billion. And several banks are selling some of their bad loans, mainly to foreign firms,[94] either directly or through the Cooperative Credit Purchasing Corporation (CCPC) – this industry-owned organisation sold loans with a book value of about 1 trillion yen to foreigners in FY 1998.

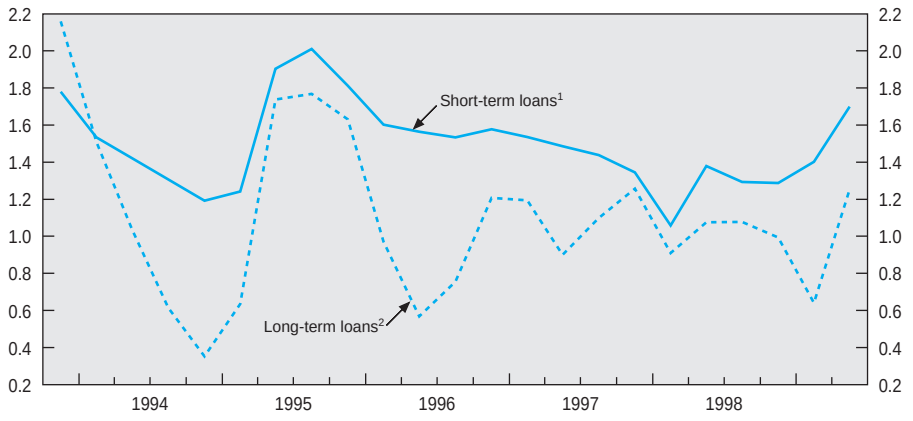
A final goal is to move the system of financial regulation to one based on "clearly articulated rules and more transparent administration". To this end the FSA tightened capital adequacy rules on securities companies in April 1999. Previously, all credits required 10 per cent capital coverage, but henceforth the requirement will be for 6 to 25 per cent, depending on the borrower's credit rating. But Tokyo's reputation as a global financial centre took another blow when some securities companies, misunderstanding the rules of the Japan Securities Dealers Association, as modified in December 1998, made so-called "price adjustment deals" whereby bonds were transacted at prices that were more than 2 per cent from market levels.[95] In the banking sector, some institutions (led by Fuji Bank) have recognised the value of transparency and are even competing in that

dimension. The goal of enhanced transparency will be served by the introduction of mark-to-market accounting, scheduled for March 2001 (a year later for non-financial companies).

The big question for the longer term is whether Japan's financial system is now assessing and pricing credit risk properly, since the absence of any credit-risk evaluation culture was to a large extent responsible for the NPLs problem in the first place. If not, then either the recapitalisation will have to be repeated when a further round of problem loans is incurred at the next cyclical downturn and banks' balance sheets cannot sustain the necessary write-offs, or more banks will have to be closed. Appropriate lending practise means extending credit and pricing it with due consideration of default risk, rather than merely based on the availability of collateral overwhelmingly in the form of real estate.[96] Since bank profitability has long been inadequate and competition from disintermediated sources is increasing, that inevitably means higher spreads on loans for most borrowers. There is little sign of widening margins in the sector in recent quarters (Figure 26), although they have at least been fairly well maintained through the recession, despite the obvious impact on pricing power. However, in some quarters banks are believed to be not discriminating sufficiently in their pricing decisions based on assessed risk. This judgement may be consistent with figures

Figure 26. **Bank lending margins**

Percentage points



1. Excess of average rate on short-term loans and discounts less 90-179 day CD new issue rate.
2. Excess of average rate on long-term loans and discounts less five-year bank debenture rate.

Source: Bank of Japan and OECD.

on banks' loans disaggregated by interest rates (Table 16): in recent years there has been increasing concentration of loans with rates of 1 to 3 per cent and shrinkage of those with less than 1 per cent and more than 3 per cent. Furthermore, banks have shown very little tendency to shift the structure of their loans based on risk considerations. Comparing end-1998 figures with those from five years earlier (Table 17), there is an amazing sense of similarity. Loans outstanding have moved away from firms in the manufacturing, trade, finance and other service sectors and towards those in construction and real estate, local governments and individuals.[97] For the most part these shifts would appear to be based more on keeping clients afloat than on evident profit-maximising considerations. However, some observers believe that the recently announced merger of three large commercial banks will lead to the application of tougher terms on borrowers, especially in problem sectors such as construction.

But the proper assessment of credit risk has, until fairly recently, been an unattained objective not only for financial institutions but for markets as well. Until very recently, even Japan's securities markets were inexperienced in judging such risks, given the restricted access to them[98] and the resulting lack of defaults historically. It is only very recently that signs of change can be detected. For example, it was only late in 1998 that Japan experienced its first ever default leading to an actual loss on a straight bond offering (the case of Japan Land Development which defaulted on 50 billion yen in straight bonds and a further 7.1 billion yen of convertible bonds). The markets' sensitivity to risk can also be judged by the yield spreads over either government bonds or over private triple A rated issuers (Figure 27): those spreads widened substantially with the

Table 16. **Loans and discounts of domestically licensed banks[1] by interest rate**

End of year shares in per cent

	1994	1995	1996	1997	1998	July 1999
Interest rate						
Less than 1 per cent	1.3	7.0	9.7	6.9	5.6	13.4
1 to 2 per cent	0.8	24.4	25.5	32.0	33.6	29.3
2 to 3 per cent	6.7	30.9	33.5	38.2	40.9	39.7
3 to 4 per cent	45.0	18.7	18.3	13.5	12.4	12.1
4 to 5 per cent	27.9	9.4	6.5	4.9	3.5	3.7
5 to 6 per cent	9.0	3.9	2.7	1.9	1.1	1.0
6 to 7 per cent	4.4	2.5	1.6	1.0	0.7	0.5
7 to 8 per cent	2.6	1.5	0.9	0.6	0.4	0.4
8 to 9 per cent	1.3	0.9	0.6	0.5	0.4	0.3
9 to 10 per cent	0.4	0.3	0.2	0.2	0.2	0.1
More than 10 per cent	0.6	0.5	0.4	0.3	0.3	0.3

1. Banking accounts only.
Source: OECD and Bank of Japan.

Table 17. **Loans and discounts classified by borrower**

End of year

	1993	1998
Total (billion yen)	477 642	484 091
Shares in per cent		
Manufacturing	16.0	14.3
Construction	6.2	6.5
Utilities	1.2	1.0
Transport and communication	3.4	3.6
Wholesale and retail trade, eating and drinking places	16.8	15.6
Finance and insurance	10.0	9.3
of which: Securities brokers and dealers and insurance institutions	0.4	0.8
Real estate	11.4	12.8
Other services	15.2	14.8
Local governments	1.0	1.9
Individuals	16.1	18.4
Overseas yen loans and domestic loans transferred overseas	1.8	1.3
Memorandum item:		
Small enterprises[1]	69.0	69.2

1. Including individuals and domestic loans transferred overseas.
Source: Bank of Japan.

Figure 27. **Yield spreads of corporate over government bonds**

Source: Japanese Securities Dealers Association, "Over-the-Counter Standard Bond Quotations".

advent of the banking crisis but have, since March, reversed much of that widening. Similarly, issuers with lower stock prices were faced with higher yield spreads during the recession (Muromachi, 1999). Finally, investors are certainly aware that there will be a change of regime and increased credit risk as from April 2001 when the "payoff system" is implemented: the relevant jump in risk premia is more than 100 basis points. While some voices have been heard calling for a delay of the implementation date for this new regime, such a delay would only weaken the pressure for the other structural changes that go hand in hand with this reform, especially those in the area of risk management, accounting and disclosure.

III. Fiscal policy and medium-term public sector issues

Expansionary fiscal policy and its consequences

Another fiscal package in the context of an increasingly expansionary stance

Fiscal policy became increasingly expansionary during FY 1998. A turning point was the April fiscal package (see the previous Survey), and another package followed in November. The government formally announced a freeze of the Fiscal Structural Reform Law[99] in the November package and moved to a still more expansionary stance. The package consisted of three parts: first, stabilisation of the financial system and anti-credit-crunch measures to eliminate downside risks (described in Chapter II); second, policies to stimulate demand; and, finally, measures to assist in the recovery of the Asian crisis countries.[100] The total increase in expenditures amounted to nearly 18 trillion yen (3.6 per cent of GDP) on a project cost basis (Table 18). The package also included personal and corporate income tax reductions worth more than 6 trillion yen. Out of 5.9 trillion yen of additional credit and guarantees provided by government affiliated institutions, 3.4 and 2.5 trillion yen were to be allocated to smaller and larger SMEs, respectively. In contrast with the April version, the November package did not include local governments' own social infrastructure projects (those unsubsidised by central government) (Table 19), in view of their severe financial condition. In response to a significant increase in the unemployment rate, employment measures were strengthened to meet safety net and human capital improvement objectives (see Chapter V). It was also decided that shopping vouchers (coupons for purchases at local stores) would be distributed, despite scepticism expressed by many observers. In addition, the package contained new initiatives in the form of projects in information and communication, transportation, welfare and other new technologies.

These measures were incorporated in the revised Fiscal Investment and Loan Programme (FILP) in November and the third FY 1998 supplementary budget[101] in December. As for social infrastructure projects, 6.8 trillion yen[102] out of the total 8.1 trillion yen was included in the supplementary budget, while the rest was to be integrated into the initial FY 1999 budget. An additional amount of

Table 18. **Fiscal measures in the November 1998 package**

Trillion yen

	Fiscal measures	November package	Third FY 1998 supplementary budget allocation
Social infrastructure investment	Public investment	8.1	4.0
	Telecommunications network, and science and technology	–	(0.9)
	Social welfare, medical care and education	–	(0.6)
	Environment and new energy	–	(0.5)
	Improvement of distribution, transportation system and industry competitiveness	–	(0.4)
	Revitalisation of rural areas	–	(0.4)
	Inducement of private investment through the redevelopment of downtown districts	–	(0.3)
	Disaster prevention		(0.4)
	Disaster relief	0.6	(0.5)
Others	Measures for credit crunch	5.9	1.0
	Measures for promotion of residential investment	1.2	0.2
	Measures for employment	1.0	0.1
	Regional promotion coupons	0.7	0.8
	Measures for Asian crisis	1.0	0.1
Total		17.9	6.2
Memorandum items			
Tax reductions, etc.	Permanent income tax reductions	4.0	–
	Corporate tax reduction	More than 2.0	–

Source: Economic Planning Agency and Ministry of Finance.

Table 19. **Fiscal packages in the 1990s**

Billion yen, project cost base

	Total amount	Social infrastructure investment					Tax cuts	Others
		Public works		Building and equipment	Public works by local government	Sum of 2-5		
		General public works	Disaster relief					
	1	2	3	4	5			
28 August 1992	10 700	3 400	500	550	1 800	6 250	0	4 450
13 April 1993	13 200	3 640	530	1 150	2 300	7 620	150	5 430
16 September 1993	6 200	1 000	450	0	500	1 950	0	4 200
8 February 1994	15 300	3 590	0	610	300	4 500	5 850	4 900
14 April 1995	4 600[1]	205[2]	718[2]	154[2]	0	1 077[2]	0	3 543[2]
20 September 1995	12 800	3 930	700	910	1 000	6 540	0	6 270
24 April 1998	16 700	4 500	200	1 500	1 500	7 700	4 600	4 350
16 November 1998	23 900	5 700	600	1 800	0	8 100	6 000	9 800
Cumulative	103 400	25 965	3 698	6 674	7 400	43 737	16 600	42 943

1. OECD estimate.
2. Budget figures, rather than project cost figures.

Source: Economic Planning Agency and Ministry of Finance.

2.2 trillion yen was allocated to government institutions (through FILP). Most prominently, about 1.4 trillion yen was provided to the Japan Development Bank to fund medium-sized companies,[103] *i.e.* companies not legally defined as SMEs, and to (unusually) finance the repayment of corporate bonds at maturity. About 0.6 trillion yen was for the Export-Import Bank's loans to affected Asian countries. With regard to other credit crunch measures, 0.6 and 0.1 trillion yen was disbursed from the general account to the credit guarantee system and government financial institutions for SMEs, respectively, in order to compensate for the expected increase in losses in expanding loans and guarantees. As a result of these revisions, both the budget and the FILP[104] in FY 1998 were strongly expansionary (Table 20 and 21). FILP outlays rose by more than 10 trillion yen or 2 percentage points of GDP.

The government regarded the initial FY 1999 budget as one of a series of expansionary measures that would be taken. The approach was designated as a "15-month budget": the economy was to be continually supported by fiscal stimulus from January 1999 to March 2000. The increase in general expenditures (5.3 per cent) was the highest since FY 1980. Although budgetary allocations to public works declined by over 40 per cent (Table 20), compared to the previous year's outcome, actual expenditures in FY 1999 were officially estimated to increase by more than 10 per cent due to a large amount of carry-over from the previous fiscal year.[105] The scheduled rise in pension contributions, normally implemented every five years, was frozen for the time being for fear of possible damage to consumer sentiment, while the benefits formula was permanently revised so that re-rating would be according to price changes only and would not be linked to changes in (real) wage payments (see below). The personal and corporate income tax reductions promised in the package were officially estimated to cost about 4.9 trillion yen in national tax revenues[106] once fully implemented. However, the net effect of tax cuts in FY 1999 was only about 1.6 trillion, most of which was due to corporate tax changes[107] (see Chapter IV). Housing tax cuts were expanded significantly, but are likely to have limited effects on revenues in the current fiscal year as the temporary tax incentive is exploited only gradually. The initial FILP budget is believed to supply enough funds to maintain the economic stimulus (Table 21) while a reform of the system is implemented (see below). Finally, while the government formulated a supplementary budget in July 1999 to expand measures promised in the Comprehensive Plan for Employment Activation (a plan to deal with the severe labour market situation announced in November 1998 – see Chapter V), the additional expenditure was not substantial – only about 0.5 trillion yen (Table 20).

Limited scope for future fiscal stimulus

There are many indications that the scope for further fiscal stimulus is quite limited. Bond markets reacted negatively to the prospect of large issuance

Table 20. **Central government budget**

Trillion yen

	FY 1997	FY 1998[5]				FY 1999		
	Settled	Initial budget	First supplementary budget	Third supplementary budget	Total budget	Initial budget	First supplementary budget	Total budget
Total expenditure	78.5	77.7	4.6	5.7	88.0	81.9	0.5	82.3
General spending	47.1	44.5	4.6	7.2	56.3	46.9	0.5	47.3
of which:								
Public works	11.1	9.0	2.8	5.1	16.9	9.9[6]	0.0	9.9
Other	36.0	35.6	1.7	2.1	39.4	37.0	0.5	37.4
Debt servicing[1]	15.9	17.3	0.1	0.0	17.3	19.8	0.0	19.8
Transfers to local government	15.5	15.9	0.0	–1.6	14.3	13.5	0.0	13.5
Other spending	0.0	0.0	0.0	0.0	0.0	1.6	0.0	1.6
Total revenue	59.5	62.1	–1.5	–6.6	54.0	50.8	0.0	50.8
Taxes	53.9	58.5	–1.5	–6.9	50.1	47.1	0.0	47.1
Non-tax revenues	5.6	3.6	0.0	0.2	3.8	3.7	0.0	3.7
Deficit[2]	18.5[4]	15.6	6.1	12.3	34.0	31.1	0.0[7]	31.1
Per cent of GDP	3.7	–	–	–	6.9	–	–	–
Financing:								
Construction bonds	9.9	8.4	4.1	4.5	17.1	9.3	0.0	9.3
Deficit bonds	8.5	7.1	2.0	7.8	17.0	21.7	0.0	21.7
Memorandum item:								
Net bond issuance[3]	15.1	–	–	–	41.3	–	–	–
Per cent of GDP	3.0	–	–	–	8.4	–	–	–

1. Including amortisation payments on existing debt.
2. The deficit is equal to the net issuance of government bonds (Financial Revenue Bonds).
3. Increase in outstanding stock of government bonds. Differs from footnote 2 by the amount of changes in cash balances.
4. Although the deficit was 19 trillion yen (the difference between total expenditure of 78.5 and total revenue of 59.5 trillion yen), 0.5 trillion yen was drawn from reserves and, therefore, the final deficit was only 18.5 trillion yen.
5. The second supplementary budget provided only additional government guarantees to borrowing by the Deposit Insurance Corporation of 33 trillion yen.
6. Including special reserves for public works worth 0.5 trillion yen.
7. Although the first-round increase in the deficit was 0.5 trillion yen (the difference between total expenditure of 0.5 trillion yen and total revenue of 0 trillion yen), 0.15 trillion yen was drawn from a contingency fund and the remainder from reserves and, therefore, there was no impact on the deficit.

Source: Ministry of Finance and Bank of Japan.

Table 21. **Fiscal Investment and Loan Programme (FILP)**

Trillion yen

	FY 1997	FY 1998		FY 1999
	Settlement	Initial	Revised	Initial
Disbursement				
Total	46.9	50.0	60.4	52.9
General FILP	34.9	36.7	43.1	39.3
Portfolio investment	12.0	13.3	17.3	13.6
of which:				
Housing Loan Corporation	6.4	9.9	9.9	10.1
Housing and Urban Development Corporation	0.9	1.0	1.1	1.0
Japan Development Bank (1)	1.5	1.4	2.9	2.6
Hokkaido Tohoku Development Finance Public Corporation (2)	0.2	0.1	0.3	0.4
(1) + (2) (Japan Policy Investment Bank)[1]	1.7	1.5	3.2	2.9
People's Finance Corporation (3)	3.2	3.8	3.8	3.8
Environmental Sanitation Business Finance Corporation (4)	0.3	0.4	0.4	0.4
(3) + (4) (National Life Finance Corporation)[1]	3.5	4.2	4.2	4.2
Japan Finance Corporation for Small Businesses	1.6	2.0	2.2	2.2
Export-Import Bank of Japan (5)	1.4	1.1	2.2	1.9
Overseas Economic Co-operation Funds (6)	0.5	0.4	0.5	0.7
(5) + (6) (Economic Co-operation Bank)[1]	1.9	1.5	2.7	2.7
Japan Highway Public Corporation	2.2	2.0	2.3	2.1
Metropolitan Expressway Public Corporation	0.4	0.4	0.4	0.3
Hanshin Expressway Public Corporation	0.4	0.3	0.3	0.3
Honshu-Shikoku Bridge Authority	0.3	0.2	0.2	0.1
Local governments	8.4	7.6	10.3	7.7
Japan Finance Corporation for Municipal Businesses	2.0	1.7	1.9	1.7
Postal Savings Special Account	7.5	8.0	12.0	8.5
Pension Welfare Service Public Corporation	4.2	4.8	4.8	4.7
Postal Life Insurance Welfare Corporation	2.0	2.5	2.5	2.0
Others	3.4	2.4	2.5	2.3
Revenues				
Total	51.7	57.8	71.4	52.9
Industrial Investment Special Account	0.1	0.1	0.5	0.1
Trust Fund Bureau Fund	41.8	48.1	61.1	43.7
Postal Savings	15.6	11.4	n.a.	11.5
Employee/National Pension Fund	7.7	6.0	n.a.	4.3
Funds repaid and others	18.5	30.7	n.a.	27.9
Postal Life Insurance Fund	7.0	7.1	7.1	6.6
Government-guaranteed bonds and loans	2.9	2.5	2.7	2.5
Memorandum item:				
Purchase of newly-issued national government bonds	4.8	7.8	11.0	–

1. The institutions in parentheses were established in October 1999.

Source: Ministry of Finance.

of central government bonds. Financial positions have deteriorated at each level of government, and the social security surplus, albeit substantial, merely masks an underlying deterioration in its finances. These are reflected in a rapid rise in both the deficit and indebtedness of general government, which might be exacerbated by potential liabilities of the wider public sector.

Bond market reactions

The fiscal stimulus measures have resulted in a significant increase in the central government's deficit-bond issuance, which was made possible by the above-mentioned freezing of the Fiscal Structural Reform Law but encountered a negative market reaction. New Financial Resource Bond issuance in FY 1998 was the largest ever – 34 trillion yen after the supplementary budgets, more than double the initially planned level (Table 20). This will be followed by further issuance of securities worth about 31 trillion yen in FY 1999. As a result, the share of spending financed by bond issuance reached the record level of 38.6 per cent in FY 1998 and is set to remain at around 38 per cent in FY 1999. The significant increase in government bond issuance started to strain the bond market in autumn/winter 1998-99, when long-term interest rates rose sharply (see Figure 19 in Chapter II), especially following the Ministry of Finance's (MOF) announcement in late December of a planned termination of government bond purchases by the Trust Fund Bureau (TFB). Those rates started to decline in mid-February, in response to announcements of a further easing in monetary policy, a larger share of mid-term bonds in total issuance, and a resumption of the TFB's purchases in February. This episode of rising long-term interest rates from December to February shows the limited room for further fiscal stimulus, although it was exaggerated by an element of surprise to investors.

The episode also demonstrates the TFB's substantial role in the market. The TFB still plans to purchase only enough bonds (2.8 trillion yen) in FY 1999 to offset those in its portfolio that are maturing.[108] Overall, the TFB is planning to buy nearly 12½ trillion yen less in government securities than it did in FY 1998, while 10.3 trillion yen more will be sold to the private sector, mainly in the form of bills and 2-year bonds. Reduced TFB purchasing mainly reflects the increased allotment of funds to government financial institutions, according to the revised FILP. In addition, it is necessary for the TFB to increase liquidity in preparation for the expected increase in repayment to the Postal Savings Special Account, when it is confronted by a net outflow recently estimated by the Ministry of Post and Telecommunications (MPT) at 31 trillion yen in FY 2000-01.

The government bond market also suffers from a measurable lack of liquidity because of the large share held to maturity by public financial institutions and the small share held by foreign investors (also discouraged by the presence of a securities transactions tax until recently). The result is that bid-ask spreads, even for the benchmark 10-year maturity, are 7 basis points, compared

Table 22. **Basic market liquidity measures in five large OECD countries**

	Canada	Italy	Japan	United Kingdom	United States
Bid-ask spread					
Fixed coupon:[1]					
2 years	2	3	5	3	1.6
5 years	5	5	9[4]	4	1.6
10 years	5	6	7	4	3.1
30 years	10	14	16[5]	8	3.1
Volume outstanding (a)[2]	285	1 100	1 919	458	3 457
Yearly trading volume (b)[3]	6 243	8 419	13 282	3 222	75 901
Turnover ratio (b/a)	21.9	7.7	6.9	7.0	22.0

1. The table shows the bid-ask spreads of on-the-run issues, given in one-hundredth of a currency unit for the face amount of 100 currency units.
2. The figures are as of end-1997, in $ billion, converted at the exchange rates of end-1997 (US$ 1 = Can$ 1.43 = L 1 770 = Yen 130, £1 = $1.65).
3. The figures are for the 1997 calendar year, on a two-way basis.
4. 6-year bonds.
5. 20-year bonds.

Source: Bank for International Settlements (1999*a*).

to around 3 basis points for corresponding US Treasuries (Table 22). Increased liquidity in the government bond market could result from the increase in volume: JGBs are expected to account for about 30 per cent of gross issuance of total OECD government bonds in the next few years. Increasing liquidity should contain volatility and might potentially make a small contribution to curtailing rises in the long-term interest rate. MOF has decided to increase the variety of bonds' maturities (the first regular 30-year issue was made in September), restrain the increase in 10-year bond issuance and improve the disclosure of bond issuance schedules. Consideration is also being given to the introduction of STRIPs (special bonds whose coupons are separable from the principal).

Local governments pushed to the limit

Each local government's budget is planned, using the Local Public Finance Programme (LPFP) formulated by the Ministry of Home Affairs (MHA) as a guide, and is revised several times during the fiscal year in response to changes such as central government supplementary budgets. According to the Ministry of Home Affairs, local budgets as of its September 1998 survey[109] (Table 23) reflected the April fiscal package and the first FY 1998 national supplementary budget in their own plans, including an increase in unsubsidised public works of about 1.6 trillion yen. However, the level of (unconsolidated) public works undertaken without central government funding was still lower than the year-earlier figure as of September (Box 2). The November 1998 package should be

Table 23. **Local public finance programmes (LPFP) and local budgets**[1]

Fiscal year, yen trillion

	1997			1998		1999
	LPFP	After supplementary budgets (unconsolidated) (September)	Settlement (consolidated)	LPFP	After supplementary budgets (unconsolidated) (September)	LPFP
Total expenditures	87.1	109.8	97.7	87.1	112.9	88.5
of which:						
Public works	30.9	33.1	27.7	29.1	33.1	29.3
Unsubsidised	20.1	18.3	15.5	19.3	17.5	19.3
Debt servicing	9.6	10.4	10.3	10.5	11.0	11.4
Total revenues	74.9	95.0	85.8	76.1	97.8	77.2
General revenues	55.2	54.9	54.4	56.6	57.0	57.4
of which: Local taxes	37.0	36.3	36.2	38.5	37.2	35.3
Matching grants	13.3	15.3	14.3	13.0	15.8	13.2
Others	6.4	24.8	17.1	6.5	25.1	6.6
Deficit	12.1	14.8	11.9	11.0	15.0	11.3
Per cent of GDP		2.9	2.4		3.0	
Memorandum items:						
Debt increase (net)[2]			10.6		16.7	
Per cent of GDP[2]			2.1		3.4	

1. Consolidated numbers and LPFP are adjusted for transfers among local governments while unconsolidated numbers are not.
2. Actual numbers at the end of fiscal year.

Source: Ministry of Home Affairs.

Box 2. Unsubsidised investments and local governments

While the component of local public works that is subsidised (by the central government) used to be larger than its unsubsidised counterpart, the ordering was reversed in FY 1988 as the former was restrained due to a decline in the central government's coverage of costs in the 1980s as part of the fiscal consolidation effort. Since then the central government has continued to emphasise unsubsidised public works in central government policies, as seen, for example, in the Basic Plan for Public Investment (1990, and revised in 1994) and in past fiscal packages. In the early 1990s, unsubsidised investment was larger than envisaged in the Local Public Finance Programmes, partly because of additional works in response to the fiscal packages. The central government encourages unsubsidised projects through authorisation of local bond issuance and disbursement of local allocation tax revenues (see below). Bonds issued for unsubsidised projects accounted for about 45 trillion yen or 40 per cent of all local bonds outstanding in March 1998.

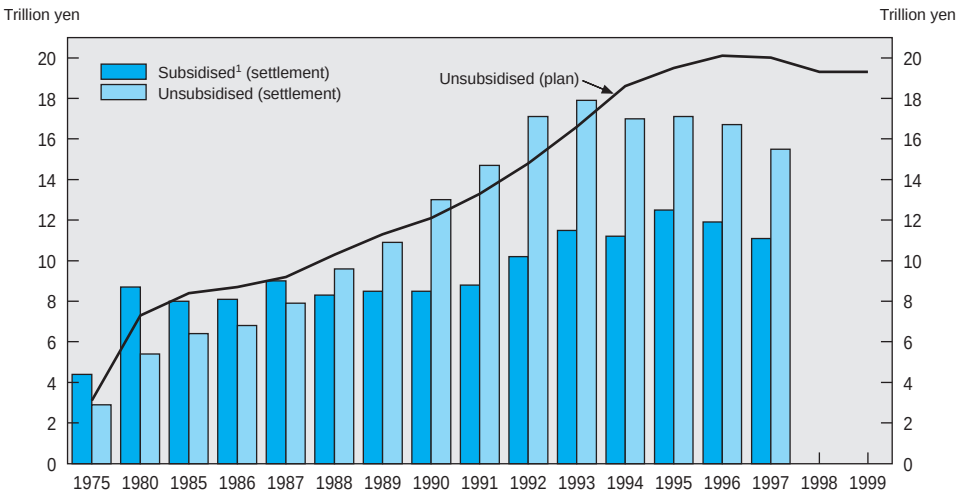
incorporated in the December and/or later modifications to local budgets because the Economic Planning Agency's (EPA's) follow-up report to the November package in April 1999[110] made it clear that local governments had taken all the necessary budgetary measures. Therefore, the final outcome of *total* local public works in FY 1998 might be larger than implied by the September 1998 figures in view of additional projects subsidised by the central government. With this, the aggregate local government deficit and debt increase in FY 1998 are sure to have been greater than the 3.0 and 3.4 per cent of GDP estimated in September, despite a steady increase in transfers to local governments.[111] The FY 1999 LPFP presented a rather conservative fiscal stance, with only a 1.3 per cent increase in total expenditures compared to the last programme – the third smallest growth rate since FY 1975. It had to balance the need to be consistent with the expansionary fiscal stance taken by the central government and the need for fiscal consolidation due to the severity of local governments' financial situations. It maintained the same amount of unsubsidised public works despite the smaller settlement numbers than envisaged in past LPFPs, with only a marginal increase in subsidised projects. Priority was placed on financial measures for SMEs and welfare spending, along with attempts to curb other expenses.

Nonetheless, the room for further fiscal manoeuvre at the local level would appear to have been exhausted. As poor economic performance continued and local governments' bonds outstanding increased, they had diminished resources to allocate to unsubsidised projects. The share of debt servicing in untied revenues (the sum of local taxes and lump-sum transfers from central government) reached more than 15 per cent, a critical threshold set by the MHA, in 56 per cent of the 3 279 local governments in FY 1997. Another piece of evidence showing local governments' financial difficulties is that the TFB decided to allow earlier repayment of its high interest-rate loans to some local governments meeting certain conditions in FY 1999. Indeed, in the context of continuing declines in unsubsidised public works since FY 1993, there has been a growing gap between plans and realisations for that component; by FY 1997 that shortfall probably reached more than 4 trillion yen[112] (Figure 28).

Burgeoning deficits of general government

Overall, the OECD estimates that the expansionary fiscal actions are likely to have increased the structural deficit of general government by 1½ percentage points of GDP in both calendar years 1998 and 1999 (Table 24). Because weak economic activities are estimated to have pushed the deficit up by a further 1 percentage point in 1998, the deficits are expected to deteriorate by a cumulative 4¼ percentage points of GDP in the two years, reaching 7.7 per cent of GDP in 1999, easily the largest among OECD Member countries. Net interest payments on government debt are expected to have increased gradually to above 1¼ per cent of GDP in 1999.

Figure 28. **Public works by local governments**



1. Subsidisation refers to cost-sharing with central government.
Source: Ministry of Home Affairs.

Table 24. **General government deficits**

Calendar year, per cent of GDP

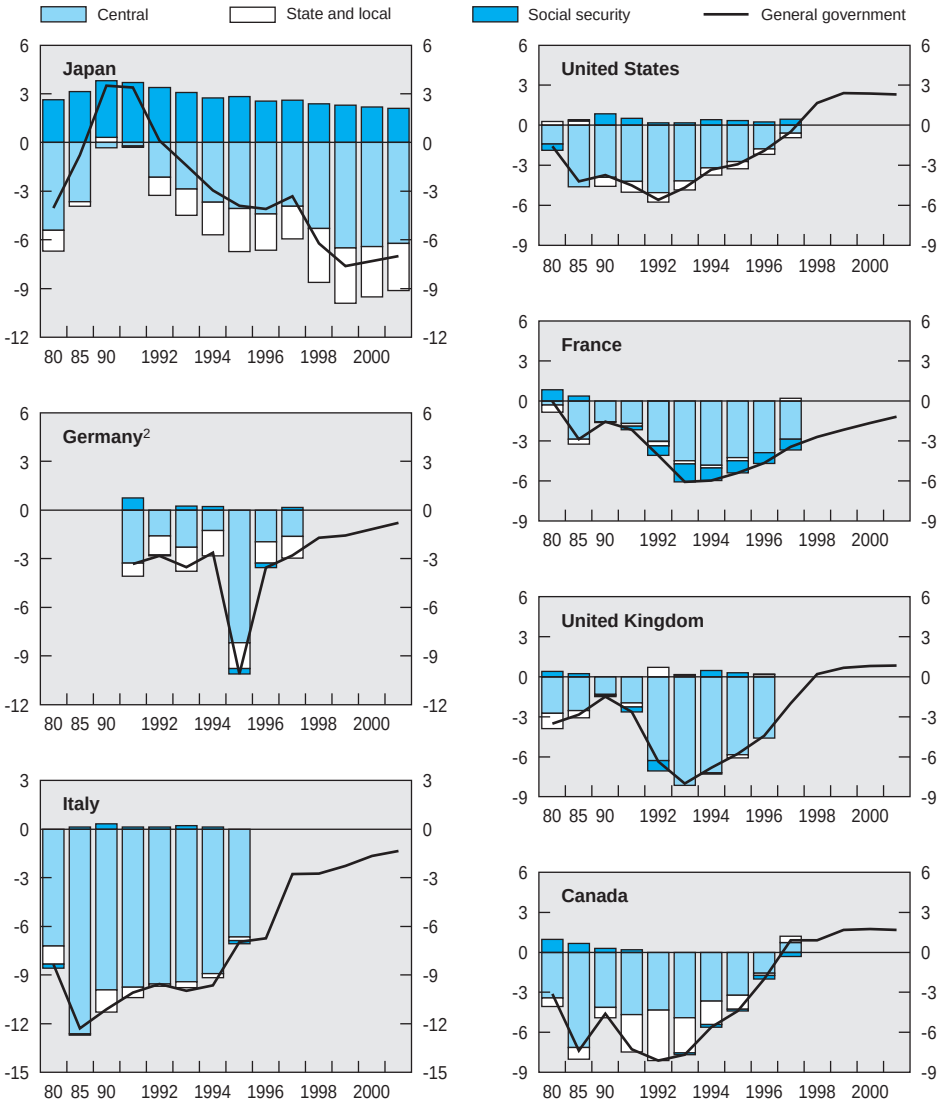
	1995	1996	1997	1998	1999	2000	2001
Cyclically-adjusted net lending	–3.1	–4.4	–3.6	–5.1	–6.7	–7.0	–6.3
Cyclically-adjusted net primary balance	–2.5	–3.7	–2.6	–4.0	–5.4	–5.6	–4.9
Net lending	–3.6	–4.2	–3.4	–6.0	–7.6	–7.9	–7.2
Net primary balance	–3.1	–3.5	–2.4	–4.8	–6.3	–6.5	–5.8

Source: OECD.

Examination of the sectoral distribution of Japan's deficits, in comparison with other countries', reveals several noteworthy points (Figure 29). First, the central government's deficits were not too large until FY 1997, but deterioration has been rapid since then. Second, local governments have larger deficits than in other countries, reflecting their stabilising role.[113] Third, the social security funds are in gradually declining but still sizeable surplus. Intra-governmental transfers seem helpful to understanding the sectoral distribution of deficits. The transfers

Figure 29. **Net lending: an international comparison**[1]

% of GDP



1. The data for 1998 to 2001 (1997 to 2001 for the United Kingdom) are OECD preliminary projections for *Economic Outlook 66.*
2. Includes as a capital transfer item the takeover of debt from the East German Housing Fund and the Treuhandanstalt of about 6 to 7 percentage points of GDP.

Source: Economic Planning Agency and OECD, *Annual National Accounts.*

are summarised in Figure 30. Central government provides heavy financial support to the other two sectors, amounting to 7¾ per cent of GDP in FY 1997. The development of capital transfers from central to local governments reflects the central government's fiscal consolidation efforts in the 1980s and its stabilisation efforts through local governments in the 1990s. Social security funds also receive significant transfers from central and local governments, amounting to 2¾ per cent of GDP in FY 1997, almost equal to the size of that sector's surplus. Their balances *vis-à-vis* households have actually been in deficit since the latter half of the 1970s, growing to over 1½ per cent of GDP in FY 1997 (Table 25). These deficits have been fully compensated by property income from portfolio investments, even though the income itself has been declining in recent years. A look at the disaggregated balances by account suggests that the health insurance system is already facing a serious problem, especially in the scheme for the elderly[114] and national health insurance (mainly for the self-employed). The deficit of the national pension system has also been large and growing. Because the national health insurance scheme is operated through health funds run largely by municipalities, its deficit has affected their financial situation. While an increase in the portion of bills paid by patients in September 1997 seems to have improved the scheme's financial condition, the effect has been quite modest.[115]

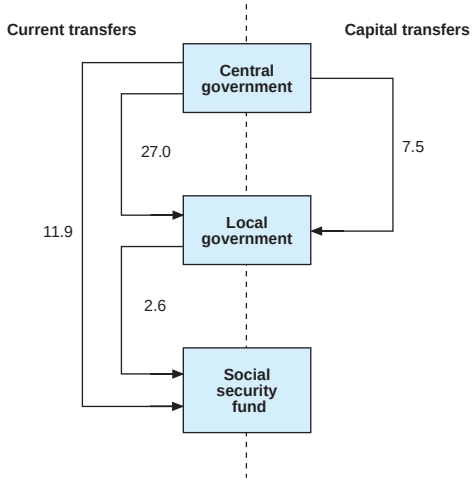
Balance sheets of general government and the wider public sector

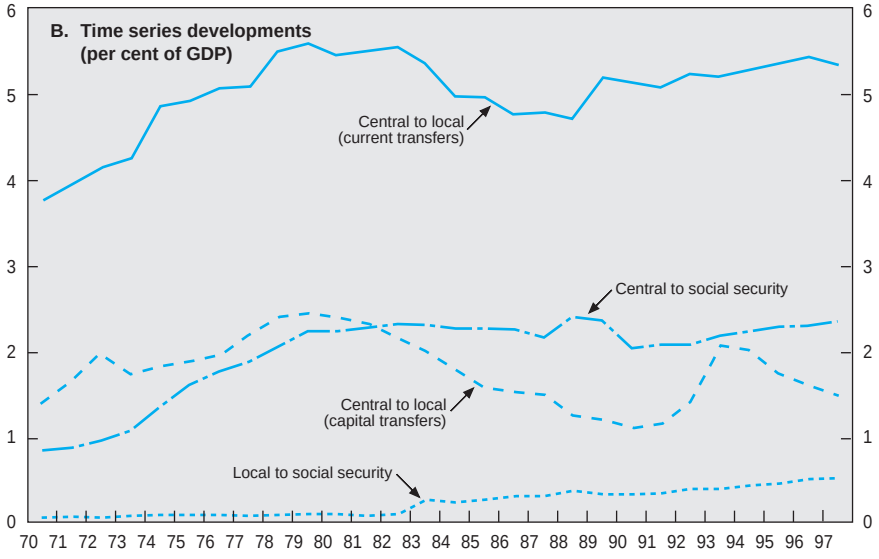
Because deficits have been so large, real GDP growth so weak and inflation largely non-existent, the path of indebtedness has been explosive, despite very low nominal interest rates. While there are various possible debt measures (Figure 31),[116] with large differences in levels, the recent rise is common to all variants. The gross debt based on the OECD definition (line B) reached over 90 per cent of GDP in FY 1997 and is expected to have increased by 25 percentage points, to 115 per cent of GDP by end-FY 2000. Another factor to be taken into account is public institutions' debt. As observed in the transfer of 26.9 trillion yen in debt of the Japan National Railway Settlement Corporation and the National Forestry Special Account to the general account in 1998 (see below), the debts of public institutions (outside the general government) represent a potential liability to the general government.

Financial and non-financial public institutions held about 550 and 25 trillion yen in gross assets in March 1998, respectively (Table 26). Together with the general government, they hold about 24 per cent of the nation's total assets, a record level and 9 percentage points higher than at the cyclical trough in 1989 (Figure 23 in Chapter II). While the financial institutions hold no net debt by definition, the net debt of the non-financial institutions amounted to 115 trillion yen (23 per cent of GDP). About 70 per cent of the debt of non-financial institutions was accounted for by central government-related institutions. The assumption of the above-mentioned debt by the general account of central

Figure 30. **Intra-governmental transfers**

A. Intra-governmental transfers (trillion yen, FY 1997)





Note: The arrows in the top panel show major transfers among three sub-sectors. The transfers not shown above are assumed to be negligible in the calculation.
Source: Economic Planning Agency and OECD.

Table 25. **Balance of social security funds by account**

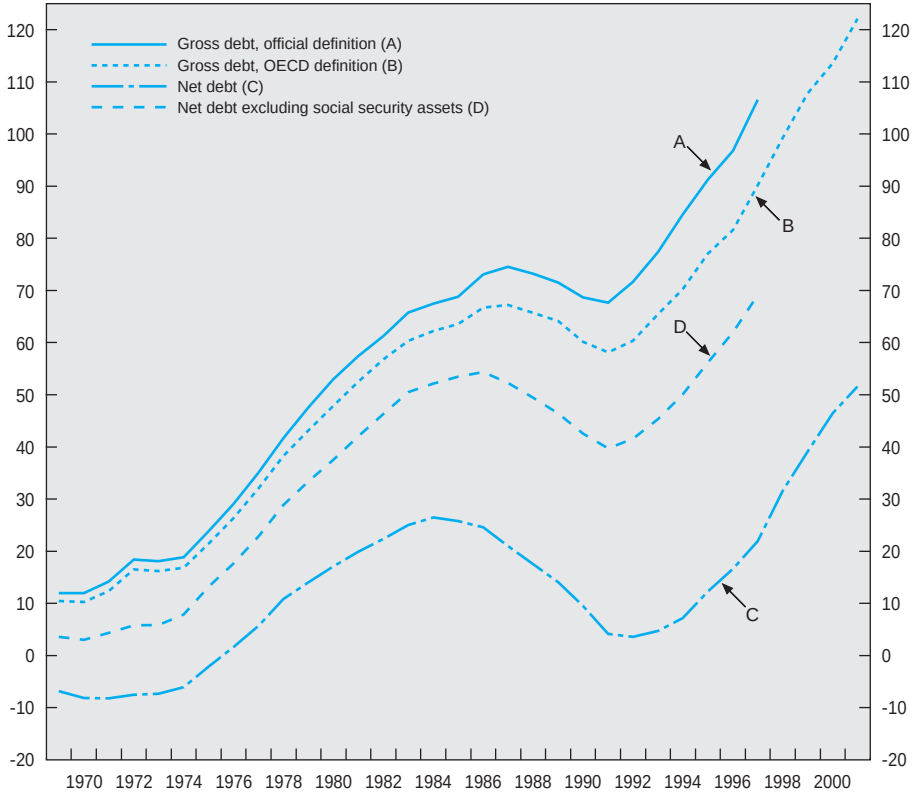
Trillion yen, fiscal year

	1970-74 average	1975-79 average	1980-84 average	1985-89 average	1990-94 average	1995	1996	1997
Pensions								
Contributions, total[1]	2.36	6.50	11.70	17.21	25.77	31.46	32.52	33.81
Benefits, total[1]	0.96	5.05	10.83	17.36	24.98	31.22	32.74	34.12
Balance[1]	1.39	1.46	0.88	–0.16	0.79	0.24	–0.22	–0.31
GDP ratio, per cent	1.37	0.77	0.32	–0.04	0.17	0.05	–0.04	–0.06
of which:								
Welfare pension	0.87	1.43	1.40	0.90	2.65	3.66	3.69	3.40
National pension	–0.08	–1.03	–1.71	–2.71	–4.05	–5.76	–6.30	–6.95
Mutual benefit associations	0.42	0.61	0.56	0.65	0.29	0.33	0.42	1.29
Pension funds	0.17	0.44	0.68	1.01	1.90	1.99	1.95	1.93
Health								
Contributions, total[2]	2.09	5.07	8.06	10.84	14.53	15.93	16.36	16.90
Benefits, total[2]	2.60	6.73	10.98	14.86	19.65	23.15	24.31	24.40
Balance[2]	–0.51	–1.66	–2.92	–4.02	–5.12	–7.21	–7.95	–7.49
GDP ratio, per cent	–0.50	–0.87	–1.06	–1.11	–1.10	–1.47	–1.58	–1.48
of which:								
Health insurance run by Central government	–0.11	–0.36	–0.23	0.59	1.05	1.05	1.03	1.42
National health insurance	–0.53	–1.49	–1.90	–1.50	–1.90	–2.36	–2.41	–2.25
Medical services for the elderly	0.00	0.00	–1.38	–4.68	–6.71	–8.46	–9.23	–9.65
Mutual benefits for associations	0.02	0.00	0.08	0.33	0.58	0.57	0.58	0.58
Health insurance run by private mutual associations	0.11	0.21	0.52	1.23	1.85	1.97	2.06	2.39
Employment[3]								
Contributions, total	0.35	0.57	0.87	1.18	1.37	1.25	1.27	1.30
Benefits, total	0.33	0.78	1.20	1.11	1.29	1.91	2.02	2.19
Balance	0.02	–0.21	–0.32	0.07	0.08	–0.65	–0.74	–0.89
GDP ratio, per cent	0.01	–0.11	–0.12	0.02	0.02	–0.13	–0.15	–0.18
Social Security Fund, total[4]								
Contributions	5.06	12.77	21.67	30.50	43.47	50.33	51.86	53.73
Benefits	4.12	13.21	23.95	34.39	47.04	57.42	60.23	61.87
Balance	0.94	–0.44	–2.28	–3.89	–3.57	–7.09	–8.37	–8.14
GDP ratio, per cent	0.92	–0.23	–0.83	–1.08	–0.77	–1.45	–1.66	–1.61
Memorandum item:								
Transfer from central and local government GDP ratio, per cent	1.12	2.03	2.44	2.60	2.50	2.74	2.79	2.86
Property income to GDP ratio, per cent	0.73	1.00	1.45	1.94	1.97	1.99	1.87	1.81

1. Includes pension account of Seamen's insurance.
2. Includes health insurance account of Seamen's insurance.
3. Sum of employment insurance account of labour insurance and unemployment account of Seamen's insurance.
4. Includes disaster and accident-related schemes and children's allowances.

Source: Economic Planning Agency.

Figure 31. **Various definitions of Japan's government debt**
FY 1969-2001,[1] per cent of GDP



1. The data for FY 1998 to FY 2001 are OECD estimates.
Source: Economic Plannig Agency and OECD.

government should mean a reduction of about a fifth in the total debts held by non-financial institutions, *ceteris paribus*. About 3 500 local public *enterprises* are managed by local governments in separate special accounts in their budget system and include businesses such as transportation (local buses, for example) and hospitals. Their debt of 2.3 trillion yen seems to be attributable to losses mainly from local transportation and hospitals.[117] About 1 700 other local public *corporations*, those for which data are available, have debts of about 13.6 trillion yen. They were established by local governments and are engaged in prior land acquisition for public works, social housing, toll-road construction, etc. They

Table 26. **Detailed balance sheet for general government and public institutions, March 1998**

Official presentation, trillion yen

	Central	Agencies	Local	Social security	General[1]	Public non-financial institutions	Public financial institutions
Gross assets							
Total	89.2	70.4	42.3	238.3	427.4	24.7	553.1
As per cent of GDP	17.7	13.9	8.4	47.2	84.6	4.9	109.5
of which:							
Cash and deposits	3.1	43.0	27.4	38.4	111.9	5.8	4.6
Short-term government securities	6.4	0.0	0.0	0.0	6.4[2]	0.0	2.5
Long-term bonds	0.1	6.9	0.0	16.6	23.6	0.3	185.3
Government	0.1	3.0	0.0	5.2	8.4	0.1	*125.0*[3]
Public corporations	0.0	0.9	0.0	6.3	7.3	0.0	*41.3*
Other	0.0	2.9	0.0	5.1	8.0	0.2	19.1
Corporate equities	0.5	0.0	0.1	0.1	0.8	1.6	0.6
Loans	12.1	14.4	9.1	7.2	40.5	2.5	335.2
Life insurance	0.0	3.9	0.0	18.2	22.1	0.0	0.0
Transfers from general government	4.1	0.0	0.0	0.0	4.1	0.0	0.0
Trade credit	0.0	0.0	0.0	0.0	0.0	11.8	0.0
Other	62.9	2.2	5.7	157.7	217.9[1]	2.7	24.9
Deposits with Trust Fund Bureau	9.9	0.8	0.0	150.4	*161.0*	*0.1*	0.0
Investments	26.1	0.0	5.5	3.7	24.9	0.1	1.2
Non-life insurance	0.0	0.0	0.2	0.0	0.2	0.0	0.0
Foreign trade credits	1.0	0.0	0.0	0.0	1.1	0.0	0.0
Foreign claims	0.8	0.3	0.0	1.9	2.9	0.0	23.6
Other	25.0	1.1	0.0	1.8	27.9	2.4	0.0
Gross liabilities							
Total	345.0	68.0	136.5	1.5	538.2	139.9	553.1
As per cent of GDP	68.3	13.5	27.0	0.3	106.6	27.7	109.5
of which:							
Deposits	0.0	0.0	0.0	0.0	0.0	0.0	241.7
Short-term government securities	36.2	0.0	0.0	0.0	36.2	1.0	0.0
Long-term bonds	271.7	0.6	43.4	0.0	315.7	55.9	26.3
Government	271.7	0.0	42.8	0.0	314.5	3.5	0.0
Public corporations	0.0	0.6	0.0	0.0	0.6	42.2	23.7
Others	0.0	0.0	0.6	0.0	0.6	10.1	2.6
Loans	31.1	56.3	92.9	1.5	179.4	69.4	1.9
From public sector	31.1	56.0	81.9	1.5	168.2	51.7	1.9
From private sector	0.0	0.3	11.0	0.0	11.2	17.7	0.0
Life insurance	0.0	0.0	0.0	0.0	0.0	0.0	106.5
Transfer from general government	0.0	0.0	0.0	0.0	0.0	0.4	3.8
Trade credit	0.0	0.0	0.2	0.0	0.2	0.0	0.0
Other	6.0	11.1	0.0	0.0	6.6	13.2	173.1
Deposits with Trust Fund Bureau	0.0	0.0	0.0	0.0	0.0	0.0	161.3
Investments	0.0	11.1	0.0	0.0	0.6	13.2	6.9
Non-life insurance	0.0	0.0	0.0	0.0	0.0	0.0	2.2
Foreign debts	1.1	0.0	0.0	0.0	1.1	0.0	0.0
Other	4.9	0.0	0.0	0.0	4.9	0.0	2.7
Net liabilities	255.8	–2.4	94.2	–236.8	110.8	115.1	0.0
As per cent of GDP	50.7	–0.5	18.7	–46.9	21.9	22.8	0.0

1. General government is not the sum of its components in all cases because of consolidation.
2. The numbers underlined are removed in the calculation of the consolidation of general government and public non-financial institutions.
3. The numbers underlined and italicised are removed in the calculation of the consolidation of general government and public financial and non-financial institutions.

Source: OECD calculations based on Economic Planning Agency.

have sometimes been reported to face difficulties selling developed land to the private sector: their reservation prices are often higher than the market prices owing to a decline during their development phase. However, the so-called "third sector" is not included in these local public corporation figures. Therefore, the consequences of their poor performance (see Box 3) are not included in this balance sheet exercise.

The public financial institutions sector includes various special accounts (such as postal savings) as well as other government financial institutions. A major concern pertains to the quality of those assets held by the latter. Although the overall non-performing loan ratio of government institutions was 0.97 per cent, slightly higher than the previous year but still lower than the corresponding figure of major private banks (4.04 per cent, in March 1999), valid worries remain for the following reasons. First, the ratios for SME-related institutions are fairly high: in particular, Japan Finance Corporation for Small Business's ratio increased by about ½ percentage point to 3.52 per cent. They could worsen further as a result of expansion in their lending as part of policy action to ease the credit crunch. Second, deterioration of Hokkaido-Tohoku Development Finance Corporation's asset quality is beginning to be visible, as a result of the poor performance of two large-scale projects, even though it still has a relatively low rate – 2.09 per cent.

The consolidation among general government and financial and non-financial government institutions is likely to give some ball-park estimate of maximum risk to the public sector. Total gross debt in March 1998 was 538 trillion yen or 107 per cent of GDP for general government according to EPA's official presentation. It would be raised to about 880 trillion yen (174 per cent of GDP) on a consolidated public-sector basis (adjusted for underlined items on Table 26) or, excluding financial institutions, to about 625 trillion yen (129 per cent of GDP). But even these numbers exclude a variety of contingent liabilities, some of which are well known and others less widely recognised. Public pension liabilities are no doubt the most prominent of these implicit debts, although the recent draft reform (see below) would diminish the value of that liability significantly. But the 1998-99 explosion of credit guarantees could ultimately add to the public sector's indebtedness. Finally, the capital injections into private banks through the Deposit Insurance Corporation might also deteriorate its balance sheet.

The medium-term fiscal challenge

The OECD projects that the general government deficit will amount to 7.9 per cent of GDP in the year 2000 (Table 24). This can be decomposed into interest payments of 1.4 per cent and a net primary balance of 6.5 per cent of GDP, respectively. About 1 percentage point of the net primary balance is estimated to be attributable to cyclical factors, while the rest, about 5½ per cent of GDP, is judged to be structural. It is important to get an early start on correcting

Box 3. Bankruptcies in the third sector

While "third sector" companies are not precisely defined in legal terms, they are generally considered to be local public corporations. According to the MHA, there were 9 344 local public corporations in March 1996, a figure which has been growing by more than 10 per cent per year for the last 30 years. About 55 per cent of these corporations were jointly established by the public and private sectors. About 55 per cent of these, *i.e.* about 30 per cent of all local corporations (about 2 800), are based on the Commercial Code. A private bankruptcy information service company (Tokyo Shoko Research) estimated the number of "third sector" companies in financial difficulty at about 100 in FY 1998, an average of one in eleven of those based on the Commercial Code, set up from 1986 to 1995 and engaging in land development, sightseeing, amusement parks, etc.[1] It should be noted that central government has encouraged "third sector" projects since the late 1980s.[2] According to the information company, the number of such corporations' bankruptcies reached 10 in 1997 and 25 in 1998 and is expected to increase further. Local governments are often heavily involved in these companies through taking the initiative in their formation and providing financial (capital and subsidies) and human resources. Banks and construction companies, general contractors in particular, are the most frequent private partners. Local governments have sometimes been forced to bear the entire costs of "third sector" corporations' bankruptcies; this has triggered lawsuits by residents in some cases. This raises two questions (Mitsuhashi *et al.*, 1999). The first is how can heavy involvement of local governments in corporations' businesses be justified when they seem unrelated to the public interest (such as running amusement facilities). Local governments (or governors) sometimes provide, in effect, loan guarantees on borrowing from banks. The second is why residents have to pay the bills. Disclosure of information pertaining to the operations of such enterprises is clearly insufficient, and close monitoring by local Diets cannot be conducted under the current legal framework. For example, unless the governments' share of the firm's equity is more than 25 per cent, the companies are subject to monitoring neither by local Diets nor by audit committees, apart from residents' disclosure requests.[3] Insufficient outside monitoring has led to ineffective discipline of management. In May 1999, the MHA provided local governments with guidelines as to set-up, responsibilities, disclosure and bailout of "third sector" companies in order to improve the situation. It remains an open question whether these guidelines will be effective in overcoming the serious problems in this sector.

1. A different private information service company (Teikoku Data Bank) reported 99 "third sector" companies were insolvent, 163 close to insolvent, 197 in stable condition and 385 lacking in disclosure out of its sample of 844 "third-sector" companies.
2. The law for utilising the private sector's know-how in public projects and the law for the promotion of resort development were implemented in 1986 and 1987, respectively, thereby encouraging local governments to undertake a "third-sector" approach.
3. As for companies that are more than 50 per cent owned by local governments, the governors have to make a report to the local Diets. Only those corporations with more than 25 per cent public ownership are subject to audit.

the situation, given that the gross (net) debt is projected to reach 114(45) per cent of GDP in 2000 (in gross terms almost equal to Italy's figure, the worst among OECD Member countries) and to continue to grow rapidly in the absence of correction. It seems reasonable to assume that the period of budgetary improvement should be over within a decade in order that the equilibrium debt ratio not be too burdensome. Therefore, the discussion here is based on assessment of what might be required by the year 2010[118] (Table 27).

Clearly, the elimination of so substantial a (structural) primary deficit will be challenging. The task will be even more exacting because of an expected increase in transfer payments as a result of the ageing of the population. As for pensions, the Ministry of Health and Welfare (MHW) has published a long-term projection of future benefits and contributions of National/Welfare Pension systems, assuming the revisions to the current schemes in the draft law (see below)

Table 27. **Needed fiscal correction by 2010: an illustration**

Per cent of GDP

Primary structural balance in CY 2000[1]		–5½	
Expected annual increase in gross health care spending (before reform)[2]		–2	
Expected annual increase in gross pension spending (including the effects of the draft 1999 reform)[3]		–1¼	
Necessary fiscal consolidation by 2010 (A)[4]		–8¾	
	Pessimistic assumptions	Central assumptions	Optimistic assumptions
Effects of possible correction measures (B)	1¼	5	8¾
Of which: Health care savings[5]	¼	1¼	2¼
Government investment reduction[6]	1	3	5
Public employment reduction[7]	0	¾	1½
Additional necessary consolidation (from tax reform, raising pension contributions, etc.) **(A – B)**	7½	3¾	0

1. OECD projection based on *Economic Outlook 66*.
2. Derived from a September 1997 Ministry of Health and Welfare projection.
3. Derived from a 1999 Ministry of Health and Welfare calculation.
4. Note that the primary structural balance and the additional consolidation required for debt stabilisation would exceed these figures unless real economic growth reaches the level of the average real interest rate on government debt, something which has generally not been the case since the 1970's. For example, if the growth rate were to fall short of the real interest rate by ½ percentage point, then stabilising net (gross) debt at 90 (150) per cent of GDP in 2010 would require a primary structural surplus of ½ (¾) per cent of GDP, rather than a zero balance. If the shortfall were 2½ percentage points, the required surplus would reach 2¼ (3¾) per cent of GDP.
6. The central assumption here is based on a reduction in health care outlays of 2 percentage points per year from the 5.6 per cent calculation made by the Ministry of Health and Welfare in September 1997.
7. The size of these reductions is suggested by international comparisons published in the previous OECD *Economic Survey of Japan*.
8. The magnitude of these savings follows from the discussion in the text.

Source: OECD.

are implemented. The Ministry's projection implies that the net (gross) increase in pension payments from general government might reach ¼ (1¼) percentage point of GDP. Health care costs will also be boosted by ageing effects. While a reform in the system is expected to be implemented from April 2000 (see below), there is as yet no agreement on its overall outline, much less its details, so that it is impossible to estimate to what extent the reform will affect the pace of future increases in expenditures. The only available calculation by the Ministry dates from September 1997 and is rather mechanical: nevertheless, it showed that the increase in national health care expenditure might amount to 23 trillion yen per year (including patients' self-payments and possible effects of the launch of long-term care insurance in April 2000) in current prices over the next decade. This implies that the cost of the increase borne by general government might be about 20 trillion yen, boosting the share of public health spending in GDP by around 2 percentage points. Overall, ageing effects on pension and health care costs could therefore raise primary deficits by about 2¼ (3¼ in gross terms) percentage points of GDP.

Thus, the predicted primary structural deficit in 2010, abstracting from changes in non-transfer spending and taxes/contributions is some 8¾ per cent of GDP. However, merely eliminating this deficit will not be sufficient unless real economic growth reaches the level of the average real interest rate on government debt, owing to debt dynamics. Because the real interest rate on debt is likely to be greater than the real GDP growth rate, as the experience of the last 20 years shows, the debt-to-GDP ratio would explode without a primary surplus. In order to stabilise gross debt at 150 per cent of GDP, the required primary surplus[119] to prevent the "snowballing" of interest payments could be as much as 3¾ per cent of GDP, assuming for illustrative purposes that potential growth is 1 per cent and the real interest rate 3.5 per cent. Alternatively, if the objective is to stabilise the net debt level at 90 per cent of GDP, the corresponding required primary surplus would be 2¼ per cent of GDP. These calculations suggest that the size of fiscal consolidation required only to prevent the debt snowballing would in any case be massive, possibly reaching double-digit figures as a per cent of GDP cumulatively over the coming decade (including the expected increases in social security contribution rates to finance the pension system, which are estimated to amount to around 1 per cent of GDP). It should be noted that the calculations of the size of the required consolidation are sensitive not only to assumptions regarding potential growth and interest rates but also to the level at which debt is eventually stabilised: thus, the later the process gets underway and the more gradual is the approach to debt stabilisation, the larger the required primary budget surplus would be.

How might such a huge cut be achieved? As for the social security system, there seems much more room for improvement in the area of health care than pensions. If the annual growth rate of national health care expenditure can be

reduced by 2 percentage points from the 5.6 per cent contained in the official projection because of the effects of the coming reform, it would improve the balance by 1¼ percentage points. Another promising way may be to cut public investment – see the previous Survey (OECD, 1998). If the ratio of public investment to GDP can be cut to a level comparable to those in France and the United States, about 3 per cent, it could save about 5 percentage points of GDP. Although public works are being reviewed to enhance their efficiency (see below), a welcome step in itself, the impact on spending seems likely to prove marginal, compared to the huge cut to be achieved. For example, the Ministry of Construction and other related ministries are trying to complete the review by the end of FY 1999 and reduce the cost of public works by at least 10 per cent as a result. The impact may be at most ¾ percentage point of GDP, even if local governments make similar efforts. As for government consumption, because the current level is already the lowest in G7 countries, there seems limited room for any significant cuts, with one exception. The government plans to cut the number of central government officials by a quarter over the next 10 years as part of the administrative reform efforts (see below). That would be worth only about ¼ percentage point of GDP. Given that the number of local government officials is four times as great as that of the central government, however, the maximum effect would be around 1¼ to 1½ percentage points of GDP, if local governments take similar measures. Nevertheless, with a smaller civil service, public services would likely have to be delivered using more non-labour inputs. Thus, it would be seem prudent not to count on savings of more than 1 percentage point of GDP from pruning the ranks of government employees.

Although asset sales do not directly affect structural deficits, they do reduce the equilibrium debt level. According to the MOF, as of March 1998 the central government held real assets worth 60 trillion yen: land, equipment and buildings accounted for 32, 15 and 9 trillion yen, respectively. However the government has estimated that only about 1.6 trillion yen (0.3 per cent of GDP) of the land, seemingly the most marketable asset in the portfolio, is easily disposable, because most is currently used for administrative activities. In September, the Ministry investigated and classified government land holdings. According to the published result, the marketable government land amounted to 2.45 trillion yen in total at the end of 1998.[120] As for financial assets, disposal of which does not change net debt, the government still holds about 59 per cent of shares of NTT (Nippon Telegraph and Telephone).[121] However, legal requirements permit it to dispose of only a further 26 per cent of the total (4.1 million shares), worth about 5.4 trillion yen (1 per cent of GDP) at recent prices. Hence, it seems reasonable to think the maximum revenue from asset sales may be 10 trillion yen, 2 per cent of GDP, which is quite tiny compared to the debt outstanding. On the other hand, as discussed earlier, there are risks that substantial amounts of debt outside general government could push up the debt level of general government, thereby also raising the required primary surplus.

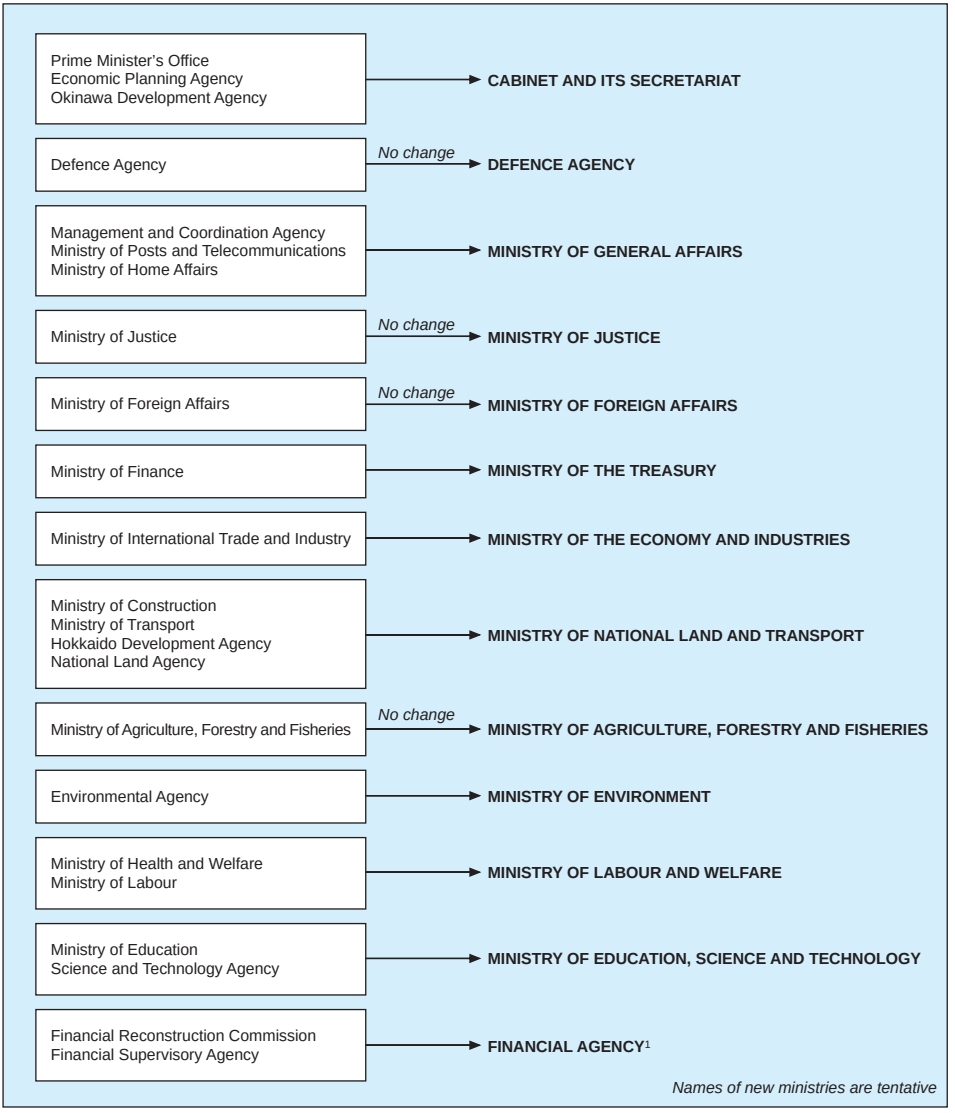
Summing up, even if the necessary fiscal correction may be as little as 8¾ per cent of GDP, the measures considered in the present exercise will probably be insufficient, saving some 5 per cent of GDP (with a range of uncertainty of 3¾ percentage points on each side). Although the wide margins reflect the rough nature of the estimate, it still gives important indications: the remaining gap is likely to be large, probably 3¾ per cent of GDP even without debt-dynamics considerations, and it would therefore seem unavoidable that tax revenues be raised by a substantial amount in order to restore the government's financial health (see Chapter IV); it also underlines the importance of stepped-up efforts to improve public sector efficiency, not least to minimise the risk of a greater gap needing to be filled by yet larger increases in tax revenues.

Public sector reforms

Reform of public administration and institutions

Many efforts are being made to enhance the transparency and efficiency of Japan's central government and to transfer more power to local governments. Following the principles outlined in the 1998 Administrative Reform Bill, in June 1999 the Lower House passed 17 bills to reorganise the 22 existing government ministries and agencies into ten ministries, two agencies and a new cabinet office (Figure 32). The government aims to move to the new system within no more than five years, and possibly as early as January 2001. As a result of the restructuring according to responsibilities, eleven Ministries and Agencies, whose responsibilities are considered as only supplementary or redundant will be reorganised into four Ministries and Agencies. Despite this appreciable effort, some of the overlapping responsibilities that have long been criticised for inhibiting efficiency still remain.[122] The future MOF will see all of its regulatory powers transferred to the future Financial Agency (to be formed in July 2000), except for a role in policy planning for bank failures and crisis management. An economic and fiscal policy advisory council, to be established under the new cabinet office, will survey and discuss general economic/fiscal management and basic budget policies. The authority of the Prime Minister to propose basic principles on national policies at cabinet meetings will be clearly stated in the Cabinet Law. Also a deputy-minister system will be introduced in the public service. As part of the reorganisation of the central administrative system, the government plans to grant corporate status to 89 state-run institutions (mostly research institutions) already earmarked for change and intends to transform these institutions into semi-autonomous agencies. The transformation of national universities is also under discussion. In order to trim the structure of the central government the number of bureaux in central ministries and agencies will be reduced from 128 to 96, the number of government advisory panels from 211 to 90 (and total panel seats to

Figure 32. **Central government reorganisation**
Scheduled for 2001



1. July 2000.
Source: *Nihon Keizai Shimbun* and OECD.

1 800 from the current 5 300[123]) and central government employees by 25 per cent in ten years.[124] But whether public-sector costs will actually be cut by corresponding amounts depends on difficult future decisions, as yet untaken.

Some measures have also been implemented this year with a view to slimming the size of the broader public sector outside the administration. Reorganisation of public institutions was put in place based on Cabinet decisions taken in 1997. Five government financial institutions and three non-financial institutions were reorganised into three banks and one corporation. Their responsibilities have been reallocated. For example, the Urban Development Corporation, a successor to the Housing and Urban Development Corporation, will leave the provision of public housing and housing land to the private sector. The Development Bank of Japan, a successor to the Japan Development Bank, will concentrate on loans for regional development projects and living-related infrastructure provision; it will gain the responsibility for lending in related fields from other public institutions, but should limit its involvement in loans to industry to a minimum: this would seem difficult at the moment because of the persistent problems in the banking sector. As a result of the reorganisation, FILP will provide funds to 48 government institutions, seven fewer than at present. As well, the Bank of Japan has made efforts to improve its management of business operations based on the principles of the new Bank of Japan Law. With regard to increasing the efficiency and transparency of the Bank's operations, it has taken measures in relation to disclosure of information, faithful fulfilment of their duties by the Bank's executives and staff, revision of its remuneration standards and efficient management of its property, which includes selling and scaling down the houses for general managers of its branches and disposing of all Bank-owned recreation facilities for its employees. Finally, a reform of the postal service is likely to proceed in two steps. In January 2001 the implementation section of the state-run operations of mail, postal savings and insurance services will be shifted to a "postal services agency", which is scheduled to be granted independent fund management authority from April of that year. Then in 2003 a postal public corporation, which will be allowed increased management freedom, will take over the "postal services agency".

The government has begun to reassess its role in the oil industry after last year's controversy regarding the use of public funds to subsidise unprofitable exploration companies.[125] The Ministry for International Trade and Industry (MITI), through the Japan National Oil Corporation (JNOC), subsidises exploration in an attempt to reduce reliance on oil produced by other countries. JNOC funds up to 70 per cent of companies' exploration costs,[126] whether the projects find oil or not. This policy has given companies little financial incentive to seek high returns on investments or assets and has created huge debts in the sector. MITI already decided in September 1998 to dissolve 27 loss-making oil exploration companies that receive financial support from the JNOC, in an effort to restructure

the government-affiliated corporation, which is saddled with massive bad loans. According to the plan presented by MITI's Agency of Natural Resources and Energy, liquidation of 13 of these oil exploration companies will start by FY 2000, while the other 14 have already announced that they will voluntarily cease operations. A 10-year programme to recover the losses from the liquidation of the 27 companies will be put into force by selling stocks of oil exploration companies with good business performance over the counter or by listing them on the stock exchange. Nonetheless, according to MITI estimates, even if restructuring measures are implemented, JNOC's cumulated losses would still be substantial unless the price of crude oil increases.

On the privatisation front Japan undertook one of the world's largest public share offerings of 1998: in December the offering of the fourth tranche of NTT, representing some 6.3 per cent of its outstanding shares, raised around $6.6 billion and reduced the government stake to 59 per cent. In July 1999, the government sold 48 000 shares to NTT pursuant to NTT's share repurchase plan. Another tranche is currently being readied for sale. Following the Cabinet decisions on the rationalisation of public enterprises taken in 1995 and 1997, the international telecommunications company KDD was privatised in 1998. Teito Rapid Transit Authority, which operates subway lines in the Tokyo metropolitan area and is jointly owned by the central and Tokyo metropolitan governments, will be corporatised by 2003 and privatised in 2007 at the earliest (after completing the programmed construction of new lines). The seven Japan Railways companies, already corporatised, will be privatised as soon as possible, and the generating company Electric Power Development Co. Ltd. will most likely be privatised in FY 2002.

Important steps have been taken to improve the transparency of government actions. In May 1999 the Diet enacted the Law Concerning Access to Information Held by Administrative Organs. The Law enables individuals and corporations, both at home and abroad, to request those in charge of central administrative bodies to disclose public administrative documents.[127] Following a request, the organisation should reply within 30 days if it will or will not disclose the documents. In case of refusal, the applicant can appeal to a nine-member committee set up within the Prime Minister's office; if still unsuccessful, the applicant can institute a lawsuit in one of Japan's eight district courts. The law will not be applied directly to public enterprises: for them a separate scheme is believed to be necessary. In parallel the government established, effective 1 April 1999, a formal procedure inviting public comments when ministries and agencies adopt new regulations or modify old ones. This should serve to enhance the average quality of bureaucratic oversight.

Although doubts have been cast on the efficiency of public expenditures, especially public works, some progress has been made in raising quality levels. First, project evaluation has been improved, following the Prime Minister's

directive in December 1997 to: *a)* introduce a re-evaluation scheme for public works already under way and *b)* adopt cost-benefit analysis in initial decision-making. As a result, relevant ministries reviewed those projects still having distant completion dates, heard opinions from third parties and ultimately decided in March 1999 to stop 26 projects.[128] Cost-benefit analysis has been used in virtually all areas as one element in the decision of whether or not to adopt a project, and the results have been disclosed. Furthermore, a more integrated approach to each area is being tried: that is, whereas up to now the analysis has not been applied to the question of a change in resources between areas (roads, sewage, etc.), just to the relative assessment of different projects of the same type, more global trade-offs are now to be considered. Second, the government succeeded in placing priorities on some projects in the budget planning process, thanks to more severe resource constraints and special allocations of funds to some projects. For example, in FY 1999 the notoriously rigid allocation of public works funding among ministries changed by 2.3 percentage points, which is still marginal, but greater than the 0.4 percentage point two years ago. Finally, a law to introduce a Private Finance Initiative scheme passed the Diet in July. Public financial supports such as investment and loan guarantees initially included in the draft law were eliminated during the Diet's deliberation, thereby limiting the risk of moral hazard. The government is trying to avoid repeating the mistakes observed in the "third sector" by making detailed contracts and clarifying risk-sharing between the government and the private sector.

As for the government procurement market, following a proposal by the Central Construction Industry Council of February 1998, the Ministry of Construction is to introduce an evaluation scheme to assess public works projects based not only on the bid price, but also on the quality of the proposed project. The new scheme may be introduced as early as FY 2000, although the Ministry launched a pilot scheme earlier this year. Should the test prove successful, the Ministry plans to start a full-scale follow-up evaluation scheme in FY 2000. The results of the new evaluation scheme are expected to help the Ministry improve its management of public works projects and cut costs. The MHA also decided in February 1999 to let municipalities adopt a comprehensive evaluation method that considers not only price but quality and technology standards as well. Central government ministries and agencies already use this method for such items as computers, electronic-communication gear and medical equipment. The administrative reform plan will also introduce a new system to evaluate the administrative policies of each ministry and agency. Under the new system each ministry would evaluate and review the newly adopted projects, the progress of public works and any reasons for delay. Ministries would have the right to stop projects if they are found to be unnecessary following a formal evaluation process, and evaluation results would be disclosed to the public. In addition to individual ministries, the

future Ministry of General Affairs is to evaluate and review each ministry's policies to ensure strict objectivity of evaluation, for example by using third-party reviewers.

The relevant ministries and institutions have started to prepare for the FILP reform scheduled for 2001. A major change will be the abolition of the obligation for the Postal Savings Special Account and the Welfare and National Insurance Special Accounts to deposit their funds with the TFB. Because their deposits account for about 40 per cent of total available resources (Table 21), this reform should ultimately have a significant impact on the system,[129] although it is expected to take more than 10 years until the effects are fully felt. Roughly speaking, the market will replace these special accounts as sources of funds to the TFB and therefore play an important role in allocating funds from these special accounts to government institutions. Therefore, the relevant parties are studying how to manage their funds in the market, on the one hand, and how to finance their business from the market, on the other. Regarding the former, all the requisite legislation to ensure adequate portfolio management is expected to be approved in the coming months. The new legislation will permit a wider range of investment instruments. As for the latter, those institutions that need to raise more than a certain amount from the market will be able to issue their own bonds (FILP institution bonds), while other institutions might rely on FILP bonds issued by the TFB because they would otherwise face high liquidity premia. The FILP institution bonds may be a kind of asset-backed security (ABS), the return on which would depend only on the projects in which the institutions are engaged. For example, the Housing Loan Corporation (HLC) is studying how to structure such bonds based on their loan assets[130] and planning to use them as a supplementary source of funds to borrowing from the TFB as from FY 2000. The issuance of such securities by HLC may facilitate the development of ABS markets. That might then help banks to dispose of their non-performing loans[131] and/or increase their capital ratios, through reducing their loans outstanding,[132] and evaluation methods for real estate to converge to a present discounted value method, which may contribute to increasing liquidity in the real estate market. Another initiative included in the reform is to introduce net present value analysis of the costs of proposed policies, dubbed Policy (Subsidy) Cost Analysis. The first such estimates have just been released on a trial basis. Activities of five institutions were chosen as a sample from many diversified FILP-related institutions. These are mainly financial institutions: the Housing Loan Corporation, People's Finance Corporation, Export-Import Bank, Japan Road Corporation and Chubu International Airport Corporation Ltd, which accounted for about a third of the total outlays of the initial FY 1999 FILP. The estimates cover the entire period until these institutions' projects are completed, from 23 to 43 years, depending on the institutions. The total cumulative future liability was estimated to amount to about 4.9 trillion yen on a discounted basis. This is more than six times larger than

the annual operating subsidies they received in the initial FY 1999 budget. The net present value analysis should help the Diet to judge the appropriateness of schemes that require little immediate cash outlays, but larger commitments in the future.

Reform of local government financing

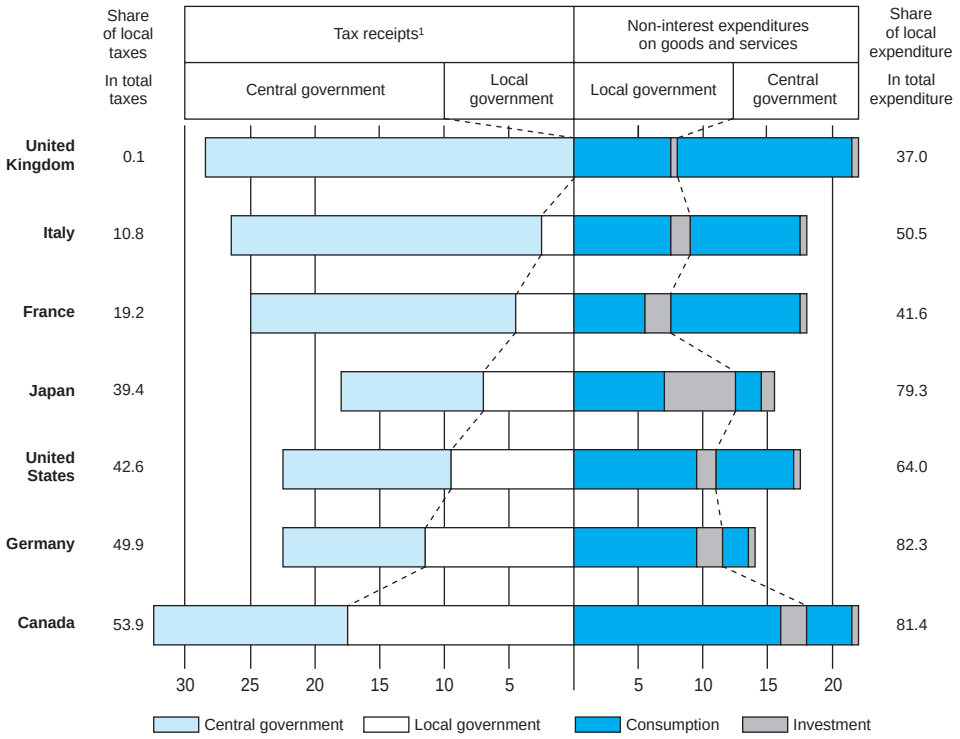
Even though Japan has a fairly unitary system of government, local governments' activities have a large presence in the economy: their revenues and expenditures are comparable to those of countries with federal systems such as the United States, Canada and Germany (Figure 33). Because local tax revenues are insufficient to finance local governments' expenditures, transfers from central to local governments are of great importance. A similar situation exists in other large countries with unitary systems, especially the United Kingdom and Italy. An overview of inter-governmental relationships in Japan is presented in Figure 34, showing various flows from the central government. According to the FY 1999 Local Public Finance Programme, local taxes are expected to account for only about 40 per cent of total local revenues. Transfers from the central government consist mainly of a lump-sum grant (*chiho kofuzei kofu-kin*) and so-called matching grants (*kokko shishutsu-kin*). The former could be described as a revenue-sharing scheme between central and local governments rather than a *tax*, despite its Japanese name (*zei*). However, it will henceforth be called local allocation tax (hereafter, LAT), following the traditional usage. While LAT is an untied transfer, *kokko shishutsu-kin* are earmarked for funding particular expenditures.

The current local tax system should also be reviewed from the viewpoint of promotion of decentralisation, one of the most important current policy objectives in Japan (see below). Local governments have few degrees of freedom in setting local taxes. The framework is stipulated in a national law.[133] Local governments have limited discretion with respect to their own tax rates: standard rates are set in the national law. They are allowed to levy their own taxes not stipulated in the law but only with the agreement of the MHA. Accordingly, there are few local authorities that take such measures, except for taxes on corporations.[134] Therefore, local tax revenues are almost an exogenous variable in their budget decisions.

Because inter-governmental transfers are made explicitly or implicitly for the purpose of income redistribution, a significant disparity arises in costs and benefits among residents, depending on their locations. The ratio of benefits to costs, which is measured by regional incidence of the ratio of spending over tax payments, is estimated to reach more than 2.5 in some prefectures (Hayashi *et al.*, 1997) (Figure 35).[135] This implies a possibility that the current system may allow "poor" regions to spend financial resources inefficiently because an increase in expenditure does not always mean an increase in tax liability.[136] While this could

Figure 33. **Local and central government shares in receipts and expenditures**

Per cent of GDP, 1997 or most recent year

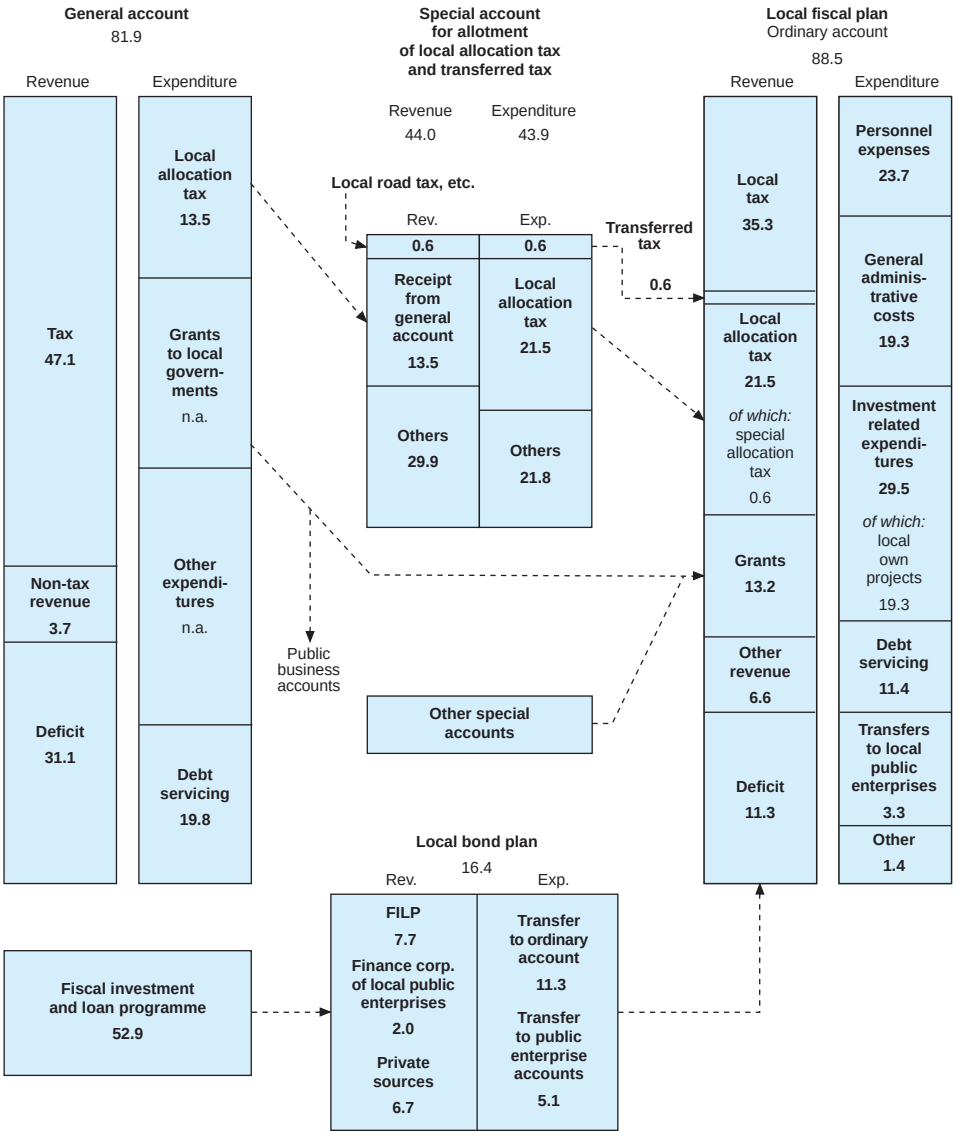


1. Indirect and direct taxes, excluding social security contributions.
Source: Economic Planning Agency and OECD, *Annual National Accounts.*

be regarded as a cost of "solidarity," the current system needs to be re-examined, considering the severe government financial situation and the principles of local autonomy.

While LAT is intended to ensure that each local government can provide uniform public services (such as education, welfare and national land development, which they are obliged to provide, with legally established standards set by the central government), regardless of differing financial capacities among them, there are several problems with it. The first problem concerns local

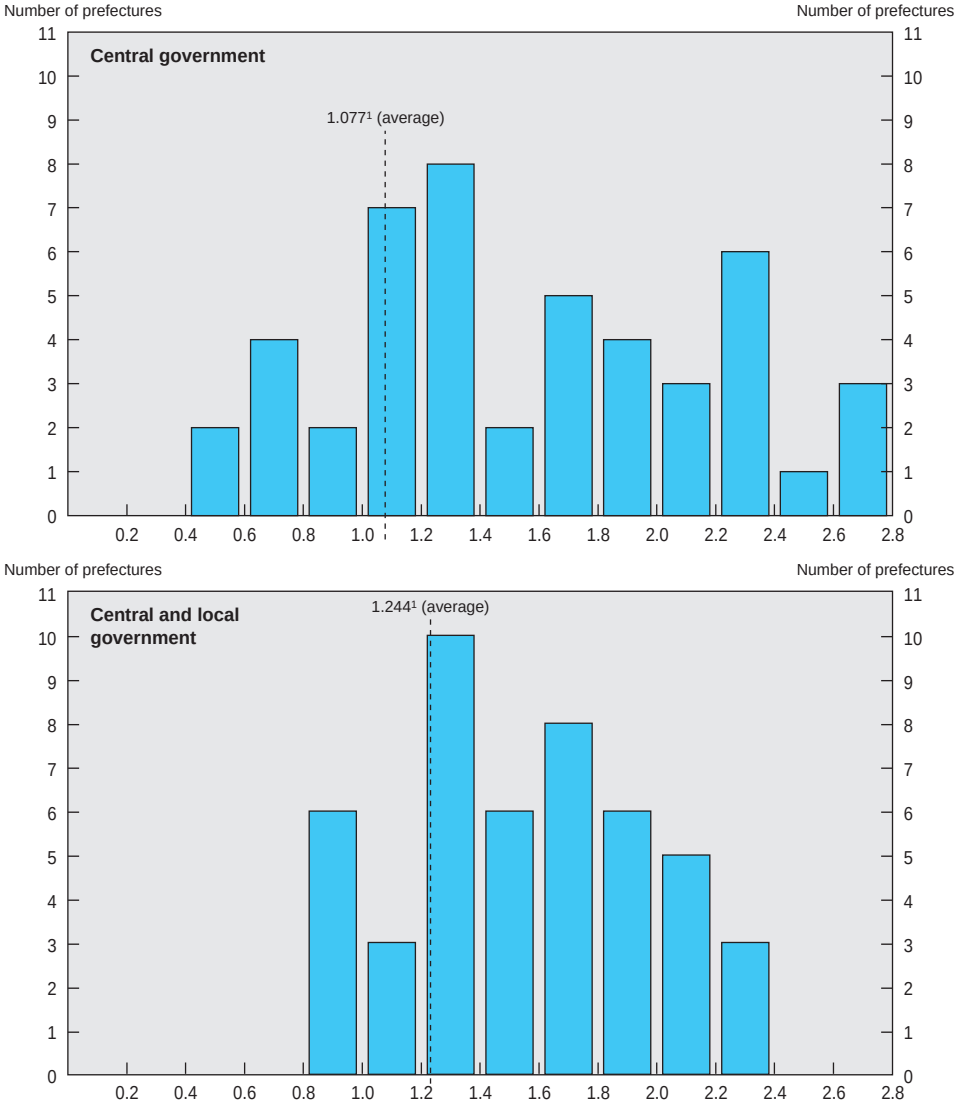
Figure 34. **Inter-governmental financial transfers to local governments**
FY 1999 initial budget, trillion yen



Source: Ministry of Finance and Ministry of Home Affairs.

Figure 35. **Regional distribution of costs and benefits of government activities**

Ratio of benefits to costs, FY 1992



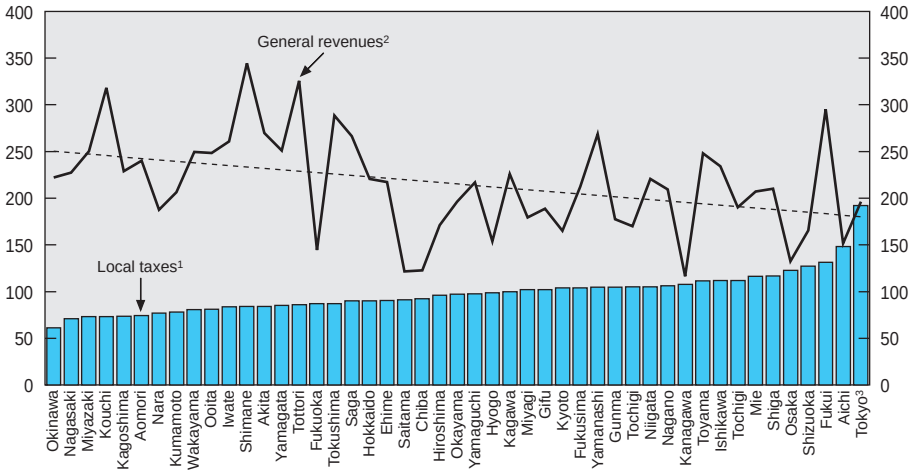
1. Ideally, the average should be equal to one. However, because some revenues (such as property income) are not taken into account, the average ratio is greater than one.

Source: Hayashi *et al.* (1997).

governments' tax-raising incentives. Comparing the situations before and after distribution of LAT (Figure 36) shows LAT's possibly too strong equalising effect: many "poor" prefectures become wealthier, in the sense that they are guaranteed more financial resources on a per-capita basis than others, than rich prefectures after receiving LAT. This, along with small "reservation ratios"[137] (see Annex I), is likely to reduce local governments' incentives to raise tax revenues. Second, LAT is likely to reduce the "tax price" of spending (the marginal tax increase due to an increase in public expenditure) significantly, thereby providing a strong incentive to spend. Suppose a local government undertakes a spending project supported by a grant from the central government. Although the local authority will have to pay a part of the cost not covered by the grant, the burden will be reduced in two ways: the authority's share is directly included in the calculation of standard fiscal needs; and a part of the amount financed by bond issues is also counted in the calculation. Marginal costs to local governments are therefore greatly reduced. Third, the targeted level of financial resources, which is decided by the Diet every year, may be well above the national minimum. In recent years, only Tokyo out of

Figure 36. **Effects of the local allocation tax**

Thousand yen per capita, FY 1997



1. Excludes transfers from prefectural to municipal governments.
2. Includes local allocation and local transferred taxes.
3. Adjusted for transfers between Tokyo metropolitan government and ward governments.

Source: Ministry of Home Affairs.

all 47 prefectures has been judged to be wealthy enough not to receive LAT by the MHA. This can be questioned in view of the fact that the total amount of local tax revenue in Japan is fairly large.[138] Finally, LAT seems difficult to justify if spillover effects of local public goods and mobility of residents are taken into account (Ihori, 1996). For example, low income does not always imply low utility if the residents can free ride on the benefits of local pubic goods provided by neighbouring authorities. Many municipal authorities seem too small to ignore these factors.

One way to promote decentralisation would be to transfer part of the tax base from central to local governments and to limit financial transfers between the two to a minimum. Such decisions should take into account a set of broad considerations, such as the reassignment of roles of government at different levels as well as the uneven geographical distribution of revenue sources. The resulting pattern of local governments' revenues and expenditures would reduce disparities in the benefit/cost ratio and encourage each local government to raise and use tax revenues more efficiently. In particular, as for LAT, the targeted fiscal needs should be reduced, and the "reservation ratio" should be raised, coupled with the reduction of national government obligations to local governments. More freedom should be allowed local governments to set their own tax schedules. These measures all together are expected to facilitate inter-jurisdictional competition through "voting with one's feet" and/or peer pressure (yardstick competition),[139] which is currently almost non-existent in Japan. At the same time, mergers of municipalities are desirable in order to achieve a sufficient scale of administrative capacities, a goal promoted by the MHA. The Ministry plans to simplify the calculation of LAT, following decentralisation, and improve its transparency, as promised in the Decentralisation Plan (see below). These are desirable first steps, which might encourage local governments to engage in "administrative innovation": the current approach probably results in excessively uniform public service delivery (Kaizuka, 1995).

Progress in decentralisation

Decentralisation has gained momentum since 1995. In response to a series of recommendations made by the Decentralisation Promotion Committee, the government introduced the Decentralisation Promotion Plan in May 1998, and a new law for this purpose was submitted to the Diet. The law intends to reallocate administrative responsibilities between central and local governments and strengthen local governments' managerial capacities through mergers of municipalities. The reallocation is to be conducted while reducing central government's interventions. A new arms-length rule is to be set up between central and local governments, and monitoring by local Diets is to be strengthened. For example, under the system of administration by proxy (*kikan inin jimu*), a local governor is

regarded as part of the central government, and administrative work is processed under the control of the central government, so that monitoring by a local Diet and local auditor committees is limited. This system would be abolished by the law and all administrative work classified either as local governments' own work (*jichi jimu*) or legally entrusted work from the central government (*hotei jutaku jimu*). Administrative responsibilities such as urban planning are to be left to local governments.

As for local public finance, in July 1997 the Decentralisation Promotion Committee recommended rationalising matching grants, improving management and securing local governments' own revenues. Because responses to the third recommendation were treated as only a medium-term goal, the draft law incorporates only the first two measures. First, local governments are to enjoy more freedom in issuing bonds, which used to require the permission of the MHA in the case of prefectures and that of a prefecture governor in the case of municipalities.[140] Second, local governments are to be allowed to institute their own taxes without the permission of the Minister. However, prior consultation with the Minister will still be necessary in both cases, and reaching agreement with the Minister will also be required concerning local governments' own taxes. The Committee also reviewed public works and other expenditures. Although the government decided the second Decentralisation Plan in March 1999, the effects are yet to be seen because most of the measures are to be implemented in FY 2000 after examination by the relevant ministries in FY 1999.

Mergers of municipalities took place frequently from 1953 to 1961 (Table 28). The mergers were driven by an increase in the financial burden of junior high schools and the heath insurance system weighing on municipal governments; they were also encouraged by the central government.[141] The number has remained almost constant since then, while progress in transportation and communication technology has made jurisdictional boundaries progressively less meaningful, especially in land use and urban planning. The government's stance

Table 28. **The number of municipal governments**

	Cities	Towns	Villages	Total
October 1945	205	1 797	8 518	10 520
April 1955	488	1 833	2 885	5 206
April 1965	560	2 005	827	3 392
April 1975	643	1 974	640	3 257
April 1985	651	2 001	601	3 253
April 1995	663	1 994	577	3 234
April 1999	671	1 990	568	3 229

Source: Ministry of Home Affairs.

changed toward favouring mergers once again in the mid-1990s. The Special Law for Municipal Mergers was revised in 1995 so that residents could take the initiative in proposing mergers. As a result, 38 proposals were made by residents in the following two years, leading to formal consultations in three cases, though no mergers have occurred yet. Perhaps that should be no surprise: as Kobayashi (1998) argued, governors' leadership and preparatory work on an informal basis were important in past mergers.[142]

The MHA is planning to give a further pro-merger impetus to municipal governments, in order to promote decentralisation and to strengthen their financial capacities to deal with an expected increase in administrative costs caused by the ageing of the population. In particular, the introduction of the long-term care insurance scheme in April 2000 is likely to affect these capacities. The Special Law was revised once again in the last Diet session to introduce an initiative by a prefectural governor and to expand financial assistance. The Ministry is trying to use the new scheme to encourage mergers and set the minimum population of merged municipalities at 10 000 in its guideline. This could have a significant impact because municipalities smaller than this account for about 47 per cent of the total. The Ministry expects prefectures to draw up concrete plans about possible mergers in their own jurisdiction by the end of the year 2000 and the mergers to take place by March 2005, the termination of the law.

Public pension reform

Given the uncertain prospects for economic growth and the rapid ageing of the population already underway, the problem of pensions is serious. In FY 1998 the average birth rate dropped to an all time low of 1.38 per woman. Japan's labour force has been shrinking steadily since June 1998, and the working-age population (aged 15-64) has been dwindling for the last three years. By 2020 the working-age population is expected to have fallen by 4 million (around 6 per cent),[143] with the ratio of workers to the total population at full employment contracting at a rate of 0.5 per cent per year. The effects of this shrinkage on per-capita GDP will be considerable. In such a scenario, if no measures are taken, future generations would have to bear 2.7 to 4.4 times as great a fiscal burden as do present generations (Takayama *et al.*, 1998). The opaque future prospects for pensions may be an underlying factor behind weak private consumption – see Chapter I. A reform of the pension system is therefore both urgent and crucial.

The government-managed Welfare Pension scheme for employees will likely see a funding shortfall in FY 1999 for the first time since the programme was set up in 1942. The MHW estimates that contributions will fall short of projected benefit obligations by some 1.5 trillion yen. Contributions, including government payments, are likely to decline some 600 billion yen from the estimated FY 1998 figure to 24.6 trillion yen and outlays to swell by some 1 trillion yen, to

26.1 trillion yen. The government had been expecting shortfalls to be recorded beginning only in two years time. However, in March 1999 the government decided to suspend the planned premium increase[144] for FY 1999 for an unspecified period because of the severity of the recession.

In July 1999 an important step toward pension reform was taken when a draft Pension Reform Law was presented to the Diet. It features lower contributions for both National and Welfare Pension schemes (Box 4) than would have been required under the current law. According to actuarial calculations by MHW, contributions to the National Pension scheme would remain at 13 300 yen until FY 2004 and then be revised every year with a view to reaching 24 800 yen in FY 2025 (in FY 1999 prices). Contributions to the Welfare Pension scheme, currently 27.6 per cent of wages, would be fixed until FY 2004 and then revised every five years in order to reach 25.2 per cent in FY 2025.[145] The draft law includes a 5 per cent cut starting from FY 2000 in the part of pension benefits that is proportional to the individual's wage level (the system's second tier), an increase in the eligibility age for that pension from 60 to 65 over a twelve-year period beginning in FY 2013 and indexation of benefits to prices only, but not to wages. The government also plans to require salaried workers aged 65-69 to pay pension premiums if they earn more than a certain amount of income. Overall, these measures, however painful, are necessary to assure the long-term viability of the system. In addition, the draft law would require government contributions to the basic part of pension schemes (the first tier) to rise from one-third at present to one-half by FY 2004, provided that a steady revenue source is ensured. According to an estimate by the MHW, this would necessitate additional revenues of 2.7 trillion yen in FY 2004 (½ per cent of GDP) and 3.7 trillion yen in FY 2005.[146]

Box 4. The pension system in Japan and its recent reforms

As described in more detail in Table A1, the public pension system is a defined-benefit plan consisting of two tiers. The first tier is a basic flat rate pension called the National Pension Scheme, covering all residents (70 million persons). The second tier is an earnings-related benefit scheme called the Welfare Pension scheme, covering employees (33 million persons). Separate tier-II schemes are available to central and local government officials, private-school employees, etc. (6 million persons), while no tier-II scheme covers the self-employed and housewives (31 million persons). The government funds one-third of the total cost of the tier-I scheme, while there is no public subsidy for the tier-II schemes. As tier-III schemes, corporate pension schemes called Welfare Pension Funds are available to some employees (12 million persons). The self-employed can voluntarily join what are called National Pension Funds to get additional benefits; this opportunity is presently taken up by 0.9 million people, about 5 per cent of the total population.

In December 1998 the Economic Strategy Council, an advisory body to the Prime Minister, proposed the privatisation over a 30-year transition period of the second-tier part of the pension system. Workers would all continue to be assured a basic pension, but would be expected to divert extra payments to private pension schemes to provide additional income in their old age. MHW opposed this proposal on three grounds: that employees would suffer a double burden in the transition phase for which social consensus would be difficult to achieve (even though all solutions to problems resulting from such demographic transitions, as well as the current system, imply such a double burden on some people unless full funding is in place) (Hatta and Oguchi, 1999); that workers would no longer be protected against inflation risks (but such protection is also possible by investing in inflation-indexed bonds); and, especially, that employees of SMEs, who generally do not benefit from any third-tier company pension scheme, might find themselves with nothing but the basic national pension (however, supplementary pensions could be made compulsory).

Health care reform

Total health care expenditure has grown steadily in recent years: over the period from FY 1990 to FY 1997 it rose by an average of 5.0 per cent per year, reaching 29.1 trillion yen (5.9 per cent of GDP; in per-capita terms that represented 230 400 yen, 38.2 per cent more than in FY 1990[147]). The growth in spending on health care appears to have been reduced by the increase in medical cost-sharing by patients beginning in September 1997:[148] in fact, the rate of growth on health care spending fell from 4.2 per cent in the year ending August 1997 to only 0.7 per cent in the following 12-month period. Despite this, the outlook remains of concern, mostly because of the population's ageing, but also because of the adverse incentive effects of several aspects of the system, fee-for-service payments and excessively low co-payments in particular. In FY 2000 there will be 21.8 million residents over 65 years old (17.2 per cent of the population), while in FY 1999 corporate health insurance schemes are expected to spend 32.4 per cent of their total premium income on care for the elderly. In FY 2025 the number of elderly is expected to reach 33.1 million (27.4 per cent of the population).

The authorities intend to implement a reform of the health care system in April 2000. The four areas covered by the reform, as indicated by the Council on Health Insurance and Welfare, are the following:

- doctors' fees;
- pharmaceutical pricing;
- provision of medical services; and
- financing the care of the elderly.

Doctors' fees are on average quite low, creating incentives to earn money by excessive numbers of "acts" and medical prescriptions. The fee-for-service system is believed to be the basic cause of excessive medical expenditure, and budget constraint measures are under study. A diagnostic-related group (DRG) system is also under consideration for hospital care. Under these arrangements hospitals would receive a standardised payment for treatment of illnesses falling into a specific group. Such systems encourage hospitals to minimise costs of treatment and, in particular, limit the excessive use of diagnostic testing and drugs. However, they need to be combined with hospital financing arrangements that force hospitals to take the financial risk of cost over-runs. And ambulatory care also suffers from similar problems. The recently adopted rule of adding an extra charge for drugs according to the number prescribed was lifted for the elderly in July 1999 under pressure from the Japan Medical Association (JMA). As well, the government proposal to implement a reference price system for medicine based on the German model has been withdrawn in the face of opposition by the JMA – doctors sell medicines directly to patients[149] in contrast to practice in most other countries – and no alternative proposal has as yet received official support. A new pricing system might help to overcome some important problems of the drugs market, such as their very high share in total health care expenditures. Another separate problem, however, is that importation and development of innovative products are discouraged,[150] while there is overproduction of new medicines only marginally better than those already available but sold at excessively high prices. With regard to the provision of medical services, the Council suggested distinguishing between acute and chronic diseases (which should result in an increase in specialisation of treatments and hence improved efficiency) and disclosing medical information to patients. Restructuring the insurance scheme for the elderly is the final major issue. At present, some 30 per cent of the medical costs incurred by the elderly is financed by taxes, while the remaining 70 per cent is covered by the contributions of employees' insurance schemes in addition to the small co-payment made by elderly patients.[151] The Council proposed that the elderly should shoulder at least 10 per cent of their health-care costs through increased co-payments. Two plans to reform the scheme are under examination. One is to finance elderly health care only by taxes, separating them from the insurance scheme. The other would basically maintain the *status quo* and retain them in the insurance scheme. The former is probably preferable in that it avoids the complex transfers between schemes currently required.

A long-term care insurance programme will be implemented in April 2000. The programme is expected to reduce significantly the medical costs for the elderly, who at present are often hospitalised in acute care beds even when they do not need a specific medical treatment.[152] The MHW estimates that about 2.8 million people (2.2 per cent of the population and 12.8 per cent of the over

65 population) will need long-term care in FY 2000 and that the number will reach 5.2 million in FY 2025 (4.3 per cent of the population and 15.7 per cent of the elderly). The new insurance scheme will be managed by local governments and had been expected to be financed as follows: users are expected to cover 10 per cent of total costs; premiums paid by people who are 40 or older will contribute 50 per cent of the estimated funding of 4 trillion yen per year, the central government will contribute 25 per cent, while government at the municipal and prefectural levels will divide the remaining 25 per cent equally between them. However, in October the ruling coalition decided to freeze such premium payments, at least for those 65 and over, for a period of up to one year. This was later modified to shorten the freeze to six months but to offer a 50 per cent reduction for a further year and to ease the burden on the 40-64 age group by offering assistance to their corporate health insurance societies. They also decided to make cash payments of up to 100 000 yen ($1 000) per year available to family members caring for the elderly and to lower the co-payment from 10 to 3 per cent of costs for low-income individuals. Under the previously decided scheme, premiums would vary according to the geographic area, being greater in western prefectures (where there are more institutions with elderly people receiving long-term medical care) than in eastern ones. The national average monthly premium, as calculated by MHW, was expected to be 2 885 yen ($25) but with a range of 1 400 to 6 200 yen; earlier estimates were for an average of only 2 500 yen per month. In order to narrow payment disparities between municipal governments, one-fifth of the amount covered by the central government would be distributed among local governments based on the ratio of people 75 years and over to people 65 years and over, and the ratio of elderly people with low incomes to total population of elderly people. A person will be eligible for long-term care only once approved by a panel of doctors, nurses and professional caregivers. The launching of this new scheme will open up opportunities for business and employment in this and related areas.[153] However, while shifting elderly patients into long-term care beds will produce savings, the most cost-effective strategy is to keep the elderly independent for as long as possible, and the new programme will also provide in-home support for those that require only minimal assistance. Nonetheless, developing home care will be a challenge, especially in rural areas.

Finally, a reduction in the number of active workers and a trend increase in their average age in a number of declining industries is contributing to a deterioration in the finances of many corporate health insurance societies. More than half of the some 1 800 societies nation-wide that have their own health insurance have deficits. To this point MHW has decided whether or not to allow such societies to disband on a case-by-case basis. It is now seeking to clarify the standards to be applied to such approvals.

IV. The tax system: a need for comprehensive reform

Unlike at the time of the expansion of the welfare system during the 1970s, Japan cannot expect to rely on high growth rates and favourable demographic developments to finance the projected substantial increase in social security costs in coming decades. Moreover, public debt can be used to smooth taxes over time only to a limited extent, and this room for manoeuvre has already largely been used up. Notwithstanding the ample scope for improving the efficiency of public expenditure, in particular in the areas of public works and administration, Japan thus faces a substantial public finance problem. Rather than being revenue-neutral, any change to the tax system must therefore aim at increasing revenues considerably. To obtain this at minimum costs in terms of distortions to economic activity and without being in conflict with distributional objectives will require improvements in both the efficiency and equity of the tax system.[154] The current fiscal situation of high deficits and increasing government debt adds to the need for fiscal consolidation, while the economic recession has made the starting point for reform more difficult. On the other hand, tax pressure (that is, the ratio of taxes to GDP) in Japan is lower than in most other OECD countries (Figure 37), and the severity of the fiscal situation is by itself promoting broad public acceptance of the need to increase taxes.

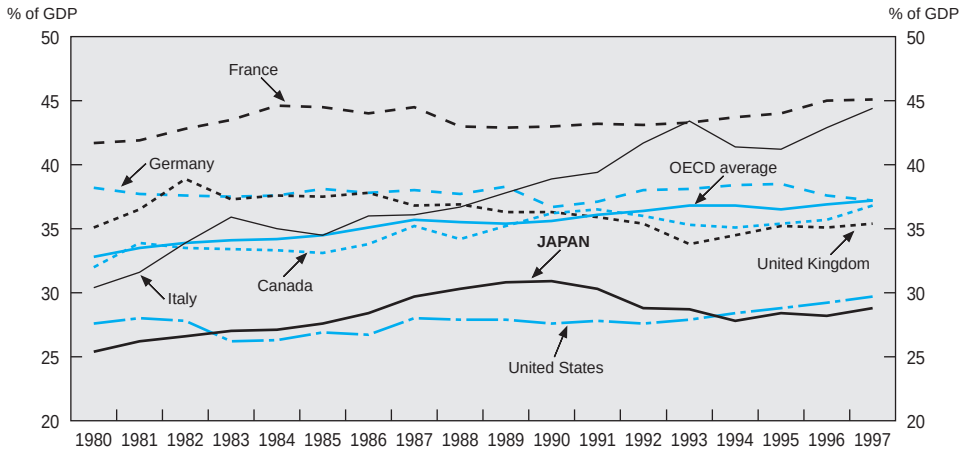
This chapter first describes the major forces shaping tax policy in Japan, then outlines the main statutory and economic features of the tax system as it stands today. Several options for strengthening the system are then discussed, and a number of specific policy recommendations are outlined in a concluding section.

Considerations influencing tax policy

The costs of ageing require significant tax hikes over the next decade

Dependency ratios[155] in Japan are projected to increase much more rapidly than in any other major country between now and 2025 (Table 29).

Figure 37. **Total tax revenues**[1]



1. General government revenues, including social security contributions. Note that numbers may not be fully comparable across countries (see Box 7) and over time due to changes of national accounts from SNA68/ESA79 to SNA93/ESA95. See OECD, *Revenue Statistics 1999* for more details.

Source: OECD, *Revenue Statistics.*

Table 29. **Dependency ratios**

	United States	Japan	Germany	France	Italy	United Kingdom	Canada
Elderly[1]							
2000 level	21	27	25	28	28	27	20
2000-2005 change	0	4	3	0	3	0	0
2000-2025 change	12	20	11	13	15	9	16
Total[2]							
2000 level	69	60	59	70	59	69	63
2000-2005 change	–3	3	2	–2	1	–1	–2
2000-2025 change	10	21	8	10	11	8	15

1. Number of elderly (65+) as per cent of working age population (20-64 years of age).
2. Number of elderly (65+) plus number of young (20-) as per cent of working age population.

Source: United Nations, *World Population Prospects 1950-2050,* 1996 revision.

Besides the implied increase in pension and health care spending, the question is also how and to what extent the ageing process is going to influence overall economic activity, notably through decreasing labour supply and household savings and the derived effects on investment and productivity.[156] Although

calculations of future fiscal requirements are inherently imprecise and reliant on a number of simplifying assumptions, including the amount of spending cuts, it does not appear unrealistic to assume that Japan's tax-to-GDP ratio (including social security contributions) will have to increase substantially over the next ten years or so in order to ensure long-term fiscal sustainability, defined as stabilising the government debt-to-GDP ratio.[157] Chapter III outlines various scenarios for spending and taxes towards 2010. As the dependency ratios are projected to continue to rise after 2010 the requirements for tax and contribution increases are likely to be even higher in the very long run – at least until 2025.

A number of options are available for designing a sufficient increase in the tax burden as discussed below. One non-recommendable option would be to leave things basically as they are, in which case the risks of spiralling government debt will increase over time. This would eventually lead to soaring rates of interest (as expectations of a monetary bail-out increase) and a collapse of private-sector confidence as expectations of higher inflation, higher taxes and/or lower public pensions start rising. There is thus little doubt that the economic costs associated with postponing fiscal consolidation will grow over time and eventually become prohibitive. The macroeconomic costs of raising taxes in a timely fashion will be much smaller although probably not insignificant (Box 5). This should also be seen in the light of suggestions put forward by for instance Tachibanaki (1997) that the public sector in general, and tax policy in particular, have had only a small influence on the overall performance of the Japanese economy since the Second World War.

Japan has accumulated public social security assets to a greater extent than any of the other major economies,[158] but the public pension system, which is partially funded, needs to raise contribution rates in the future. To make matters worse, the assets held by the public pension system – of which a significant part is invested in public infrastructure projects (through the Fiscal Investment and Loan Programme) – are often of uncertain value, in part due to their lack of liquidity. Even under fairly optimistic assumptions on income growth, fertility rates and the rate of return on the pension system's assets, the government's official estimates suggest that total pension contributions would have to rise by almost half over the next two decades in order to support currently legislated benefits and restore balance in the system (see Chapter III).[159]

From a revenue-raising perspective, the increase in social security contributions needed to ensure the solvency of the public pension system can be seen as one of several ways to fill the financing gap caused by the ageing population. Another solution would be for the government to provide a larger subsidy to the pension system and finance this by direct or indirect taxes, as was proposed this summer.[160] More radical solutions for improving the financial situation of the pension system would include gradually switching to a defined contribution

Box 5. Economic impact of major tax hikes

The challenge of raising tax revenues as a percentage of GDP substantially over a relatively short time span (a decade or so) can be viewed in the light of the experiences of other OECD countries. Since 1960, a total of 19 episodes can be identified where OECD countries have raised taxes substantially[1] within a span of 9-12 years (Table 30). Half of these episodes largely took place in the 1960s in the Northern European countries, whereas the Southern European countries went through similar developments in the 1980s. The most recent episode was the tax increase in Greece during 1981-92. To what extent the increasing taxes as such have influenced overall economic performance is difficult to gauge, since cross-country empirical evidence concerning the relationship between taxation and aggregate growth, investment and employment is not always conclusive, and results are generally not robust to changes in underlying assumptions. Some studies find that higher taxes hamper growth via crowding-out of private activity and increased distortions to investment (including in human capital and R&D), savings and work incentives, while others find growth effects to be close to zero.[2] The economic impact of taxation also depends on how taxes are raised (the structure of taxation) and how the revenues are spent. As shown in Table 30, however, countries have generally under-performed in terms of key macroeconomic variables during tax-raising episodes. Average annual growth of GDP has been 0.7 percentage point lower than average OECD growth (with only 3 episodes out of 19 experiencing higher growth rates than the OECD average), and annual private investment growth has been 1.5 percentage points lower on average. Employment creation has also lagged behind: total employment growth fell short of the OECD average in ⅔ of the episodes and private employment creation was even more substandard. The average change in the total employment ratio was thus 1.7 percentage points lower than the OECD average and that of private employment was 4 percentage points lower.

1. Defined here – rather arbitrarily – as an increase in the total tax to GDP ratio of at least 7.5 percentage points.
2. For a recent survey see Leibfritz *et al.* (1997). This study suggests that, based on a simple cross-country regression analysis, an increase in the tax/GDP ratio of 10 percentage points could lower annual growth rates by around ½ percentage point. However, estimates of this kind are very sensitive to underlying assumptions, and both lower and higher growth effects are found in other studies. Many recent studies generally find long-run growth effects close to zero – see for instance Mendoza *et al.* (1997), Hendricks (1999), Engen and Skinner (1996), Slemrod (1995), and Milesi-Feretti and Roubini (1998).

system in which the risks would be transferred from corporations to employees, or to move to a fully funded, actuarially fair defined benefit system (requiring the transitional generation(s) to pay twice). A pertinent consideration is which instruments, including taxes, could bring about a more balanced burden across generations. According to several analyses, more recent generations are placed at a considerable disadvantage compared with their predecessors.[161]

Table 30. **Tax increasing episodes in OECD countries**

Percentage points

		Increase in tax/GDP ratio	Real annual GDP growth compared with OECD average	Real annual growth in private investment compared with OECD average	Change in employment ratio compared with OECD average[1]	Change in private employment ratio compared with OECD average[2]
Canada	1965-74	7.8	0.2	0.1	–1.3	–1.8
Belgium	1960-71	10.2	–0.6	–2.4	–4.9	–3.3[3]
	1971-82	8.9	–0.3	–2.9	–4.2	–5.4
Denmark	1960-69	10.3	–1.1	–0.8	–7.6	–8.2[4]
	1969-80	9.8	–1.4	–3.8	–2.0	–8.7
Finland	1966-76	10.2	–0.1	–1.0	–3.5	–5.3
	1980-91	10.0	–0.6	–2.6	2.2	–5.0
Greece	1981-92	9.4	–1.0	–2.9	2.9	–2.8
Ireland	1965-76	8.6	0.1	2.1	–3.8	–9.6
Italy	1979-88	10.0	–0.4	–1.5	0.9	–0.0
Luxembourg	1970-80	14.0	–1.0	..	–2.1	..
Netherlands	1960-72	9.8	–0.5	..	..	..
Norway	1960-72	8.7	–1.1	–2.9[5]	–6.8	–5.9[5]
Portugal	1979-91	9.5	0.3	0.5	12.1	4.5
Spain	1979-89	11.1	–0.1	0.4	–2.7	–5.0
Sweden	1965-76	12.6	–1.3	–1.9	3.5	–3.6
	1976-87	7.8	–1.1	–2.0	5.3	0.3
Switzerland	1967-77	9.5	–2.0	–4.4	–6.3[6]	–6.7[6]
United Kingdom	1973-82	7.7	–1.6	0.1	–13.2	–2.1
Average of above episodes		9.8	–0.7	–1.5	–1.7	–4.0

1. Employment ratio = total employment divided by working age population (15-64 years).
2. Private employment ratio = total minus government employment divided by working age population (15-64 years).
3. The period covered is 1962-71 instead of 1960-71 for the change in private employment.
4. The period covered is 1962-69 instead of 1960-69 for the change in private employment.
5. The period covered is 1962-72 instead of 1960-72 for the annual real growth in private investment and for the change in private employment.
6. The period covered is 1971-77 instead of 1967-77 for the change in total and private employment.

Source: OECD.

The starting point is difficult, and reform requires careful timing

The timing and implementation of a revenue-enhancing reform is another difficult issue for tax policy. The weakness of economic growth has led to a sharp deterioration of government finances since 1990, a development which has been compounded by a series of fiscal packages entailing an accumulated net stimulus of some 8 per cent of GDP. The deficit has thus soared to an unprecedented level of nearly 8 per cent of GDP in 1999, and net government debt has increased almost exponentially since 1992 to reach its current level of around 40 per cent of GDP. This calls for corrective fiscal action to be taken sooner rather than later. Against this concern stand short-term considerations of stabilising domestic demand. As illustrated in Box 6, the effects on real GDP from increasing taxes varies somewhat across instruments, but economic activity is likely to be negatively affected over at least a five-year horizon after the implementation of higher taxes. Moreover, the causes of the recent economic crisis are widely believed not to be rooted in the tax system – although a very light taxation of land holdings was partly responsible for the hike in land prices until 1990 and triggered the introduction of the so-called land value tax in 1992 (see below). These observations notwithstanding, reform of the tax system should be considered as one of the key elements to lay the foundation for renewed prosperity. This is not only because of the likely adverse impact on confidence of failing to map out concrete plans for ensuring fiscal sustainability, which must inevitably include tax reform, but also because of the risk that future taxation influences current economic activity and equity more than in the past, given the need to increase the overall tax burden substantially.

Devolution of fiscal autonomy is a high priority reform area

Another important factor influencing tax policy is the intended delegation of fiscal autonomy to local governments and the way in which their expenditures are financed. In general, an efficient financial architecture relating national and local governments should ensure that local governments have adequate incentives for pursuing strict expenditure and tax discipline and that their revenues are not excessively volatile. Although there is room for improvement, Japan does not stand out as having particularly inappropriate financing arrangements between the national and local governments (see Chapter III). However, with respect to the volatility of revenues, there is a problem of having prefectures relying excessively on one of the most volatile tax components of all, namely corporate income taxes.

Box 6. Short-term economic impact of tax reform

In order to illustrate the short-run impact on the budget and the real economy from various tax increases, a simulation exercise has been carried out on the OECD's macroeconometric model, INTERLINK, for the three major regions (Table 31). In each of three cases (direct personal taxes, social security contributions and indirect taxes) taxes have been raised by 1 per cent of GDP relative to baseline assuming unchanged nominal exchange rates and real interest rates. In all three regions, real GDP and employment are negatively affected by the tax shocks in the short run as higher taxes depress real household disposable income and hence private consumption. Inflation subsequently starts to slow as the result of the weaker activity. To some extent this offsets the negative impact of the fall in household income on consumption and contributes to improving the international competitiveness of the economy and to increasing its net exports (given the exchange rate assumption). In the case of a rise in social security contributions and indirect taxes, the deceleration in inflation is slower than for income tax increases, and hence it takes a longer time for the economies to recover. As a result, the improvement in the fiscal balance and the reduction of government debt are on average bigger when direct taxes are raised than when either social contributions or (to a lesser extent) indirect taxes are increased. The simulations suggest that real GDP will recover fairly slowly, with the impact for Japan being negative for five years for direct tax increases and nine years for rises in indirect taxes and social security contributions. The improvement of the budget and reduction of the debt are substantially larger in Japan than in the two other regions for all three cases, and the difference between indirect and direct tax increases on the budget is also smaller. This is mainly due to a different response to the tax shocks in wage/price formation: social contributions do not play as large a role in wage setting as in the United States and the major euro countries (as the incidence is to a larger extent on workers in Japan), and nominal rigidities in prices are also smaller. Of course these simulations are illustrative only, since they do not capture any structural effects from the different tax changes. It is remarkable, however, how much lower the costs of tax increases are in Japan than in Europe or the United States – measured as the negative effect on real activity compared with the improvement in the government's financial position. Another outcome of the simulation (not shown in the table) is that tax increases in Japan have only small effects on activity in other regions: the short-term impact from a tax increase in Japan on real GDP in the United States and Europe is less than 10 per cent of the impact on Japan's own GDP. A 10 percentage point increase in Japan's tax-to-GDP ratio could thus reduce the level of real GDP in the United States and Europe by 0.3-0.6 per cent depending on the instrument used. Over the medium term the other regions could even benefit as their terms of trade improve.

Another consideration related to timing is that tax changes may have different real economic effects according to whether they are (perceived as) permanent or transitory. Moreover, the private sector may respond to tax changes as they are announced rather than at implementation. Watanabe *et al.* (1999) examined the response of consumer spending to tax changes over the 1975-98 period and found that the effects of permanent changes far exceeded those of transitory changes. They also found that the bulk of consumers' response came at the time of implementation of the tax change.

Table 31. **Model simulations of tax reform**

Difference from baseline in per cent

	Euro area			Japan			United States		
	After			After			After		
	1 year	5 years	10 years	1 year	5 years	10 years	1 year	5 years	10 years
Household direct taxes									
Real GDP	–0.7	–0.6	0.4	–0.5	–0.1	1.2	–0.6	–0.1	0.3
Government balance	0.8	1.3	2.5	0.9	1.5	3.2	0.9	1.1	1.5
Government debt	–0.4	–2.4	–7.5	–0.6	–4.4	–12.8	–0.4	–1.8	–4.8
Employers' contributions to social security									
Real GDP	–0.5	–1.6	–0.3	–0.8	–1.0	0.4	–0.6	–1.0	0.2
Government balance	0.4	–0.4	0.7	0.4	0.4	2.3	0.4	0.0	0.6
Government debt	–0.4	–1.2	0.7	–0.4	–2.1	–3.8	–0.5	0.0	–0.5
Indirect taxes									
Real GDP	–0.9	–1.1	–0.4	–0.9	–0.8	0.2	–0.9	–0.5	–0.4
Government balance	0.2	0.3	1.1	0.7	1.2	2.4	0.4	0.6	0.9
Government debt	–0.3	–1.4	–4.8	–0.5	–4.8	–11.4	–0.5	–1.1	–2.8

Note: All simulations have been run assuming unchanged real interest rates and fixed exchange rates. The shock is an *ex ante* rise in taxes or social contributions.
Source: OECD Interlink simulations.

Main features of the tax system[162]

Separate income taxation and narrow bases

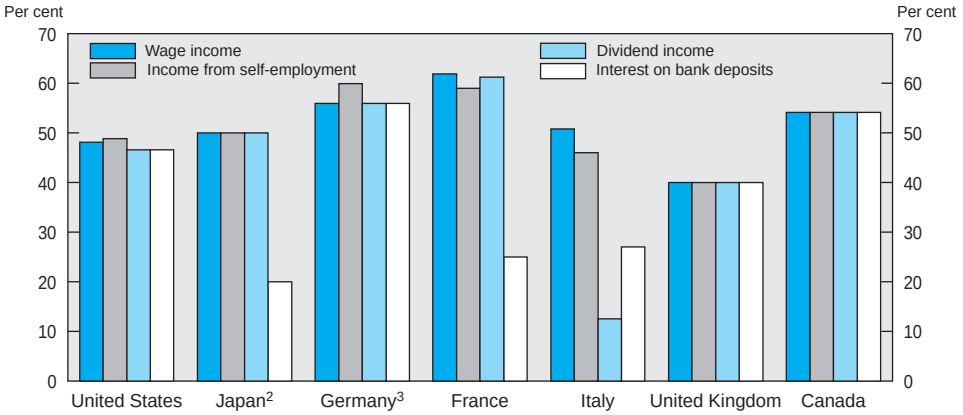
The current tax system has been shaped by a history of fairly drastic, although gradual, changes to the tax structure since World War II: from the *comprehensive income tax system* designed by Dr. Carl S. Shoup in the late 1940s,[163] to an approximation of an *expenditure tax system* (which practically exempted savings from taxation) and further on to today's system which can probably best be described as a hybrid of the two. Reforms over the past decade have not only aimed at improving efficiency and equity, but have to a large extent been reactions to cyclical developments and the fiscal challenges posed by the ageing of the population. The reform strategy has relied on achieving an appropriate mix of taxation of income, consumption and property. In some respects, the reforms have contributed to improving the tax system – notably by introducing a value-added tax and reducing marginal tax rates for individuals and corporations. On the other hand, the base for personal income taxation has been substantially narrowed, and many of the tax changes in recent years have been temporary, thereby diluting their effectiveness both from a structural and demand-management perspective.

The personal income tax system identifies ten different kinds of income that are taxed at different effective rates. Japan generally applies marginal personal rates for top income earners close to the level found in the United States and somewhat below rates in Germany and France (Figure 38). Retirement and capital income are taxed separately from other types of personal income at lower rates.[164] A large range of allowances imply that the base for personal income taxation is narrow, and income brackets are also very wide. The corporate tax rate has recently been reduced from 50 to 41 per cent and has thus come down to a level not far from that of many other OECD countries. This has clearly made the corporate tax system more robust to potential base erosion. Japan applies a modified classical system for dividend payments, whereby tax is levied both at the corporate and personal levels and a partial dividend tax credit is given in the personal income tax. The VAT system has few exempted goods and services compared with many other OECD countries, but the base is eroded by a tax-exempted sales threshold of 30 million yen, far exceeding those of other OECD countries and possibly also by a so-called simplified regime for small firms.[165] The standard VAT rate of 5 per cent is the lowest in the OECD among nations with a VAT.

Taxes are high on capital and low on consumption

Japan has traditionally levied high taxes on capital (primarily property and corporate income) and low taxes on consumption. This is reflected in the

Figure 38. **Highest all-in tax rates for top income earners**[1]
1998



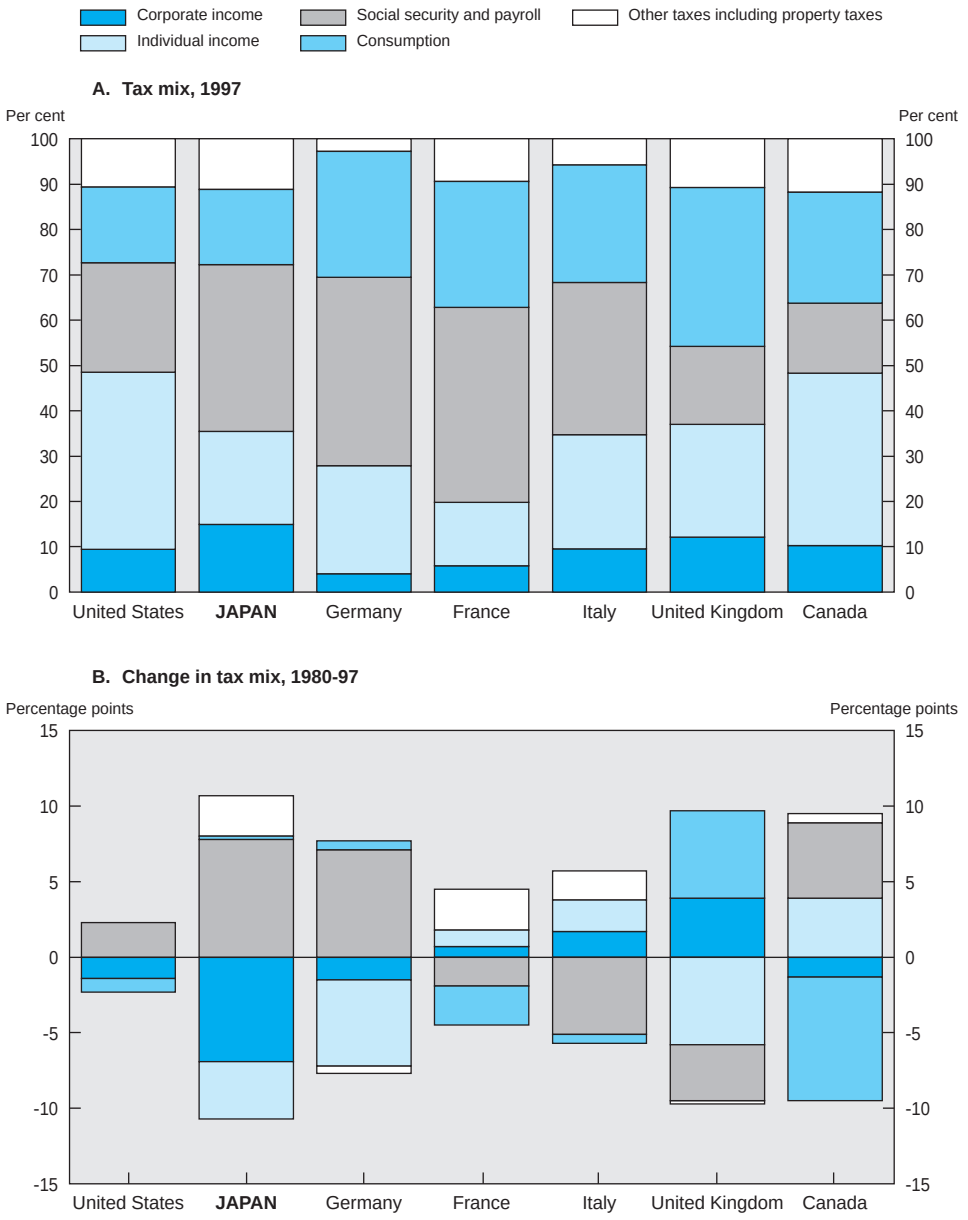
1. The all-in rates include central and sub-central government taxes as well as social security contributions where these are not capped.
2. Japanese data are for FY 1999. Tax on dividends depends on the size of payment. Tax credit not included.
3. Income from self-employment applies only to business income which is subject to the local business tax ("Gewerbesteuer").

Source: OECD Tax Database.

actual mix of tax revenues compared with other countries (Figure 39, Panel A). With the recent reductions in corporate taxes and given likely future increases in social security contributions and consumption taxation (see below), this pattern may change somewhat over the coming years. Personal income taxes have a small weight in the tax mix despite relatively high statutory rates (until the rate cut in 1999 to 50 per cent the top marginal rate stood at 65 per cent) – the main explanation being wide brackets and generous allowances. The latter have even gained in importance over the past decade and have reached a level where some 15 to 20 per cent of all employment income earners do not pay income tax at all. Germany is the only country among the major seven that has a higher tax exempted threshold than Japan, but since social security contributions are also much higher, more revenues are raised from lower income groups in that country.

A remarkable shift has taken place since 1980 in the Japanese tax structure, with a sharp increase in the share of social security contributions and a similar reduction in corporate and individual income tax shares (Figure 39, Panel B). Japan is thus the country among the major seven that has seen the biggest change in its tax mix. The share of consumption taxes has not changed despite the introduction of the VAT in 1989 and the initial low share. The share of

Figure 39. **Tax mix in major OECD countries**



Source: OECD, *Revenue Statistics.*

Box 7. Comparison of tax to GDP ratios

Making international comparisons of the tax mix is inherently difficult due to differences in the composition of GDP; the relative role of tax and transfer systems in providing a given benefit to individuals or corporations; the use of regulatory instruments as opposed to taxes; the tax treatment of public transfers; and the way the self-employed are organised and how they are recorded in the tax accounts. Two features are potentially relevant when comparing Japan with other countries. One is the relatively low indirect tax burden in Japan. Since GDP, in contrast with national income, includes indirect taxes, Japan's tax-to-GDP ratio tends to be biased upwards compared with countries with higher indirect taxes (all else being equal). However, the impact is small since cross-country differences in tax levels are almost invariant to whether taxes are stated relative to national income or GDP.* The other feature is the organisation of the self-employed. The corporate tax share is biased upwards by the previously existing incentives for the self-employed to incorporate themselves, forming so-called "quasi-corporations", or (from 1974) to be taxed as corporations under the "deemed corporation" scheme. The favourable tax treatment of the self-employed was abolished as of 1993 to make the personal income tax system more equitable, but some compensation was given to the self-employed by raising their allowance. The impact on the corporate tax share from these arrangements compared with other countries where the self-employed are taxed as individuals is not entirely clear but is likely to be small. For instance, the income tax payments from the 3 million individual business income tax payers amounted to less than 1 trillion yen or 0.2 per cent of GDP in 1996 (National Tax Administration, 1997). Assuming the incorporated self-employed pay the same or even less taxes, their total tax bill should be smaller since they constitute only around 1 million corporations (Ishi, 1993).

* See for instance Ministry of Finance (1999*a*).

revenues from property taxes, including the inheritance tax, is not particularly small in Japan and has increased somewhat since 1980, thanks especially to soaring land prices during the "bubble" period.

Another way to compare tax mixes between countries is to look at effective tax rates, *i.e.* how much revenue is actually collected from various tax bases. The advantage of using actual tax payments as opposed to statutory rates is that they capture the overall effects from rates and tax reliefs as well as from compliance, which is not reflected in the statutory rates. Moreover, they provide a more accurate measure of the average tax burden on different bases than simple tax-to-GDP ratios. The problem is that the definition and measurement of tax bases corresponding to the different types of revenues are often severely flawed.[166] Another problem is how to interpret data for effective tax rates, since these measure neither distortions (marginal effects) nor the incidence of the tax, *i.e.* who finally bears the burden. Most of these *caveats* relate to measuring

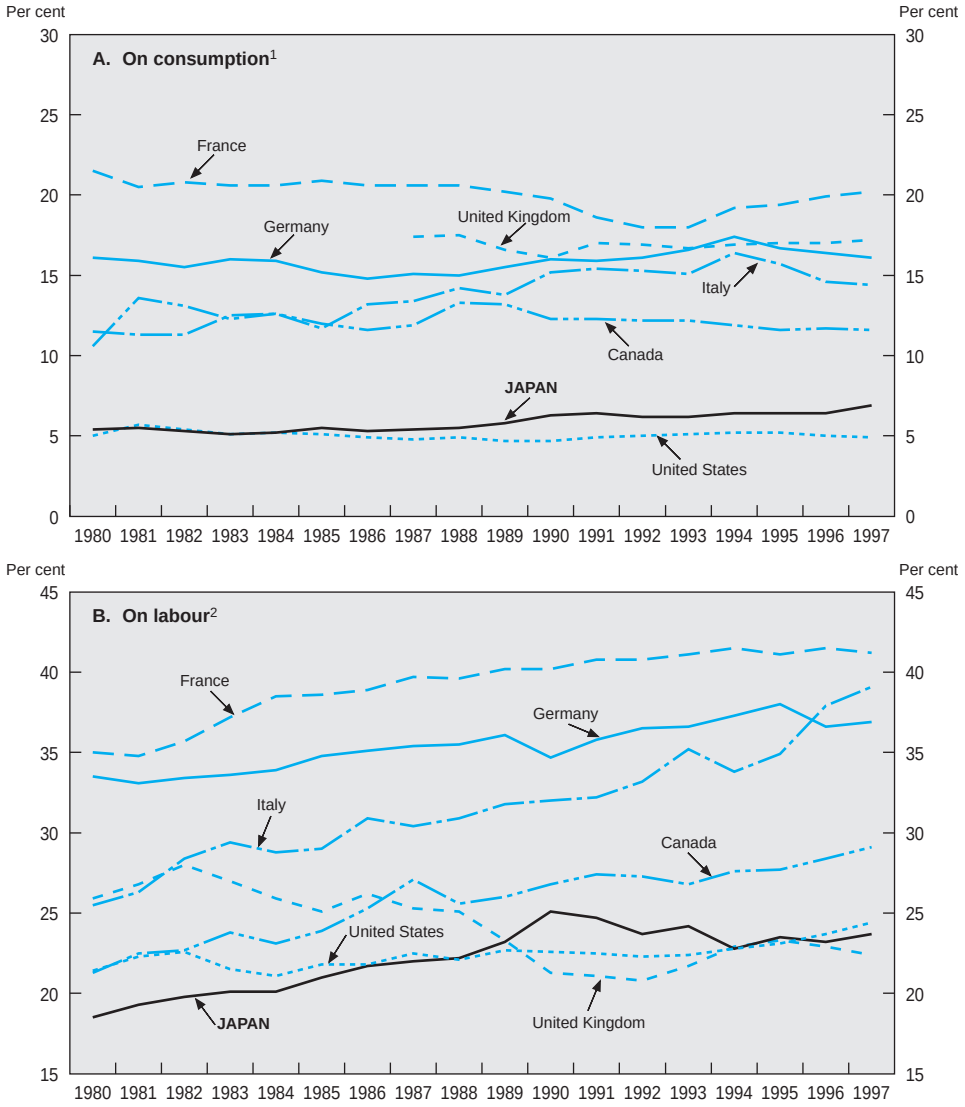
effective rates on capital and corporations, however, and it is generally found that the methodology is better suited to measuring effective tax rates on labour and consumption (Figure 40). Japan appears to apply fairly low rates on both these bases compared with other major countries. On the other hand, the ratio of corporate tax payments to GDP is higher in Japan than in any other OECD country, except for Luxembourg. Tajika and Yui (1997) also found evidence – based on balance sheet information for the corporate sector – that effective corporate tax rates in Japan have significantly exceeded those in the United States since the early 1980s.[167] By 1993, the last year covered by the study, the rate was 45 per cent in Japan against 35 per cent in the United States. However, the cuts in Japanese corporate tax rates in FY 1998 and FY 1999 are likely to have narrowed the gap in effective corporate rates significantly.

Overall incentives to work, save and invest are not heavily distorted

The generous allowances in the personal income tax system – in combination with the wide income tax brackets – imply that average personal tax rates are generally low in Japan. Indeed, for an average production worker the average tax wedge has been fairly stable and lower than in the other major economies since 1980 (Figure 41, Panel A). This also applies when taking into account the effect from indirect taxes on the price level and thus the real disposable income of the employee (Figure 41, Panel B). The tax reductions of FY 1998 and FY 1999 have left average tax rates even lower. Taxpayers with an annual income below 12 million yen – which is around 95 per cent of all personal taxpayers – face average tax rates, including social security contributions, below 20 per cent (Ministry of Finance, 1999*b*). Marginal tax wedges, *i.e.* the difference between an increase in the cost to employers and the additional take home pay of the employee, are also low in Japan for most family and income circumstances (Figure 42).[168]

The total labour market distortion caused by taxation is given by the combination of marginal tax rates and labour supply elasticities. Since empirical estimates generally tend to find low elasticities (Box 8), the tax-induced distortion to the Japanese labour market is probably low in most income segments. However, for certain groups, in particular second earners in households, implied marginal rates can be significant (due to phasing-out of dependent spouse deductions and phasing-in of social security contributions). Since the labour supply elasticity of this group also tends to be high, it cannot be ruled out that a substantial distortion exists with respect to the labour market participation decision of these persons. The problem of disincentives for secondary earners' labour market participation is not, however, unique to Japan but exists in many OECD countries.

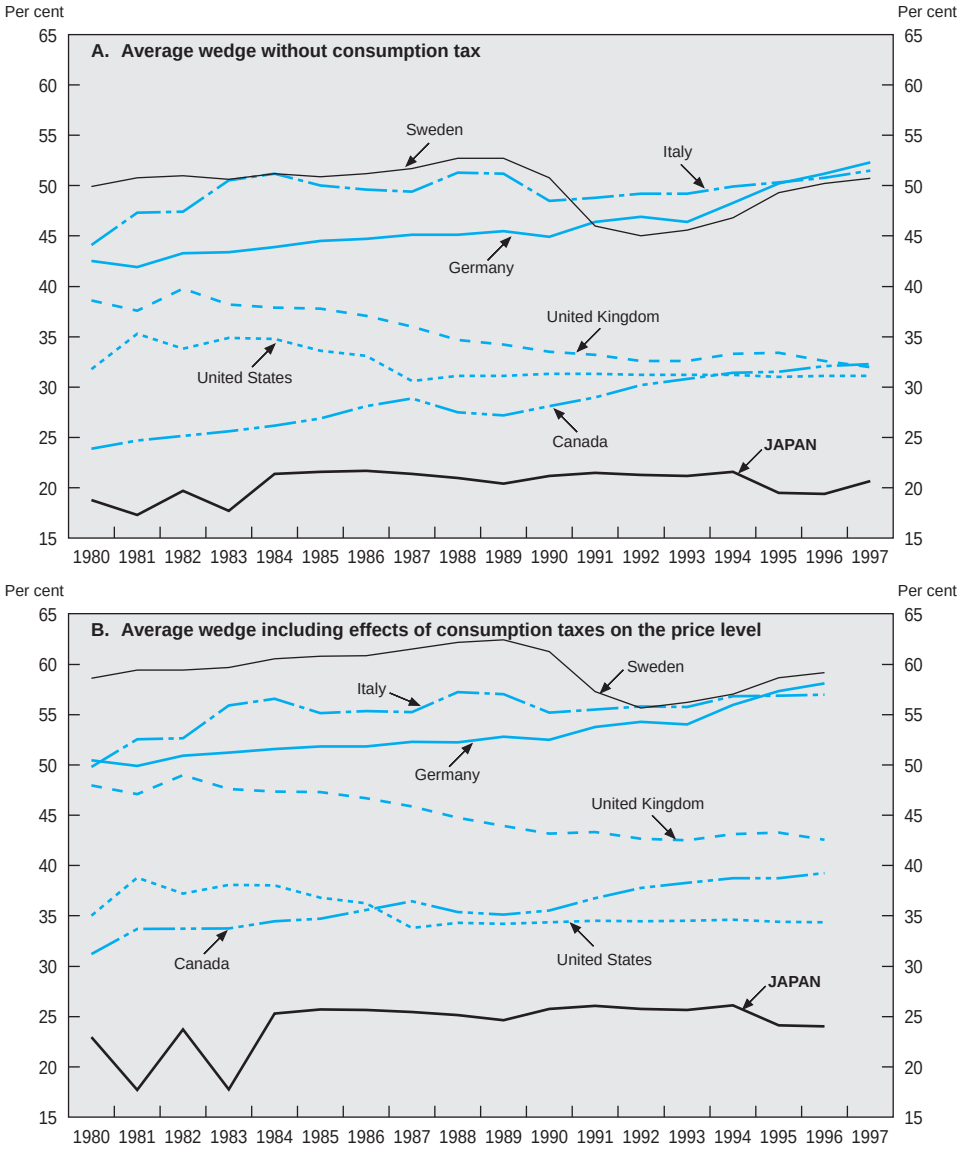
Figure 40. **Average effective tax rates**



1. General taxes on goods and services plus excises, divided by private and government non-wage consumption excluding these taxes.
2. Individual income taxes paid on wages (including tax on self-employed labour income) plus payroll taxes and social security contributions, divided by wages and salaries (including imputed wages of self-employed), and including employer's contribution to social security and to private pension schemes.

Source: *OECD Revenue Statistics, OECD National Accounts* and OECD calculations.

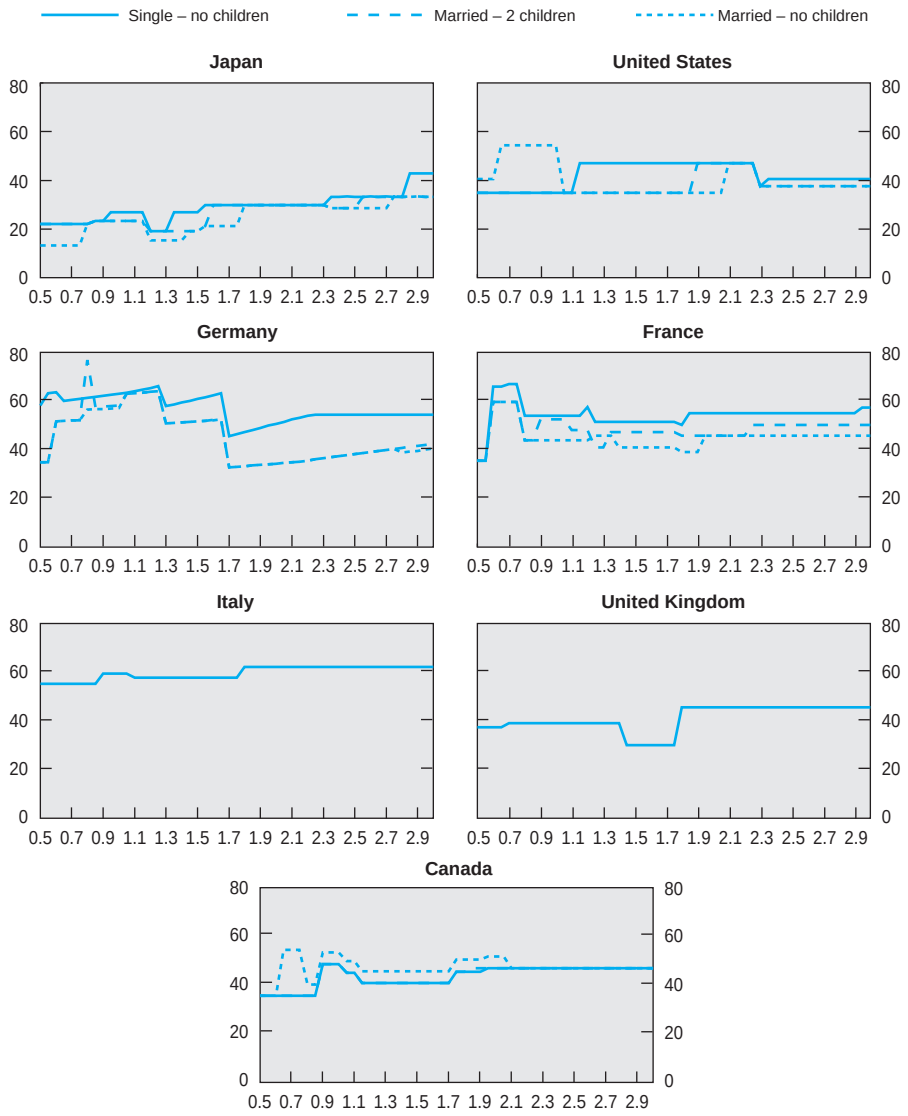
Figure 41. **Average tax wedge for a production worker**[1]



1. Employees' social security contributions and personal income taxes as a percentage of gross labour costs (incl. employers' social security contributions). At 100 per cent of an Average Production Worker (APW) income. Panel B takes into account price level effects of consumption taxes.

Source: OECD, *The Tax/Benefit Position of Employees.*

Figure 42. **Marginal tax wedges on labour income**[1]
At multiples of average production worker earnings, 1997



1. Marginal tax rate, covering employees' and employers' social security contributions and personal income tax, with respect to a change in gross labour costs, by family type and wage level, assuming spouse earns 0% of APW earnings.

Source: OECD Tax equations.

Box 8. Labour supply elasticities

The total labour market distortion is given by the combination of tax wedges and elasticities of labour demand and supply.* These elasticities are shaped by a host of factors, including the wage-bargaining framework, labour market policies and the degree of competition in product markets. Most cross-country studies find that labour supply elasticities are low and often insignificant for primary earners, whereas they tend to gain in importance for secondary household earners, typically married women. This pattern also appears to exist in Japan, although evidence is scarce (see Tachibanaki, 1997). One composite measure of the interaction of labour supply and demand elasticities is the extent to which tax increases get shifted on to wages (and thus push up unemployment). Tyrväinen (1995) finds evidence that the extent of tax shifting in Japan is higher than in the United States, but lower than in Germany. Japan is thus in an intermediate position together with France and Italy. However, according to Alesina and Perotti (1997), unit labour costs are virtually unaffected by tax increases (direct tax plus social security contributions) in countries with low labour market centralisation, including Japan.

A particular problem is the labour market participation decision by dependent spouses. The phasing-out of the spousal allowances as well as the introduction of pension contributions when spouses earn more than 1.3 million yen annually creates high marginal tax rates for the household when the spouse enters the labour market or decides to increase working hours. The elasticity of Japanese married women with respect to own wages and family income is around 1.1 (much higher than what is normally found for primary earners) and -0.2, respectively. These elasticities are roughly similar in magnitude to estimates found for France and Germany, but higher than those found in the United States (Tachibanaki, 1997).

* An increase in after-tax wage income will have both a substitution effect, which raises labour supply, and an income effect, which lowers it. The relevant labour supply response for measuring distortions to economic decisions is the substitution effect (the so-called compensated labour supply elasticity), which indicates the impact of marginal taxes on the choice between labour supply (and thereby the consumption of goods) and leisure. The income effect, on the other hand, reflects the effects of average taxes on disposable income.

The overall low marginal tax rates also suggest that the return to investment in human capital is not unduly distorted by taxes in Japan. Another notable feature is that the progressivity of the tax schedule unfolds relatively smoothly compared with many other countries (Figure 42), implying a virtual absence of problems related to unemployment and poverty traps – which are often caused by abrupt changes in tax rates along the income schedule. A final aspect related to marginal income tax rates is that Japan, despite its earnings test for elderly workers receiving pensions, is in no sense an outlier compared with other major countries with respect to the implicit taxation of elderly workers for staying another year on the labour market (Table 32). Overall, it seems safe to conclude

Table 32. **Average implicit tax rates on continued work due to old-age pension systems**

1995

	Postponing retirement from 55 to 64	Postponing retirement from 55 to 69
Australia	0	6
Austria	34	47
Belgium	23	33
Canada	6	11
Denmark	0	5
Finland	22	33
France	14	42
Germany	14	23
Ireland	14	17
Italy	79	79
Japan	**28**	**26**
Netherlands	13	14
New Zealand	9	15
Norway	15	22
Portugal	4	25
Sweden	18	22
Switzerland	0	10
United Kingdom	5	10
United States	12	17

Source: Blöndal and Scarpetta (1997).

that the tax system, unlike those in many other OECD countries, does not pose major obstacles to a high utilisation of labour force potential. This conclusion is underpinned by the fact that Japan has one of the highest labour market participation ratios in the OECD, although this is due to a combination of a very high male participation and a female participation ratio closer to the OECD average. Participation rates of the elderly (55-64 years old) are also relatively high and hence so is the average effective retirement age.

Non-neutralities in corporate finance, private savings and the land and property market

Incentives to save and invest in physical capital also benefit from the low overall level of taxation. To what extent the tax burdens for corporations and individuals influence aggregate investment levels is a question of *a)* how much taxation changes the cost of capital and *b)* to what degree the cost of capital determines investment. According to Japanese evidence, neither of these two linkages is particularly strong, and hence there is no reason to believe that taxation as such has had a strong influence on the aggregate level of private investment in Japan.[169]

The tax system is neutral towards corporate financing decisions if a given pre-tax flow of corporate profits produces the same after-tax income in the hands of the ultimate investor, whether the return takes the form of interest payments, dividends or capital gains. This requires that the combined corporate and personal tax burden is equal across financing instruments. Most OECD tax systems favour debt finance since corporate interest payments – as opposed to dividends – are deductible from the corporate tax base and because effective tax rates on personal interest income are often low. This is also the case for Japan, where interest income is taxed at a final withholding rate of 20 per cent, capital gains generally at 26 per cent and dividends are taxed in a "mixed" fashion: large dividend payments are taxed as part of total personal income and smaller payments are taxed at withholding rates of 20 or 35 per cent depending on the size of the payments.[170] The overall marginal effective tax wedge[171] across corporate financing and investment instruments is relatively high as is the standard deviation, which is a rough measure of non-neutrality (Table 33).[172] Although these numbers should be interpreted with caution, they indicate that distortions to corporate financial decisions are at a level with those of France and Canada and considerably higher than in more neutral systems, such as those of the United Kingdom and Germany. In particular, the numbers suggest that debt financing in Japan enjoys an even bigger tax advantage over equity and retained earnings than what is the case in the other major countries. With respect to investment, buildings are taxed at relatively high marginal rates, which is mainly due to a long depreciation period (65 years against 39 years in the United States). The calculations do not take into account special depreciation measures in the countries shown, such as the two accelerated depreciation schemes in Japan (additional depreciation and increased initial depreciation).[173]

Since Japan, like many other OECD countries, taxes capital gains less heavily than dividends, corporations typically also have an incentive to retain their earnings. This tendency is reinforced by the mixed taxation of dividends which encourages firms to limit their dividend payments to individual shareholders. This, in turn, could hamper the reallocation of funds from mature companies to their more innovative and fast growing counterparts, although it should be stressed that dividend payments between domestic corporations are tax exempted. The mixed taxation of dividends also implies inequities on the side of the investor, since: within the same fiscal year, an individual taxpayer may not pay the same tax rate on dividend income deriving from different companies (conflicting with neutrality); and two taxpayers receiving the same total amount of dividend income may not be taxed at the same rate (conflicting with horizontal equity).

Savings incentives are provided through several channels in the tax system. *First*, personal capital income is taxed at relatively low effective rates. Even though dividends are taxed more heavily than interest income and capital gains

Table 33. **Marginal effective tax wedges in manufacturing in major OECD countries**[1]

Excluding wealth taxes
Per cent, 1998[2]

	Sources of financing[3]			Physical assets[4]			Overall weighted average[5]	Standard deviation
	Retained earnings	New equity	Debt	Machinery	Building	Inventories		
United States	1.9	5.0	1.7	1.5	2.8	2.7	2.1	1.3
Japan	**3.5**	**5.6**	**0.3**	**1.4**	**4.1**	**3.4**	**2.6**	**1.9**
Germany	1.5	1.0	1.4	1.0	1.7	2.1	1.4	0.4
France	3.7	6.9	0.8	2.2	3.8	3.8	3.0	2.1
Italy	2.3	2.6	0.6	1.0	1.8	3.3	1.7	1.0
United Kingdom	2.2	2.8	1.8	1.7	2.1	3.1	2.2	0.6
Canada	4.3	5.0	1.3	2.2	4.1	4.8	3.3	1.5
G7 average[6]	2.8	4.1	1.1	1.6	2.9	3.3	2.3	1.3

1. These indicators show the degree to which the personal and corporate tax systems scale up (or down) the real pre-tax rate of return that must be earned on an investment, assuming that the household can earn a 5 per cent real rate of return on a demand deposit. Wealth taxes are excluded. See OECD (1991) for a discussion of this methodology.
2. For Japan, the FY 1999 rules are applied.
3. Calculated using the following weights: machinery 50 per cent, buildings 28 per cent, inventories 22 per cent.
4. Calculated using the following weights: retained earnings 55 per cent, new equity 10 per cent, debt 35 per cent.
5. The weighted average uses weights indicated in footnotes 3 and 4.
6. Simple averages.

Source: OECD.

owing to double taxation, the tax wedge on dividends is not excessively high compared with other major countries (Table 34). The low overall taxation of savings should, however, be viewed in light of the significant changes made over the past decade to increase taxation of personal capital income from virtually no taxation at all. *Second*, privileges are given to the savings of the elderly and to those of workers saving for housing, which are tax exempt up to certain amounts. *Third*, and more importantly, pension savings are given preferential treatment in basically three ways: contributions for the mandatory schemes are exempted as are contributions for individual voluntary pension plans and life insurance (up to a certain amount); accumulated earnings in pension funds are tax exempt,[174] and benefits are taxed very lightly, thanks to significant tax reliefs for pension payments, whether these are lump-sum or annuities (see Annex II for details). The tax treatment at each of the stages in pension savings (contributions, fund earnings and distribution of benefits) can be summarised as "near exempt"-exempt-"near exempt" (E*EE*) compared with taxed-lightly taxed-exempt (TT*E) on ordinary savings, where E* indicates near zero taxation and T* low taxation of capital income.[175]

Most other OECD countries also have tax-privileged arrangements for pension savings, although the amount of tax relief given varies considerably (Table 35). A few countries tax earnings as they accumulate (Australia, Denmark, Luxembourg, New Zealand and Sweden). Two countries which have recently reformed their taxation of pensions, Australia and New Zealand, have both moved toward less generous systems, at least partly with the aim of raising revenue, but also with the aim of improving the efficiency and equity of the tax system as a whole. New Zealand has taken the boldest steps, making all contributions taxable, taxing all fund income (with no allowance for inflation) and then leaving all pensions untaxed (TTE). This puts pension savings on the same basis as savings in an ordinary interest-bearing account.

As for most other countries, there is no solid evidence for Japan that preferential treatment of savings, including pension savings, leads to higher aggregate levels of private savings, not to mention national savings (taking into account the loss of tax revenue from preferential regimes which, *inter alia*, lower public savings). Annex III explores the links between taxation and savings in more detail. One robust conclusion, however, is that tax incentives have an effect on the *composition* of savings – in the case of Japan with a bias towards pension savings and housing as against investment in equities.

Property and inheritance taxes are other areas of significant tax-induced non-neutralities. The tax system increases the attractiveness of land as a long-term investment tool, thereby discouraging an efficient use of land. This distortion is most clearly observed in the case of farmland in urban areas. It is mainly

Table 34. **Dimensions of capital taxation in selected OECD countries**

A. Interest, profits, dividends, capital gains and wealth taxes

	Highest tax rate on interest income	All-in corporate tax rate	Total dividend taxation[1]	Taxation of capital gains	Wealth tax	
					Rate	Threshold[2]
Japan	**20***	**41**	**71****	**26**[3]	**0**	**–**
France	25	33	66***	26	0.5-1.5	103
Germany	56	58[4]	49***	0	0	–
Norway	28*	28	28***	28	0.7-1.1	2
Sweden	30*	28	50**	30	1.5	15
United Kingdom	40	31	48	0-40	0	–
United States	47*	35	68**	0-20	0	–

B. Inheritance taxes

	Spouses			Children			Valuation rules: fair market value unless otherwise noted
	Rate	Lower threshold[2]	Upper threshold[5]	Rate	Lower threshold[2]	Upper threshold[5]	
Japan	10-70	160 or half of the inheritance (whichever the higher)	2 000[9]	10-70	60	2 000[9]	Special valuation for small lands; residential (up to 200 m^2) and business (up to 330 m^2)
France	5-40	7	256	5-40	7	256	n.a.
Germany	7-30	45	3 772	7-30	30	3 757	First 36 million yen of business property not assessed
Norway	0	–	–	8-20	2	7	n.a.
Sweden	10-30	5	15	10-30	1	11	Business capital 30 per cent of substance value
United Kingdom[6]	0	–	–	40[7]	47	–	
United States[8]	18-50	82	475	18-50	82	475	Special-valuation for farms and closely-held businesses

Note: Tax rates are in per cent and thresholds are in million yen. Exchange rates used are: 1 US$ = 130.9 yen; 1 DM = 74.4 yen; 1 FF = 22.2 yen; 1 NKr = 17.3 yen; 1 SKr = 16.5 yen; 1 £ = 216.7 yen (average of daily rates, 1998).

* Tax deductibility of interest payments for housing mortgages. For Japan, there is a tax credit for interest payments on housing mortgages.

** Denotes taxation of dividends both at corporate and investor level without partial credits at investor level.

*** Denotes full imputation at investor level for taxes paid at corporate level.

**** For one-payment dividends both at corporate and investor level with partial credits at investor level.

1. Taxation at both corporate and investor level.
2. When tax rates above 0 become effective.
3. Until FY 2001 investors can opt to be taxed by 1.05 per cent of total proceeds instead.
4. Retained profits.
5. When the highest rate becomes effective.
6. Inheritance tax is dropped upon transfers of: *i)* interest in an unincorporated business; *ii)* controlling interest in a trading company and shares in unquoted trading companies. Other transfers of business may have a 50 per cent reduction of rates.
7. Applying to property passing on death. The rate is halved for property transferred *inter vivos* into a discretionary settlement or a closely-controlled company.
8. Federal tax schedule.
9. Applicable to the legal share of inheritance of each inheritor.

Source: OECD; International Bureau of Fiscal Documentation (1998), *European Tax Handbook 1998;* United States Internal Revenue Service.

Table 35. **Tax treatment of pensions in major OECD countries**

1998

	Contributions	Fund earnings	Benefits
United States	E/T (Employer contributions are not taxed. Employee contributions are taxed)	E	T* (50-85 per cent of benefits taxed as earned income. Lump sums are subject to an averaging provision. A small additional allowance is given to persons above age 65 and a small tax credit is given to elderly with very low incomes)
Japan	E* (A part of employee contributions for qualified pension schemes may not be deductible)	E	E* (The allowances for retirement and annuity income effectively exempt pensions, both lump sums and annuities, from income taxation)
Germany	E/T (Employer contributions are not taxed, except for contributions to direct insurance and pension funds. Employee contributions are exempt up to a certain limit)	E	T*/E* (Payments made out of relief funds and on the basis of direct pension commitments are taxed as ordinary income after an allowance of 40 per cent (up to a certain limit). Taxpayers older than 64 years and deriving income which is not considered income from prior employment are also granted an allowance of 40 per cent (up to a certain limit). In the case of annuity payments from pension funds and life insurance, only interest earned after pension payments have begun is subject to tax, while lump-sum payments are exempt)
France	E	E	T* (Liable to ordinary income tax but subject to a 10 per cent basic deduction and a 20 per cent "additional deduction"(up to certain limits) Life annuities are only partially subject to income tax with the taxed share ranging from 30 to 70 per cent dependent on the recipient's age)
Italy	E	E	T* (60 per cent of annuity benefits are taxed – however, for the complementary pension the base is 87.5 per cent of the annuity)

Table 35. **Tax treatment of pensions in major OECD countries** *(cont.)*

1998

	Contributions	Fund earnings	Benefits
United Kingdom	E*/T* (All contributions to personal and occupational pension schemes are eligible for tax relief, subject to limits both on the level of contributions as a proportion of earnings and on the amount of earnings)	E	T*/E* (Pension income is taxed as employment income subject to an age allowance for persons above age 64. The allowance is reduced for higher-income individuals. At the time of the commencement of the pension, part of the pension can be converted to a tax free lump-sum payment. The income from a purchased life annuity is split between an income and capital element on an actuarial basis with only the income element being taxable)
Canada	E	E	T* (Old age security basic pension is taxed whereas the Guaranteed Income Supplement and Spouses Allowances are not. Tax credits are given elderly persons and to receivers of pensions. The current system will merge into a tax free Senior Benefit in 2001. The income from a purchased life annuity with no tax-assisted savings is split between an income and capital element with only the income element being taxable)

Note: T = taxed, T* = taxed lightly, E = exempt, E* = virtually exempt.
Source: OECD, Tax database; *The European Tax Handbook,* 1998; and national sources.

caused by two factors. The first is under-assessment of land value at the time of inheritance. The ratio of the valuation price to market price[176] is subject to some debate but was believed to be around 50 to 60 per cent up to the end of the 1980s, the "bubble" period (Nishimura *et al.*, 1999), and has probably only increased slightly since. The second is postponement of inheritance tax payments by farmers: if their inheritors continue to use the land as farmland for more than 20 years, they are allowed, in effect, not to pay the inheritance tax. These measures strengthen landowners' incentives to hold land and raise reservation prices accordingly (Box 9). Two important measures were taken in the 1992 land tax reform: it was decided to try to increase the land valuation for inheritance tax purposes to 80 per cent of the market price; and the requirements for postponement of inheritance tax payments by farmers were tightened.[177] Even if these measures and the falling land prices since 1990 have alleviated the seriousness of the distortion, the current tax system still supports high reservation prices.[178]

Under-utilisation of land is facilitated by low effective rates of property and city planning taxes. Although the standard statutory rate of property tax is 1.4 per cent, the effective rate – which takes into account under-valuation of the land and property – is estimated to have ranged between 0.1 and 0.3 per cent over the past couple of decades with a movement towards the high end of the interval in recent years (Figure 43). As in the case of inheritance tax, it was decided to raise the land evaluation for property tax in the 1992 reform to 70 per cent of the market value. The capital gains tax is also likely to be responsible for under-utilisation of land through two channels. First, frequent changes in tax rates might discourage the disposal of land (Table 37) (Yamazaki and Idee, 1997). Second, the structure of the tax schedule encourages long-term holding since the rate comes down in steps as the holding period increases, leaving long-term gains taxed at only half the rate applied to short-term gains. However, fairly generous deductions[179] alleviate to some extent the adverse effect on transactions posed by the relatively high tax rates on short-term capital gains.

The tax-induced incentive to hold land implies, in combination with transaction costs of up to 2 per cent of the transaction value (Kanemoto, 1997), a low frequency of land and housing transactions.[180] The government has been promoting securitisation in order to stimulate the real estate market. For this purpose, the registration tax was reduced in 1998 and once again in 1999. Another factor contributing to low liquidity is the treatment of capital gains tax: the tax is imposed when a person sells a house and buys another even if no net gain is realised. This treatment is rather rare in OECD Member countries: only Hungary follows Japan in not allowing an exemption in these cases. A scheme for capital loss carry-over for real estate was established in 1998 to alleviate the problem and was expanded in 1999 to allow joint use with the tax credit on mortgage loans.

Box 9. Reservation price of landholders and the tax system

The effects of the tax system on landholders' incentives are shown in the following examples.* Note that only "marginal" choices available to a landowner with sizeable assets are considered here: in what form should he leave his "marginal" land worth 100 million yen to his descendants? Here it is assumed that the number of descendants is always one per generation, just to simplify the story.

Example 1:

Consider the choice between land and a financial asset, say a bank deposit, in the form of an inheritance. The first strategy is to keep the land until death, leaving it to the child tomorrow. The second is to sell the land today, thereby leaving a bank deposit of 100 million yen. The top panel of Table 36 shows to what extent underassessment and high marginal tax rate increase the landowner's reservation price. As Case 2 shows, if the assessment is half the market price, as was in the "bubble" period, he is not willing to sell his land until the offer price is more than twice the market price unless he faces liquidity needs. More appropriate assessment (Case 1) and a lower tax rate (Case 3) are helpful in mitigating the problem.

Example 2:

The above problem could be worse in the case of a farmer in a metropolitan area. Consider his choices to leave his land to his child and further to his grandchild. Assuming that the first inheritance by the child and second one by the grandchild will take place 25 and 50 years later, respectively. Three strategies are available to him. First, keep the land as farmland for 50 years, thereby enabling his child not to pay inheritance tax. Second, leave the land to the descendants, but change its use. Third, sell it today and leave bank deposits to the descendants. Assume that rents from agricultural or other land use are negligible as is the acquisition price of land. The interest rate on the deposit and appreciation of land are assumed to be the same, *i.e.* the landowner is indifferent between the two in terms of the returns. Note that a part of the asset is assumed to be sold to pay inheritance taxes and that the payments are deductible from capital gains. After-tax asset values 50 years later and reservation prices are reported in the middle panel of Table 36. Comparing the three strategies, it is worth noting that agricultural use could provide a valuable tax shelter to long-term holding: the reservation price was pushed up to over 7 (Case 5). However, this could be also alleviated jointly by more appropriate assessment and a lower tax rate (Case 6).

Example 3:

The settings are the same as the previous example, except that a property tax and a city planning tax are introduced here. Case 8 may be close to the "bubble" period when the underassessment for the property tax is believed to have been about 70 per cent. An attempt to reduce it to 30 per cent should be helpful to solve the problem, but not so effective as in the case of inheritance tax.

* The examples below heavily depend on Kanemoto (1994).

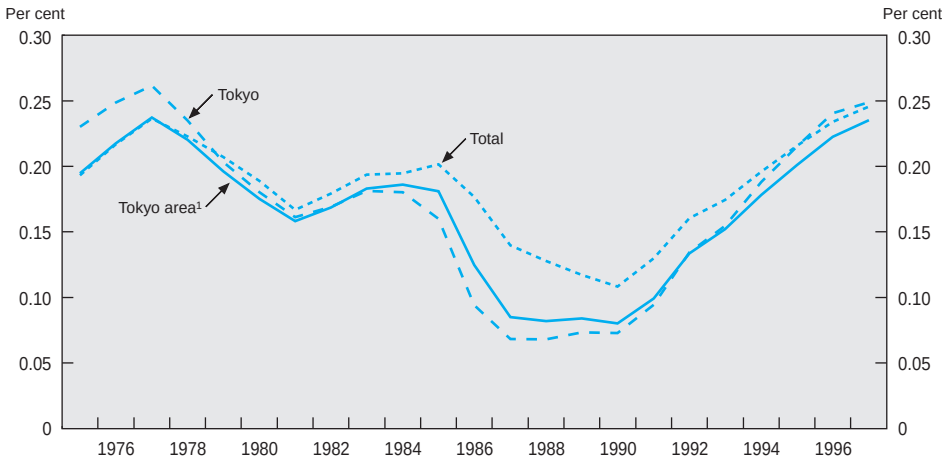
Table 36. **Tax effects on landholders' reservation prices**

	Assumptions		Strategy 1		Strategy 2		Strategy 3	
	Ratio of land evaluation to market value (%)	Tax rate (%)	After-tax asset value (yen million)	Reservation prices	After-tax asset value (yen million)	Reservation prices	After-tax asset value (yen million)	Reservation prices
Example 1[1]	Inheritance tax							
Case 1	80	70	33	1.47	22	1		
Case 2	50	70	48	2.17	22	1		
Case 3	80	50	44	1.20	37	1		
Example 2[2]	Inheritance tax							
Case 4	80	70	143	4.89	63	2.15	29	1
Case 5	50	70	211	7.22	137	4.69	29	1
Case 6	80	50	195	2.91	117	1.75	67	1
Example 3[3]	Property tax[4]							
Case 7	0	1.7	143	4.89	63	2.15	29	1
Case 8	30	1.7	111	3.79	49	1.67	29	1
Case 9	70	1.7	78	2.69	35	1.18	29	1

1. The calculations in example 1 are as follows: Strategy 1: the after-tax asset value (ATV) = 100*(1 – ei*ti)*(1 - tc); Strategy 2: ATV = 100*(1 – ti)*(1 – tc), where ei = the ratio of land evaluation for inheritance tax to market value, ti = inheritance tax rate, tc = capital gains tax rate (26%). Therefore, the reservation price (RP) of Strategy 1 is equal to (1 – ei*ti)/(1 – ti).
2. The calculations in example 2 are as follows: Strategy 1: ATV = $100*(1 + r)^{50}*(1 – ei*ti)*(1 – tc)$; Strategy 2: ATV = $100*(1 + r)^{50}*(1 – ei*ti)^2*(1 – tc)$; Strategy 3: ATV = $100*(1 + r)^{50}*(1 – ti)*(1 – tc)$, where r = interest rate and appreciation rate of land (3%). Hence, RPs are calculated as follows: Strategy 1: RP = $(1 – ei*ti)/(1 – ti)^2$; Strategy 2: RP = $(1 – ei*ti)^2/(1 – ti)^2$.
3. The calculations in example 3 are as follows: Strategy 1: ATV = $100*[(1 + r)*(1 – ea*ta)]^{50}*(1 – ei*ti)*(1 – tc)$; Strategy 2: ATV = $100*[(1 + r)*(1 – ea*ta)]^{50}*(1 – ei*ti)^2*(1 – tc)$; Strategy 3: ATV = $100*[(1 + r)*(1 – ea*ta)]^{50}*(1 – ti)*(1 – tc)$, where ea = the ratio of land evaluation for fixed asset tax to market value, ta = sum of fixed asset tax rate and city planning tax rate. Hence, RPs are calculated as follows: Strategy 1: RP = $(1 – ea*ta)^{50}*(1 – ei*ti)/(1 – ti)^2$, Strategy 2: RP = $(1 – ea*ta)^{50}*(1 – ei*ti)^2/(1 – ti)^2$. In example 3, ei and ti are set equal to 80 and 70 per cent, respectively.
4. Including city planning tax.

Source: Kanemoto (1994) and OECD.

Figure 43. **Effective property tax rates**
Fiscal year



1. Tokyo area is Tokyo, Kanagawa, Chiba and Saitama prefectures.
Source: Economic Planning Agency and OECD.

Tax effects on tenure choice are limited

There are a number of favourable tax treatments for housing investment such as reduced property and real estate acquisition taxes if houses satisfy certain requirements. However, these tax privileges do not seem to have as large an impact on housing investment as the low interest rate loans provided by the Housing Loan Corporation; these have been 2 to 3 percentage points lower than market rates.[181] The tax system also treats owner-occupied houses more favourably than rental houses through various measures such as the exemption of imputed rents and the tax credit given for part of the outstanding amount of mortgage loans. On the other hand, rental housing enjoys other advantages such as the deductibility of interest payments and depreciation from income.[182] These measures seem to offset each other, resulting in small differences in the cost of capital between rental and owner-occupied housing (Iwata, 1997). However, wealthy people enjoy lower costs of capital of rental house construction, as the combination of the deductibility of interest payments and the tax shelter provided by the capital gains tax in the case of sales at later dates could save a further ¾ to 2 percentage points.[183] Overall, tax effects on the tenure choice are limited compared with countries where interest payments on housing loans are

Table 37. **Changes in capital gains tax rates for individuals**

	1982-86	1987	1988	1989-91	1992-94	1995	1996	1997	1998	1999
Tax on short-term holdings										
Marginal tax rate (per cent)[1]										
Special treatment of holding for less than two years		65[2]								
Ordinary treatment	52[3]									
Tax on long-term holdings										
Eligibility of long-term holding	Over 10 years	Over 5 years								
Marginal tax rate (per cent)[1]										
Bracket										
0 to 20 million yen	26			26	39	32.5	26		26	26
20 to 40 million yen										
40 to 60 million yen	Progressive rate[4]			32.5		39	32.5			
60 to 80 million yen									32.5	
Over 80 million yen							39			

1. All tax rates shown here include local residents' tax.
2. The amount of tax payment is decided by the greater of: *a)* separate taxation by a 65 per cent rate or *b)* an additional payment in the case of comprehensive taxation times 1.2.
3. The amount of tax payment is decided by the greater of: *a)* separate taxation by a 52 per cent rate or *b)* an additional payment in the case of comprehensive taxation times 1.1.
4. Half of the gains are added to other incomes and taxed in a comprehensive way.

Source: Ministry of Finance.

deductible from income. Although the introduction of such a measure may reduce significantly the costs of capital of owner-occupied houses, the benefits would probably be concentrated on wealthy households.[184] Equity considerations thus seem to explain the government's decision this year to temporarily expand the existing tax credit scheme, rather than introducing deductibility of interest expenditure (see Chapter III).

Income taxes are almost linear but pre-tax distribution is also even

The salient egalitarian features in Japanese society are reflected in a relatively even income distribution both before and after taxes and transfers (Table 38).[185] Although the distribution in both pre- and post-tax terms has widened a bit since the early 1980s, it is still among the most equitable in the OECD, apart from the Nordic countries and the Netherlands. Since the pre-tax income distribution is already fairly compressed, the total amount of redistribution is relatively low: post-tax and transfer income is only 22 per cent more equally distributed than its pre-tax and transfer counterpart. This amount of redistribution is roughly the same as that of the United States, but lower than that of Germany and Italy (30-35 per cent). It is much below Sweden and other small European countries, where the redistributive impact of taxes and transfers lowers inequality by around half. The income measure used for calculating the Gini coefficients does not include fringe benefits. These tend to increase with income and hence raise inequality. More importantly, wealth is not included in the distributional measures presented here. The "bubble" economy years spurred a debate about this aspect of distribution (which also formed part of the decision to raise taxes on land holdings in 1992). For instance, one study found that adding estimated imputed rent from land to personal income would have raised measured inequality in pre-tax income by 16 per cent (based on 1990 numbers).[186] The fall in equity and land prices in recent years has presumably attenuated inequality in the distribution of wealth.

Looking at various measures of *statutory progressivity*[187] a somewhat mixed picture emerges. Measured by the distance between the average tax rate for an average production worker (APW) and the top statutory rate of 50 per cent, Japan has by far the most progressive system of the major economies (Figure 44, Panel A). This measure, however, does not take into account the distance between the APW and the point where the top rate sets in, which is particularly extended in Japan (where the top rate only starts to kick in at four times an APW). When measured at the margin, *i.e.* how much net income increases when gross labour costs rise by 1 per cent, progressivity in Japan appears to be relatively low (Figure 44, Panel B). This is primarily a reflection of the smoothly progressing marginal rates. In short, the statutory system applies low progressivity to income earners making small incremental steps up the income ladder, but potentially a very high progressivity for individuals making big jumps on the earnings scale.[188]

Table 38. **Redistribution through taxes and transfers**

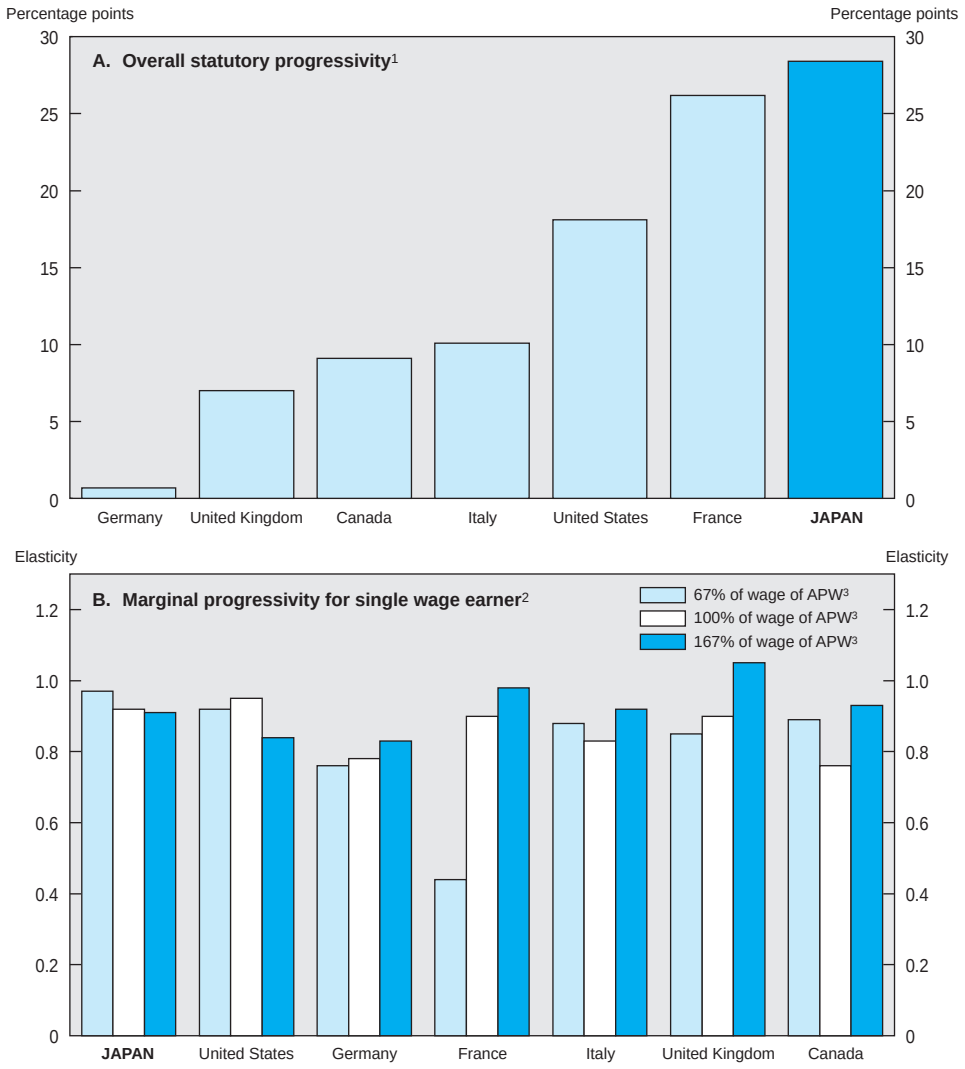
Equivalence scale elasticity = 0.5

	Gini		
	Before taxes and transfers (1)	After taxes and transfers (2)	% changes due to taxes and transfers (2)/(1) – 1
Australia, 1993/94	46.3	30.6	–33.9
% changes, 1975/76-1993/94	36.6	5.2	
Belgium, 1995	54.5	29.9	–48.4
% changes, 1983-95	–	2.3	
Canada, 1994	–	28.4	–
Denmark, 1994	42.0	21.7	–48.3
% changes, 1983-94	11.2	–4.9	
Finland, 1995	39.2	23.1	–41.0
% changes, 1986-95	11.4	9.1	
Germany, 1994	43.6	28.2	–35.3
% changes, 1984-94	1.2	6.4	
Italy, 1993	51.0	34.5	–32.4
% changes 1984-93	20.8	12.7	
Japan, 1994	34.0	26.5	–22.0
% changes, 1984-94	14.0	4.9	
Netherlands, 1994	42.1	25.3	–39.8
% changes 1977-94	14.2	11.8	
Sweden, 1995	48.7	23.0	–52.9
% changes, 1975-95	17.2	–1.0	
United States, 1995	45.5	34.4	–24.5
% changes, 1974-95	13.1	10.0	

Note: The Gini coefficient is a measure of income inequality; the lower the coefficient the more equally are incomes distributed.
Source: Oxley *et al.* (1997) and Burniaux *et al.* (1998).

Actual progressivity encompasses the effects of differences in take-up rates of tax allowances and tax credits across different income groups. Tax statistics illustrate how the income tax system is almost linear, except for the top and bottom end of the income curve (Figure 45, Panel A).[189] The tax statistics do not, however, include tax planning – the shifting of income into low taxed bases – or outright evasion. These effects are to some extent captured by income and tax payments reported by households in survey data. Although not directly comparable with the tax statistics, these data seem to reveal a slightly lower progressivity in direct tax payments (Figure 45, Panel B). Including social security contributions and implicit VAT payments,[190] which are both near-proportional or slightly regressive, the progressivity of the total tax burden diminishes further.

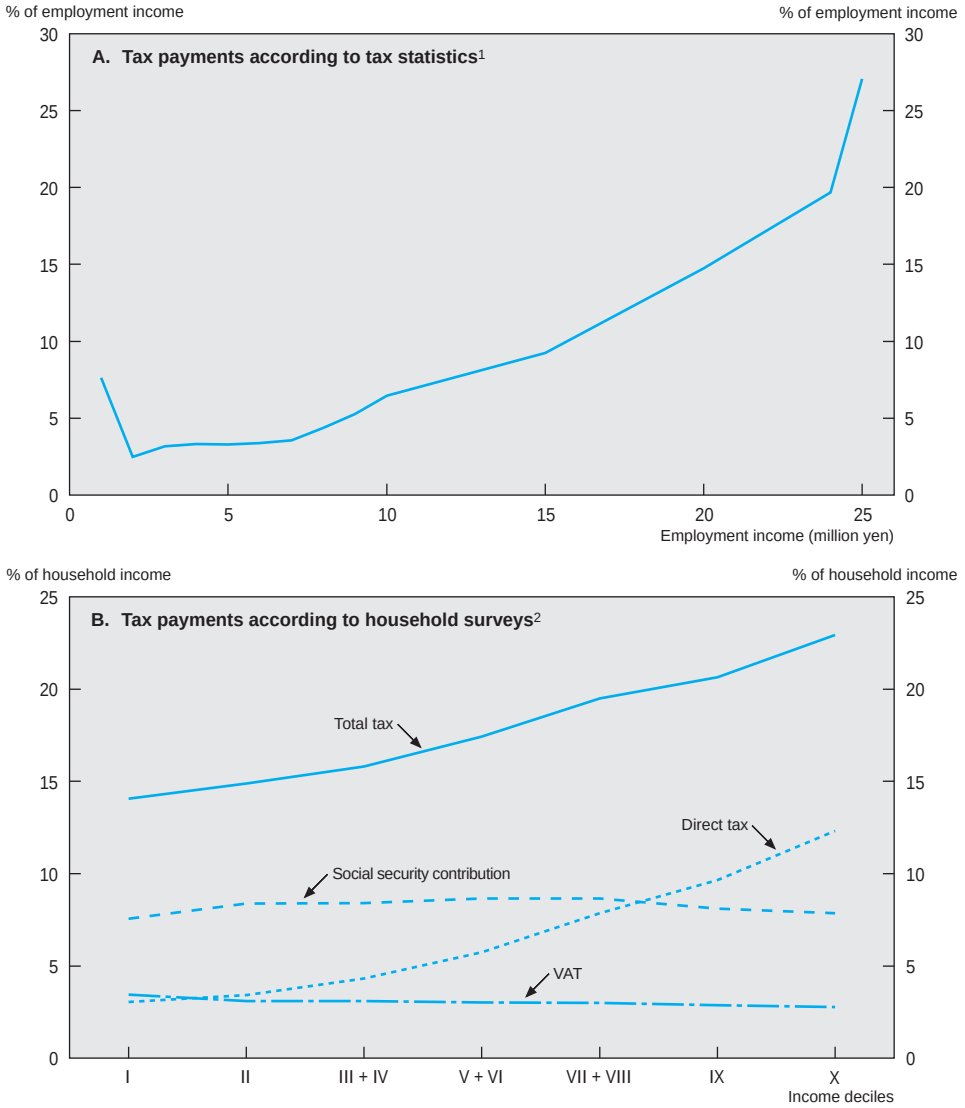
Figure 44. **Overall and marginal progressivity in major OECD countries**



1. This measure of progressivity is based on the difference between the top rate paid and the average production worker's (APW) rate. The figures for the APW are for the year 1997, while the figures for the top bracket earner are for the year 1998 (1999 for Japan).
2. Increase in net income after a one per cent increase in gross labour costs. Net income is gross earnings minus employees' social security contributions and personal income tax. In a proportional system the elasticity would equal 1. The more progressive the system, the lower the elasticity.
3. APW: Average production worker in manufacturing.

Source: OECD, *The Tax Benefit Position of Employees* and the OECD tax database.

Figure 45. **Actual tax payments by income group**
1998



1. Actual tax payment per income group in per cent of employment income (national employment income tax only).
2. Including local taxes.

Source: Statistics Bureau, Management and Coordination Agency, *Monthly Report on the Family Income and Expenditure Survey,* December 1998 and National Tax Administration, *Actual Condition of Private Sector Employment Income.*

Table 39. **Revenue losses due to special tax measures, FY 1999**

Billion yen

Measures related to savings	367
of which:	
Deduction of life casualty insurance premiums	285
Exemption of interest income on small deposits owned by elderly, etc. *(maruyu)*	69
Others	13
Measures related to investment	1 707
of which:	
Tax credit for acquisition of dwellings for individuals	551
Special depreciation measures	349
Tax credit for incremental R&D	29
Tax free reserves for corporations *(jumbikin)*	125
Others	653
Total	2 074

Source: Ministry of Finance.

Tax expenditures are moderately sized

Tax expenditures can be defined broadly in terms of departures from a "normal" tax structure. The precise definition differs somewhat among countries according to what is perceived as a "normal" structure. In Japan, the bulk of allowances, including the tax relief for retirement income, are defined as "normal" rather than "special", and hence tax expenditures are rather limited. For FY 1999, the total revenue loss due to so-called special measures amounts to 2 trillion yen, or 0.4 per cent of GDP (Table 39). Most tax expenditures in the personal system arise from the tax credit for individuals' acquisition of dwellings and the deduction of contributions to pension plans and life insurance companies. The *maruyu* allowances for small savings of the elderly, etc. are negligible. By far the most important tax expenditure in the corporate system is the accelerated depreciation schemes, including the scheme for information technology investment. Tax-free reserves for corporations constitute only a very small part of total tax expenditures. Since the tax-free reserves, the corporate tax credits and the special depreciation schemes together represent a revenue loss of less than 0.1 per cent of GDP, the scope for broadening the corporate tax base is rather limited.

Scope for strengthening the system: increasing revenues, efficiency and equity

The basic issue for Japan is how to maintain the overall low distortions resulting from the tax system while at the same time increasing revenues

sufficiently to meet future spending and deficit-reduction needs. This requires improvements in both efficiency and equity, first and foremost by broadening tax bases. While not an easy task, given the different political and economic interests involved, there appears to be broad public and political acceptance of the need to raise revenues in order to maintain the basic public welfare benefits. The timing and instruments are still debated, and successful tax reform is likely to involve a broad range of measures so as to spread the burden of higher taxes widely across different groups of society. The reform process should be facilitated by the ample scope for base broadening that already exists within the current system – in particular in the personal income tax system. A political consensus seems to be emerging that increasing VAT revenues should finance the brunt of the costs related to ageing, but no concrete steps have been taken since the VAT rate increase in 1997. Moreover, the scope for administrative improvement would be exploited more easily if Japan were to introduce a taxpayer identification number (TIN) system in line with the practice in most other OECD countries.[191] This issue is currently high on the agenda and is gaining increasing political support. These and other potential areas for strengthening the tax system are discussed in more detail below.

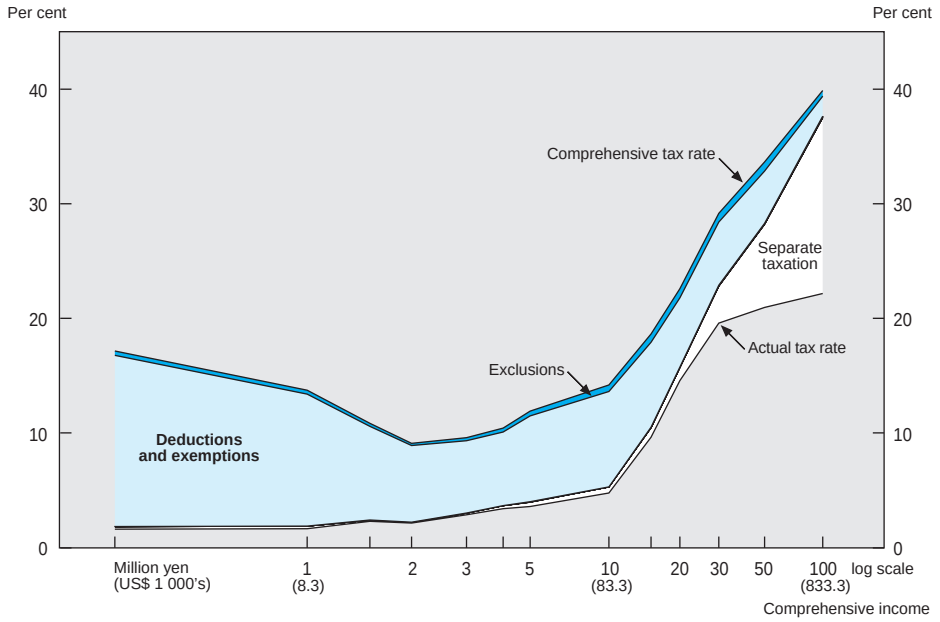
Base broadening is required if tax wedges are to be kept low

The rate and bracket structure of the personal income tax system implies that economic incentives are less distorted in Japan than in most other OECD countries. However, the same does not hold true for horizontal equity, the sense of fairness among taxpayers and the revenue-raising capacity of the system (*i.e.* how much more revenue is raised by a 1 percentage point increase in rates) due to generous allowances and tax credits. There is thus a substantial revenue potential in broadening the base for personal taxes, which, *inter alia*, could be used to finance a fairly large share of the incremental ageing costs or lowering income tax rates further if the public finance gap is filled by other means. A significant broadening of the personal income tax base would reduce the risk of having tax wedges – and hence economic distortions – rising excessively as the need for more revenue materialises. In the case of the ordinary and special allowances for dependent spouses, a substantial reduction could also help mitigate the adverse incentives for labour market participation faced by this group. However, broadening the tax base is a highly sensitive issue, given the vested interests involved, and so far the government has not been successful in broadening the base or even reversing the trend of ever-increasing allowances.

The impact of personal income tax reliefs on revenues and progressivity can be illustrated by comparing the actual tax paid by different income categories with the tax that would have had to be paid under a comprehensive income tax system (where all allowances, deductions and credits are abolished and personal

capital income[192] is taxed at the ordinary progressive income tax schedule). Figure 46 shows the difference, considering the national income tax only and assuming unchanged behaviour of the taxpayers. Several interesting patterns emerge. First, the comprehensive tax rate exceeds the actual by 8 to 18 percentage points, depending on the income group. Second, whereas low-income earners receive a significant relief of their average tax payment through the various allowances, the very high-income earners (with incomes above 20 to 30 million yen) receive a relief of similar relative magnitude through the separate

Figure 46. **Comprehensive versus actual tax rates by income group**
Fiscal year 1996



Note: Tax payments are for national income tax only. Comprehensive income is taxable income plus allowances and exemptions. It does not include unrealised capital gains, fringe benefits or imputed rents. The figure does not take estimated under-reporting of income into account.
Deductions and exemptions include around 20 kinds of ordinary reliefs encompassed in the income tax law, *e.g.* basic deduction, deduction for spouse and dependents, for medical expenses, etc. Exclusions concern the part of interest income excluded from taxation (maruyu), special deductions for retirement, timber and occasional income, etc. "Separate taxation" measures the effect of taxing income from interest, dividends, retirement and capital gains together with other personal income. Not shown in the figure is the effect of tax credits, *i.e.* credits for dividends, foreign taxes, incremental R&D expenditure and acquisition of dwellings. These credits have only a negligible effect on taxation. In 1996 they amounted to 0.11 per cent or less of comprehensive income for all income groups. With the expansion of housing tax credits introduced as of FY 1999 the effect has increased, but these credits are still only marginal to the total loss of tax revenues caused by tax relief.
Source: Hiromitsu Ishi, *The Japanese Tax System,* 3rd edition (forthcoming, Spring 2000).

taxation of capital and retirement income. Those in between are relatively less favoured by the tax reliefs. Third, the tax credits for dividends and foreign taxes as well as the exemption of interest income of the elderly are of very limited importance in reducing total tax payments.

The various tax reliefs imply a somewhat lower progressivity of the tax system, especially at the high end of the income scale, whereas progressivity for incomes between 10 and 30 million yen is barely affected. The total revenue loss stemming from the tax reliefs defined in Figure 46 amounts to around 30 trillion yen or 6 per cent of GDP, of which some 5 per cent of GDP is due to tax allowances (1996 figures). If local personal income taxes, which are levied on almost the same income base as national taxes, were also made "comprehensive", an additional 2 to 3 per cent of GDP could be added to tax revenues.[193] Since allowances have increased significantly since 1996, it is not unrealistic to assume that the tax reliefs, even excluding the separate taxation of personal capital income, could amount to somewhere near 10 per cent of GDP. It is, of course, important to bear in mind that an elimination of tax reliefs would change taxpayers' behaviour and thus offset some of the revenue gains. In particular, higher taxation of capital income could lead to more evasion and avoidance among high-income earners. Elimination of the employment income deduction and basic allowance would discourage labour market participation at the lower end of the income scale, whereas an abolition or reduction of the deductions for dependent spouses could stimulate labour supply. Finally, the figure makes the important point that allowances are a costly mechanism of redistributing income, since high-income groups benefit significantly from the reliefs. In fact, as the tax value of deductions increases with the tax rate, higher-income groups receive a larger tax relief per yen worth of deduction. It follows that any objective for the distribution of net disposable income could in principle be met at a much lower cost through targeted (means-tested) transfers or tax credits to low-income groups, while reducing general deductions substantially. The *caveat* is then that targeting normally implies high effective marginal tax rates for persons leaving the targeted group – a problem that can be mitigated by gradually phasing-out the transfer or credit as income increases but is normally hard to circumvent, especially where the pre-tax income distribution is compressed, and budgetary costs are therefore considerable.

Any reduction in the allowances of the personal income tax system would be difficult to carry out without strengthening the effective taxation of the self-employed. The self-employed (persons earning personal business income) are allowed to split their incomes by paying salaries both to other members of their families and to themselves, resulting in a lower tax burden (Ishi, 1998). Also adding to the sense of unfairness among wage earners is the possibility for the self-employed to deduct some of their private consumption as necessary business expenses. From a revenue-raising point of view, the largest problem is not

the direct loss of revenue incurred by taxing the self-employed at low effective rates, but the indirect loss due to allowances made to wage earners to compensate for horizontal inequities (Box 10). One option could be to move from an income tax base to a broader and less easily manipulated base for the self-employed, for instance value added.[194] A less radical option is to strengthen surveillance and enforcement of tax payments by the self-employed, so that at least they are not able to "get away" with evasion so easily. It should be emphasised that the problem of effectively taxing the self-employed at the same rate as wage earners is inherently difficult and thus contributes to perceived inequities in almost all OECD countries. Furthermore, the problem in Japan has been somewhat reduced since the 1960s for the simple reason that the share of the self-employed in total employment has declined sharply and is now close to the average of the other major economies.

Box 10. Inefficiency of tax reliefs

Japan's different tax allowances are based on policy objectives of encouraging home ownership, stimulating family formation, etc. The employment income deduction was introduced to improve horizontal equity between wage earners and the self-employed, who are allowed to deduct so-called "necessary" expenses, including expenses for clothing, etc. Likewise, the deduction and special deduction for dependent spouses have been instituted to narrow the gap between wage earners and the self-employed with respect to their opportunities for income splitting: whereas the latter are able to shift a part of their income to family members, employees do not have this option. A reason for introducing the special deduction for spouses was also the so-called "part-timer question": spouses were given incentives not to earn incomes above a certain threshold in order for the household not to lose the entire regular spouse deduction. However, instead of remedying this "kink" in the married woman's labour supply curve by gradually phasing out the existing allowance, an additional allowance (the special allowance for spouses) was introduced (in 1987).

Even though allowances and deductions may be for legitimate social and political purposes, it is often questionable whether these objectives are achieved in the most efficient way (if at all), and these objectives should also be seen against the social cost of higher tax rates. The gradual increase in personal allowances and deductions not only makes the base smaller, thereby necessitating – all else being equal – higher rates and thus more distortions; it also contributes to a more complex system and reduces equity. Widening of the tax brackets is a better way of alleviating tax burdens at certain points of the income scale, since it retains horizontal equity and also mitigates the problems of bracket creep and bunching of income in short time spans (which in particular tend to put people with a long education at a relative disadvantage, since they risk paying higher marginal taxes than people with the same lifetime income earned over a longer time span).

The potential for base broadening is also found elsewhere. Social security contributions are levied primarily on ordinary salaries, whereas bonuses and retirement lump-sum payments have traditionally been (almost) exempted.[195] This has serious consequences for the base, since these payments constitute a considerable part of total remuneration of employees in Japan.[196] The exemptions imply that social security rates have to be raised unduly to restore actuarial balance in the social security system and that employees with the same income may be treated differently depending on the ratio of bonus to ordinary salaries. Even though benefits are also based on ordinary salaries it would be preferable to treat all wage income in a similar fashion for social security purposes (Horioka, 1999). Another exemption is that of dependent spouses who do not contribute to the pension system but are eligible to the basic flat-rate benefit and survivors benefits. This causes a redistribution from single workers and double-income couples to couples with a dependent spouse. Moreover, it gives an incentive for dependent spouses not to participate in the labour market, since they would have to contribute once their earnings exceed a certain (low) level.

The VAT base could be moderately broadened by a substantial lowering of the tax-exempt threshold, whereas an elimination of the simplified regime would probably only add marginally to the base.[197] Even though administrative simplicity calls for some minimum threshold of sales to be taxed, there seems to be no economic or administrative justification for extending the tax exemption as far as is the case in Japan. Furthermore, detrimental effects on equity and compliance are likely to increase in proportion to any increase in the VAT rate. On the other hand, the potential for base broadening should not be exaggerated: total sales by exempt enterprises are estimated by the Ministry of Finance to be less than 3 per cent of total taxable sales. In addition, since tax-exempt enterprises cannot deduct VAT paid on purchases, the overall revenue loss is probably rather limited. The absence of invoicing may make the system less robust to evasion. However, the single tax rate and the requirements to keep trade documents (receipts, business invoices, delivery statements, etc.) to be allowed to deduct purchases imply that enforcement is probably not much hampered compared with invoice systems. The question of invoicing would become even more trivial if access to the tax exempt and simplified regimes were tightened.

Another potential area for base broadening is environmental taxes. As with most OECD countries Japan levies excises on a number of energy products, mainly unleaded petrol, diesel and electricity consumption, as well as on motor vehicle registration. In 1995 the total revenues from these taxes amounted to less than 2 per cent of GDP (or 6 per cent of total taxes), which is higher than in the United States but lower than in most other OECD countries. As discussed in Chapter V, a greater reliance on environmental taxes could serve several purposes since Japan both needs more tax revenue and is faced with major problems of fulfilling its Kyoto commitments on reducing greenhouse gas emissions. The

extent to which environmental taxes can raise revenues should not be overestimated, however, in part because the idea of such taxes is to reduce emissions and hence revenues. Even in countries with higher environmental taxes, like the Nordic countries, the share of revenue from these taxes to total tax revenue does not exceed 10 per cent.

The VAT rate should be increased, while income tax rates may be lowered further

Even if bases are broadened there would probably still exist a financing gap – especially if the revenues from base broadening in the personal tax system are used for lowering income tax rates in the personal and corporate system. A simple, efficient and fairly equitable way to close any remaining gap would be to raise the VAT rate. The VAT has several advantages from a taxation viewpoint. It is neutral *vis-à-vis* savings decisions and is relatively difficult to avoid or evade for all groups, thus meeting the criteria of horizontal equity better than income taxes (as long as it is broadly based as is the case in Japan).[198] Moreover, raising the VAT will be one way of ensuring higher effective taxation of the elderly, thereby contributing to a fair sharing of the tax burden across generations.[199] However, it would not be advisable to try and earmark additional VAT revenues for the costs related to ageing (Box 11).

Japan's VAT system has so far worked well compared with those of many other OECD countries. The effective VAT rate is fairly close to the standard rate, indicating that revenue losses due to the high tax threshold, the simplified regime and non-compliance are lower than those experienced in many other countries (Table 40). If this level of effectiveness can be sustained under increased VAT rates, the revenue potential would be high: about ½ percentage point of GDP additional revenue is raised per 1 percentage point increase in the VAT rate.

Evidently, an increase in the value added tax could give rise to legitimate distributional concerns, since the immediate effect is to tax households with low savings rates – typically lower income households – at the highest effective rates. Over a longer time horizon, however, VAT is in effect a proportional tax on lifetime income or intergenerational income. Based on survey data for worker households it is possible to get a grasp of which income groups are facing the highest burden from VAT payments. As expected, the highest-income households pay a lower percentage of their disposable income in VAT (Figure 47), but the difference among income levels is not very significant: VAT amounts to around 3.4 per cent of the lowest income deciles' disposable income and 2.8 per cent of that of the highest income decile. When correcting for household size, the difference between income deciles is virtually eliminated, since higher income households are also the largest.

Box 11. Earmarking of VAT revenues for social security

There has been some discussion in Japan of whether increases in VAT revenues should be earmarked for future public pension obligations. Politically it may be easier to make the electorate swallow VAT increases if these are referred to as "earmarked" for specific contingent public expenditure. However, earmarking is normally not the most efficient way to manage government finances. Even though earmarking in principle allows a closer link between those who pay the tax and those who benefit, there are several disadvantages. First, it reduces the flexibility of policy makers to adjust spending policies as expenditure needs change over time. Second, if revenues are more buoyant than the expenditure for which they are targeted, it is difficult to avoid extending the programme beyond its original objectives. Revenues from a specific tax should thus be spent in areas where the returns to society are highest, including on reducing other, more distortive taxes. In particular, setting up a government fund for the additional VAT revenue would not be recommended. It is a much better option to use any additional revenues to bring down government debt, since a large gross government debt makes the budget more vulnerable to interest and exchange rate shocks and since the returns to government assets are often lower than payments on liabilities. Besides potential problems of governance and moral hazard linked to government funds, there are also costs related to managing the assets in such a fund, just as it adds more complexity and less transparency to the overall management of the government's net financial position. Finally, the accumulation of earmarked revenues risks blurring perceptions of what agents are entitled to receive in terms of pensions and other benefits, *i.e.* the question of ownership can become non-trivial.

Tax treatment of savings – in particular pension savings – should be more equitable

Equity considerations may call for personal capital income to be taxed more in line with other forms of income – for instance by taxing such income at the ordinary progressive personal income tax schedule in a comprehensive system.[200] As shown in Figure 46 above, the bulk of the tax subsidy implied by separate taxation of capital income – which roughly amounts to some 0.7 per cent of GDP – is appropriated by very high-income earners. However, concerns over efficiency and compliance may call for a continuation of the current separate tax regime, since these bases are particularly sensitive to tax changes, and even more so as international markets become increasingly open. The effective taxation of capital gains is difficult to enforce, but a taxpayer identification number system could help to alleviate evasion in this area. A further development of the separate system to a pure "dual" income system could help to improve neutrality across savings instruments, in particular by aligning tax rates on capital income from all sources, including corporate income and capital income of the self-employed.[201]

Table 40. **Effectiveness of value added taxes**

1997

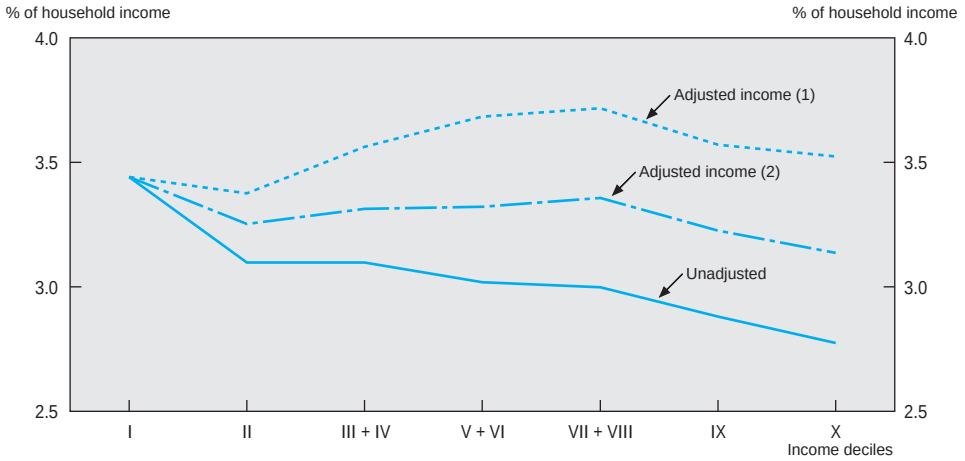
	Value added tax revenues in per cent of GDP	Standard rate,[1] per cent A	Effective VAT rate,[2] per cent B	Effective VAT rate in per cent of standard rate B/A
United States	n.a.	n.a.	n.a.	n.a.
Japan[3]	**1.8**	**5.0**	**3.1**	**89**
Germany[4]	6.6	16.0	11.5	77
France	7.9	20.6	14.7	71
Italy[5]	5.7	20.0	9.8	52
United Kingdom	6.9	17.5	10.8	62
Canada[6]	2.5	7.0	4.3	61
Australia	n.a.	n.a.	n.a.	n.a.
Austria	8.2	20.0	15.3	76
Belgium	7.0	21.0	12.5	60
Czech Republic	7.1	22.0	12.6	57
Denmark	9.8	25.0	22.2	89
Finland	8.2	22.0	18.1	82
Greece	7.5	18.0	11.6	64
Hungary	7.9	25.0	14.4	58
Iceland	9.4	24.5	17.3	71
Ireland	7.7	21.0	17.4	83
Korea	4.3	10.0	8.5	85
Luxembourg	5.7	15.0	..	..
Mexico	3.1	15.0	4.7	32
Netherlands	7.0	17.5	13.2	75
New Zealand[7]	8.8	12.5	15.2	122
Norway	8.8	23.0	21.7	94
Poland	8.4	22.0	..	..
Portugal	7.9	17.0	14.4	84
Spain	5.8	16.0	9.9	62
Sweden	7.3	25.0	14.5	58
Switzerland	3.4	6.5	5.7	88
Turkey	6.5	15.0	9.9	66
OECD average[8]	6.7	17.7	12.5	73
G7 average[8]	5.3	14.4	9.0	69
EU average[8]	7.3	19.4	14.0	71
Dispersion OECD				
Range (maximum-minimum)	7.9	20.0	19.1	90
Standard deviation	2.1	5.7	5.0	18
Coefficient of variation	0.3	0.3	0.4	0.2

Note: n.a. = Not applicable; .. = Not available.

1. As of 1 January 1998 (except for Germany).
2. The effective VAT rate is VAT revenues divided by the base (*i.e.* consumption excluding VAT).
3. Fiscal year basis (Q2/97-Q1/98). The standard VAT rate was raised from 3 to 5 per cent on 1 April 1997. However, the bulk of VAT payments in FY 1997 were based on the 3 per cent rate since the payments are lagged 2 months and are based on tax payments in previous periods (business operators with tax payments in the previous year exceeding 5 million yen pay VAT 4 times a year, others pay twice a year. Moreover, business operators can choose tax periods of 1 year or 3 months). It is assumed that the "effective standard VAT rate" was 3.5 per cent in FY 1997.
4. The standard VAT rate was raised from 15 to 16 per cent on 1 April 1998.
5. The standard VAT rate was raised from 19 to 20 per cent on 1 October 1997. It is assumed that all 1997 payments were based on the 19 per cent rate.
6. Fiscal year basis (Q2/97-Q1/98).
7. Fiscal year basis (Q3/97-Q2/98). The general sales tax at standard rate is levied on 60 per cent of the value of the supply for long-term stay in a commercial dwelling, which may partly explain why the effective VAT rate exceeds the standard rate.
8. Simple average over available countries.

Source: OECD, *Revenue Statistics* and *Consumption Tax Trends,* and OECD calculations. See also Carey *et al.* (forthcoming).

Figure 47. **VAT payments based on household surveys**
1998



Note: Adjusted income takes into account differences in household size. Adjusted income (1) makes a simple correction for the number of individuals in the household, whereas adjusted income (2) corrects for households economies of scale by applying an equivalence factor of 0.5 (*i.e.* corrects for the square-root of the number of individuals in the household).
Source: Statistics Bureau, Management and Coordination Agency, *Monthly Report on the Family Income and Expenditure Survey,* December 1998 and National Tax Administration, *Actual Condition of Private Sector Employment Income.*

A number of issues relate specifically to the taxation of pension savings. When pension contributions are exempted from taxation there is a strong case for taxing benefits, and *vice versa*. By far the most important tax on employees' pensions in the OECD is that on benefits. Two issues are relevant here, the appropriate treatment of lump-sum payments and whether pension income more generally should receive preferential treatment compared with wage income. In several countries, including Japan, Australia, Ireland and the United Kingdom, lump-sum payments are taxed more leniently than annuities. For some countries, like Japan, this is an accepted part of the tax system and relates to the long-standing practice of making lump-sum payments to employees at the point of retirement. This lump-sum payment has traditionally been one of the main means of providing income security after retirement, together with public pensions and individual savings. However, since lump-sum payments and pension income from occupational schemes are in fact substitutes for each other, there is no *a priori* reason to give one option tax preference over the other. Taxation of lump sums requires some adjustment, however, in order to avoid a marginal rate well in excess of the recipient's expected average marginal rate during retirement. One possibility is to

apply an averaging provision (as is already the case for timber income and fluctuating and extraordinary income).

Since Japan is among the countries taxing pension savings least heavily, it would also appear appropriate – not only due to the horizontal inequity this causes, but also given the financing needs and the generational imbalances – to investigate the possibilities of higher taxation in this area. However, although the relative income position of retirees in Japan is better than in many other OECD countries (Table 41), indicating that their ability to pay taxes is not particularly low, higher effective taxation of pension benefits is a politically delicate matter and should be designed carefully with other policies to take into account any adverse effect on income distribution and potential poverty problems among retirees. The recent proposal to index public pension benefits to prices rather than net wages would have the same effect as higher taxes on benefits. Changes in taxation should also, in any case, only be implemented gradually over a fairly long time horizon in order to avoid abrupt changes in individuals' standards of living. A point of departure could be to reduce the deductions for retirement and annuity income.

Taxing the earnings of private pension funds as they accumulate would not help alleviate intergenerational imbalances, since the cost would be borne primarily by the younger generations, but it could contribute to improving neutrality across savings instruments.[202] If such a change were made, it would be vital to maintain or even enhance the compulsory elements of the retirement income system so as to avoid the moral hazard problem of people not providing adequately for their retirement in order to obtain social assistance benefits. Another problem is that many of the private pension trusts are severely under-funded. A tax on these funds would either place a higher burden on corporations or require the defined benefits of the occupational schemes to be adjusted accordingly. A pertinent issue is also the specific design of the earnings tax. Only real earnings should be taxed for long-term savings commitments of this type so as to preserve the real value of the savings.[203] For Japan, however, the most straightforward option would be to reinstate the tax on the assets of tax-qualified plans, which is suspended for FY 1999 and FY 2000, and extend this tax to other pension funds as well. The revenue potential of taxing pension savings as they accumulate is likely to be small, probably below 1 per cent of GDP on an annual basis,[204] and the contribution to closing the gap in government finances would therefore be modest.

Corporate restructuring in the short and long term could be promoted more strongly

The recent reductions in the marginal corporate tax rate have to some extent mitigated the non-neutrality across financing and investment instruments

Table 41. **Relative disposable income by age of the household head**

Ratio in per cent and changes in percentage points

	Age of head of household			
	Young head	Prime-age head	Older working-age head	Retirement age head
Australia, 1993/94	101.1	101.4	110.9	68.2
Changes 1975/76-1993/94	–4.4	1.4	1.3	–5.7
Canada, 1990	87.7	101.5	111.9	87.3
Changes 1985-90	–1.8	0.4	0.5	–0.4
Denmark, 1994	89.6	105.9	117.3	73.4
Changes 1983-94	–10.9	–1.5	10.7	4.7
Finland, 1995	80.3	106.8	114.5	78.1
Changes 1986-95	–7.6	0.3	6.4	1.1
Germany, 1994	78.5	100.9	113.0	89.3
Changes 1984-94	–1.9	–1.5	0.6	4.3
Italy, 1993	92.1	98.1	109.9	84.7
Changes 1984-93	–4.8	1.5	–2.7	2.9
Japan, 1994	75.9	94.2	120.7	93.1
Changes 1984-94	–6.0	–0.9	3.6	–0.8
Netherlands, 1994	85.2	100.8	114.0	87.5
Changes 1977-94	–5.9	5.0	–2.2	–8.9
Norway, 1995	78.0	107.3	117.3	73.7
Changes 1977-94	–11.5	0.6	4.6	4.0
Sweden, 1995	73.3	104.2	125.8	89.3
Changes 1975-95	–15.8	–5.3	12.2	16.5
United States, 1995	75.0	101.5	120.0	91.9
Changes 1974-95	–9.5	0.9	1.8	6.4

Note: Relative income by age head is the ratio of the relative household disposable income of the households concerned to the average for the entire population. Thus, for Japan, the relative average income for households with a retirement age head was 93 per cent of total average income in 1994. This ratio has decreased by 0.8 percentage points over the 1984-94 period.

Source: Oxley *et al.* (1997).

and have brought Japan more into line with some of the other large economies. A lowering of corporate taxes tends to decrease the marginal effective tax rates on new equity and retained earnings while increasing the rate on debt.[205] However, even though tax considerations are only one of the determinants of corporations' financial structures, the tax system still gives firms a relatively strong incentive to rely on debt financing (see Table 33). This has adverse consequences both on a microeconomic level (how to ensure sufficient risk capital to new ventures and a level playing field between small and large enterprises) and on a macroeconomic level: business cycles tend to be amplified, since corporations are more vulnerable to cyclical downturns and shifts in banks' lending standards, not to mention

outright credit crunches. One indication of this exposure is the high debt/equity ratio in the non-financial corporate sector.[206]

It is difficult to quantify the extent to which distortions to corporate financing have led to an inefficient allocation of capital and hence lower growth. Although the effect should not be exaggerated (*cf.* Tachibanaki, 1997), it could nevertheless be an objective for the longer term to analyse options for a tax system in which the incentives for corporate finance are more neutral. Several options are available, including: aligning personal taxation of various types of capital income, limiting interest-rate deductions at the corporate level, lowering corporate taxes further and/or moving towards an imputation system (Box 12).

Policies for encouraging restructuring in the corporate sector in the short run are also appropriate, since a significant amount of capital and land is currently locked into unproductive use for a variety of reasons, including taxation. The

Box 12. Imputation versus a classical system

Under a full imputation system dividend income would only be taxed once since the shareholder receives a tax credit for the corporate tax payment. However, even though such a system would be more neutral with respect to financing decisions (debt versus equity), it is not a panacea. First, the double taxation of dividends does not necessarily affect overall corporate investment, depending on whether the marginal investment is financed out of debt, retained earnings or new equity. Only in the last case is the level of investment distorted by the taxation of dividends. Second, corporate income taxes may be partly or fully shifted.* Third, the investor may be *de facto* exempted from dividend income tax. Fourth, there are severe practical problems related to changing the corporate tax system to a full imputation system, not least with respect to international tax agreements (see for instance Messere (1993)). There is no conclusive evidence among the OECD countries as to which systems lead to better economic performance (if they affect economic performance at all), and there has been no clear tendency to move in one or the other direction. The issue of imputation is thus not confined to the interaction between corporate and personal tax rates but should be considered in a more general framework. In any case, it is probably safe to conclude that double taxation is not a major problem for large, internationally oriented companies, since these firms have access to many different financing sources. However, for small firms relying on domestic capital markets the double taxation of dividends can be a potentially serious problem.

* Who eventually pays the corporate tax bill is an open question: employees, consumers, shareholders or all owners of capital. Furuta (1965, 1970) found in a classic study for Japan a significant degree of tax shifting into higher prices. This conclusion has not been unchallenged, however, and is probably less valid today anyway since markets are becoming increasingly open.

government has already taken measures to promote restructuring. The period for which firms are allowed to carry forward losses has already been extended as of July 1999 – from 5 to 7 years[207] – although only up to and including FY 2002. Judging whether further extension of the loss carry-forward period should be considered requires more analysis of its impact on corporate restructuring as well as on the book-keeping and tax review period which, according the basic tax policy orientation of the authorities, should match that of the loss carry-forward period. Another instrument for promoting corporate restructuring would be the implementation of a fully consolidated tax treatment for holding companies. The Japanese government is considering the introduction of such a scheme. Both measures will imply budgetary costs, but these should be weighed against the need for revitalising the corporate sector. The government has also moved to increase the support given to new ventures through the tax system because of the importance of reversing the declining trend in enterprise creation. This will give at least some impetus to medium-term growth prospects, although at the cost of some loss of neutrality.

Land and property taxation – distortions should be reduced to promote more efficient use of land

The combination of a high and non-neutral taxation of estates, a low holding tax on property and the structure of capital gains taxes causes important distortions to land and property prices. To alleviate the detrimental effects on land use, the inheritance taxes on land, in particular farmland, should be aligned with the taxation of other assets, *i.e.* the base should be broadened. This could also pave the way for reducing marginal rates, which in turn would reduce price distortions further. The benefits of improved neutrality achieved by such a change – more efficient land use, often in areas where land is in high demand, and more transactions – would come at a "cost" of basically removing a significant tax shelter from a very small group of urban farmers. Raising the valuation of land and property for property tax purposes would also be desirable from a local government point of view (Box 13), but it is much less important in terms of price distortions than improving the inheritance tax system (Nishimura *et al.*, 1999). Moreover, higher holding costs could encourage investment in projects yielding early cash flows, resulting in excessively small-scale developments (Kanemoto, 1997). Therefore, measures to increase property taxation should be accompanied by other incentives such as an increase in the maximum allowed coverage ratio of building capacity to lot size. The structure of capital gains taxation, which was established in the late 1980s to choke off short-term land speculation, also encourages long-term holding.[208] With speculation much reduced compared to the bubble period, consideration should be given to assessing whether the current structure of capital gains taxation is sufficiently neutral over the holding period (*e.g.* to what extent the special deduction – see footnote 179 – offset

Box 13. Increasing the valuation for property taxation

The property tax should be regarded as the "price" of infrastructure provision by local governments. The low effective tax rate is believed to discourage local governments from attracting housing projects because they will not raise sufficient tax revenue to cover necessary infrastructure expenditures.* A remarkable feature of the 1992 reform was the introduction of a national land value tax. This could be regarded as a counter measure by the central government to local governments' unwillingness to raise the effective rate of property tax (Kanemoto, 1994). That is, they were encouraged to raise the effective rates for fear that the tax base once "monopolised" by them might be used up by the central government. Although the national land value tax was suspended indefinitely in 1998, mainly in order to reduce the tax burden on companies, it served a useful purpose by successfully encouraging local governments to raise holding costs – a tendency which has been reinforced by the current severe financial situation of local governments. In this sense, the triennial reassessment of land valuation for the calculation of property tax in the year 2000 should move assessed values toward market values and thus constitute an important step in the right direction.

* Local governments often ask developers to bear a sizeable portion of the costs of infrastructure provision based on their guidelines for residential developments (*takuchi kaihatsu yoko)*. See Asami (1994) and Ministry of Construction (1998*b*) for detailed discussions.

the impact of relatively higher rates on short-term gains).[209] Finally, transaction taxes have been reduced recently, but there may be some scope for further simplification. Overall, the taxation of land should be subject to a review so as to promote its appropriate use.

Tax administration – a taxpayers identification number should be introduced

As in many other OECD countries Japan seems to face particular problems of taxing the self-employed, including farmers, on an equal footing with wage earners. As valid estimates of underreporting of income are inherently difficult to obtain – in particular on an internationally comparable basis – it is not possible to give any quantitative indication of whether Japan fares better or worse than other countries on this account. The horizontal inequity between these different groups is primarily due to the fact that taxes on wage earnings are withheld at the company level, whereas taxation of the self-employed is based on self-assessment. Evidence for Japan indicates that the tax gap – the difference between income reported to the tax authorities and actual income – has generally been higher for self-employment income than for wage income, although the difference seems to have narrowed in recent years. Ishi (1993), for instance, finds

Box 14. Blue and white returns

The self-assessment system is basically divided into two systems: white and blue returns. The blue return system was introduced around 1950 based on the recommendations of Dr. Shoup and is considered by the NTA as a fundamental requirement for efficient tax administration of self-assessed income. The main objective of the blue return is to encourage small and medium-sized businesses to keep a minimum set of accounting records. To promote the system, significant tax advantages are offered to corporations and individuals opting for the blue return: *a)* taxpayers filing a blue return are not subject to reassessment as long as errors are not found in their accounting books; *b)* they are allowed a special blue return deduction; *c)* they are also allowed to deduct reasonable amounts for wages paid to family members working in the company; *d)* finally, they are allowed to use special tax-free reserves (*e.g.* reserves for bad debts, losses due to price fluctuations, etc.), special depreciation rules as well as loss carry-forward and carry-back options. White return taxpayers are not granted these benefits, but are not obliged to keep books and records.

that while the period up to around 1980 was characterised by large discrepancies, the tax gap for both agricultural and other self-employment income declined sharply throughout the 1980s and almost reached the level of wage income by the early 1990s.[210] This development is attributed partially to a sharp increase in the share of farmers filing Blue Returns (Box 14), as well as the general decline of the agricultural sector to a modest size (from 12.9 per cent of total employment in 1980 to 7.0 per cent in 1997) and the tendency for merging small-size farms into larger units with modernised business management. However, the share of self-employed taxpayers filing Blue Returns has not increased at all since 1980 (Table 42) and remains at a modest level of around 50 per cent (despite the tax incentives for filing Blue Returns and the continued restructuring of the agricultural sector). The Blue Return diffusion ratio for corporations, which is not immediately comparable to that of the self-employed since corporations are required by the commercial code to keep detailed accounting records, appears to have stabilised at around 90 per cent.

Even if the problem of underreporting has been mitigated in recent years and is likely to continue to decline as the agricultural sector shrinks, there seems to be scope for further improvements in administration and tax structures to enhance horizontal equity. On the administration side, introducing a taxpayer identification number system (TIN) would be warranted, not only to facilitate monitoring of taxpayers, detect underreporting of income, enforce compliance and strengthen the perceived fairness of the system by wage earners, but also to improve tax procedures more generally to the benefit of citizens having contact

Table 42. **Blue Return diffusion ratios**[1]

	Number of Blue Return individuals (thousand)	Number of White Return individuals[2] (thousand)	Blue Return diffusion ratio for individuals[3] (per cent)	Number of Blue Return corporations (thousand)	Blue Return diffusion ratio for corporations[3]
1950	111	–	4	145	48
1955	519	–	32	347	68
1960	579	–	33	458	70
1965	794	–	33	653	75
1970	1 610	–	48	919	83
1975	2 421	–	53	1 305	88
1980	3 208	–	52	1 612	90
1985	3 812	–	51	1 834	90
1990	4 245	942[4]	52	2 242	89
1995	4 548	1 058	50	2 501	93
1996	4 593	1 054	50	2 538	93
1997	4 609	1 046	51	2 582	92
1998	n.a.	1 013	n.a.	n.a.	n.a.

1. Under the Blue Return system, a corporation or an individual who conducts a business, may obtain the approval of the tax authorities to file a tax Return using a special form printed on Blue paper. These taxpayers are required to maintain books and keep continuous accounting records that meet prescribed standards. In Return, the taxpayer is entitled by law to a variety of tax benefits.
2. The 1984 tax reform introduced an obligatory bookkeeping system and related provisions for non-Blue Return taxpayers (White Return taxpayers), which imposes less stringent bookkeeping requirements on the taxpayers.
3. The diffusion ratio for individuals shows the number of Blue Return taxpayers out of total business taxpayers with a tax liability. The ratio for corporations shows the number of Blue Return corporations out of the total number of corporations. The diffusion ratio up to 1965 includes corporations in liquidation but excludes them from 1970 onward.
4. 1991.

Source: National Tax Administration (1997).

with the tax authority. Japan is one of only five OECD countries not applying such an identification number, and the experiences of other countries suggest that a TIN is indeed an important and effective tool for the tax administration. A TIN would also pave the way for a more efficient taxation of capital income, not least capital gains.[211] Should Japan choose to move towards a comprehensive income tax system, *i.e.* incorporating capital income into the ordinary income tax system, a TIN will also be indispensable to make the system work effectively. The prospect of implementation of a TIN appears to have improved recently, in part due to the introduction of a new pension numbering system and the proposed introduction of a resident numbering system by the Ministry of Home Affairs. It should be emphasised, though, that the TIN is not by itself a guarantee for improved administrative efficiency – this depends on the information return system in which it is embedded.

Local governments: autonomy should be increased and prefectures' tax base made more stable

The municipalities' share of total local tax revenues amounts to 60 per cent, with almost all revenues coming from relatively stable bases such as individual income taxes and property tax (Table 43). Prefectures, on the other hand, are much more reliant on corporate income taxes. The revenue from the enterprise tax, which is their largest single revenue source, is highly sensitive to the business cycle. Current financial difficulties faced by prefectural governments, such as Tokyo, Osaka and Kanagawa, are partly due to a large fall in revenue from this source. Assuming that stabilisation policy is the proper responsibility of central government, local tax revenues should be based on more stable revenue sources. The government's Tax Council is currently considering how to change local corporate taxes, notably the enterprise tax, so as to select a broader and more stable base than corporate income. Such a step would also mitigate the inequity between loss- and profit-making firms, which encourages companies to look for ways to reduce their tax burden. According to tax statistics, approximately 60 per

Table 43. **Local tax revenues**

FY 1997, billion yen

	Prefecture tax	Municipal tax	Total of local taxes
Inhabitants tax	4 209	9 704	13 913
Individual income	2 808	7 172	9 980
Corporate income	954	2 532	3 486
Interest income	448	–	448
Enterprise tax	5 100	–	5 100
Individual	271	–	271
Corporate	4 829	–	4 829
Local consumption tax	807	–	807
Automobile tax	1 705	–	1 705
Light oil delivery tax	1 331	–	1 331
Real property acquisition tax	731	–	731
Property tax	8	8 822	8 830
City planning tax	–	1 326	1 326
Tobacco tax	248	799	1 047
Other taxes	809	557	1 366
Total	14 948	21 208	36 156
Memorandum item:			
Income-based taxes [1]	9 135	9 209	18 344

1. Sum of inhabitants tax (excluding per capita or per establishment payments) and enterprise tax.
Source: Ministry of Home Affairs.

cent of corporations nation-wide report losses and hence do not pay any corporate income tax. This leaves 40 per cent of corporations to shoulder the total corporate income tax burden even though all companies enjoy the benefits of the same public services.[212]

Several alternatives are available in choosing a more stable base for local corporate taxation, such as the capital stock, the wage sum or floor space. One particularly appealing option could be to tax value-added at the level of the firm (wage payments plus interest payments plus depreciation plus rents plus profits). First, such a tax would circumvent the problems of "cascading" (paying tax on taxes) inherent in other types of taxes based on sales. Second, it could be designed so as not to distort unduly the substitution between capital and labour inputs as could be the case for other options. Third, it would give reasonable assurance that the tax base would be growing in line with expenditure so that the need for frequent changes to the system would be reduced. Italy has recently introduced a tax of this type at the local level (Box 15). It could be argued that rather than creating a new corporate tax on firms' value added, it may be somewhat easier just to expand the already existing VAT and compensate the local governments accordingly through intergovernmental transfers.[213] Such a move, however, could limit local autonomy, even if this is not necessarily the case, and would also to some extent be in conflict with the idea of linking local taxes and benefits for corporations: the VAT payments by consumers may be distributed differently by prefecture than the distribution of enterprises receiving the services provided by local government. Furthermore, exporting companies do not pay VAT at all. A more serious concern of changing the base is how the incidence is distributed (for instance, taxing value added on the production side rather than on the consumption side) and how local governments' incentives are influenced

Box 15. Italy's new local corporate tax

Since 1 January 1998, both resident and non-resident companies in Italy have been subject to a "regional tax on productive activities" (IRAP) on their Italian source income. For manufacturing companies, IRAP is imposed at a rate of 4.25 per cent on the net value of production, which is calculated by subtracting the cost of production from the value of production. Certain deductions are not allowed, however, such as labour costs and interest expenses. The IRAP is thus equal to a value added tax at the level of the firm, except that depreciation is not taxed. This generates a tax bias in favour of capital *vis-à-vis* labour. Different rates of IRAP apply to banking institutions and insurance companies, and special rules for calculating the base also apply to such institutions.

in terms of competing to attract consumers, producers or wage earners by lowering tax rates. A closely related aspect is how to allocate tax revenues obtained from companies residing in more than one locality. In any case, any change in the base requires careful consideration and should also be related to the way the intergovernmental transfer system works and local governments' scope for setting their own taxes (which is currently very limited).[214]

Strategies for the future

Two broad options exist with respect to reforming the tax system so as to increase revenues, efficiency and equity. Basically, Japan can choose to move toward a comprehensive income tax system or a system of dual taxation of labour and capital income – in both cases assuming a significant increase in value added taxes and social security contributions as well as reduced personal allowances. In practice, the comprehensive income approach would require marginal personal tax rates to be lower than the current 50 per cent in order not to deviate too much from capital taxation in other countries. Such a lowering of rates could be financed by a reduction of personal allowances. The distributive profile of such a shift is uncertain: on the one hand, higher-income groups would face lower marginal rates; on the other hand, these groups also benefit the most from the income tax allowances and the separate taxation of personal capital income. A comprehensive system would entail a higher degree of horizontal equity but would risk being more vulnerable to base erosion and would also require administrative changes. The dual system could maintain the current structure of personal income tax rates basically as it is, in combination with one flat rate being levied on all capital income, including on corporate income. It has the advantage of being more flexible in terms of adjusting capital taxation in response to changing circumstances, for instance increased tax base mobility. It can also basically be carried out within the current administrative framework. Another issue is how the burden of the ageing society and the existing government debt should be distributed between generations. Clearly, the younger generations would be better off with any increase in taxes that alleviates the burden they would otherwise face because of increasing government debt. Both increased consumption taxes and reduced personal allowances, especially for retirees, would spread the burden more evenly across generations, while higher social security contributions and taxation of pension savings would be less balanced.

Conclusion and recommendations

The Japanese tax system applies relatively low marginal tax rates on most economic activities, which, in combination with moderate tax elasticities of the bases, indicates that the overall distortion from the tax system (the excess

Box 16. Options for reform

- *Tax reform* should cover a sufficiently broad range of measures ("package approach") to make all groups contribute to the inevitable tax increases. Transparency, including pre-announcement of reform, is important in order to stabilise expectations. Gradual implementation of measures may also be warranted.
- *Tax bases should be broadened substantially*. In the personal income tax system allowances should be reduced, including those for dependent spouse and dependent children. Efforts to tax the self-employed should be strengthened. It is imperative for any reduction in allowances in the personal income tax system that the feeling of unfairness among wage earners is not increased, and this requires strict enforcement and/or a broader base for taxing the self-employed. The base broadening in the personal income tax system could be used to finance part of the costs of ageing. The base for social security contributions should be broadened to include bonuses and lump-sum retirement payments. The VAT base could be moderately broadened by reducing the tax-exempt threshold. The potential room for base erosion created by the simplified regime should continue to be monitored carefully. Japan should also consider ways to broaden the base by implementing environmentally related taxes.
- *Increasing the VAT rate* gradually over a number of years (to 12-15 per cent) should be one of the key financing mechanisms for the costs related to ageing. However, earmarking of the revenues should be avoided. Introduction of the invoice method may be considered but does not appear essential.
- *Taxation of pension savings* should be stepped up in effective terms, at a minimum perhaps by reducing the indexation of retirement and annuity income allowances. Considerations should be given to levying a tax on earnings of pension savings as they accumulate (reinstate tax on assets of pension funds or introduce tax on real earnings). The tax treatment of lump-sum pensions and annuities should be made equal.
- *Social security contributions* should be increased as projected in the draft 1999 pension reform, but additional ageing costs should be borne by other taxes, in particular VAT.
- *Corporate taxation* is not in need of substantial reform but could be enhanced with a view to improving neutrality across financing and investment instruments. Incentives for corporate restructuring should be strengthened by implementing a consolidated tax treatment of holding companies.
- *Tax administration* should step up the efforts to control evasion – in particular among the self-employed. The introduction of a taxpayer identification number system would contribute to enhancing administrative efficiency.
- *Taxation of land and property* should focus on alleviating price distortions by aligning the inheritance tax treatment of land – in particular for agricultural use – with that of other assets. Re-examination of the inheritance tax rates should be considered as the base is broadened, and whether the current structure of capital gains taxation is neutral over holding periods should be carefully assessed. Further simplification of the transaction taxes should be considered. Overall, the taxation of land should be subject to a review so as to promote the appropriate use of land.
- *Local government taxes* – in particular at the prefectural level – should be made less volatile and more equitable among tax- and non-tax paying firms by broadening the local corporate base, for instance to a tax on corporate value added.

burden) is probably modest compared with other OECD countries. In particular, it appears that neither the labour force potential, including human capital formation, nor aggregate savings and investment decisions are unduly hampered by taxation. Corporate taxation has previously been fairly heavy, giving rise to a high overall effective taxation of capital, but effective corporate rates have recently been reduced by 9 percentage points and are now in line with, or only slightly above, those of many other OECD countries. The VAT has been a success in terms of its effectiveness in raising revenue and in contributing to balancing revenues from different bases. However, the high tax-exempt threshold implies an unnecessary erosion of the base. A main concern is the non-neutrality caused by differential treatment of different corporate investment and financing sources, but an even more serious matter is the vast allowances and tax credits given in the personal tax system. These preferences dilute the tax base and thereby sacrifice revenue; complicate administration and increase overall compliance cost; distort incentives for some groups on the labour market; and cause horizontal inequities. Effective taxation of the self-employed could be strengthened by further stepping up enforcement or by shifting the base for taxation from income to value added. Property taxes, in particular the inheritance tax, distort the prices of property and land and should be made more neutral through base broadening and perhaps also some reduction in rates. Tax administration would benefit from the introduction of a taxpayer identification number system. Local government finances would be improved by a shift in the base of local corporate taxation to a value added type tax.

Under current central projections it is not unrealistic to assume that Japan's tax-to-GDP ratio, including social security contributions, has to be increased by at least 3½ percentage points over the coming decade or so to finance the cost of ageing and merely stabilise government debt. Besides the rise in pension contributions scheduled by the draft 1999 pension reform as well as other social security contribution increases and spending cuts, additional revenues will have to be raised, mainly from increased personal income and consumption taxes, assuming that taxation of corporations and property are already at, or close to, their limits. Increased environmental taxes could probably only contribute marginally to financing the gap. A rough calculation shows that substantially reducing the reliefs in the personal income tax system could give an additional revenue of some 5 per cent of GDP,[215] and that increasing the VAT rate by 10 percentage points and broadening the base would give approximately the same amount. How exactly the tax increase should be composed between social security contributions, income taxes and consumption taxes is an open question. However, two firm conclusions can be drawn from the preceding analysis. First, a significant part of the required revenue increase should come from increasing the VAT rate and base. Second, that the personal income tax base should be broadened significantly, which in turn requires another hard look at how to ensure more

horizontal equity between wage earners and the self-employed. This would require a reversal of the recent trend of narrowing the base. Even though exemptions can serve some economically and socially legitimate goals, the alternative – to raise rates – would almost certainly be even more harmful to society, since distortions tend to rise disproportionately with rates. Box 16 summarises the total set of policy recommendations derived from the analysis.

V. Structural reforms to revitalise the economy

Structural policies and institutions are now widely seen as crucial in determining economic performance. It is therefore necessary that they be constantly checked as to their appropriateness to the evolving social and economic context. This process exists in all OECD countries, and the resulting reform proposals are debated and steadily implemented. In Japan there has been widespread agreement for some years now that the needed changes must cover a wide range of areas and that they must be profound in order for the economy to regain some of the dynamism it was renowned for in earlier decades. Many of the significant changes in the areas of banking and the public sector (including public pensions and health care) have been discussed in earlier chapters. Reforms in several other areas will be reviewed in this chapter. A summary of measures taken and an assessment thereof is provided in Table 44. Together they paint a picture of an economy in gradual transition from heavy regulatory and bureaucratic control to greater reliance on market-determined outcomes.

Financial sector reform: ongoing liberalisation to enhance competition

While the authorities have been preoccupied with ensuring that financial markets and institutions do not hinder the likelihood of economic recovery, at the same time they have continued to pursue the liberalisation agenda that was set a number of years ago and was popularly described as the "Big Bang". In general, these changes remove unjustified restrictions on entry, and by enhancing competition they are expected to expand consumer choice, cut costs, improve the quality of service and deepen markets. But, according to some observers (Ogawa, 1999), there is a long way to go, with precious little progress in the last decade in the products and services offered, staffing and promotion practices and use of labour-saving technology. Most of the Financial System Reform Law took effect in December 1998 (in the form of new Investment Trust and Securities and Exchange Laws, among others). Its key features were to remove the ban on over-the-counter sales of investment trusts (mutual funds) by banks, end the restrictions on off-exchange trading of listed securities, replace the system of licensing securities

Table 44. **Recommendations for further structural reform**

Based on previous and current surveys

Previous Surveys recommendations	Action taken since 1998	Current Survey assessment recommendations
I. Reform the financial sector		
i) Recapitalise the banking system	Ongoing. Second round of injections in March 1999 and initial injections for 4 regional banks in September 1999. Two large banks nationalised.	Not enough conditionality imposed, *e.g.* specific ROE objectives should be required.
ii) Shore up the life insurance sector	Inspections underway to ascertain their solvency. Prompt corrective action (PCA) procedures expected next year.	Relax rules preventing downward adjustment of guaranteed returns. Also move to raise the resources of the Policy Protection Corporation and set premia based on risks. PCA powers helpful.
iii) Tighten regulatory oversight and boost transparency and disclosure	New Financial Supervisory Agency established. Provisioning standards raised. Consolidated accounting being implemented.	Continue to deal aggressively with systemic weaknesses, especially in auditing.
iv) Complete the "Big Bang" liberalisation by March 2001	New Investment Trust and Securities and Exchange Laws took effect December 1998. New Financial System Reform Law took effect October 1999.	Do not delay implementation of "pay-off system" in April 2001. Remove all compartmentalisation and restrictions on issuance and purchasing of securities.
II. Enhance product market competition		
i) Accelerate deregulation	Deregulation programme expanded in March 1999. Move away from prior intervention to retroactive supervision.	Implement these measures as early as possible. Other actions, such as auctioning airport landing slots, would be helpful.
ii) Further strengthen competition policy	Consumer Contract Law proposed.	Implementation of the Law would be welcome.
	Co-operative-based cartels may be banned.	Such cartels have little justification.
Allow victims of unfair trade practices to go directly to court	No action.	Same as previous Survey.

Table 44. **Recommendations for further structural reform** *(cont.)*

Based on previous and current surveys

Previous Surveys recommendations	Action taken since 1998	Current Survey assessment recommendations
iii) Take steps to encourage competition in utilities. Lift fare controls in transport sector	Tariffs to be set freely except for price caps.	Unclear if competition for local telephone services will rise with NTT holding company structure. Allow private-sector entry in local services. Set network access charges based on lowest foreign rates.
iv) Eliminate use of supply/demand adjustment mechanisms	Expected by FY 2001 in taxis. Planned in February 2000 for domestic airlines and inter-city buses. Considered for local bus services.	Progress needs to be consolidated by actual implementation.
III. Promote industrial restructuring and boost technology and innovation		
i) Update the Commercial Code	In process. Mergers based on stock-swaps to be permitted. Minority holders to be obliged to tender their shares.	Ensure enough true outsiders on boards, especially on auditing committee. Recognise corporate officer systems. Facilitate stock-option schemes.
ii) Revise the patent law	Industrial Revitalisation Law legislated. Researchers to be accorded intellectual property rights on government-sponsored research.	New Law should be made permanent. Avoid need for government approval of business restructuring plans.
iii) Increase emphasis on basic research and improve efficiency of public technology spending	On-going efforts to make the system more flexible, competitive and open.	Continue these efforts and evaluate the achievement of the Basic Plan after 2000.
iv) Encourage more start-ups and venture business	Numerous tax and lending programmes since the November 1998 Emergency Economic Measures.	Limit personal liabilities in case of failure.
v) Streamline bankruptcy procedures	Composition Law to be reformed.	Greater simplification and reduced cost would be helpful.

Table 44. **Recommendations for further structural reform** *(cont.)*

Based on previous and current surveys

Previous Surveys recommendations	Action taken since 1998	Current Survey assessment recommendations
IV. Liberalise the agricultural sector	A new Basic Law enacted in July 1999.	Boosting self-sufficiency may be achieved only by incurring substantial costs.
	Little progress in reducing producer support although a move away from price support is planned.	Reduce producer support. Extend the range of products for which support shifts from prices to direct income aid. Encourage farm consolidation through changes in land and tax policies. Remove any direct support for the agricultural co-operatives.
V. Use market-based instruments to improve environmental outcomes	Automobile Acquisition Tax already based on fuel efficiency and other automobile-related taxes may be similarly changed. Fees being considered for household garbage disposal.	Make greater use of the tax system and market pricing more generally to achieve official objectives.
VI. Boost public sector efficiency	Administrative Reform Bill passed.	May not relieve overlapping responsibilities. Cost savings unclear.
	Public institutions being reorganised.	
	Bank of Japan to restructure its operations.	
	Postal service to be corporatised.	Accelerate implementation. Allow private mail delivery.
	Retrenching oil exploration operations.	Accelerate this process.
	Privatised KDD, further sales of NTT shares, plans moving forward for selling other transport and power producers.	Retain public ownership only where there is a clear economic advantage.
	Increasing cost-benefit and project analysis.	Good progress.
	Private Finance Initiative scheme passed.	Avoid moral hazard risks.
	Broadening criteria for awarding public procurement contracts.	Accelerate the adoption of this change.
	Reforming FILP.	Bring greater public scrutiny to bear on FILP-financed spending.
	Reforming local government financing and promoting decentralisation.	Improve incentives for local governments to tax more and spend less by making LAT less equalising. Favour municipal mergers.

Table 44. **Recommendations for further structural reform** *(cont.)*

Based on previous and current surveys

Previous Surveys recommendations	Action taken since 1998	Current Survey assessment recommendations
VII. Reform the pension system		
i) Overhaul the public component to put it on a sustainable basis	Draft Pension Reform Law presented to the Diet in July 1999.	A reasonable compromise which should be legislated without delay. However, contribution base could have been broadened.
ii) Ensure the corporate pension system is adequately funded	Disclosure required in FY 2000; shortages to be funded by FY 2015.	Allow employers to contribute cross-held shares, along with their voting rights, for far-reaching corporate governance benefits.
iii) Propose basic framework law	No action.	Same as in previous Surveys.
iv)		Require pension fund managers to seek maximum returns for their beneficiaries under a "prudent man" rule.
VIII. Reform the health care system		
i) Institute a long-term care insurance scheme	Set to go into operation in April 2000.	Do more to even out regional disparities in premia. Provide modest subsidies to family-based care.
ii) Modify the basic health insurance system	Still under discussion.	Reduce pharmaceutical consumption; encourage move away from fee-for-service payments.
Broaden contribution base	No action.	Same as previous Survey.
Avoid longer-term oversupply of physicians	No action.	Same as previous Survey.
Avoid incentives for overprescription of medicines and excessive numbers of "acts"	DRG-type system being considered for hospital financing.	Move to Bungyo system and stop direct dispensing of drugs. Doctors' incomes could be supported by raising average fee levels in exchange for constraints on the number of acts.

Table 44. **Recommendations for further structural reform** *(cont.)*

Based on previous and current surveys

Previous Surveys recommendations	Action taken since 1998	Current Survey assessment recommendations
IX. Increase the flexibility of employment and improve actual labour market policies		
i) Allow temporary employment agencies to play a larger role	Done *de jure* but not *de facto*.	Removing the regulatory prohibition should be programmed. Consider extending the one-year limit on such work periods.
ii) Make corporate retirement allowances portable	Defined contribution pension plans being considered.	Such a move would deepen the mid-career labour market.
iii) Permit private job placement firms to play a more important role	Done.	The exclusion of port and construction workers needs justification.
iv) Evaluate the effectiveness of all active labour market policies	No action.	The importance of this recommendation has grown with the increase in spending on such programmes.
v)	Comprehensive Plan for Employment Activation.	Ensure that the government jobs are indeed only temporary. Justify regional differentiation of subsidies or remove it.
X. Increase labour force skills		
i) Improve the quality and availability of training outside of firms	Training subsidies being offered to 60 000 people.	Providers should be certified.
ii) Improve the education system	Minor higher education changes set for April 2000.	Same as previous Survey.

Source: OECD.

firms with registration requirements alone, expand banks' disclosure obligations, subject insider trading to stricter penalties, allow banks to trade over-the-counter securities-based derivatives and begin the process of removing the walls between banks, securities and insurance companies businesses. Price competition in the life insurance sector began in April 1999, and prices have indeed diverged, albeit only to a small extent.[216] At that point the Financial Supervisory Agency (FSA) relaxed the rules for the firewalls that banks must erect between their banking and securities operations. However, it will be March 2001 before banks are allowed into the insurance business at all and then it will only be for homeowner coverage, despite some discussion of bringing forward the date and broadening the range of risks to be opened up. Plans are also afoot to allow the major commercial banks to enter the trust banking field directly (rather than through subsidiaries) as early as FY 2001 (although some minor areas may be reserved for the trust banks). This would speed bank restructuring.

In the next stage of "Big Bang", since October 1999, insurance companies were allowed to establish bank subsidiaries and banks to issue straight bonds and to set up brokerage subsidiaries; previously, these had been limited to corporate bond underwriting, but most city banks look set to enter this newly contestable business. Brokerage commissions on all securities transactions were also fully deregulated as from October; at this point, it looks as though the industry will split into full-service and discount components, but, in any case, profitability in the sector is likely to be severely hit, despite the expected benefit in terms of increased trading. Finally, in January 2001 the health insurance market (the so-called "third sector") will be opened to all insurance firms.

The new law also provided for mandatory, industry-financed funds in the securities and insurance industries to protect clients and policy holders. These cover 100 per cent of clients' cash and securities kept in the securities firms and most insurance liabilities until 1 April 2001, when the coverage will drop to 10 million yen per brokerage client and 90 per cent of voluntary reserves paid in to insurance firms by their policy holders. As a result the Japan Investors Protection Fund was established on 1 December 1998 by members of the Japan Securities Dealers Association with funding of around 33 billion yen. But most foreign firms as well as some domestic ones were unwilling to join this venture (in part because they felt that it failed to impose a strict separation of clients' and own funds until April 1999 and also because they felt it unfair that they should have to participate in the losses that might result from the failure of Yamaichi Securities the year before[217]) and set up a rival fund (the Securities Investor Protection Fund) with assets of around 10 billion yen. It is to be hoped that the two funds will eventually merge in order to reduce administrative overlap and fully exploit risk-pooling benefits. The insurance industry also set up the Policy Protection Corporation, a successor to the Policy Protection Fund set up in 1996, whose assets were exhausted by the 1997 Nissan Mutual Life collapse. It will be funded

by company contributions based on premium income and valuation reserves in order to reach a level of 400 billion yen over about 10 years. This risks being an inadequate target: a substantial portion is already likely to be spoken for by the recent failure of Toho Mutual Life.

In December 1998 the Ministry of Finance and the FSA eased the requirements to start investment trust operations, which at the time numbered 54 and which continue to struggle to attract new funds.[218] Shortly after, the Ministry decided to expand the range of eligible institutional investors of privately placed bonds (so that they can be bought by some non-financial firms), annual issuance of which has been around 500 billion yen heretofore. The ban on bond stripping may also be lifted as early as FY 2000. Finally, non-banks have been able to issue bonds since May 1999, so long as they have at least 1 billion yen in capital and adhere to the same bad-debt disclosure standards as banks. All these changes are based on the belief that well-informed agents can make their own investment decisions without blanket, government-imposed restrictions.

The Tokyo Stock Exchange (TSE) is under persistent strong pressure for change: for example, the number of foreign firms listed, 127 as recently as 1991, had shrunk further to 45 earlier this calendar year,[219] and in FY 1998 the total number of shares outstanding of all listed firms fell for the first time since World War II. This is the result of a tripling in delistings – mainly resulting from the bank failures – and in share retirements: share buybacks were facilitated by an amendment to the Commercial Code that permitted firms to use capital reserves for this purpose, and the number of firms engaging in such actions jumped almost five-fold to 249. Shrinkage from these sources exceeded expansion from share splits, new listings and new issuance by listed firms. Also, while overall trading volumes continued to rise in FY 1998, the value of transactions fell 9.8 per cent, the sharpest drop since the 1992 plunge. Furthermore, in June the US NASDAQ and the Japanese firm Softbank announced their intention to form an automated trading system in Japan. Recognising the actual and potential competition, the Exchange has reacted in several ways in a bid to boost its competitiveness. After allowing large-lot after-hours trading as from November 1997, in January 1999 it relaxed listing requirements: for example, firms not paying dividends are now allowed to list. In April it ended floor trading, taking on the role of a clearing house, and in October it abolished the limit on the numbers of regular members. The resulting entry will add further to the competition that will result from the end of the fixed-rate commission system. The Exchange's Securities Policy Committee recommends that it go private, apply its listing criteria more stringently, integrate its sections, end the requirement for foreign firms to submit financial results in Japanese (which help raise routine costs to about 20 million yen per year) and engage in greater co-operation with other bourses in Asia. Changes such as these seem crucial to reviving its prospects.

As in previous years a substantial amount of corporate restructuring and consolidation is going on among financial institutions. The most noteworthy move this past year was the announcement in August that three of the major banks – Industrial Bank of Japan, Dai-Ichi Kangyo and Fuji – would form a joint holding company in the autumn of 2000, with a view to fully merging their businesses by early 2002 in order to consolidate their operations. The merged entity would be the world's largest bank. Within five years 150 branches would be closed, and around 6 000 jobs (17 per cent of combined employment) would be eliminated. In addition, mergers have also been agreed between two other pairs of banks – Sumitomo and Sakura and Asahi and Tokai. The first is expected to lead to a workforce reduction of 9 000 (30 per cent) by 2004 and a closure of 32 foreign and 151 domestic branches. If these targets are actually achieved, they will go some way toward removing the overbanking problem in Japan. But consolidation has not been limited to major banks. The total number of deposit-taking institutions in Japan fell to 842 in September 1998, down from 900 in March and 1 110 some 20 years earlier. Most of this decline has taken place among credit co-operatives, especially those called *shinkumi* (which are often in poor condition and will not come under the regulatory umbrella of the FSA until FY 2001). Mergers and acquisitions are also numerous among securities firms, many of which are being bought by larger banks in preparation for the new competitive market paradigm. And the fifteenth largest life insurance firm has recently been effectively taken over by a Canadian insurance company. Finally, three large non-life insurers are planning to merge by April 2002 with the hope of cutting their combined workforce by 3 000 (nearly 20 per cent).

Financial deregulation and restructuring have been stimulated by liberalisation of cross-border capital flows and financial services, which reached its final stage with the last set of amendments to the Foreign Exchange and Foreign Trade Law. A natural consequence of these moves is an internationalisation of Japanese financial markets and an expected increase in the use of the yen as an international currency. The authorities have been concerned about the international standing of the yen, especially in the light of the birth of the euro this year. A panel of the Ministry of Finance issued a report[220] this spring (Council on Foreign Exchange and Other Transactions, 1999), which urged government officials to attempt to persuade Japanese firms to increase the share of their settlements in yen, given that currently only some 35 to 40 per cent of Japan's exports and 20 per cent of its imports are denominated in yen.[221] Surely, however, if firms are avoiding the use of their domestic currency, it must be because the associated costs are higher. The only solution can be to lower those costs to minimum levels by financial deregulation, for example. In this context the government also decided to abolish the securities transactions and bourse taxes on 31 March 1999 and to eliminate the call provision on all government bonds issued as from 1999. In order to improve transparency and liquidity, the government also decided to

abolish the withholding tax on non-residents' interest receipts on government securities as from September. Consideration is being given to issuing a five-year bond[222] and, as mentioned above, to allowing bonds to be "stripped". Such actions will help to bring down transactions costs and safeguard the yen's role as a major global currency.

Corporate reform: enhancing flexibility and accountability

Changes in corporate governance are also being implemented with a view to raising profitability among the competing goals of corporations. Cross-shareholding and other forms of stable shareholding are continuing to be unwound (Kuroki, 1999), and those acquiring the unwound shares are foreigners,[223] who are more interested in profitability than had been the sellers.[224] The number of directors is being slashed,[225] and more outsiders are being appointed, although thus far the auditing function remains dominated by insiders. In the expected future amendments to the Commercial Code, it would be desirable to include a requirement that at least half of the auditing committee be true outsiders (never before employed by the firm or any of its affiliates) in order to fully benefit from outsider presence. Take-overs are still rare, but, significantly, a prominent firm in the telecommunications industry was the target of an unsolicited but ultimately successful acquisition by a British rival (IDC by Cable and Wireless). Such outcomes should serve to increase the role of the markets in disciplining corporate performance. Shareholder pressure remains weak, although recent legal reforms facilitating shareholder suits have apparently had some effect.[226]

But more needs to be done. If pension management rules were changed to require their managers to invest funds in the interest of plan members, as was done in the 1970s in the United States, then they too would begin to press for improved returns. The potential impact of such a change would be enhanced by the recent decision to allow firms to make up their corporate pension shortfalls by contributing their cross-holdings along with the voting rights (see below). Executive compensation remains excessively focussed on base pay (still 63 per cent of the total), with virtually no long-term incentive remuneration.[227] Similarly, stock-option schemes are still rare.[228] Performance-based pay would encourage managers to take reasonable risks, thereby yielding more attractive returns.

The government has just made a number of important legislative changes to ease corporate restructuring through a new Industrial Revitalisation Law that will remain in effect only until March 2003. The new law also provides low-interest loans and guarantees and tax incentives such as an extended period for loss carry-forward (seven rather than five years), a reduction in Registration and License Tax (due when assets are transferred) and accelerated depreciation allowances for firms whose business restructuring plans are approved by the

government. It also facilitates management and employee buy-outs and eases regulations governing corporate divestiture. The effectiveness of mergers and acquisitions will be enhanced by amendments to the Commercial Code permitting such transactions through stock swaps without incurring transfer taxes and by obliging minority shareholders who oppose the transaction to sell their stakes to an acquiring firm once the majority agrees to the transaction.[229] Other future plans call for bankruptcy law reform, the introduction of consolidated taxation and the harmonisation of accounting practices with international norms. Yet besides the costs involved, the question of possible moral hazard must be raised: some firms have already undertaken restructuring without benefiting from these measures, which will be available to those who have shown themselves to be laggards.

Other measures that are intended to promote technological innovation were also included in the law. Following on the recent revision to the patent law described in last year's Survey, the government has decided to begin to hand back to private-sector researchers the intellectual property rights on research they have conducted on its behalf in order to encourage them to exploit its potential and thereby strengthen industrial competitiveness.[230] The bill also reduced patent fees for Technology Licensing Organisations, creatures of last year's Law for Promoting University-Industry Technology Transfer.

In addition, the law included measures to assist business start-ups through the provision of credit insurance without collateral and zero-interest loans. These firms are likely to be the source of most future employment growth and a rising share of investment.[231] Similar assistance to venture-type business will be expanded. The government is also planning to join with the private sector to set up a series of venture-capital funds to provide cash to promising start-up companies, while private venture-capital companies are to put up half of the 20 billion yen in funding. The funds are to be set up by the Small and Medium Enterprise Agency by the end of March 2000. The 10 billion yen in capital which the Small and Medium Enterprise Corporation (SMEC) will provide is limited to be no more than half of each venture-capital fund and no more than 1 billion yen per fund. The maximum duration of such assistance is to be normally 12 years, although extensions of up to 3 years will be allowed. Management of the funds is to be left to private venture-capital companies. The government expects the venture capital companies to participate in the management of the start-ups by providing board members and advice. It hopes that the public funding will encourage the private sector to take more investment risks. To be sure, the culture of risk-taking has not been as well developed in Japan as in English-speaking countries, for example. But whether "force-feeding" the venture-capital sector will generate the hoped-for returns is unclear.

Corporate pension reform: filling the hole

The situation of third-tier corporate pension plans is critical. Dissolutions of these Welfare Pension Funds have occurred with increasing frequency in recent years, and in FY 1998 18 failures were registered (around 1 per cent of the total number of funds).[232] The problem of under-funded pension plans is becoming more prominent and is affecting both small and large companies. Until 1997, corporate pension funds had long been obliged by government regulation to guarantee a 5.5 per cent annual return on premiums paid. But as the stock market remained stagnant until recently and interest rates are at historic lows, the actual returns on most corporate pension funds' investments have been hovering around 4 per cent in recent years, requiring firms to make up for the gap. Moreover, new accounting rules, to be introduced in FY 2000, require companies to disclose any underfunding of their expected obligations for retirement allowances and pension payments. The new rule allows 15 years to fund such shortages. According to a report by NLI Research Institute, the total shortage of funding among listed companies will reach 56 trillion yen on a consolidated basis;[233] Goldman Sachs' estimate is 80 trillion yen, and that only for those firms on the first section of the Tokyo Stock Exchange. In order to give companies a range of options in making up shortfalls in their employee pension fund reserves, the Ministry of Health and Welfare (MHW) plans to allow companies to contribute stock (including cross-shareholdings) and other securities directly to employee pension funds in FY 2000. Permitting the contribution of cross-held shares will not only enable firms to cover fund shortages, but will also help to reduce cross-shareholding. The transfer of cross-held stocks to pension funds implies the transfer of voting rights as well. As mentioned above, this could have profound implications for the system of corporate governance in Japan, with the possibility of powerful return-seeking funds effectively disciplining corporate management.

The government also hopes to introduce a Japanese version of US 401(k) *defined-contribution* plans in FY 2000, as an option in addition to the conventional *defined-benefit* schemes in the third tier of the pension system. Usually, such funds are invested in several financial products, including employers' stocks, and are tax-free. Workers decide how much of their salaries are put into the fund up to a ceiling,[234] employers make matching contributions, which are counted as expenses, and funds are taxed only when withdrawn. The new system will be advantageous not only for firms but also for workers, because it will better serve individual preferences and pensions will become portable, removing a major barrier to the development of a mid-career labour market. On the other hand, the employee assumes the investment risk, and therefore the available pension at retirement is not predictable. The tax treatment to be applied to such a scheme is under examination. In any case, whether or not to introduce 401(k) plans for workers will be up to individual firms, and defined-benefit plans will probably still play a major role, especially in large firms (as they do in the United States).

Another important point of the reform is the lifting of most of the regulation concerning employee pension funds, which constitute the main source of funds for third-tier corporate pension plans and enrol about 12 million employees, one-third of all salaried workers in the private sector. In an attempt to make the management of corporate pension plans more efficient, MHW has proposed to scrap the requirement that a company must have at least 50 billion yen in its pension fund before it is allowed to run the fund by itself, rather than hand over its management to a trust bank, a life insurance firm or an investment advisory company. Regulations concerning the investment of funds, mainly in government bonds, would also be eliminated. With the proposed revisions, corporations would be able to invest as much as they choose in stocks and investment trusts under a "prudent man" rule.[235]

Labour market policies: cyclical reactions and structural reforms

Worsening labour market conditions and the government's response

Despite its advantages,[236] the Japanese major-company employment system, characterised for decades by lifetime employment and seniority-based wages and promotions, is increasingly seen as inappropriate to the current context of slower and more variable growth and inconsistent with the need for boosting returns on invested capital in order to provide savers with the income required for their retirement. As seen in Chapter I, the unemployment rate is now approaching 5 per cent – some 3.4 million people – despite steady shrinkage in the labour force and a further rise in the discouraged worker rate (Table 45). The Ministry of Labour estimated in its 1999 White Paper that the structural component of unemployment was around 3.2 per cent of the labour force in 1998 with the remaining 0.9 per cent attributable to lack of demand. The major reason for the rise in joblessness in 1998 was growing demand deficiency, but the structural unemployment rate was also on an upward trend because of rising mismatch along age, industry, occupational and regional lines. Rising mismatch is reflected in an increasing average duration of unemployment spells,[237] as outflow rates are shrinking. Those unemployed following redundancies represent a growing share of total joblessness: they now account for more than a third, up from a 1997 average of less than a quarter, and for nearly half of all separations. One favourable sign is that employees, especially younger workers, are switching jobs more readily than in the past, deepening the mid-career job market.

Relative labour market outcomes for men have worsened. New job creation for full-time, indefinite-duration positions has shrivelled, leading to a decline in the number of so-called "regular employees".[238] Since nearly 80 per cent of aggregate male employment is in this job category compared to only 64 per cent for women, the deterioration in labour market conditions has been more heavily

Table 45. **Some structural labour market indicators**

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999
Discouraged workers (per cent of labour force)	1.4	1.5	1.3	1.1	1.0	0.9	0.9	1.0	1.3	1.4	1.5	1.7	1.9	2.2
Share of unemployed out of work (per cent)														
More than 6 months	39.0	38.7	40.5	37.1	38.0	36.0	35.0	33.8	35.1	38.2	40.2	40.9	39.0	44.4
More than 12 months	17.1	18.8	20.2	18.5	19.0	17.6	15.3	15.3	16.5	18.1	19.6	20.9	20.7	22.4
Total job separations in past years (per cent of employment)	7.9	9.1	9.3	9.0	7.3	8.7	8.8	8.8	9.3	10.1	10.1	10.5	10.9	12.0
Share of separations resulting in unemployment (per cent)	18.4	19.9	16.0	13.9	12.9	12.5	13.7	16.2	18.5	18.8	20.6	19.1	20.7	23.9
of which: Share involuntary	47.3	n.a.	43.4	36.5	31.3	34.8	32.6	40.2	43.4	42.5	40.7	40.3	46.1	47.2
Separations resulting in new job (per cent of employment)														
Total	3.1	3.1	4.1	4.2	3.5	4.2	4.4	4.1	4.0	4.2	4.3	4.7	4.9	4.9
Aged 25-34	4.0	3.7	5.4	5.3	4.9	5.4	6.2	5.9	5.5	5.6	6.1	6.7	6.6	6.9
New job in past year (per cent)														
Share through public job placement service	n.a.	n.a.	n.a.	7.8	6.7	6.3	5.8	7.4	9.1	8.5	9.7	8.7	9.1	10.8
Share through media advertising	n.a.	n.a.	n.a.	25.0	24.2	26.3	26.9	27.3	28.6	31.5	32.1	34.3	34.9	35.3

Source: Management and Coordination Agency, *Report on the Special Survey of the Labour Force Survey,* February (annual).

concentrated on men. In May and June 1999 their unemployment rate was 0.7 percentage point above the female rate, a reversal of the differential observed in the decade to the mid-1990s. This relative shift in outcomes in favour of women has been magnified by the government over the past year: its employment levels have been stable, but there has been a shift of more than 2 percentage points of its total work force towards women. Men have also suffered because they are more heavily employed by the largest firms (with over 1 000 employees), which have been recently responsible for about half of the decline in dependent employment because of their more intensive restructuring efforts (see Chapter I). Similarly, they are employed to a greater extent in the manufacturing and transport and communication sectors, where employment fell 2.6 and 3.5 per cent, respectively, in the year to June. Parallel occupational structure shifts have also been unfavourable to men.

The government has reacted to the increasingly severe labour market conditions both by implementing general economic stimulus measures and by actions focussed on the labour market itself. The traditional Japanese focus on subsidising the wages of workers at struggling firms (see below) was clearly no longer sufficient, and the government supplemented that programme in two steps. First, in November 1998 it formulated a set of Emergency Economic Measures, including a Comprehensive Plan for Employment Activation, in order to create large numbers of jobs. The Plan was to implement various projects (discussed below) worth 1 trillion yen on a project cost basis, of which only 0.3 trillion yen represented new money provided in the third supplementary budget for FY 1998. Second, in June 1999 the government expanded the measures taken previously, following the recommendations of the Competitiveness Commission, with the objective of creating more than 700 000 jobs (fully worked out by individual measure). They were enacted in the first supplementary budget of FY 1999 in July at a cost of over 0.5 trillion yen. The government measures taken can be summarised as follows:

– *Development of new business, support for start-ups by individuals and promotion of venture business in order to create employment opportunities.* A revised law was implemented in January 1999 to these ends. It also aimed to encourage existing companies that have abundant managerial resources, including technological and human resources, to make aggressive efforts towards the development and expansion of new lines of business based on this abundance, possibly leading to the spin-off of these new ventures. As discussed in the earlier section on Corporate Reform, the government has also boosted assistance to start-ups, building on the public lending scheme carried out through the Japan Small Business Finance Corporation. The government will pay half of the first year's salaries, up to a ceiling of 3 500 000 yen ($29 000), for a maximum of six employees at start-up companies in any industry.

- *Creation of over 300 000 temporary jobs by central and local governments in areas such as education, welfare, environment, etc.* This will be encouraged by a special grant, which is worth around 200 billion yen, and is mainly provided to local governments' employment programmes. The grant is limited to two years, since this employment programme is meant to be a transitional measure while the promoted new industries gear up. However, it may prove politically difficult to actually bring these jobs to an end once the beneficiaries grow used to the additional services provided.
- *Establishment of a fund to subsidise employers in hiring middle-aged or elderly workers in regions of high unemployment.* These subsidies would be available in areas where the unemployment is higher than 5.4 per cent in two consecutive quarters in order to create 200 000 jobs.
- *Support for new jobs in 15 new and growing fields* (as defined in the Action Plan for the Economic Structure Reform of 1997 and including health care, environment, information and communications, biotechnology and the medical system), for example, through further deregulation. As well, subsidies of about 100 billion yen will be disbursed from the aforementioned fund to employers hiring middle-aged or elderly employees earlier than scheduled, aiming to create 150 000 jobs.
- *Expansion of a subsidy programme to facilitate mobility of middle-aged and elderly workers*, with an objective of creating 70 000 jobs.
- *Provision of subsidies to the training of both unemployed and employed wishing to improve their skills.* Since December 1998 the government has covered up to 80 per cent of the tuition costs of any of 3 400 job-related courses whose duration extends from three months up to about a year for anyone who has worked full-time for at least five years. It also now pays full tuition and unemployment benefits for those attending courses of up to six months' duration at an authorised professional school. This is expected to involve about 60 000 people. Careful assessment of the effect of these provisions should be carried out so as to ensure they are providing incremental value.

The government has also boosted the generosity of its Employment Adjustment Subsidy (EAS) programme. EAS consist of financial subsidies to companies to overcome temporary demand shortfalls through suspension of operations, training and secondment, as well as to alleviate "frictions" such as changes in employee residence and acquiring new skills, and have played a major role in stabilising the job market. In the past the government has provided subsidies of half of wages for employees of major firms, while for SMEs the help from the state has been two-thirds of wages. But for the year to September 1999 those shares were raised, and the maximum duration was extended from one to two years (in the case of secondment to another firm). The effect of this as well as the

worsening labour demand situation was expected to raise the budgeted FY 1999 cost of the programme to 61 billion yen (about half a billion dollars), four times more than it had been in FY 1997. Rightly fearing labour hoarding, the government has also decided to try to limit the scheme's availability – by making more difficult the renewal of regional eligibility – so that the focus would remain on temporary adjustment. Nonetheless, the programme is not well suited to a reduction in labour demand that is likely to prove of more than cyclical duration.

Longer-term structural reforms

There have also been a number of policy changes of a more structural nature. For example, a revised law governing dispatched workers will be implemented by the end of 1999, with the intent of lifting the ban against hiring such workers in all but certain job categories. While the previous law allowed temporary (agency) staffing in just 26 job categories (including secretaries, software developers and interpreters), the revision allows it in all jobs, in principle, except for harbour, construction and security work. But the legal restriction has been replaced with a similar regulatory prohibition whose removal has not yet been programmed. Moreover, in principle, except for the 26 categories where dispatched workers had already been allowed (where a maximum duration of three years is permitted), the new law, following the examples of France and Germany, limits the work period to one year. The time limit is aimed at discouraging companies, anxious to reduce labour costs, from replacing permanent employees with dispatched or other temporary workers.[239] If the contract for such workers is extended beyond one year, the labour minister is supposed to recommend that the company hire the person as a regular employee, while the temporary job placement agency would be subject to a penalty of up to 300 000 yen. Workers registered with temporary job-placement agencies in Japan numbered about 856 000[240] (*i.e.* 1¼ per cent of the labour force) in FY 1997, and their number is rising about 10 per cent per year. Analysts expect that a liberalisation would double the value of the temporary-job-placement market to more than 2 trillion yen per year within a few years. While the question of allowing employers more flexibility in the nature of the contracts they offer has an element of social choice, international experience demonstrates that the existing restrictions are costly in terms of overall employment levels.

The private job-placement business was also liberalised under recently passed legislation, in principle allowing private companies to handle all job types except port and construction workers. While there has been a trend improvement in the share of all new job matches in which the *public* job placement service has been involved (Table 45 above), the added competition should improve the average quality of job matches.

Revisions to the Equal Employment Opportunity law, passed in 1997, took effect in April. They are designed to enforce equal treatment for men and women. They require that management bear responsibility for preventing sexual harassment at work and that employers advertise jobs equally to men and women. Revisions to the Labour Standards Law, also effective in April, abolished regulations that restrict women from working overtime or late at night.

Finally, the unemployment insurance benefit system will be reviewed to place priority on middle-aged or elderly people with supposedly more urgent needs (even though youth rates have also risen sharply); changes, as yet undecided, may be enacted in the next ordinary Diet session. Its financial situation has deteriorated: the number of recipients rose to more than a million people in FY 1998,[241] a jump of around 17 per cent. With contributions falling in line with aggregate compensation, the deficit was expected to balloon from 370 billion yen in FY 1997 to around 1 trillion yen in FY 1999.

Deregulation: further progress toward more market-determined outcomes

The overall strategy

Regulatory reform has been a central element in the economic structural reform programme underway since 1995. Its goals are to complete the move to a model of what will be largely *market-led* growth. A sustained effort has begun to reduce economic intervention in many sectors, among them large retail stores, gasoline imports, telecommunications and financial services, and consumers have already seen positive results. Deregulation has boosted domestic demand,[242] but there is much remaining potential in this dimension, as prices are still much higher than in other major economies, and Japan remains far below average in attracting pro-competitive foreign direct investment.[243] Following the 1995-97 deregulation programme, the 1998-2000 deregulation plan was approved in March 1998 and revised in March 1999. The total number of items was raised from 624 to 917, and the need for promoting reform to improve economic outcomes was stressed anew. In particular, the 1999 plan calls for the establishment of new rules in accordance with the 1997 shift in administrative style from the conventional type characterised by *prior intervention*, which has restricted free and fair market competition, to one based on *retroactive supervision*, with a greater role for the judicial system. As indicated by the OECD's recent *Review of Regulatory Reform in Japan*, the challenge is to deepen and speed up the regulatory transition already underway, as "rapid and forceful action will … contribute to establishing a new basis for sustainable long-term growth" (OECD, 1999*a*).

The government plans to increase reliance on market forces in setting public utilities' tariffs. The Price Stabilisation Policy Council is planning to promote a pricing reform with the introduction of competition and disclosure of

information. Tariff-setting would henceforth be free, subject only to price caps rather than fixed administratively based on the full-cost principle,[244] and cost efficiency would be promoted through yardstick competition, which was already introduced in 1996 in the municipal natural gas industry. EPA will draft the guidelines for information disclosure by the end of FY 1999. But several potential problem areas will remain even after the planned reforms. First, the question remains whether it is feasible to retain ministerial regulatory oversight as well as policy-making functions and still reap the full benefits of competition, or whether regulatory bodies independent of policy-making functions should be established. Second, a number of sectors will still not be subject to market forces. Prime examples are the services provided by local authorities. Their prices have generally continued to rise in recent years, while those supplied by central government have experienced declines. Water is one such case: prices vary across districts by a factor of about ten, which is probably more than can be attributed to the admittedly enormous variance in natural advantages. Private provision might prove to be more efficient. A third potential problem is with sectors that might be deregulated but where there are barriers to effective competition other than regulation. For example, supply in the airlines industry is effectively constrained by a lack of airport slots (see below). A fourth issue is the efficient setting of access charges to networks (in gas, electricity and telecommunications). Finally, for sectors that will be only partially deregulated (electricity and gas, in particular), the need to separate the operational and financial dimensions of the regulated part of the business from those of the deregulated segment has not been resolved.

Sectoral developments

Significant reforms of the *electricity* sector are expected over the next two years. Japan has easily the highest industrial electricity prices of any IEA member country; this has put a heavy burden on its economic performance economy-wide. That prices are unnecessarily high is demonstrated by the significant fraction of industrial demand that is satisfied by in-house generation:[245] in 1998 that share was 26.4 per cent in Japan, the fourth highest among OECD countries, whose mean share was only 15.4 per cent. The government has accordingly set a target to reduce prices to internationally comparable levels by FY 2001. That seems an ambitious (albeit somewhat ambiguous) target, both in light of the factors that contribute to the high costs of electricity in Japan[246] and the limited nature of the reforms envisaged. The government's 1997 Action Plan for Economic Structure Reform identified efficiency through competition as a basic principle for reform in the electricity sector. Competitive tendering for power generation, which has been open to independent power producers (IPPs) since a 1995 liberalisation, together with the experience of in-house generators, has shown that competition can significantly reduce the costs of generation. Now the Ministry of International

Trade and Industry (MITI) plans to open the market further as from 21 March 2000 to allow IPPs to compete with the existing ten private, regional, vertically integrated utilities in bidding for new capacity coming on stream as from 2008. While this should have a favourable effect in the longer term, it will have no impact by the 2001 horizon. In order to meet its objective, MITI is proposing partial retail liberalisation (allowing the largest users to choose their suppliers) in order to increase the downward pressure on prices for the larger users.[247] That increase could be substantial if groups of users are allowed to join together in their purchasing. But further substantial regulatory changes are seen to be necessary to make such competition work effectively and to reduce prices for the bulk of users. These include (OECD, 1999*a*):

- modifying price regulation mechanisms to increase producer incentives to cut costs;
- introducing peak-load pricing based on time and seasonal factors;
- creating markets for trading electricity;
- reducing concentration in markets for power from their current oligopolistic level, both through expanding the geographic scope of markets (by requiring strengthened transmission connections between utilities' service areas, and – if necessary to achieve the government's major policy goals in the sector – by increasing the number of sizeable generating companies through divestiture) and reducing the incentive and ability to discriminate in providing transmission access through vertical separation of generation from transmission and regulating access to the transmission grid.

In the absence of regulatory or other means of strengthening interconnection, utilities can continue to protect their near-monopoly status behind the barrier of limited transmission capacity. In order to make the regulatory regime credible and predictable, it should be administered by a regulatory authority independent both of the policy-making functions and of the utilities, and accountable through transparent procedures.

The main development in Japan's *telecommunications* market this past year was the division on 1 July 1999 of NTT's operations into two regional carriers and a long-distance and international carrier under the umbrella of a holding company. Efforts to restructure NTT have been at the heart of Japanese telecommunications reform for a long time, but it remains to be seen whether the NTT break-up will promote the promised infrastructure competition in basic services with the holding company structure (OECD, 1999*a*): it is not in the shareholders' interests for the holding-company to allow NTT East and West to enter each other's market for such services. In addition, under the new law, neither regional company can enter long-distance markets. However, the split might encourage more competition in new services.

The primary alternatives to local fixed telephony service are cellular mobile and cable TV (CATV) services.[248] On the one hand there is reason to be hopeful concerning the competitive impact of cellular because, in 1997, mobile telecommunications services represented nearly 40 per cent of the entire telecoms market, the highest share recorded in the OECD (OECD, 1999*b*). Nearly one in three inhabitants was a subscriber, a penetration rate exceeded only in the Nordic countries. As for the other source for competition for local fixed telephony, earlier regulatory constraints have resulted in a very low penetration rate for CATV (11.3 per cent of households at the end of 1996) as compared with other OECD countries. However, catch-up seems to have been occurring. Another noteworthy change occurred in November 1998, when the Ministry of Post and Telecommunications (MPT) introduced a tariff notification system that allows providers to determine their own prices without obtaining approval from MPT (except for NTT's basic local services). This decision is a milestone for the liberalisation of the telecommunications market, since it finally makes effective price competition possible. At the same time a price cap was introduced on local calls and the monthly line charge, based on a pre-set reduction from the overall development of consumer prices. Finally, the authorities have decided to make a number of other significant pro-competitive changes to telecoms policies. They have agreed to submit legislation to amend the Telecommunications Business Law to the next ordinary session of the Diet in the spring of 2000 to introduce a new system of setting inter-connection rates (based on long-run incremental costs), which are much higher than those in some other markets. In addition, MPT has already taken such pro-competitive measures as follows: to allow foreign firms to use leased lines in order to bypass the international settlement system; to eliminate all restrictions on foreign investment in telecommunications carriers[249] (except NTT) and in the CATV business. Once these changes have been implemented, it is quite likely that the price gap between Japan and other countries for telephone services will diminish.[250]

Important developments have also occurred in the transportation sector, which looks set to become subject to substantially more price competition in the next two years. Since 1997 a ceiling price for *rail services* has been set by the full-cost method, with price flexibility allowed so long as the ceiling is not exceeded. Following the introduction of a zonal pricing system for *taxis* in January 1998 there has been some new entry[251] and welcome signs of an increase in competition (because of a relaxation of the Ministry of Transport's criteria) as well as a trend toward payment of drivers by commission. However, the demand-supply adjustment clause may not be eliminated until FY 2001. This should ensure that remaining rents are competed away. In 1996 a 25 percent price flexibility band beneath a cap set on the basis of all suppliers' costs was introduced in the *domestic airline* business; because of the industry's concentrated structure and continued entry controls, though, that flexibility has led to only a limited amount of actual fare

competition. A bill is before the Diet to end all demand and supply regulation in the industry from February 2000. The Ministry of Transport (MOT) plans to start awarding new and some existing landing and departure slots for domestic airports based on such factors as carriers' seat-occupancy rates, on-time records, accident rates and fare competitiveness. The new system will take effect next spring and cover several major airports. This might facilitate the entry of new carriers, which can currently gain slots only when new runways are constructed, although it remains to be seen how new entrants will be considered when measuring historic performance. The timing is opportune because two new firms have recently entered the market. However, auctioning the slots would do more to promote efficiency and would be more transparent. Finally, MOT is considering how to remove demand-supply adjustment regulation in the *local bus service* market by a FY 2001 target date; it will be abolished in February 2000 for *long-distance bus services*.

Japan's *traditional retail industry* has come under renewed pressure to drive down prices, with dual challenges emerging in the form of discount stores and aggressive foreign retailers/franchisers. Retail markets have long been dominated by a few domestic manufacturers, who control more than 60 wholesalers and around 20 000 shops through exclusive selling arrangements. While such strong business ties have maintained relative stability in the market, they have also reduced competition and allowed high prices to be sustained. But the severe consumption slump, along with the easing of restrictions on openings under the Large-Scale Retail Store Law, has touched off changes in the industry and made it easier for foreign firms to enter the Japanese market. Lately foreign companies have apparently been bypassing wholesalers and conducting direct transactions with manufacturers to minimise costs. This direct-procurement policy has two objectives: one is to reduce costs, such as commission fees for trading companies, and the other is to be able to respond quickly to fashion trends and evolving consumer needs. Although wholesalers are trying to react, a continuing decline in their number seems inevitable: according to MITI there were 391 000 wholesalers in 1997, down about 20 per cent from 1991.

Consumer and competition policy: rebalancing the rules

In order for markets to serve their purpose in the new paradigm, consumer sovereignty will have to be safeguarded. This will require changes in both consumer and, to a lesser extent, competition policies. The need for further strengthening of consumers' rights legislation is indicated by an increase in the number of consumer complaints in recent years.[252] In January 1999 the Social Policy Council compiled a report on the outline of a proposed new law, tentatively called the Consumer Contract Law, which is intended to provide

transparent minimum civil rules for the prevention and the smooth settlement of disputes over consumer transactions. A final report is expected later this year. The resulting draft law, which will be a special law of the Civil Code, should be submitted to the Diet in the ordinary session of FY 2000. The legislation envisages two basic rules for contracts: the *first* would enable consumers to void contracts concluded under any of the following conditions: 1) the contracts did not provide enough information to consumers on important provisions needed for consumers in deciding to sign up; 2) the business institution involved gave the consumer false or misleading information; 3) the consumer was forced to sign up under threat or similar circumstances. The *second* is aimed at nullifying contract provisions that are unduly detrimental to consumers. Furthermore, the Council report urged the government to provide for: simple legal proceedings friendly to consumers, so that these rules are strictly observed in consumer transactions; reinforcement of the out-of-court settlement system; and measures to improve consumer education and information-provision systems. The new law is regarded by the government as more comprehensive than the equivalent European Union directive (adopted in 1993) which lacks the first basic rule. To the extent that these changes in the law set out consumers' legal rights, their effectiveness will depend on other reforms to improve access to the courts and to increase the availability of legal representation.

The government is continuing its efforts to reduce the scope of exemptions from the Antimonopoly Law. MITI is considering abolishing one of the most significant remaining exemptions, for cartels of smaller businesses formed through commercial and industrial co-operatives. In June 1999 a MITI advisory panel started deliberations on banning the cartels, hoping to submit legislative revisions to the Diet in the near future.

Agricultural policy: revising the framework

The Uruguay Round Agreement for Agriculture brought an acknowledgement from government and farmers of the need for fundamental changes in agricultural policies in Japan. The government established an Investigative Council on Basic Problems Concerning Food, Agriculture and Rural Areas in 1997 to review current policies and to establish, after 35 years, a new Agricultural Basic Law. In July 1999 that new Basic Law was enacted. The four basic philosophies that will direct future policy, as presented in the new Basic Law, are:

- to guarantee food security and a stable food supply based on domestic agricultural production;
- to enable agriculture and rural areas to fulfil their multifunctional roles;[253]

– to promote sustainable agricultural development to maximise the potential of Japanese agriculture;
– to ensure comprehensive development of rural areas.

In accordance with the Law, revised policies relating to food, agriculture and rural areas are being developed. Major changes in *food* policy consist of setting an explicit target for the food self-sufficiency ratio[254] as a guideline for production and consumption and formulating a policy that focuses on consumers' needs, such as through the introduction of an improved food labelling system.[255] Whether targeting any particular self-sufficiency ratio can be consistent with a shift to a greater focus on market-based outcomes and consumers' interests is doubtful. Food security can be achieved at lower cost by globally diversifying production sources. But in any case only measures that do not distort prices could form even a second-best solution.

As for *agricultural* policy the leading idea is to foster efficient and stable agricultural management through improvements in farming productivity and demand-driven production. In particular, the government plans to abolish price support for milk and dairy products, including butter and skim milk powder, in FY 2001, while for soybeans and sugar it intends to decide the future direction of price policies before the end of 1999. As shown in Table 46, Japan's Producer Support Estimate (PSE) has declined only modestly over the last decade, in particular due to a decrease in market price support.[256] It would be preferable to wind down all price support schemes and direct the savings to direct income support for small farmers and for programmes to enhance their human capital. New entry of corporations into agriculture is to be allowed,[257] but such corporations will have to meet some (as yet unspecified) requirements and will have in general to have a non-profit nature in order to acquire land,[258] much as privately-owned hospitals are also supposed to abjure from earning profits. Direct payments to farmers in hilly and mountainous areas are expected to start from FY 2000 to revitalise agriculture in these districts, compensate for disadvantages in production conditions and help secure the multifunctional roles of agriculture.[259]

However, the sector continues to suffer from exceedingly small scale, with little improvement over time. The average amount of land per farmer in the United States is some 126.5 times as great as the 1½ hectares in Japan (Table 47), and even European levels are some 20 to 40 times that found in Japan. Price- and income-support policies helped push average agricultural incomes per household up 24 per cent in the first half of the 1990s, much faster than other incomes, encouraging farmers to remain in the sector, especially when such income can be supplemented by off-farm sources.[260] The number of farmers has been declining only around 2 per cent per year since 1994. Given the age structure of the farm population – 44 per cent are over the age of 65 years and a further 26 per cent are

Table 46. **Support to agricultural producers**

US$

	1986-88	1991-93	1996-98	1997[1]	1998[1]
Japan					
Producer support estimate (billions)	52.1	55.6	55.6	52.6	49.1
Percentage of gross farm receipts[2]	65	58	63	61	63
Per full-time farmer equivalent (thousands)	14	18	23	22	21
Per hectare of agricultural land (thousands)	9.8	10.8	11.1	10.5	9.8
Nominal assistance coefficient[3]	2.85	2.40	2.69	2.53	2.72
OECD					
Producer support estimate (billions)	246.6	292.0	259.0	245.5	273.6
Percentage of gross farm receipts[2]	41	39	33	32	37
Per full-time farmer equivalent (thousands)	12	14	11	10	11
Per hectare of agricultural land (thousands)	0.2	0.2	0.2	0.2	0.2
Nominal assistance coefficient[3]	1.69	1.65	1.50	1.47	1.59

1. Provisional.
2. Valued at farm gate prices including budgetary support. This is the percentage PSE.
3. Ratio of PSE to the value of gross farm receipts (excluding budgetary support) at world market prices.

Source: OECD (1999), *Agricultural Policies in OECD Countries, Monitoring and Evaluation,* Paris, May.

Table 47. **Agriculture-related indicators**

	1960	1970	1980	1990	1995	1998
Agricultural production						
Yen billion, 1990 prices	n.a.	7 189	6 875	8 382	7 772	7 575[3]
GDP ratio, per cent	n.a.	3.8	2.4	1.9	1.7	1.5[3]
Employment						
Thousand persons	11 960	8 110	5 060	3 920	3 270	3 170
Ratio to total employment, per cent	26.8	15.9	9.1	6.2	5.1	4.9
Farmland area (1 000 ha)	6 070	5 800	5 460	5 240	5 038	4 905
Agricultural households (1 000 units)	6 060	5 340	4 660	3 840	3 440	3 291
Farmland per household (hectare)	1.00	1.09	1.17	1.36	1.47	1.49
Agricultural household income (yen 1 000)						
Total	449	1 592	5 594	8 399	8 917	8 139
Agricultural income	225	508	952	1 163	1 442	1 167
Working household income (yen 1 000)						
Total	502	1 391	4 254	6 323	6 850	7 067
Self-sufficiency rate[1] (per cent)						
Grains	82	46	33	30	30	28
Total of calories	79	60	53	47	42	41
Agricultural co-operatives[2] (yen billion)						
Deposit outstandings	n.a.	5 934	26 846	56 108	67 573	68 439
Loan outstandings	n.a.	3 008	11 358	14 390	19 663	21 434

1. Fiscal year.
2. End of fiscal year.
3. 1997 figures.

Source: Economic Planning Agency, Management Coordination Agency, Ministry of Agriculture and Bank of Japan.

over 55 – the relevant population will decline sharply in coming years unless they are replaced by their offspring or other younger cohorts. The result will either be that land will be shifted into other uses or that farm consolidation will occur. However, land and tax policies provide barriers to new entry, to consolidation and especially to shifting land to non-farm uses; jointly they boost its value enormously.[261] Fixed asset tax and inheritance tax advantages are described in Chapter IV. But, more importantly, local committees, whose members are elected by farmers, limit the transfer of farmland to other uses as well as entry by newcomers: total farmland has declined at only about a ¾ per cent annual rate in the 1990s.[262] The government also buys substantial amounts of land for public works, which brings about favoured capital gains tax treatment. Owners will therefore rationally hold land at the margin because there is a modest chance that the land will be re-zoned or purchased by the government, bringing handsome capital gains.[263] The result has been slow progress in consolidation.

Rice is easily Japan's most important agricultural product. Although at the time of the Uruguay Round Japan accepted the substitution of tariffs for quotas on agricultural products such as yams and peanuts, it refused tariffication for rice. Instead the government promised that it would import an agreed amount of rice, equivalent to a set ratio of Japan's domestic rice consumption – the so-called minimum access formula. According to this formula, Japan was obliged to import an amount of rice equivalent to 4 per cent of annual domestic consumption in 1995 and to increase imports by 0.8 percentage point every year until the import share reached 8 per cent in FY 2000.[264] However in December 1998, the Japanese government notified the WTO that Japan would cease to apply this special treatment (under Annex 5 of the Uruguay Round Agreement on Agriculture) in respect of rice from 1 April 1999. The quantitative restriction on rice imports was abolished on that date, and tariffs are now levied on imports that exceed the minimum access amount.[265] Minimum access opportunities are to be increased in FY 1999 and 2000 by only 0.4 percentage point of consumption per year, which is the same rate as other tariffied products, so that Japan can limit imports to 7.2 per cent of base period (FY 1986-88) internal consumption in FY 2000 (instead of 8 per cent). The Food Agency will continue to be the manager of rice under the minimum access commitment. But, from FY 1999 private firms will be allowed to import rice beyond the quantity stipulated by the minimum access formula.

The agricultural co-operatives (*nokyo*) have also been of concern. As in other countries, they undertake agriculture-related business, but their main business is in financial (banking and insurance) operations, which provide about 40 per cent of their profits. With few profitable opportunities to lend in agriculture[266] and few fund management skills, most deposits have been either invested in unprofitable endeavours (such as the *jusen* mortgage lenders) or merely re-deposited in other institutions, mainly affiliated central organisations (58 per cent of total assets). The spreads earned on such re-deposited funds have shown a

shrinking trend, from about 1.5 percentage points up to the mid-1970s to around 0.5 point in the 1990s (Maki, 1997). The result is low returns overall, as well as substantial non-performing loans (7.3 per cent of total loans as of the end of FY 1997, compared to 11.1 per cent for the commercial banking system) and persistent problems of insolvency.[267]

The government's policy is to restructure the system to remove one of its three layers – local, prefectural and national. The restructuring of agricultural financial institutions should be accomplished by the year 2000, with the integration of the prefectural institutions into the national Norinchukin Bank (*Norin Chuo Kinko*). Otherwise the government is encouraging the amalgamation of individual co-operatives to achieve a certain minimum efficient scale. To this end a minimum capital requirement of 100 million yen will be implemented in April 2001. The government-supported objective of the industry is to cut the number of co-operatives from 1 580 in April 1999 to 530 over the next two years.[268] At the same time it is hoped that the number of employees of co-operatives will fall from recent levels of around 350 000 to 300 000, still about the same as the total number of farmers – an average of 308 000 in 1998. Surely, if the sector does not benefit from any special protection, then its future will be determined by its quality-adjusted cost and price performance. This should be the government's ultimate objective.

One other aspect of the agriculture sector that is worthy of mention is its well recognised relationship with the environment. Budget outlays on government programmes that aim primarily at promoting environmental conservation and reducing the adverse environmental effects of agriculture were increased from 14.9 billion yen in FY 1997 to 15.7 billion yen in FY 1998. These programmes include financial support for local governments to promote environmentally friendly farm management through, for example, reducing excessive use of fertiliser and pesticides. Government investments aimed at improving rural infrastructure, such as roads and sewerage, and to encourage the establishment of industries in these areas fell from 454 billion yen in FY 1997 to 394 billion yen in FY 1998.

Environmental protection: putting some of the pieces in place

Protecting the environment is a multi-faceted challenge. In this brief section attention will be focused only on some of the major recent developments. According to the Kyoto protocol Japan is committed to reducing greenhouse gas (GHG)[269] emissions by 6 per cent, compared with 1990, during the first commitment period between 2008 and 2012. Even though an OECD study suggests that Japan's burden is not as onerous as that faced by other OECD countries and regions,[270] this target will not be easy to achieve, especially as GHG emissions in

FY 1997 were already about 8½ per cent *above* 1990 levels.[271] However, progress is now being made, as these 1997 figures show a 0.2 per cent decrease of GHG emissions over the previous year due to a decline in demand for oil by electricity companies following an increased use of nuclear power (up 5.6 per cent, because of new plants and an improved utilisation rate) and an unusual increase in water power (13.3 per cent). A decline in demand for electricity is also, however, attributable to the stagnation in economic output. Approximately 90 per cent of carbon dioxide emissions produced in Japan are energy-related, and consequently energy conservation is considered one of the keys to avoiding global warming without obstructing economic growth.[272] Another key is to reduce greenhouse gas emissions by changing the fuel mix toward nuclear power, hydro, geothermal and new energies.[273] The government plans to fulfil its commitment through a combination of shifting toward non-fossil (especially nuclear and renewable) energy on the supply side and energy saving through conservation (see below) and changes in life style on the demand side. Supplementing voluntary measures with market-based instruments might increase the chances of meeting the nation's commitments.

Japan aims to reduce the trend in energy consumption growth from the rate of over 3 per cent per year seen in the mid-1990s (which is above the rate of economic growth) to 0.1 per cent per year. The government's other major demand-side strategy is to steadily implement conservation measures that are economically and technologically feasible. Oil is still a critically important energy source, accounting for nearly 55 per cent of primary energy supply. Because of this, oil accounts for about 65 per cent of CO_2 emissions. On the supply side the official goals are:

- to achieve the target of 480 TWh for nuclear power generation in 2010, increasing the proportion of electricity supply from nuclear plants from 35 per cent in FY 1996 to 45 per cent in FY 2010;
- to triple present supply from new energy resources;[274]
- to reduce the dependence on oil, decrease the use of coal and promote the introduction of natural gas[275] in order to shift to a more environmentally acceptable supply structure, while maintaining a stable and economically efficient supply of fossil-based energy;
- to promote the diffusion of heat storage air-conditioning systems and gas coolers.

Inducing changes in life style plays an important role as well. An important application takes the form of reduced taxation of environmentally friendly cars (see Chapter IV). Cuts have already been implemented on the Automobile Acquisition Tax (AAT).[276] Consideration is being given by the Ministry of Transport and the Environment Agency to lowering possession taxes on cars and mini-cars

and weight taxes on cars with higher fuel efficiencies, while raising taxes on cars with lower fuel efficiencies. In the same vein, the Tokyo Metropolitan Government is considering instituting tolls on heavily travelled routes.

In July 1999 the government passed a new law that requires companies to report the release of chemicals and pollutants into the environment. Known as the Pollutant Release and Transfer Register (PRTR), this environmental information management system is likely to have a significant impact on the business practices of Japanese companies and, more speculatively, on their abatement efforts. The PRTR system requires Japanese companies and industries that release large quantities of pollutants to report the emissions of many different kinds of chemicals, including by-products. The data will then be sent to the Ministry of International Trade and Industry (MITI) and to the Environment Agency, which will compile and tabulate it according to industry types and localities. The public, upon request, will be able to gain access to emissions data on specific factories. The PRTR system may also encourage Japanese companies to seek environmental efficiencies in all phases of their operations.[277] The revised Energy Conservation Law, which came into effect in April 1999, sets stricter target values for energy conservation standards and establishes rules for penalising non-compliance. In particular, the baseline for energy conservation standards is the performance level of the product with the highest energy efficiency on the market (the "top-runner" method).

Another piece of legislation passed in July 1999 is designed to reduce the emissions of dioxin (a carcinogen), which are believed to be higher than those in other OECD countries. Emissions will be more tightly regulated in regions with a high concentration of incinerators, the primary source of such emissions.

There have also been further developments on the recycling front. The Containers and Packaging Recycling Law is moving towards its second and last implementation step due in April 2000. In the interim MITI has just announced quantitative targets for recycling product packaging by manufacturers. As well, the Electric Appliance Recycling Law will come into force in 2001. And a bill that will place the onus on companies that demolish houses and buildings to recycle scrap lumber was presented to the Diet in May 1999. There will be no public subsidies to cover the costs for additional recycling mandated by the new laws. An auction-based system will be set up by the Japanese Containers and Packaging Recycling Association. In a related matter, the Health and Welfare Ministry is considering the idea of charging fees for collecting and disposing of household garbage, so as to minimise the production of dioxin when such waste is burned and to limit the need to find appropriate landfill sites. Some cities have already adopted such schemes and have managed to cut garbage generation by half by instituting a two-price tariff for mandatory garbage bags. Finally, relaxing waste disposal regulations has been proposed in order that companies not primarily engaged in such

activities may nonetheless make use of such waste materials. One example is steel makers who are willing to use plastic beverage bottles as fuel in their blast furnaces.

Commercial and external relations: coping with persistent tensions

Japan is the country with the largest trade and current account surpluses in the world. This has led some observers to charge that Japan's markets are closed to imports and that only by applying consistent pressure will openness increase and the surpluses subside. But the evidence regarding overtly protectionist government policies is scarce: for example, Japan's average tariff rates and recourse to non-tariff barriers have been lower than in other major trading zones for the past decade at least (Table 48).

For the past eight years Japan has also been the world's largest creditor nation, with net external assets in 1998 totalling 133.3 trillion yen (some 27 per cent of GDP), up 7 per cent from a year earlier. The nation's gross external assets (that is, the sum of assets held overseas by the government, companies and individuals) at the end of December 1998 stood at 345.1 trillion yen, down 3 per cent from the previous year. Meanwhile, gross external liabilities declined more than 8 per cent to 211.9 trillion yen. Japan is also the world's largest bilateral aid donor: its total net Official Development Assistance (ODA), which comprises grants and low interest loans to developing countries, has been over one trillion yen (over $9 billion) in recent years. At the May 1999 APEC ministerial meeting, Japan also offered to help its Asian neighbours recover by partially guaranteeing their sovereign bond issues for two years beginning 1 October through a $3 billion guarantee fund at the Asian Development Bank and a much larger amount at the Japanese Export-Import Bank. On the question of delayed repayment of ODA Japan has pursued various actions to lessen the burden for countries incapable of repayment, while simultaneously helping them to fulfil their responsibilities. In the past, the cumulated effect of these measures was equivalent to debt cancellation of over 80 per cent. Pursuant to the HIPCs Initiative agreed at the Cologne G7 Summit in June 1999, however, Japan will appropriate funds to allow 100 per cent forgiveness of its bilateral ODA loans outstanding to the 41 HIPCs (heavily indebted poor countries). Japan's actions have mainly involved offering grants for debt relief to poorer countries: for instance in FY 1997 it committed $233 million to 14 countries to cover interest and principal due it.

US-Japan commercial relations

The United States is Japan's largest trading partner. According to Japanese data, Japan's trade imbalance with the United States hit 6.7 trillion yen

Table 48. **Tariff and non-tariff barriers to trade in the Quad countries**

	Late 1980s			1993			1996		
	Average tariff trade[1]	NTBs		Average tariff trade[1]	NTBs		Average tariff trade[1]	NTBs	
		Frequency[2]	Coverage[3]		Frequency[2]	Coverage[3]		Frequency[2]	Coverage[3]
United States	4.4	25.5	16.7	4.7	22.9	17.0	5.2	16.8	7.7
Japan	**4.2**	**13.1**	**8.6**	**3.6**	**12.2**	**8.1**	**3.4**	**10.7**	**7.4**
European Union	8.2	26.6	13.2	8.4	23.7	11.1	7.7	19.1	6.7
Canada	8.7	11.1	5.7	8.4	11.0	4.5	12.1	10.4	4.0

1. Weighted by each country's own value added.
2. Share of national tariff lines affected by NTBs.
3. Share of imports subject to NTBs.

Source: OECD (1997*a*).

($51 billion) in 1998, up 33.3 per cent from the previous year, while US exports to Japan fell 4.1 per cent to 8.8 trillion yen ($67 billion). US data show a larger fall in exports to Japan but a smaller rise in Japan's bilateral trade surplus. While bilateral trade tensions with the United States have persisted, the two nations have been able to co-operate more constructively than during most of the past several decades, and most contentious issues have been successfully dealt with.

The resolved disputes

Many disputes have been resolved. Following a controversial Law of the State of Massachusetts, barring companies *doing business with Myanmar* (formerly Burma) from bidding for public contracts in Massachusetts, Japan and the European Union called in September 1998 for the establishment of a World Trade Organisation (WTO) dispute panel. Although the US Court of Appeals has ruled that law to be unconstitutional, the State of Massachusetts has appealed to the Supreme Court. WTO will keep its investigation going until November 1999. The National Foreign Trade Council (NFTC), which represents 580 companies including many of the biggest US multinationals, claimed that 346 companies were affected, and 58 of them were Japanese. During an annual review meeting of the Japan-US Insurance Talks held in Washington in April 1999, the Japanese government reaffirmed that the so-called *"third-sector" insurance market* (covering medical insurance, personal accident insurance, etc.) would be liberalised in January 2001 now that all the pre-conditions agreed with the United States have been met. In July 1999 the Ministry of Agriculture, Fisheries and Forestry decided to allow imports of five additional varieties of apples, which the United States had requested the Japanese government to approve. In March 1999 the WTO Dispute Settlement Body adopted reports that concluded that Japan's varietal testing requirement is not in conformity with its obligations under the WTO Sanitary and Phyto-Sanitary (SPS) Agreement.[278]

The disputes still outstanding

The most important conflicts have occurred in the *steel* industry. According to the Japan Iron and Steel Federation total steel exports to the United States increased from 2.7 million tonnes in 1997 to 7.0 million tonnes in 1998 (up 160 per cent). In particular, exports of cold-rolled steel increased from 250 000 to 371 000 tonnes and exports of steel plate products from 5 600 to 305 000 tonnes. MITI believes that the surge in exports was purely a result of increased demand: in fact demand for steel in the United States grew from 88 million tonnes in 1991 to 138 million tonnes in 1998. However, in the first eight months of 1999 such exports fell back to around 3.5 million tonnes at an annual rate. This represents some 2.8 per cent of the US market, down from around 5 per cent last year, and similar to the 2 to 3 per cent range seen for the rest of the decade to this point. Following a series of eight anti-dumping complaints filed by US steel makers

against a wide range of imported steel products that began in 1997, the US Commerce Department announced final duties of between 18 and 67 per cent on imports of hot rolled steel from Japan in April 1999. In June 1999 the US International Trade Commission confirmed that hot-rolled steel from Japan was being sold at unfairly low prices, injuring US companies, thereby clearing the way to impose the duties for the next five years. Duties have also been approved in two other cases. Officials from Japan and other countries have harshly condemned these measures as protectionist: about 80 per cent of Japan's steel exports to the United States are covered by US anti-dumping orders of current trade investigations. In October 1999, Japan decided to request bilateral consultations with the United States in accordance with WTO rules to deal with a number of issues pertaining to the hot-rolled steel case. In preparatory documents for the forthcoming WTO conference in Seattle (see below), the Permanent Mission of Japan to the WTO expressed "concerns about the abuse of anti-dumping measures" in general and insisted on "removing ambiguity and excessive discretion inherent in the [WTO] Anti-dumping Agreement". Following the request of the Japanese government, in July 1999 the WTO established a panel to examine the US Anti-Dumping Act of 1916.[279] The Act aims to prevent imports and sale of goods on the US market at prices below those applied in the country of production.

Another sector of long-standing tension is *flat glass*. Japan's flat-glass market is the world's second largest and was valued at $4.5 billion in 1998. In a meeting held in June 1999 between officials of the Japanese government and the US government, the US delegation claimed that the 1995 Measures by the Government of Japan and the Government of the United States of America Regarding Flat Glass had not been as effective as hoped. The Japan Fair Trade Commission had issued a report in May 1999 that found no problems in this market in respect of the Antimonopoly Act. Since the Measures were implemented, the foreign share of the flat-glass market initially increased to a high of 14.7 per cent from 7.9 per cent in 1994. The US share tripled to 5.8 per cent. In FY 1998, however, while the foreign share dropped to 14 per cent, the US shares plunged to 3.9 per cent, the lowest level since 1995. In the US government's view this poor result has been caused by unfair competitive practices and increasing ties between Japanese flat-glass manufacturers and distributors, developed *via* manufacturers' downstream vertical integration. Japan attributes the decline to deflationary price-setting by domestic producers, which has eliminated the price advantage of foreign flat glass, as well as their stable supplies and better services. In fact, in the first half of 1999, the foreign share rose to 15.4 per cent and the US share to 4.0 per cent.

In May 1999 during a meeting between officials of the two governments, the US delegation urged that, in accordance with the 1992 Measures Related to Japanese Public Sector Procurements of Computer Products and Services,

transparent *procurement procedures for computer equipment* be settled so that foreign companies can have greater access to information and more opportunities to bid on Japanese government contracts.

With reference to Japan's slow progress in deregulating the *telecommunication sector*, the US government has long argued that the high inter-connection rates for the use of the Nippon Telegraph and Telephone Co. (NTT) network, which dominates the domestic industry, serve as a virtual barrier for entry into the Japanese telecommunication market. Most recently, US officials have also expressed their dissatisfaction over the details of the plan to move to setting those rates based on long-run incremental costs. The US government also argued that prohibiting the use of leased lines by foreign carriers was not envisaged in the Japanese government's promises to the WTO. Japan's Telecommunications Business Law stipulates that broad service providers, known as Type I carriers, must establish telecommunications facilities, while if service-providers were allowed to lease fibre optics, switchboards and other facilities they could have gained easier entry into the market. NTT defended its position saying that it had cut the fees charged to competitors to connect to NTT's local network by 40 per cent since 1994. Recently NTT and NTT's successor companies announced that their procurement decisions would henceforth be non-discriminatory with respect to supplier origin. The US government accepted the decision while bilateral annual meetings to discuss systemic and other issues will be held for two years. The United States government had wished to renew the expiring bilateral equipment arrangement on equipment procurement by NTT, after its reorganisation in order to ensure that Japan's $35 billion market for telecom equipment remain open to foreign suppliers.[280] Japan's position was that the procurement arrangements, which have been in effect since 1981, were no longer necessary.

Over the past year Japan has been considering what for it would be a rare accusation: that its trading partners are *dumping* some of their products on its markets. At the end of 1998, five of the biggest Japanese petrochemical companies called for the imposition of anti-dumping tariffs on acrylonitrile[281] imports from some US, Korean and South African firms. The import share in the Japanese chemical market is currently 36 per cent but is expected to rise, and the five Japanese companies are suffering serious financial losses because of harsh price competition. Japan currently imposes a 4.6 per cent tariff on imported acrylonitrile, while the US levies a 9.6 per cent duty. November 1998 also marks the first time the Japanese government has submitted a formal complaint in high-level trade talks about *excessive regulation in the US car parts market*. Japan called for the deregulation of that market, charging that the American Automobile Labelling Act, which forces companies to list the country of origin of parts used in cars, unfairly discriminates against foreign carmakers. It also called for the loosening of US Corporate Average Fuel Economy regulations, which require foreign car-makers to submit separate statistics on the fuel economy of cars imported into the

United States, because they are allegedly inconsistent with the WTO non-discrimination principle. While the Labelling Act is applied to both US and foreign companies, it is said to impose a particular burden on foreign (including Japanese) car-makers because of the excessive costs involved in tracing the origin of parts in their home countries. Finally, the dispute over *harbour practices* that began in September 1997 when the US Federal Maritime Commission (FMC) imposed surcharges on the Japanese shipping carriers, has still not been fully resolved. Those surcharges had already been suspended when agreement was reached to improve those practices. But the FMC still requires US and Japanese carriers to file reports on those practices in order to watch matters closely. Furthermore, Japan has asserted repeatedly that the imposition of surcharges violated the Treaty of Friendship, Commerce and Navigation between the two countries, a view with which the United States disagrees.

Japan-EU commercial relations

With reference to the European Union (EU), despite the few disputes that have cropped up, there have recently been more areas for discussion and co-operation. On the deregulation front, the parties have been working since 1994 on a wide range of regulatory reform issues both in Japan and the EU, and there has been some progress. The parties have also been discussing mutual recognition co-operation since 1995 and have recently begun formal negotiations with a view to concluding a bilateral agreement. The main elements of an accord were reached earlier this year; and the definitive Agreement, concerning in particular electrical appliances, telecom equipment, good manufacturing practices of drugs and good laboratory practices for chemicals will be presented for legislative approval. The parties are working together on the agenda of the next WTO round. A new energy efficiency law, approved in April 1999, elicited an EU complaint because the legislation allegedly discriminates against EU vehicles and lacks transparency. Bilateral discussions have been held, and an announcement of their results is expected soon. In 1998, the EU also initiated a WTO dispute settlement procedure regarding Japan's management of its tariff quotas for leather and its subsidies to its leather industry. The EU believes they exceed WTO stipulated levels. Finally, the long-standing restraints on EU car imports from Japan are scheduled to be eliminated at the end of 1999.

Japan's commercial relations with other nations

Japan and Korea have settled the terms of a new bilateral fisheries pact: the major issue concerned access for Korean boats to Japan's exclusive economic zone, especially boats fishing for snow crab. Japan and Korea are also working together on investment agreements, standards and certification, intellectual property rights and the next WTO trade liberalisation round.[282] In July 1999 Korea's

ban on selected Japanese products (mainly cars and consumer electronics) was lifted, some 21 years after it was imposed. Finally, in February 1999, following Japan's request, the WTO established a panel to examine Canada's discriminatory measures regarding its auto import tariffs (under the terms of the 1965 Canada-US Auto Pact). Press reports in October indicated that the panel's interim decision was that these measures are in violation of the Uruguay Round agreements on goods and services.

Future developments

A new round of global trade liberalisation negotiations is expected to be formally inaugurated at the third ministerial meeting of the WTO scheduled for the end of November 1999 in Seattle. At present the world's major trading powers are largely divided into two camps. Japan and the EU are calling for negotiations to cover many issues and sectors[283] and advocate the principle of a "single undertaking", which would require WTO members to accept agreements on a wide range of areas as one package, believing that more benefits could be achieved for all members. The United States is concerned that such an approach will prolong negotiations beyond the agreed three years; its highest priority is on the so-called built-in agenda of agriculture and services (required to be on the agenda under the terms of the Uruguay Round agreements).

Notes

1. The Economic Planning Agency's (EPA) Survey on Corporate Activities of major non-financial companies reported that expected future growth rates for the next three (five) years declined steadily to 0.8 (1.2) per cent per year in FY 1999 from 2.2 (2.3) per cent in FY 1995.
2. The job-offer-to-seeker ratio, an alternative inverse measure of slackness in the labour market, continued to decline to 0.47 in the fourth quarter.
3. According to the Special Survey of the Labour Force Survey, the share of job losers who exit the labour force has tended to decline ever since the mid-1980s, although it picked up in 1999.
4. According to a survey by the Ministry of Education, only 60 per cent of all the new university graduates in March 1998 got a job, a record low. In addition, about a fifth of all new graduates could neither find jobs nor enrol in graduate schools, having no concrete plans following their graduation.
5. At first sight, the decline in compensation looks small relative to the drop in employment and in compensation per worker. This is because the employment concept relevant to compensation is *dependent* employment, whose fall was only 0.4 per cent, and because different government agencies are responsible for the data. Nonetheless, it would appear that labour compensation in 1998 has been overestimated.
6. Because 9 to 10 years is the normal lifetime of refrigerators, vacuum cleaners, etc., consumers may need to replace these goods bought around 1990, the so-called bubble period. However, the ratio of the stock of durable goods to disposable income rose to a record level (34 per cent) in 1997, increasing consumers' reluctance to buy them unless forced to do so because of break-downs.
7. Nakagawa (1999) argued that an increase in various perceived risks encouraged households' precautionary saving. Low-income, middle-aged and elderly people's consumption was affected by job insecurity; concerns about the future of the pension system worried young people; and worries about the costs of health care restrained the outlays of the elderly.
8. The number of houses per household has been greater than 1 since 1968 and rose to 1.133 in 1998 from 1.120 in 1993, according to the Housing and Land Survey.
9. Historically, the ratio has fluctuated in the range of 0.7 to 0.8. It registered its highest value, 0.78, in 1997.
10. This is the maximum amount of borrowing from banks under the constraint that monthly payments be less than a certain per cent of disposable income. The index, calculated assuming a ceiling of 25 per cent of monthly income, improved by about 6 per cent in 1998.

11. Affiliates of Japanese companies expanded their exports to Japan, utilising the currencies' depreciation. Fukao (1999) estimated export-oriented affiliated companies in the rest of Asia increased their exports in FY 1997 by about 12 per cent, and, furthermore, companies that had previously been oriented to supplying domestic markets changed their strategies and almost doubled their exports.

12. While net investment income from Thailand, Indonesia and Malaysia plummeted, that from the Philippines and Korea increased: on balance, the income from these countries decreased by 14.5 per cent in 1998.

13. This is based on notifications to the Ministry of Finance, a requirement under the Foreign Exchange Law. That is why the number is different from the balance of payments figure of 0.4 trillion yen.

14. The balance-of-payments figures include additional investments not covered by notifications to the Ministry of Finance. Fukao (1999) argued that the financial support reflected Japanese companies' long-term commitments to these countries, such as customer relations and skill formation of employees.

15. Because the inflation figures on a year-on-year basis in January to March 1998 were still affected by a rise in consumption tax rate in April 1997, the fiscal year data (April to March), which abstract from this factor, are a better indicator and are used throughout this paragraph.

16. Fresh food prices soared by 19.4 per cent (year-on-year) in the fourth quarter of 1998 because of bad weather. The CPI rose by 0.2 per cent in FY 1998, if fresh food is included.

17. According to Shiratsuka's (1999*b*) most recent estimate, the Japanese CPI suffers from a 0.9 percentage point upward bias in total. Out of this figure 0.7 percentage point is attributable to new products or quality changes. The rest is equally attributable to new outlets and lower level substitution, respectively; upper level substitution is negligible. Compared to the US case (Advisory Commission to Study the Consumer Price Index, 1996), the effects of new products or quality changes are larger (0.6 percentage point for the United States), the total bias is smaller (1.1 percentage points) because the effects of upper and lower level substitution are estimated to be lower (0.15 and 0.25, respectively). Shiratsuka's (1999*b*) estimated overall bias is larger than other estimates for Germany (Hoffman, 1998), United Kingdom (Cunningham, 1996) and Canada (Crawford, 1998): 0.75, 0.75, 0.5 percentage point, respectively. This is largely due to a comparatively large bias coming from new products or quality changes – see Table 1 of Shiratsuka (1999*a*). However, his estimates should be considered as offering only a crude idea of the possible size of the upward bias, given that there have been few investigations of this kind and he adopted sometimes rather bold assumptions. Nevertheless, the bias itself could be significant, because the current CPI does not include new products such as personal computers, for example. The Management and Co-ordination Agency, the agency responsible for producing the CPI, intends to include personal computers, mobile phone charges and other new products and services in the basket next year at the time of the regular review conducted every five years. The Agency will also consider the introduction of a new scheme to add new commodities to the basket between reviews and improvement of ways to select representative stores in each area. A revised treatment of short-term bargain sales might also be examined.

18. A comparable basket in the WPI also showed a price decline of 1.1 per cent for domestically produced consumer goods for domestic demand in contrast to the gap between overall CPI and WPI movements.
19. It is also necessary to move into the new houses by December 2000 in order to qualify for these credits.
20. Housing starts with loans from the HLC accounted for about 30 per cent of all housing starts in 1998. In particular, their share in owner-occupied houses was about 50 per cent.
21. The Ministry held the rate increase to 0.2 rather than the 0.65 percentage point that would have been implemented according to a usual rate-setting formula.
22. The share of those for sale that were sold increased from 71.7 per cent in second half of 1998 to 80.0 per cent in the first half of 1999.
23. This increase is also partly accounted for by increased imputed rents as a result of increased residential investment and by the estimated increase in the number of households.
24. The current balance fell in the first half of 1999 to 2.7 per cent of GDP due to a deterioration in the investment income and official transfer balances, and a sharp rise in import volumes.
25. Manufacturers' inventories in July 1999 were almost 20 per cent lower than in the previous peak at February 1998 and the lowest since December 1994.
26. This resiliency is reflected in a tendency for their fixed investment outlays to rebound faster than overall business investment in the recovery. For example, while most investment intentions surveys completed in June or earlier showed declines in FY 1999 for SMEs greater than the average, the Ministry of Finance's figures for the first quarter of the fiscal year show the smallest firms have experienced easily the smallest fall in the last four quarters.
27. Taking a slightly longer-term perspective on sectoral patterns based on deviations from trend of labour's share in output, it is clear that by this measure manufacturing is the sector with the greatest number of excess workers, although construction, distribution and transportation also have substantial numbers (Bank of Japan, 1999*a*), possibly because these sectors have historically been more sheltered from competition.
28. Traditionally, SMEs suffered from unsatisfied labour demand, as workers preferred the higher wages and job security offered by larger employers.
29. A recent example may be illustrative. A major department store chain (Mitsukoshi) announced 600 "early retirements" in February at a cost of 15.5 billion yen – an average of 25.8 million yen (over $200 000) per person.
30. This estimate for 1998 was derived by Goldman Sachs simply by extending sectoral labour productivity trends beyond 1991, when excess labour was assumed to be near zero. Other private economists have published estimates of 3.0 to 4.5 million workers.
31. This figure was derived by calculating a time series for the equilibrium labour's share – that share which results in the Bank of Japan's dispersion index for excess labour being zero – based on a regression of the dispersion index on its own lag, a time trend and the actual share. Then, given compensation, the excess labour figure is derived residually. The results show that labour has been in excess supply since around 1993, that it was virtually eliminated at the recent 1997 cyclical peak, since which time it has surged to the level reported in the text.

32. The EPA uses two methods to determine this excess. The simplest is to take the GDP gap in per cent and multiply by the capital stock, based on the assumption of fixed factor proportions. Using their gap estimate of a little over 3 per cent of GDP gives an estimate of 35 trillion yen at the end of 1998. The second approach used was to estimate a regression model for the Bank of Japan's dispersion index for excess capacity (including a time trend, real capital costs and the ratio of a weighted average of actual and expected output three years ahead to the capital stock) and, by setting it to zero, to derive an equilibrium value for the output-capital ratio term, which, when combined with the assumption that the disequilibrium is corrected entirely on the capital side, gives the 41 trillion yen figure. The two methods generate time series that are fairly collinear, although the second is even more cyclical than the first. It should be noted that the size of the excess estimated by the second method was even larger in the 1993 recession than it was in 1998, both in absolute terms and, to an even greater extent, as a share of the capital stock.

33. For example, Masahide Owada of Dai-Ichi Mutual Life Insurance Company estimates that capital under-utilisation in the third quarter of 1998 reached 115 trillion yen, of which 87 trillion was attributable to demand weakness and 28 trillion to excessive investment. Work done at Goldman Sachs, JP Morgan and the Japan Centre for Economic Research reaches similar conclusions. The most pessimistic assessments go as far as claiming 30 per cent of the capital stock to be in excess. But a careful analysis by Mikihiro Matsuoka, formerly of the Daiwa Institute of Research, points to excess capacity beyond the business cycle of 43 to 71 trillion yen (1990 prices), 7½ per cent of the equilibrium capital stock and equivalent to 9 months' investment.

34. The average annual loss in capital productivity since 1970 in Japan has been over 4 per cent.

35. A recent study by the Japan Development Bank showed that the ratio of debt to cash flow for SMEs – 12.5 in FY 1997 – was double that in the first half of the 1980s.

36. The biggest change is the move to consolidated financial reporting. This will be introduced in stages over a three-year period beginning in FY 1999. The implication is that companies will focus on and report not just parent-only results but results inclusive of all subsidiaries, and the definition of what constitutes a subsidiary will be broadened to include all enterprises that the parent actually controls (rather than basing classification just on share ownership), as demonstrated by provision of employees or funds. Note that this will greatly diminish the incentive to transfer excess human resources to a subsidiary as well as the ability to hide group losses or debts. Another change that will be implemented in 2001 is the move to market-value (rather than book value) based accounts, in accordance with the International Accounting Standards determined by the International Organisation of Securities Commissions in December 1998.

37. Some companies have sold off certain operations. Others have sold office buildings. At least a few have been constrained by Japan's rigid land-use regulations. For example, Cosmo Oil owns 330 000 square metres of land in Yokohama that is unused because the prefectural government refuses to rezone it for residential or commercial use. The Ministry of Construction hopes to ease such rules in order to encourage urban redevelopment, especially of former factory sites within cities. In what should be the first comprehensive overhaul of city planning laws in 30 years, the Ministry plans to present a bill to the Diet next year, so that new rules can be implemented in FY 2001.

38. Since these were first allowed in 1994 nearly half of all listed firms have changed their articles of incorporation to allow them, but only about 18 per cent of these have actually proceeded with any.

39. One interesting case is Toshiba. Its share price soared earlier this year after it announced serious steps to restructure its business lines and cut both head count and management pay by 20 per cent. It also set a target of raising return on equity to 10 per cent and reducing the debt/equity ratio from 1.88 to 1.0, along with an unwinding of its cross-shareholdings.

40. On average, 1 925 listed firms foresee a 3.1 per cent decline in FY 1999 sales, compared to an 8.7 per cent fall the previous year. Pre-tax profits, which fell 70 per cent last year, are expected to surge by 532 per cent this year, mainly thanks to financial sector outcomes. Excluding such firms, profits are expected to rise 12.5 per cent after a fall of 21 per cent last year.

41. Among the announcements made in the first quarter of the year, cost-savings goals averaged only 1.4 per cent.

42. It is noteworthy that these figures are widely regarded as more meaningful than those on legal bankruptcies themselves. In FY 1998 there were only 2 948 such events, only about one-quarter the number of companies going out of business. The fact that most failures take place out of the court system reflects the heavy costs and cumbersome nature of Japan's bankruptcy law. The Ministry of Justice plans to introduce amendments to the Composition Law at the next Diet session, so that SMEs can have the courts begin the reorganisation procedure before the firm actually fails.

43. The exposition in this paragraph is based largely on insights from Robert Alan Feldman of Morgan Stanley Dean Witter Japan.

44. Heightened risk aversion was manifest in: further widening of yield spreads on both bank debentures and corporate bonds over comparable government bonds (see Figure 27 below); and a shift of volumes in the call market from uncollateralised to collateralised transactions once the problems with the Long-Term Credit Bank began to surface towards the end of June. Specifically, in the third quarter of 1998 uncollateralised call contracts outstanding fell by 2.8 trillion yen (9.7 per cent), while their collateralised counterparts rose by 1.9 trillion yen (20.1 per cent). However, some of that would appear to be seasonal – see Figure 20.

45. These concerns were heightened as the Bank wrote off 16.5 billion yen of equity in the Resolution and Collection Bank in February 1999 and indirectly to Nippon Credit Bank (80 billion yen in contributions to the New Financial Stabilisation Fund that provided it equity). Special lending to ensure financial system stability under what is now Article 38 of the Bank of Japan Law has taken place in the cases of Yamaichi Securities (which was finally declared bankrupt in June 1999, with 480 billion yen in liabilities to the Bank, but it has not yet been decided who will bear its losses) and several regional banks. In addition, officials of the Bank have been increasingly concerned about the enormous loans they have outstanding to the Deposit Insurance Corporation as part of the banking system cleanup (6.7 trillion yen or around $55 billion at the end of FY 1998, about 8.5 per cent of its assets and 1.4 per cent of GDP) (Bank of Japan, 1999*b*), which could limit its ability to conduct needed open market operations. The DIC used the funds borrowed from the Bank and private financial institutions for such operations as extending loans to banks under special public management (Long Term Credit Bank and Nippon Credit Bank), capital

injection and financial assistance. Though the Bank's loans to the DIC are mostly government guaranteed, excessive growth of such loans could reduce the liquidity of its portfolio. In no other major industrialised countries does the central bank provide funds to the deposit insurance system.

46. The Bank's holdings of commercial paper receded by 10 percentage points to 32½ per cent of the total outstanding in the first quarter of 1999.

47. The implied one-year forward rates from two to nine years later, calculated from domestic-currency swaps, approximately doubled over this period, rising from less than 2 per cent at the longest horizon to around 3¾ per cent. The 1998-99 surge in bond yields did not reflect strong upward pressure on corresponding foreign rates nor, almost certainly, changes in inflation expectations. Rather it is widely believed to have resulted from supply concerns emanating from the deteriorating budget position and the increasingly critical longer-term outlook for public indebtedness. However, with the benefit of hindsight, some observers now point to the officially proclaimed turnaround in GDP growth in the first quarter as an additional explanatory factor. This pattern is consistent with the minimal impact of the narrowing interest differential on the exchange rate – see Figure 21. Around Christmas the Ministry of Finance announced that the share of the government's financing needs that would be borne by the public would be much higher than the market had anticipated. This was partly because of a reduction in net purchases by the Trust Fund Bureau (which had purchased 15 trillion yen worth of such bonds in 1998 and whose holdings had reached some 34 per cent of the outstanding stock) due to concerns about a substantial drop in funds provided by the postal savings system over the next year or two (about 80 trillion yen in deposits more than recent trends is due to mature, and estimates of how much of this will flow out of the system vary from around 20 to 50 trillion yen. In August 1999, the Ministry of Post and Telecommunications estimated the likely net outflow at 31 trillion yen.). After a severe market reaction, the Ministry announced that such purchases would resume in February and that funding would shift away from the ten-year security (to both medium-term notes and one-year treasury bills, issuance of which began in April).

48. Liquidity premiums were also rising: the yield differential between the benchmark and other bonds with similar maturities widened from the normal 5 basis points to 15.

49. In addition, the rate charged on the temporary lending facility decided on in November was lowered from 0.5 to 0.25 per cent per year.

50. Some of the decline in money-market rates for maturities of three months and more was reversed at the beginning of the third quarter, in part reflecting concern over Y2K problems. One major financial institution announced that it planned to hold massive amounts of extra liquidity in the first 10 days of next year.

51. Net foreign buying on a sustained basis began in October 1998, since which time it had cumulated to nearly 7 trillion yen (about $60 billion) by late August.

52. Since the financial crisis began in November 1997, the base money/broad money/GDP relationships have been unstable because of the malfunctioning of the system of financial intermediation. The unusual rise in broad money relative to GDP in 1998 is attributable to an increase in precautionary demand for liquidity. Households chose to hold more banknotes because of fears of bank failures and businesses more deposits because of the credit crunch. These phenomena have been unwinding this year.

53. Most recently (in the 4 May edition of the *Financial Times*), he argued that a target for annual inflation of at least 2 per cent was required. An alternative means to pushing real interest rates into negative territory has been suggested by Brown (1999). He proposes a currency reform under which the authorities would announce a currency reform to be carried out at some point in the future, at which point old yen would be converted into new yen at a certain loss. Then the central bank could levy a corresponding charge on its deposit liabilities and implement negative official interest rates without agents hoarding banknotes, since they would be subject to what amounts to a tax. Currency and deposits would not be perfect substitutes during this intervening period, with banknotes depreciating gradually. Retail prices would be differentiated with a premium charged for payments by cash, and the said premium rising as the conversion date approached.

54. For example, Article 5 of the 1947 Finances Law prevents the Bank from monetising the government debt.

55. Financing bills had traditionally been 60-day instruments bought mainly by the Bank at rates slightly below the discount rate, with little in the way of market transactions. However, the authorities plan to make the new 13-week Bills a benchmark short-term instrument. They have abolished withholding taxes on them, and weekly auctions of 1 trillion yen got underway in April. The yield was initially 0.10 per cent but has since fallen steadily, reaching 0.03 per cent by early June. The Bank is still buying some of these issues directly, with full auctioning not expected to be implemented for at least a year. The Bank's holdings have continued falling because sales of such bills under repurchase agreements got underway once again in February and public auctions began in April.

56. Repurchase agreements have become the main market operation in the Bank's arsenal since their introduction in November 1997. They involve twin operations. First, the Bank borrows government securities from financial institutions, increasing the items "government securities in custody" on the assets side and "government securities borrowed" on the liabilities side. Second, it provides cash collateral (as well as a fee), raising "cash collateral" on the assets side and "current deposits" on the liabilities side.

57. A variant of this option was espoused by former US Federal Reserve Board member Wayne Angell. He argued (*Wall Street Journal*, 14 April 1999) that the Japanese authorities should fix the dollar/yen parity for a pre-announced period of three to five years at a rate that would be consistent with a gold price that would be consistent with approximate price stability. In his view, that would be between 40 000 and 45 000 yen per ounce. That would imply a bilateral exchange rate of 140 to 150 to the dollar.

58. The figure of 100 is an early 1999 estimate of the fundamental equilibrium exchange rate – the rate that is consistent with a stable ratio of external debt to GDP in the long run – by the Institute of International Economics, as reported by the Bank for International Settlements (1999*b*). On the other hand, the figure of 163 is the latest (1998) OECD estimate of the purchasing power parity exchange rate. Regular surveys of large domestic firms by the Economic Planning Agency have most recently shown a break-even rate of about 113 yen to the dollar.

59. In a March 1999 report the Bank for International Settlements said that the carry trade may have reached $30 billion in outstanding positions through the Cayman Islands alone in early 1998.

60. The role of expected real interest rate differentials in the determination of the yen-dollar exchange rate has recently been demonstrated by Wu (1999).
61. Some observers have attributed this reticence among life insurance companies to fears of taking currency risk in view of the ongoing inspections by the Financial Supervisory Agency (see below).
62. One problem with targeting the base is that the overwhelming majority of it is comprised of currency in circulation, over which the Bank has minimal leverage. The remainder is also difficult to influence precisely, since central bank expansionary actions may just bring about an increase in banks' excess reserves or leakages to *tanshi* brokers. Milton Friedman's recommendation is to set a target growth rate for the broader aggregate M2+CDs of 8 per cent per year.
63. One member of the Board has gone to great lengths to compare the Bank's policy with popular alternatives, pointing out that its stance "involves the same degree of commitment to easy policy as does money supply targeting, but by using different semantics" (Ueda, 1999).
64. The year-on-year growth rates in the monetary base have fluctuated between 3 and 10 per cent over the last year or so, as demand has varied according to agents' views of the soundness of the financial system as well as their transactions needs.
65. For example, recent recovery rates for large private-sector banks have been around 22½ per cent.
66. For borrowers in the corporate business sector the share is only 17½ per cent and has edged up only modestly in recent years.
67. One private estimate of these effects put them at 8 000 bankruptcies and 77 000 job losses avoided.
68. Research results now beginning to circulate demonstrate the importance of the drop in asset prices, weak demand conditions and moral hazard problems in raising the probability of a banking crisis (Hutchison and McDill, 1999). However, what sets the Japanese case apart from others is its long duration, no doubt in part due to the slow regulatory response and the delay in committing public funds to bank recapitalisation.
69. The benefits of the funding included not only the ability to assure depositors that their capital was safe and to improve the balance sheets of solvent banks but also to threaten weak banks with (temporary) nationalisation even if no sound receiving bank were available to take it over as in the past under the so-called "convoy system".
70. The new authorities gained the power of "Prompt Corrective Action", which was implemented in April 1998. This was designed to mitigate the problems associated with regulatory forbearance and inadequate capitalisation that itself was systematically related to the presence of former public officials in bank management (Hanazaki and Horiuchi, 1998).
71. A small brokerage failed in November 1998, as did Toho Mutual Life in June 1999.
72. Liquidating this bank cost a total of 3.41 trillion yen, of which 1.79 to the banks who took over its healthy parts and 1.62 to the Resolution and Collection Bank to assume its non-performing loans.
73. Of the plan's total of 60 trillion yen, 25 trillion was for solvent bank recapitalisation, while the other 35 trillion consisted of 28 trillion in borrowing authority and 7 trillion in taxpayers' funds.

74. The Resolution and Collection Bank took over less than half of LTCB's non-performing loans, with face value of around 4.6 trillion yen. Thus, LTCB still has some 12 trillion in risk assets, and it looks likely that the eventual buyer (see below) will have to inject around 1 trillion yen in new capital.

75. The Bank of Japan has recently estimated that 36 per cent of all bank deposits are in accounts with balances in excess of the 10 million yen ceiling for deposit insurance coverage. Once the fact that individuals have multiple accounts is allowed for, the figure could be much larger.

76. It was probably wise that the authorities did not attempt to unify all banking supervisory functions in the FSA but continued to allow the central bank to participate in such oversight: recent US evidence shows how even in its market- (rather than bank-based) financial system information gleaned from its supervisory role has helped the Federal Reserve make better monetary policy – see Peek *et al.* (1999).

77. The importance of such information for both supervisors and investors has recently been underlined in empirical work at the Bank of Japan (Yonetani and Oda, 1999).

78. A variety of others had previously made realignment plans. Yasuda Trust was absorbed by Fuji Bank in March. At the same time Sanwa Bank and Toyo Trust combined certain business operations. Sumitomo Bank and Daiwa Securities have announced a broad-based tie-up, as did Industrial Bank of Japan, Nomura Securities and Dai Ichi Life. Asahi and Tokai Banks will pursue some sort of combination. Daiwa Bank is combining some regional banking affiliates in the Kansai area. Mitsui and Chuo Trusts are to merge in April 2000. Bank of Yokohama will close its affiliated securities company, in favour of that of Tokai Bank. Most recently, Sumitomo and Sakura Banks announced their intention to merge next year.

79. The banking sector will contract, assuming that investors will gradually shift their portfolios towards global standards and therefore away from deposits and towards riskier investments such as mutual funds at the same time as borrowers will increasingly bypass intermediated providers of capital.

80. The 15 banks undershot their end-year target very slightly for total lending but more substantially (by 0.7 trillion yen) for SMEs.

81. The range of projections across banks is from 5 to 12 years. However, analysis by Merrill Lynch indicates that these projections are excessively optimistic and that it will take an average of 14.3 years and as much as 40 years for some banks to reimburse the capital provided. In any case, banks' recurring profits are expected to continue to be below operating profits because of a greater likelihood of loan losses. Thus, while boosting loan spreads might contribute 0.2 per cent to returns on assets, loan losses are expected to wipe out the bulk of that benefit, leaving little underlying improvement in returns on equity.

82. However, there was little effort to cut expenses in line with low and stagnant profit levels, possibly because of a failure of corporate governance resulting from a lack of effective disciplinary mechanisms – see Hanazaki and Horiuchi (1999).

83. Self-responsibility is the foundation of the new system. Asset quality treatment will shift from inspector assessment to risk management. At the same time, however, the authorities decided to allow the banks to adopt deferred tax accounting, thereby permitting them to take future corporate tax credits onto their books. This created an additional 3 trillion yen in assets for the nine city banks, and as much as 500 billion yen for the biggest.

84. These estimates may still turn out to be too low: for example, ongoing inspections by the FSA have turned up 2 trillion yen more problem loans at regional banks as of March 1998 than they had disclosed.
85. The difference is over 6 trillion yen, by chance precisely the amount of the capital injection.
86. The FSA's budget for FY 1999 has been boosted, so that it has been able to raise its total staff and the number of its inspectors by 135 and 87, respectively, increases of more than a third. They can also mobilise staff from the ten local bureaux of the Ministry of Finance. The Bank of Japan also conducts on-site examinations on a contractual basis.
87. Among the top eight firms in FY 1998 this gap was worth some 1.3 trillion yen, about 7 per cent larger than in the previous year, but nearly 20 per cent less than the 1995 peak. This is eating into the amount of capital gains on securities, which had turned negative for four of these firms by September 1998, but which no doubt has since improved in line with the rebound in equities prices.
88. These commitments are required by law. In the late 1980s life insurance companies were offering returns of 5.5 to 6.25 per cent, substantial parts of which were guaranteed for more than 30 years. At the same time they were unable to find investments with such a long maturity: no government bonds had lifetimes greater than 20 years, for example.
89. The 14 major non-life firms also had a second consecutive year of falling premium income and unrealised capital gains.
90. This is the ratio of capital plus reserves plus a fixed share of unrealised capital gains on equities and land (85 per cent) to half the assessed amount at risk from all sources (insurance, promised returns, portfolio investment and management).
91. It is notable that the more aggressively the recipient bank disposed of problem loans in FY 1998, the smaller its projected increase in FY 1999 lending. Banks face some incentives not to make new loans indiscriminately, at least not without credit guarantees, since there is a risk that shareholders may launch derivative lawsuits against bank management if loans are extended to borrowers who are not creditworthy.
92. The use of discounted cash flow analysis is still not widespread in Japanese banking circles. Another reason why property has not been sold to the extent possible is that there are many pieces over which there are complex legal claims. The government recognised this early in the current recession and in 1998 even drafted legislation to create committees to work out such claims, but that legislation was never passed.
93. Not only is the RCC the recipient of all bad loans of the 40 failed banks and credit unions, but since March 1999 it has had the right to buy from functioning banks, and its current head believes that loan collection for all public financial institutions should be centralised within it. It will receive about 9 trillion yen in assets by book value from the two nationalised long-term credit banks. In September 1999 it agreed to buy 138.4 billion worth of real estate collateral tied to bad loans from 35 solvent banks and credit co-operatives for 6.7 billion yen.
94. Press reports say that foreign entities bought bad loans with a book value of around 8 trillion yen in FY 1998 for 5 to 10 per cent of that amount. Most buyers are said to plan to sell the associated collateralised real estate, possibly by auction.
95. Between December 1998 when the rule governing the practice was modified and April 1999 when it was clarified, 35 firms were involved in more than 3 trillion yen of such deals.

96. It should be pointed out that loans secured by real estate and floating mortgages have shrunk more rapidly than total loans in recent years. By 31 March 1998, Bank of Japan figures show that their share was 23.5 per cent, down from 27.9 per cent five years earlier. There has been a compensating shift toward unsecured loans (33.6 per cent, up from 32.3 per cent in 1993) and loans with a third-party guarantee (34.1 per cent, up from 29.9 per cent)

97. However, banks' loans in the form of consumer credit have fallen sharply over this period, implying that the increase in lending to individuals is for business or housing purposes.

98. Traditionally main banks were the major buyers of corporate bonds.

99. The Law, which was originally passed in late 1997, was revised in June 1998 to introduce a flexibility clause, but it still constrained deficit bond issuance, thereby limiting the government's scope for fiscal manoeuvre.

100. The adoption of the so-called New Miyazawa Plan to help Japan's Asian neighbours to recover led to a 40 per cent rise in such government loans to 910 billion yen in FY 1998.

101. The second FY 1998 supplementary budget, which passed the Diet in October 1998, only extended the government's guarantee to the Deposit Insurance Corporation's borrowing from 17 to 40 trillion yen to cover the banking system rescue plan (see Chapter II).

102. This cost was to be shared between central government (4 trillion yen) and local governments (2.8 trillion yen).

103. The Japan Development Bank Law was revised so that the Bank may now provide loans for current expenses as well as capital outlays. The Bank also received an additional injection of capital through the Special Account for Industry and Investment.

104. In addition to the November revision, the initial FILP budget was modified on three other occasions during FY 1998: twice in June (provision of an additional 6.7 trillion yen), and once in December (1.4 trillion yen), mainly in order to provide the necessary funds to implement the measures included in the April and November packages.

105. Front-loading of public works contracting was also encouraged, aiming at increasing the value of contracts signed in the first half of FY 1999 by more than 10 per cent compared to the previous fiscal year.

106. National personal and corporate income tax reductions accounted for 3.2 (including dependents' allowance of 0.3) and 1.7 trillion yen, respectively. The total amount of tax reductions promised in the package was larger because that included local tax reductions: residential (1.1 trillion yen) and local enterprise taxes (0.8 trillion yen) were cut.

107. The previously temporary lump sum personal tax cut was replaced with permanent cuts through reductions of the top marginal tax rate and of 20 per cent for others to a maximum reduction of 250 000 yen (for national income tax).

108. As a result, the government will sell 61 trillion yen out of the 71.1 trillion yen in gross government bond issuance in FY 1999 on the market, while the Postal Savings system will purchase and the Bank of Japan will roll over 3.6 and 3.7 trillion yen, respectively.

109. The local budget numbers as of September are the only officially available information aggregated over all local governments (of which there are nearly 3 300). They are released in the White Paper by the MHA in March of the following year.

110. Unfortunately, follow-up reports were not made on several of the many packages in the 1990s. This, together with the incomplete nature of the follow-up work reported, has made it difficult for outside observers to judge the extent of the actual implementation of the measures contained in the packages as well as their effectiveness.

111. The automatic reduction in the transfers due to the decline in the relevant five national tax revenues (see Annex I) was well compensated for by 1) a special increase in the transfer from the general account (0.9 trillion yen) and 2) new borrowing of the Special Account for Local Allocation Tax and Transferred Tax (2.1 trillion yen) (see Figure 34).

112. The number should be interpreted with care because the plans do not allow for either carrying forward or backward of spending.

113. The government uses the deficit of central and local governments as a policy indicator because the surplus on social security reflects the nation's relatively young age structure in the past and institutional arrangements in the pension system, and therefore it merely provides (and even then only very partially) for future pension liabilities.

114. It should be noted that elderly health benefits are financed by transfers from other schemes and that the elderly scheme itself does not receive any contributions. Therefore, a financial problem in each account in the health insurance system is masked by a large deficit on the scheme for the elderly.

115. According to the MHA, the number of local governments suffering from deficits on the National Health insurance accounts was 642 in FY 1997, down from 765 in the previous year, but still much higher than 452 in FY 1991.

116. See the previous Survey (OECD, 1998). The OECD has changed the definition of gross debt since then. The current definition corresponds to the one labelled as "Gross debt, OECD definition plus further consolidation" in Figure 22 of that Survey.

117. According to the MHA, the accumulated losses of enterprises subject to the Public Enterprise Law amounted to 3.9 trillion yen in FY 1997. More than half the losses are due to transportation business and more than a quarter to hospitals. Annual losses then result in capital transfers from local government in order to hold down their indebtedness.

118. The government decided the medium-term economic plan up to 2010 in July 1999. The plan points out that all available measures on both revenue and expenditure sides should be taken to restore the soundness of public finances and promises a concrete plan will be designed once the economy is back on a recovery path. The plan is based on the following macroeconomic scenario: average annual real and nominal GDP growth rates of about 2 and 3½ per cent, respectively, and an unemployment rate of 3½ to 4½ per cent in 2010. The OECD's assessment is less optimistic: real growth might average only about 1¼ per cent per year.

119. Debt dynamics is shown as follows, $b_t = d_t + (1 + r)/(1 + g)b_{t-1}$, where b is the debt-to-GDP ratio; d is the primary-balance-to-GDP ratio; r is the nominal interest rate; g is the nominal growth rate. In the case where the (nominal) interest rate is greater than the (nominal) growth rate, the system is unstable: the debt-to-GDP ratio would follow an explosive path once the economy is outside the equilibrium. The required primary surplus to stabilise the debt-to-GDP ratio at time t is calculated as $d_t = (g - r)/(1 + g)\ b_t$.

120. The proceeds from land sales were 230 billion yen in FY 1997 and 88 billion yen in FY 1998.

121. Although the government also holds JT (Japan Tobacco) shares, the current holding is required by a law and is not disposable.

122. As examples: while the building of sewage systems for residential areas falls under the jurisdiction of the Ministry of Construction, the Ministry of Agriculture, Fisheries and Forestry is responsible for the construction of sewage systems for farms. Also the divisions between the Ministry of Health and Welfare, which oversees nursery schools, and the Ministry of Education, which is responsible for kindergartens, will continue to exist even under the new legislation.

123. This will be done also by limiting the number of panel members to 20 in principle, and 30 at its maximum per panel.

124. In FY 1998 the number of full-time employees in local governments fell to 3¼ million (down 0.5 per cent on the previous year), while those in central government dropped to 845 000 (down 0.9 per cent on the previous year). In both cases these were the largest year-to-year declines in staff levels since records began in FY 1975.

125. In fact Japan's oil sector is lagging behind its competitors because it has failed to invest strategically in productive oil fields or generate enough return on those assets to become self-sufficient. The ratio of after-tax earnings to fixed assets is a low 3.2 per cent in Japan, compared with an average 13.7 per cent at the big western oil groups. The industry is too diversified geographically and is said to have often made investment decisions with little concern for marketability of the resulting output.

126. It is currently involved financially with 130 projects around the world, but it pulled out of at least seven over the summer of 1999.

127. The administrative documents exempt from disclosure include: information on individuals, information that could endanger rightful profits of companies or organisations, information that could damage national security or international trust, information that could hamper prevention, investigation or prosecution of crimes, minutes of deliberations for which disclosure could hinder fair discussion and information whose disclosure could hamper proper execution of public business.

128. They decided to stop 51 projects temporarily and to shrink the size of 15 other projects in the process of preparing the FY 1999 budget.

129. Note that some limited discretion in portfolio investment by these special accounts, either directly or indirectly, has been effectively allowed for more than a decade. The discretionary share of overall FILP resources in flow terms has been increasing over time, reaching a quarter in FY 1999. But the results have not always been favourable. For example, the Pension and Welfare Corporation – the conduit for investment by the Welfare and National Insurance Special Accounts – has experienced cumulating losses (about 1.2 trillion yen as of March 1999) on its investments and will be liquidated in FY 2001, when a newly set-up fund will take over the fund management.

130. A possible obstacle to issuing the bonds is that the return on HLC's assets might be too low, as reflected in its need for substantial government operating subsidies.

131. The Resolution and Collection Corporation (RCC) is unwilling to undertake securitisation of their assets.

132. Because ABS markets have been underdeveloped (the outstanding stock in June 1998 was 219 billion yen according to the Security Dealers Association and 367 billion yen according to the Ministry of International Trade and Industry), the banks have sold their assets through bilateral negotiations in most cases.

133. Therefore, as was observed at the time of the most recent reduction in income and corporate taxes initially promised in the fiscal package in April 1998, the details of the changes were decided by negotiation between the Ministries of Finance and Home Affairs.

134. While higher rates on corporate residents tax and business tax are observed (in 46 and 7 prefectures, respectively, out of all 47 prefectures), there are none on other taxes. In addition, in 14 out of 15 cases, prefectures' own taxes are related to the production of nuclear energy.

135. The well-known large disparity of marginal productivities of social infrastructure among regions is consistent with this result. See Yoshino and Nakajima (1999) for a recent estimate of the marginal productivity of social infrastructure.

136. It can be argued that a lump-sum transfer from central to local government is equivalent to a tax cut for the locality of the same amount (Bradford-Oates theorem). However, empirical investigations have sometimes found failures of this prediction: the transfer tends to have a greater impact on local expenditures, the so-called "flypaper effect". See Bailey and Connolly (1998) for a recent survey. This effect could be evidence of fiscal illusion, but it has not been firmly established in Japanese data. Doi (1996) found supportive evidence in data pertaining to relatively poor municipalities. However, as Nagamine (1995) showed, the outcome might result from the exogeneity of tax revenues in local-authority decision-making. Mochida and Lotz (1999) classified ways of inter-government transfers for equalising purposes into the grants method and the solidarity method. The former is adopted more widely (including in Japan) and only intends to raise poor local authorities' revenue up to a certain national standard, while the latter removes resources from the rich authorities and gives to the poor (like in Germany (among the *Länder)*, Denmark and Sweden). Mochida and Lotz (1999) also argued that the wider use of the former method is due to fiscal illusion.

137. Reservation ratios are the share of local tax not included in the definition of standard revenues for a typical local government in the calculation of LAT.

138. Yoshida (1998) argued that a third of all 47 prefectures might be wealthy enough not to receive LAT.

139. See Besley and Case (1995) for a description of how yardstick competition among US local governments affects their policies.

140. Yoshida (1998) is dubious about the effect of the deregulation of local bond issuance because the exogeneity of local tax revenues seems to remain unchanged.

141. This was the second wave of the mergers, supported by the Promotion Law for Municipal Mergers. The first wave took place in the 1880s, *i.e.* the Meiji era, and was also driven by increased financial needs to deal with the establishment of elementary schools and the registration of soldiers.

142. Kobayashi (1998) surveyed 145 mergers that took place from 1965 to 1995 and found governors' leadership to have played a key role in 106 cases. He also argued that while residents' initiatives allow relevant parties to enter a formal consultation process, skipping preparatory work in informal consultation would be very costly. The effects of municipal mergers have been the subject of only a few empirical

studies, such as Kobayashi (1998). It seems premature to make an overall assessment. But one interesting finding by Kobayashi (1998) is that the post-merger reduction of administrative costs is slow, as is often observed in corporate mergers.

143. Note that Japan's working-age population had expanded by about 1 per cent per annum during the 1980s and early 1990s.

144. The contributions and benefits are normally revised every five years, the last revision having been in FY 1994.

145. The projections assume a wage increase of 2.5 per cent, a price increase of 1.5 per cent and a return on pension funds equal to 4.0 per cent. Note that if benefits were not cut, the contribution to the national pension scheme would have reached 26 400 yen per month in FY 2025 and that to the welfare pension scheme 34.5 per cent of wages in order to balance the two schemes.

146. The Economic Strategy Council proposed to finance all of the basic part by taxes.

147. The main reasons for such growth are ageing, technology and, more speculatively, systemic inefficiencies, with only a minor role for population growth. For instance, in FY 1996 national health care expenditures rose 5.8 per cent, of which only 0.2 percentage point was attributable to population growth. Of the remainder, about 1.7 percentage points can be explained by ageing. As elsewhere, advancing technology has no doubt pushed spending up, assisted by the mainly fee-for-service payment system.

148. Under the current law, the maximum premium rate is set at 9.5 per cent of monthly salary for policyholders covered by health insurance societies consisting mainly of employees at major companies. For those policyholders covered by government-run health insurance associations, the maximum premium is set at 9.1 per cent of monthly salary. Each company-run health insurance society sets its own premium rate (shared equally between employers and employees), while policyholders covered by the government-run health insurance associations pay premiums of 8.5 per cent of salary.

149. Only 18 per cent of all drugs were dispensed by pharmacists in 1994 (Jacobzone, 1999).

150. It is believed that a high proportion of drugs that have appeared in the US and EU markets in the past five years are not yet available in Japan. This reflects the fact that many products are not presented for official approval because of low expected profitability as well as slow official authorisation.

151. For the elderly the co-payment for outpatients is 530 yen (about 5 dollars) per visit to doctors, with a maximum of four visits per month, beyond which care is free. For inpatients the co-payment is fixed at 1 200 yen per day (about 10 dollars).

152. This is the main reason why the average stay for a Japanese general hospital patient is about 39 days, well above the OECD average.

153. The designated services are: home helpers, nurses, rehabilitation, short-term stays at facilities with nursing care, lending of nursing instruments and subsidising home improvements. Specific facilities for the elderly are special nursing homes, health-service facilities and medical facilities specialising in long term care. MHW estimates that by 2010 about 560 000 helpers will be needed.

154. Vertical equity refers to taxation according to ability to pay (higher incomes pay more). Horizontal equity is achieved when individuals with the same economic capacity (measured *e.g.* by income) are taxed to the same degree.

155. The elderly dependency ratio measures the number of retired people (here defined as those of 65 years of age and above) divided by the working-age population. The total dependency ratio measures the number of retired people plus the number of young people below working age (less than 20 years of age) divided by the population of working age.

156. These issues have been dealt with intensively in previous OECD publications; see for instance OECD (1997), Leibfritz *et al.* (1995) and Roseveare *et al.* (1996). See also Yashiro *et al.* (1996) for a thorough analysis of Japan's situation.

157. This assessment is based on the "central assumptions" scenario of Table 27.

158. Japan's social security funds held assets equivalent to 48 per cent of GDP in 1997 compared with around 10-15 per cent of GDP in the United States and even less in the other major OECD countries. See also OECD (1998).

159. Takayama (1996) shows how, under more pessimistic assumptions, the contribution rate to the earnings-related pension scheme would have to rise to as much as 40 per cent in 2025 from currently 17 per cent. An even more pessimistic projection is found in Tajika (1998). He projects a contribution rate of nearly 50 per cent in 2020 based on a doubling of the effective ratio of pension beneficiaries (including those who have contributed less than 25 years to the system and survivors) to contributors and keeping the current gross wage replacement ratio (which, however, is unlikely, given the shift in indexation in the draft reform, see Chapter III).

160. The 1999 draft pension reform schedules an increase in the central government subsidy to the basic pensions from one-third to one-half by 2004.

161. See for instance OECD (1997*b*), Takayama (1996), and Kotlikoff and Leibfritz (1998).

162. Annex II gives a more detailed exposition of the key statutory features of the tax system.

163. In a comprehensive system all income is taxed according to the same schedule – following the so-called Schantz-Haig-Simons principle.

164. The system thus bears some resemblance with the dual tax systems of the Nordic countries, where capital income (including corporate income) is taxed at a low flat rate and wage income is taxed in a separate progressive schedule. See Sørensen (1998).

165. The simplified regime allows firms with taxable sales below 200 million yen to be taxed at a certain percentage of their gross sales rather than their actual value added. The implied value added may to some extent underestimate the true base (*i.e.* the actual value added), see Ishi (1998), but the implied and taxed values were brought closer together following an adjustment of the implied values in 1997.

166. Following Mendoza *et al.* (1994), a number of studies have tried to measure effective tax rates using aggregate revenue statistics and national accounts data to approximate the relevant tax bases. However, since aggregate data are not well suited for dealing with self-employment income, depreciation, interest, cross-border returns and the treatment of losses in the corporate sector, such measures can only give a rough indication of the level of effective taxation, and there is a non-negligible risk of these measures being severely misleading (see also Carey *et al.*, forthcoming).

167. This measure of effective rates is based on income and tax data for individual corporations, thereby circumventing many of the *caveats* related to using aggregated data. For more details see Tajika and Yui (1993 and 1997).

168. Korea is the only OECD country with generally lower tax wedges than Japan.

169. Tachibanaki (1997) surveys a large amount of empirical evidence for Japan on both issues. This evidence suggests that the corporate tax system, including preferences given to corporations in the form of tax-free reserves, accelerated depreciation and investment tax credits, accounts for only a modest part of aggregate changes in the user cost of capital. When taking into account the effects from the personal tax system, some studies find a fairly low tax wedge for Japan between the pre-tax return to investment and the post-tax return to the saver; however, this wedge increased significantly after the strengthening of dividend and interest taxation in the late 1980s. Moreover, most evidence points to capacity considerations, rather than the cost of capital, as the dominating influence on investment (*i.e.* output accelerator effects). These studies allow Tachibanaki to conclude that "a large number of studies, with a few exceptions, suggest that the effect of various tax policies on investment was very minor". The Economic Planning Agency (1998) likewise concludes that even a significant cut in corporate tax rates would imply only a small change in the cost of capital. These findings largely correspond to the bulk of evidence from other countries, some of which is surveyed in Leibfritz *et al.* (1997).

170. The tax credit system for dividends implies that dividends are taxed at a maximum rate of 40 per cent for incomes below 10 million yen.

171. The marginal effective tax wedge reflects the required pre-tax rate of return an investment has to earn in order to provide a personal investor with the same after-tax return as a bank deposit earning a pre-tax 5 per cent real rate of interest. The estimates shown in Table 33 are based on the King-Fullerton methodology – see Gordon and Tchilinguirian (1998) and OECD (1991). The results should be interpreted with caution since for some investment and financing decisions the results generated may not adequately reflect incentives and the effects of taxation of these incentives. The marginal investor, for instance, may not be a fully complying individual (as assumed here) but rather a *de facto* tax-exempt entity. Other simplifying assumptions are also applied, including: perfect competition, a rudimentary treatment of financial structures and the intermediation process, absence of uncertainty, perfect loss offsetting and capital irreversibility.

172. A totally neutral system would apply the same marginal rates across all types of financing and investments, thereby leaving the standard deviation of tax wedges at zero. Tachibanaki (1997) also finds evidence of strong non-neutralities based on marginal effective tax rate analysis.

173. As explained below, however, these measures are fairly limited (amounting to around 0.1-0.2 per cent of GDP) and are likely to have only a very small effect on the cost of capital and aggregate investment in Japan. See also Tachibanaki (1997).

174. Until last year a 1 per cent tax was levied on the value of qualified plan assets. This tax has been abolished temporarily.

175. Annex III gives a more detailed exposition of different ways of taxing savings.

176. There are four prices pertaining to a piece of land. The first is a market value observed in transactions. However, it is difficult to ascertain precisely what this is, because land is a heterogeneous commodity, each piece of land having its own features, and because liquidity in the market is low. The second is an official price (*koji chika*) announced by the National Land Agency once a year based on experts' assessment. The third is a value for the purpose of levying inheritance tax. Finally, a value for imposing property tax is also available.

177. However, the area designated as "productive green land" (*seisan ryokuchi*), a requirement for deferred payment of inheritance tax, is still quite large – about 40 per cent of total agricultural land located in urban areas.

178. Nishimura *et al.* (1999) estimate that the "tax shelter" value of farmland reached a peak in 1991 of nearly 3½ times the market price (implying a reservation price of 4½ times the market price). The tax shelter value has since dropped sharply due to falling land prices and the tax changes described in the main text, but was still estimated at some 130 per cent in 1997 (implying a reservation price for farmers of 2.3 times the market value). The tax shelter is much lower for non-farmers, *i.e.* some 20 per cent in 1997. Empirical evidence suggests that the tax shelter value is a significant cause of high residential land prices.

179. For instance, there is a 30 million yen special deduction for the transfer of residential property.

180. Sales of second-hand houses amount to 100 000 to 150 000 units (owner-occupied houses only) per year in Japan, while the corresponding number (all types of houses) reaches about 4 million in the United States (Ministry of Construction, 1998*a*).

181. Iwata *et al.* (1987) estimated the effects of various measures on the cost of capital and found that the HLC loans reduced it by 0.6 to 0.7 percentage point, whereas the reduction of the property tax had a much smaller effect, around 0.1 percentage point.

182. Accelerated depreciation is also allowed for certain so-called "quality rental houses" (*yuryo chintai jutaku seido*).

183. Other tax factors could affect tenure choice as well. The inheritance tax treats rental houses more favourably than owner-occupied houses through lower asset evaluations, which are likely to be worth a decline in costs of capital by a few percentage points. Salary earners can also reduce tax liabilities by registering losses of running a rental business. These factors as well as the lower costs of capital of rental houses for wealthy households are believed to account for part of the construction boom of rental houses in metropolitan areas in the late 1980s. However, this tax advantage was reduced in 1992: the tax code was revised so that only losses incurred on buildings have since been included in the calculation of income.

184. Iwata (1997) estimated that the effects of deductibility of interest and property tax payments from income would be to cut the costs of capital by 20 -25 per cent and that households with an annual income of more than 15 million yen would enjoy sizeable benefits.

185. Here, income inequality is measured by the so-called Gini coefficient. The higher the coefficient, the less equal the income distribution. Income includes both wage earnings and capital and self-employment income.

186. See Tachibanaki (1997). The same study also found that treating all income, including imputed rent, under a comprehensive income tax system would increase the redistributional effect of taxes by 50 per cent.

187. The statutory progressivity is based on OECD's tax equations for an average production worker. A few limitations to measuring the progressiveness of the tax system based on these equations should be borne in mind. First, take-up rates for deductions and exemptions not included in the tax equations tend to increase with income (as does income in kind, *i.e.* fringe benefits, which are also not covered). Second, large groups of taxpayers do not pay social security contributions in many countries (the self-employed, retirees, benefit recipients).

188. However, persons with highly fluctuating incomes can apply an averaging provision which mitigates the progressivity.

189. This measures national income tax payments and employment income only. The reason why the low employment income group faces a relatively high tax rate is that many individuals in this group have other sources of personal income. Unfortunately, the tax statistics encompass only the national income tax. However, since local income taxes are levied on the same base, the distributional profile is likely to be close to that of the national tax. Social security contributions are not included.

190. The implicit VAT payments are calculated as the VAT rate multiplied by total consumption expenditure after deduction of expenditure on housing rent, medical care and education.

191. Japan is one of only five OECD countries not applying a taxpayer identification number system, the others being Switzerland, Germany, Austria, and the United Kingdom.

192. This includes interest income, dividends and realised capital gains but excludes unrealised capital gains and imputed rents.

193. In 1996, national personal income tax revenues amounted to 19 trillion yen and local inhabitants tax revenues to around one-half of this amount.

194. A similar approach has recently been adopted for local corporate taxes in Italy as discussed in Box 15 below.

195. A contribution on bonuses was introduced in 1996, but it is far below the contribution rate on ordinary salary.

196. It is not unusual for bonuses to amount to as much as 30-40 per cent of ordinary salaries.

197. The simplified regime was changed in 1997 so as to bring the taxable implied value added in line with the true values, and the sales threshold was lowered from 400 million yen to 200 million yen. Furthermore, based on the authorities' information, the amount of taxable sales by businesses in the simplified regime is estimated at less than 6 per cent of total taxable sales (in 1998). It would thus appear that the scope for base broadening through an elimination of the simplified regime is rather limited.

198. Japan, being an island, also largely escapes the border-trade problems facing other VAT countries.

199. Whether a VAT increase is the best way to achieve more intergenerational equity is debated among economists. For instance, a reduction of personal allowances would strike elderly workers particularly hard, as their average income and hence marginal tax rates are higher than other groups. Higher effective taxation of pension benefits would of course also improve intergenerational equity.

200. This would require interest expenditure to be deductible and the tax credit for acquisition of dwellings to be eliminated.

201. Sweden, for instance, imposes a uniform rate of capital taxation of 30 per cent. This applies not only to income from interest, dividends and capital gains, but also – symmetrically – to capital expenses, of which interest payments are the most important. The counterpart of the interest deductibility is a property tax levied on the market value of owner-occupied housing (another option would be to include imputed rent in capital income). The corporate tax rate is basically similar to the capital income rate as is the imputed capital income component of the self-employed and that of active owners of closely held companies.

202. As a side effect, this would also imply a levelling of the playing field for financial intermediaries, where pension funds and life insurance companies hold market shares that other, and possibly more efficient, intermediaries would otherwise have held.

203. This is the case for instance in Denmark, where a so-called "real interest tax" was introduced on (new) pension savings in 1982. Australia, on the other hand, taxes full nominal fund incomes, but only the real part of capital gains. This, of course, gives a strong bias in favour of assets providing capital gains rather than regular income, thereby distorting pension fund behaviour, with the bias increasing with the rate of inflation.

204. This can be illustrated by a simple calculation: assets in the occupational pension funds amount to around 13 per cent of GDP and those in private pension funds, life insurance companies and other institutional investors amount to some 80 per cent of GDP, a total of nearly 100 per cent of GDP. An asset tax of 1 per cent (as the temporarily abolished tax on tax-qualified plans) would then initially yield revenues of around 1 per cent of GDP. Assuming a real return of 3.5 per cent, a 20 per cent tax on earnings would initially yield around 0.7 per cent of GDP. These calculations do not take into account any change in behaviour or second-round effects on consumption and income tax revenues (as future disposable income and consumption would be lower).

205. The OECD's calculations suggest that lowering the corporate tax rate by 9 percentage points (for an inflation rate close to zero) has reduced the standard deviation across instruments (*i.e.* the measure of non-neutrality) by 25 per cent and the overall marginal tax wedge by 20 per cent. This is not out of line with Tachibanaki (1997), who finds that for every 10 percentage point reduction in the corporate tax rate, the overall marginal effective tax rate is reduced by around 2.5 percentage points (at zero inflation). The offset is due to lower tax values of depreciation and interest deductions.

206. The ratio is estimated at around 3 in 1997, relatively high among OECD countries. Its evolution over time is distorted by the very large swings in equity evaluation related to the bubble economy and its demise.

207. Other OECD countries apply loss carry-forward periods ranging from five years (France, Italy, Greece, Denmark, and Turkey) to infinity; see Table A3 in Annex II.

208. Although the extra corporate tax on short-term capital gains on land was eliminated in FY 1998, the tax rates for individual transactions still provide an incentive to long-term holding. Yamazaki and Idee (1997) examined the effects of capital gains taxes on the transfer of agricultural land to other uses and found lock-in effects only in rural prefectures.

209. It is debatable how effective the capital gains tax structure was in calming down speculation during the bubble years. Various capital gains taxes that are neutral to the timing of disposal have been proposed. Iwata (1989) proposed a tax on unrealised capital gains with a rate set equal to the interest rate. Hatta (1989) proposed higher rates the longer the holding period. Auerbach's (1991) idea was similar to Hatta's. Kanemoto (1994) made a comparison among the three proposals. An appropriate measure for Japan could also be to lower the rate for short-term holding and to raise it for long-term holding; such a policy was adopted in the current boom around Dublin by the Irish government (OECD, 1999*e*).

210. This study is based on the difference between tax statistics (collected by the National Tax Administration) and national income statistics (estimated by EPA, based on monthly income statistics from the Ministry of Labour). A number of earlier studies find massive underreporting of agricultural and self-employment income, but no recent studies are available to support or reject the hypothesis of some convergence in the tax gaps since the early 1990s. Anecdotal evidence, based on the National Tax Administrations auditing results, suggest that the amount of non-declared income is around 25 per cent for both agricultural and self-employment income. Moreover, since auditing is targeted toward taxpayers with suspected high evasion ratios, the sample is biased towards finding high tax gaps.

211. Some attention in the public debate has been paid to the possibility for holders of so-called discount bonds (i.e zero-coupon bonds where the entire return comes in the form of capital gains) to evade taxation. Since the holders of these bonds are not registered, a loophole for evasion *via* discount bonds exists. The extent of this base erosion should not be exaggerated, however: by the end of 1998 the total outstanding stock of these bonds was only 14 trillion yen. Assuming (for the case of illustration) an average capital gain of 5 per cent a year, of which none is declared, would amount to an annual tax revenue loss of less than 200 billion yen, or 0.05 per cent of GDP.

212. It could be argued that at least part of the costs of public services are already being paid through the fixed part of corporate inhabitants tax (see for instance the Tokyo Tax Accountants Association's view on the FY 1999 tax reform).

213. It should be noted that a shift in corporate taxation from an income-based tax to another (related) tax base could create problems for non-resident companies in their home countries if the new tax is not recognised by tax authorities in the home country as creditable against corporate tax liabilities there.

214. Chapter III offers a more detailed discussion of decentralisation and local government finances.

215. Eliminating all of the tax reliefs, but keeping the separate taxation of capital income, could yield an additional 7 per cent of GDP in national taxes and 3 per cent of GDP in local taxes. If one-third to one-half of the current reliefs are retained this would bring about a revenue increase of around 5-6 per cent of GDP.

216. One example shows monthly whole life premiums varying by up to 6 per cent from the lowest to the highest charges by the top seven firms. Competition is also heating up in the automobile insurance market in the wake of the full liberalisation in August 1998. Foreign insurers presently offer premium rates 30 to 40 per cent lower than conventional providers for low-risk drivers.

217. That failure has left 160 billion in excess liabilities.

218. Early this year the net assets of such trusts amounted to around 43.6 trillion yen, exactly where they were at end-1994 and less than one-tenth of corresponding US figures; correcting for relative GDPs, the ratios are still 8.7 and 42.8 per cent, respectively. Demand would benefit in Japan if and when the government agrees to authorise defined-contribution pension plans, much of which would presumably be invested in such funds.

219. By way of comparison there were 388 foreign firms listed on the New York Stock Exchange at the end of 1998.

220. The report made a number of observations. Japan is involved in 7 per cent of world trade, but the yen is used in only 5 per cent. The share of yen-denominated bonds in global offerings fell from over 13 per cent in 1994 to around 4½ per cent in 1997.

The yen was involved in 27 per cent of foreign currency trades in 1989 but only 21 per cent in 1998, less than the Deutschemark's 30 per cent and the dollar's 87 per cent. The yen's share of banks' outstanding foreign-currency-denominated cross-border liabilities was only 6 per cent in mid-1998, compared to 13 per cent for the Deutschemark and 47 per cent for the dollar. Finally, the yen's share of official currency reserves had fallen to 5 per cent by end-1997, and not a single country is using it as an anchor currency.

221. The corresponding figures for Germany, for example, are said to be 77 and 56 per cent, respectively.

222. The variety of maturities was extended when a thirty-year bond was issued for the first time in September 1999.

223. According to an NLI survey, in the decade to FY 1997 the proportion of outstanding shares of listed firms by market value owned by financial institutions and other businesses fell 7½ percentage points, 3 points of which was in cross-shareholding and a further 3 in other forms of stable holdings. Foreigners' share rose by more than 9 points to 13.4 per cent, with most of that rise occurring in the last few years. Goldman Sachs estimates that cross-shareholding has fallen 9 percentage points in the six years to 1998. Recently foreign investors have taken control of a number of Japanese firms, *e.g.* IDC, Japan Telecom, Japan Leasing, Daihyaku Mutual Life and Sumitomo Rubber.

224. Some evidence exists that firms that have cut cross-shareholdings have tended to outperform the market.

225. Typically listed firms are downsizing their boards from 30-odd members to around 15 in a bid to turn them into real strategic decision-making bodies, and many are adopting a corporate officer system. These officers look after the day-to-day management of the firm. Their legal positions would be clarified if the Commercial Code explicitly recognised their existence and allowed the board to delegate authority to them.

226. That effect has been constrained because directors are usually too poorly paid to be assessed much of a penalty in the event of poor outcomes and because the proceeds of such suits accrue to the firm and not to the initiator of the suit.

227. Performance-related bonuses for directors are not tax deductible in Japan, and management is not set up to measure individual performance.

228. Daiwa Securities reports that they were present in about 5 per cent of all listed firms by late-1998.

229. This should be supplemented by a change in tax law exempting such shareholders from immediate liability for capital gains taxation.

230. This reform is patterned on a similar provision in the US law named after former Senators Bayh and Dole.

231. A recent survey showed that such start-ups are planning to hire 17 per cent more employees in the annual hiring season next spring compared to a 19 per cent reduction for publicly-traded firms. They were planning a small cut in their capital outlays in FY 1999, but the cut is much smaller than the previous year's and than those of either exchange-listed or OTC-traded firms.

232. There were 18 failures in FY 1998, 14 in FY 1997 and 7 in FY 1996. In the previous 20 years there were only 19 in all.

233. The report calculated that the total annual burden over the next 15 years will be at least 2.9 trillion yen and that it will squeeze combined parent-only pre-tax profit by 38 per cent per year.

234. The draft proposal being finalised sets the ceiling at 432 000 yen per year for company employees and 816 000 for the self-employed and homemakers. For those company employees with a separate company plan the maximum would be halved or 216 000 yen.

235. This admittedly vague rule requires those with a fiduciary role to make investment decisions as any prudent person would.

236. These include higher worker morale, greater training incentives and more within-firm experience.

237. Over two-thirds of all unemployment benefit recipients in FY 1997 received benefits of 60 to 80 per cent of previous earnings for the maximum period, which varies from 90 to 300 days, depending on the beneficiaries' age and work experience.

238. One of the structural reasons for weak employment growth (such as seen among regular employees over the past year) is no doubt rigid job-protection practices. The Labour Standards Law says clearly that one month is all the notice that is required to dismiss any employee, but, according to Japanese case law, four clear conditions must be met before such an action may be taken in respect of regular employees. The result is that dismissal is extremely costly for employers, who are therefore hesitant to hire. However, while employment is protected, specific jobs are not. Thus, employees can find themselves transferred to a completely different position, sometimes one for which they have no training or experience, and quitting is the only solution if they cannot adjust.

239. Definite-duration contracts of up to one year for either part-time or full-time workers are permitted. While those of longer duration are technically legal, the restrictions applied are so severe that they are effectively forbidden.

240. This is likely to be an overestimate of the actual number of such workers due to double counting of the same workers registered with different companies.

241. The difference between the average number of recipients and the average number of unemployed in FY 1998 was over 2 million. The low coverage ratio (about one-third) is attributable to the high share of the unemployed who were not previously employed (*i.e.* housewives, new graduates, etc.) or who have exhausted their rights to benefits. However, as stated in the text above, redundancies have been a rising share of total joblessness in the recession, which should boost coverage figures.

242. According to the Economic Planning Agency, deregulation led to increased domestic demand between FY 1990 and FY 1997. The measures which affected the telecommunications, distribution and financial service sectors in particular, resulted in an increase of about 8.25 trillion yen – or 1.73 per cent – in the nation's GDP. Of the average year-on-year nominal economic growth of 2.8 per cent during that period, the report concluded that 0.45 per cent was related to deregulation measures.

243. Surveys on price differences by the Ministry of International Trade and Industry and the Economic Planning Agency show that Tokyo citizens paid 8 to 30 per cent more for purchases than did citizens in other major international cities in 1997. Japanese industries pay almost twice as much for non-manufacturing intermediate inputs compared to their competitors, and three or four times as much as those in Asian emerging countries. Regulatory barriers also contribute to very low levels of foreign direct investment (FDI) in Japan. Even though there has been improvement in recent

years, Japan remains far below average in attracting FDI compared to other OECD countries. Figures for 1998 show that FDI inflows to Japan amounted to only $3.2 billion, 0.7 per cent of the OECD total. Although Japan is the OECD's second largest economy, it is ranked only 19th with respect to FDI inflows (OECD, 1999*c*). This is seen in some quarters as a sign of the restrictive nature of the Japanese business environment, notably as a result of domestic regulation.

244. The basic idea in the price cap method is to consider two elements. One is an estimate of inflation of the costs of producing a specified service. The other is the likely increase in the operator's productivity for that service. The former is approximated by the rate of inflation in the economy (or Retail Price Index, RPI), and the latter is characterised by a factor X. The general price-cap formula is RPI – X.

245. The argument is that higher self-generation by industrial users may well be a strong indicator of avoidably high prices. Indeed, the correlation coefficient between the country's share and its rank in terms of electricity price levels in the first quarter of 1999 was a significant 0.38.

246. High electricity prices are attributable to high capital, transmission and distribution costs (pushed up for example, by high land acquisition costs, strict safety standards and remote siting), high fuel costs (due to fuel choice – subject to policy guidance – and availability, and government-imposed duties and taxes), high regulatory costs (mainly for environmental goals, but also for maintenance purposes) and low load factors (because air conditioning demand requires substantial additional peak-load capacity). For further details see OECD (1999*a*).

247. This would be the same as the 1997 reform in the natural gas industry. At that point all consumers of more than two million cubic metres per year were allowed to choose their own suppliers. An amendment to be implemented in November 1999 will drop that threshold to one million cubic metres and will allow price cuts without MITI approval.

248. As of March 1998, NTT had 99.5 per cent of the market for local fixed telephony services, as measured by the percentage of access lines, and 98 per cent in terms of revenue. Another source of competition may come from electricity producers using their security network. For example, recently NTT was pressured to set a fixed monthly charge for internet connection, but the announced tariff – 10 000 yen (or around $90) – is so high that it elicited a counter-offer of half that rate by a well-known entrepreneur in conjunction with an electrical utility.

249. Recently it allowed the take-over of another major firm in the sector by a foreign rival, despite a competing, albeit less attractive offer from NTT.

250. In August 1998 this gap was nearly 20 per cent for a typical business user and 12 per cent for an average residential user (OECD, 1999*b*).

251. While the number of taxis and taxi companies had been falling nation-wide, after the January 1998 deregulation 11 new companies were launched, and the number of taxis increased by 1 000 to 46 000.

252. The number of consultations handled by local consumer centres and the Japan Consumer Information Centre increased by over 70 per cent to 564 071 in the eight years to FY 1997. Complaints accounted for over three-quarters of the consultations in FY 1997 against five-eighths in FY 1989. A "poll on consumer problems", operated by Prime Minister's Office in January 1998, found that only 1.5 per cent of those who were dissatisfied with goods and services had consulted a consumer centre.

253. The OECD is undertaking a study into the concept of multifunctionality, but it remains highly controversial.

254. The self-sufficiency ratio has declined from 62 per cent in 1960 to 41 per cent in 1997, the lowest level among industrialised countries. The ratio has continued to decline primarily because of a downturn in agricultural production and a change in diet. So the Basic Law allows the government to set the nation's target self-sufficiency ratio. This is commonly expected to be 50 per cent in 2010.

255. In August 1999 the government decided to require that certain genetically modified (GM) foods be labelled as such, beginning in April 2001. All GM food is imported. The EU, Australia and New Zealand had already announced similar obligations. The issue is said to be one of consumer choice rather than safety.

256. However the share of gross farm receipts in the form of producer support (63 per cent in 1998) is still much higher than the OECD average (37 per cent). In 1998 the total PSE was almost the same level as in 1997, in spite of a decrease in producer prices, as the fall in world prices more than offset the depreciation of the yen.

257. So-called "agricultural production legal persons" (APLPs) were first authorised in 1962. There are four variants of such non-family landowners, but joint-stock corporations are not permitted. APLPs currently number only 5 246, but they are most often trend-setters in terms of management techniques and productivity gains.

258. The justification for this restriction is that corporate entrants should not be permitted to speculate on land. To this end it has been suggested (Okuno and Honma, 1998) that zoning regulations should be reformed so that land would be either restricted to current use for at least 30 years or would be classified as available for changing use; something like this has been in effect for land near urban areas since 1994 (see Chapter III). Companies would be allowed to purchase only farmland in the first category. The government also wishes to relax the requirements of the present special corporation system, which foresees, for instance, that more than 50 per cent of the executive board should be farmers.

259. Hilly and mountainous areas account for about 40 per cent of the total cultivated land and the population mainly engaged in farming. These areas also play a role in protecting urban residents living downstream of rivers. In general, hilly and mountainous areas have several disadvantages in cultivation, but on the other hand some farmers in those areas may specialise in high quality products.

260. Only one in eight farm households has no outside income; about one in two has more non-farm than farm income.

261. Godo (1998) estimates that, on average, land whose use is shifted sells for about 30 times the present discounted value of its return from agricultural operations.

262. Even though this is not low by OECD standards, the decline would likely have been much faster without the constraints mentioned in the text.

263. FY 1997 data show that 18.8 per cent of land formerly used for agriculture was bought by the government, while 29.9 per cent was shifted to residential use.

264. The government first imported 379 000 tonnes of rice in 1995. According to the Uruguay Round agreement, mandatory rice imports should increase by 75 800 tonnes each year and would have reached 758 000 tonnes in 2000. After the notification to WTO imports are expected to be 644 300 tonnes in FY 1999 and 682 200 tonnes in FY 2000. In Japan rice production is in surplus, and stocks of rice at the end of the 1998 crop year reached 3.44 million tonnes (about 35 per cent of annual domestic

consumption), including 0.42 million tonnes of stockpiled foreign rice (about 11.8 per cent of the total). This is the largest surplus since 1980. The government thinks that the ideal level of stocks is about 1.5 million tonnes. Ridding itself of 13 million tonnes of excessive inventories on two previous occasions (1970 and 1980) by means of discount sales to domestic processors cost it some 3 trillion yen in total ($25 billion).

265. The specific tariff on imported rice is 351.17 yen per kilogram for FY 1999 and 341.00 yen for FY 2000. Note that the government purchase price for domestic rice for the 1998 crop was reduced by 2.5 per cent (from 270.3 to 263.4 yen/tonne), while the government sales price for domestic rice was reduced by 1.5 per cent (from 287.3 to 283.0 yen/tonne). The government purchase price for imported rice for the 1998 crop was reduced by 25.1 per cent (from 64.6 to 48.4 yen/tonne), while the government sale price for imported rice was reduced by only 6.5 per cent (from 225.6 to 210.9 yen/tonne).

266. The loans and discounts of their credit federations are only 14 per cent of their total assets, compared to 63 per cent for all domestically licensed banks.

267. There were 56 insolvent co-operatives at the end of FY 1997.

268. This consolidation has been underway for a long time: there were as many as 4 300 agricultural co-operatives in December 1983. But the rate of decline has accelerated from an average of 2½ per cent per year in the 1980s to 8½ per cent in the 1990s.

269. Greenhouse gases are those that contribute to the warming of the earth's surface. CO2 (carbon dioxide) is easily the most important of them. Another is methane. In Japan's case most methane is emitted by agriculture; thus, subsidising its production has an important negative environmental externality.

270. OECD work using the GREEN model shows that the marginal abatement cost of reducing emissions to Kyoto targets – the change in real income and GDP resulting from either cost-effective domestic measures alone or supplemented by international emission trading – would be lower in Japan than in the United States, the European Union or the residual Other OECD group of countries (OECD, 1999d, Table V.2) because of the slower growth in the baseline.

271. Official figures show that among OECD countries only Denmark and Canada have a greater cut required in percentage terms to reach their Kyoto objectives given where they stood in 1995. See OECD (1999d, Table V.1).

272. Energy conservation and, more generally, environmental protection should be seen as a potential promoter of economic growth. In the next decade orders of environmental equipment from the private sector will amount to 1.33 trillion yen. Public environment-related expenditures are expected to reach 3.0 trillion yen, an 11 per cent increase this year. Major companies are making investments aimed at cutting emissions of greenhouse gases (for example, Mitsubishi Electric will spend 3 billion yen per year). Furthermore MITI estimates that environment-related business will grow significantly: the market is expected to expand from the present 15 trillion yen to 37 trillion yen in 2010 and employment from the present 0.64 million to 1.40 million.

273. New energy sources are defined in Japan as oil-alternative energy sources which have not yet been used widely, mainly because of economic constraints and instability of supply, but whose introduction is necessary in order to reduce Japan's oil dependence and to assist in environmental protection. There has thus far been no official indication as to what measures will be taken to ensure that this target will be met.

274. The government's target for the share of new energies in total primary energy supply has been set at 3.1 per cent in FY 2010, compared with 1.1 per cent in FY 1996.

275. Natural gas can be considered as an alternative means of meeting energy demand with an acceptable environmental outcome, and one of the government goals is particularly to promote the use of natural gas among fossil fuels. At present the gas grid covers only 50 per cent of the population.

276. The reductions in AAT are the following: in the case of electric, methanol-fuelled, natural gas and hybrid cars the tax was cut by 2.4 percentage points in FY 1998 and by a further 0.3 per cent in FY 1999 bringing the level from the original 3 percentage points to 0.3 percentage point for business customers and from the original 5 percentage points to 2.3 percentage points for households. As a result the advantage of buying a fuel-efficient car in terms of AAT is 0.5 percentage point for all customers. Tax advantages based on fuel efficiency seem a second-best approach, compared to taxing fuel itself.

277. International standards for environmental management were formulated by the Geneva-based International Organisation for Standardisation with its 128 member nations. As of January 1998 it had approved 11 258 standards in all, but the relevant series for the environment is ISO14001. Workplaces that seek to be certified under this standard must pass an audit by a recognised certification institution. By May 1999 2 124 certifications had been granted, far more than in other OECD countries and more than double the level of a year earlier.

278. The multilateral issue of Japanese rice imports was also resolved by Japan's acceptance of full tariffication (see the agriculture section below).

279. Japan contends that this Act is inconsistent with WTO trade rules as well as its Anti-Dumping Agreement. The EU has also instituted dispute settlement proceedings against the Act; this dispute is now pending before a separate panel. Note that this Act is *not* the one usually used in the United States to implement anti-dumping actions.

280. The concern is that without bilateral government oversight, NTT and its related companies will revert to their traditional reliance on domestic suppliers. In FY 1997 NTT's purchase of foreign equipment came to 1.85 billion yen, compared to foreign procurement by the new common carriers of 4.93 billion yen. The Japanese government still owns 59 per cent of the parent firm but claims that it is a private company and so should not be subject to government interference. But US government officials do not share this view.

281. Acrylonitrile is an industrial chemical used extensively in the manufacture of synthetic fibres, resins, plastics, elastomers and rubber for a variety of consumer goods such as textiles, dinnerware, food containers, toys, luggage, automotive parts, small appliances and telephones. It is also used in fumigants.

282. The Japanese Economic Council, which reports to the Prime Minister's Office, has even suggested the creation of a free-trade zone comprising the two Asian nations.

283. In particular, Japan would like to include in the negotiations: agriculture, services, trade-related aspects of intellectual property, anti-dumping, international competition, investment and electronic commerce rules; as well as barriers to trade in the areas of forest products, fisheries, medical equipment, chemicals, toys, and mutual recognition of standards for telecommunications equipment.

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Annex I

Institutional framework of local allocation tax (LAT)

The total availability and allocation of LAT is determined through the following steps (see Mochida (1998), for example).

(1) Total availability: supply-side

The total amount available to LAT is a pre-specified share of national tax revenues, specifically the sum of *a)* 32 per cent of income tax and alcohol tax, *b)* 32.5 per cent of corporate tax, *c)* 29.5 per cent of consumption tax, and *d)* 25 per cent of tobacco tax as of FY 1999.*

(2) Standard fiscal needs and standard financial revenues: demand-side

The total demand for LAT is the sum of the differences between standard fiscal needs and standard financial revenues of each local authority. First, the fiscal needs of local authority *i* (N_i) are calculated using the following formula:

$$N_i = \sum I_{ik} * U_{ik} * M_{ik}$$

were I_{ik} is a measurement unit, U_{ik} is a unit cost for service k, and M_{ik} is a modification coefficient. The measurement units are specified for each expenditure item; the number of school pupils for education, for example. Population and area size are often used. The unit cost is calculated by the Ministry based on the assumption of expenditures and measurement units of a hypothetical "standard" local government. The adjustment coefficient reflects the differences in unit costs coming from different natural and social environments surrounding local governments. Second, standard financial revenue is calculated as the sum of 75 per cent (80 per cent for prefectures) of local taxes and local transferred taxes. The remaining ratio, 25 per cent (20 per cent for prefectures) of the revenues, is called a *reservation ratio* and is regarded as an allowance for the uncertainty of estimation and a measure of the incentives for local tax collection.

(3) Adjustments of imbalance

The sum of demands for LAT over all local governments is now always equal to a certain portion of national tax revenue. If an imbalance results, the supply-side determines the total amount, and each allocation is changed proportionally, in principle, though some adjustments are usually made. If the imbalance is expected to continue, a change in the specified ratio of national tax revenues may be required. But it is not usually the case. Adjustments are usually made through temporary increases or decreases in transfers from the national general account, borrowing from FILP, etc., in the process of formulation of the Local Public Finance Programme.

* These shares were revised over time. The share of consumption tax revenues was raised from 24 to 29.5 per cent in FY 1997 because of a reduction of income and inhabitants taxes and the rise in the consumption tax rate from 3 to 5 per cent, etc. The share of corporate tax revenues increased from 32 to 32.5 per cent in FY 1999 and is scheduled to increase once again to 35.8 per cent in 2000 in order to compensate the revenue losses caused by a permanent business tax cut in FY 1999.

Annex II

Statutory features of the tax system and key elements of recent reforms

Main statutory features[1]

The Japanese tax system is a "separate" or "schedular" system, in which different components of personal income are taxed at different effective rates. Retirement and capital incomes are separated from other types of personal income, mainly employment and business income,[2] and there is no integration between the personal and corporate income tax systems.[3] After a recent 15 percentage point cut in marginal tax rates, personal income is now taxed at progressive rates running from 10 to 50 per cent (including local taxes). This is in line with the rate structure of other major countries (Table A2), but Japan applies somewhat wider tax brackets, implying that the highest marginal rate kicks in at a relatively high income level, *i.e.* four times that of an average production worker (APW). Interest income is taxed at a final withholding rate of 20 per cent, capital gains are generally taxed at 26 per cent, and dividends are taxed in a "mixed" fashion: large dividend payments are taxed as a part of total personal income, while small dividend payments are taxed at a final withholding rate of 20 or 35 per cent.[4]

The marginal corporate tax rate was lowered from 50 to 41 per cent in FY 1998 and FY 1999 and has thus come down to a level not far from that of many other OECD countries (Table A3). Special investment incentives are given through the increased initial depreciation regime (and the less important accelerated depreciation regime) as well as investment tax credits for R&D. Corporations may generally carry back net operating losses one year to offset them against the previous year's taxable income or may carry them forward for seven years (five years until 1999). The VAT is fairly broad in coverage – there are few exemptions and no differentiated rates (Table A4) – although the VAT-exempted threshold for sales is by far the highest among OECD countries. A so-called simplified regime applies to enterprises whose taxable sales do not exceed 200 million yen. It taxes an implied value added (which is calculated as a fixed percentage of gross sales) and is thus equivalent to a turnover tax on final sales with rates ranging from 1 to 2.5 per cent, depending on the industry. In contrast with other OECD countries, the Japanese VAT system is not based on invoices but on deduction of taxable purchases as registered in the corporate accounts. Taxation of land and property is carried out by numerous national and local taxes: five national and six local taxes are thus levied on acquiring, holding and transferring land.

Tax policy in Japan is basically designed through two bodies, the government's Tax Council and the Tax Commission of the Liberal Democratic Party, the main coalition partner. There is an implicit division of labour between the two committees in which the Tax Commission is the *de facto* policy proposing body when it comes to short-term tax policy and specific changes. The Tax Council (or the Tax Advisory Committee) – which reports directly to the Prime Minister's Office – is more of a think-tank dealing with longer-term

Table A1. **Key statutory features of the Japanese tax system, 1999**

1. Personal income tax

Tax unit: The individual

Types of income: The income tax law classifies individual taxable income into the following ten categories: Interest, dividends, real estate, business, employment, retirement, timber, capital gains, occasional, miscellaneous. Income tax is paid on the basis of the individual's aggregate income, regardless of its category. However, retirement and timber income is taxed separately under the income tax law, while interest income is taxed separately under the special taxation measures law. Dividends and capital gains are, or may be, taxed separately as well.

Withholding: The tax system is in principle based on self-assessment, but withholding is widely used, including interest and dividend income, employment income and retirement income. An optional separate withholding tax on capital gains on listed securities (*i.e.* in reality a transaction tax of 1.05 per cent on the proceeds from sales) will be abolished as from FY 2001.

Tax rates	National	Prefectural	Municipalities	All-in marginal rate for top income earners
Income tax	10, 20, 30, 37 per cent of taxable income (4 brackets)	–	–	–
Inhabitants tax	–	2, 3 per cent of taxable income (2 brackets) + 1 000 yen per cap	Standard rate of 3, 8, 10 per cent of taxable income (3 brackets) + 2 000-3 000 yen per cap	
Total				50 per cent

Allowances:

A wide range of deductions and allowances apply, of which the main categories are:

- Deduction for employment income (a fixed amount ranging from 650 000 yen to 2 200 000 yen plus a gradually declining rate scale, starting at 40 per cent deduction from gross employment income below 1.8 million yen to a 5 per cent deduction from gross employment income exceeding 10 million yen).
- Deduction for life insurance premiums and personal pension plan premiums (max. 50 000 yen per year for each).
- Deduction for medical expenses (max. 2 million yen per year).
- Basic allowance (380 000 yen per year).
- Allowance for spouse (380 000 yen per year) and special allowance for spouse (max. 380 000 yen per year). The special allowance for spouse is gradually phased out as the employment income of the spouse exceeds 650 000 yen. The deduction is eliminated when the employment income of the spouse exceeds 1.4 million yen.
- Allowance for dependants, including children under 16 (480 000 yen per year for each dependent).
- Special allowance for dependent children between 16-22 (630 000 yen per year for each dependent).
- Other deductions (including special allowance for aged persons, handicapped persons, widows, widowers and working students).

These allowances apply to the national income tax, but a number of them also apply to the local inhabitants' tax (although at lower amounts).

Table A1. **Key statutory features of the Japanese tax system, 1999** *(cont.)*

Tax credits:

– The most important tax credit is the housing loan tax credit, which allows the taxpayer to deduct from his income tax liability a certain percentage (0.5-1.0 per cent) of the loan balance remaining on his dwelling of residence. The deduction period has been extended from 6 to 15 years (as of FY 1999) and the ceiling on the loan outstanding increased from 30 to 50 million yen as of January 1999. These extended measures will apply to newly resided houses in 1999 and 2000. A maximum amount of 5.9 million yen can be credited against taxes over the 15 year period, of which a maximum of 500 000 yen can be credited per year for the first 6 years. Other tax credits include those for dividends, for foreign tax payments and for incremental R&D expenditures.
– 20 per cent proportional tax reduction (max. 250 000 yen (national) and 40 000 yen (local) per year).

Income taxed at preferential terms and taxed separately (main items):

Public annuity income: Pensions and annuities paid by the social insurance system and qualified pensions (*i.e.* employer-financed pension schemes) are taxed at preferential terms as miscellaneous income. The preference is primarily given by the "public pension benefit deduction". For people above 65 a minimum of 1.4 million yen can be deducted. For public pensions between 2.6 and 4.6 million yen the deduction is 1 million yen plus 25 per cent of the pension benefit minus 1 million yen (*i.e.* the total deduction ranges from 1.4-1.9 million yen). For public pensions between 4.6 and 8.2 million yen the deduction is 1.9 million yen plus 15 per cent of the pension benefit minus 1 million yen exceeding 3.6 million yen (*i.e.* the total deduction ranges from 1.9-2.44 million yen). For public pensions above 8.2 million yen the deduction is 2.44 million yen plus 5 per cent of the pension benefit minus 1 million yen exceeding 7.2 million yen (*i.e.* the total deduction exceeds 2.44 million yen). Besides this deduction, the allowances mentioned above also apply to pensioners (notably including a 500 000 yen special allowance for persons above 65 and with incomes below 10 million yen). In practice, pension income is thus tax exempt or taxed very lightly for most people.

Retirement income: Retirement income is taxed separately. The tax base of retirement income is 50 per cent of retirement income after special retirement deductions. The amount of special retirement deduction is 400 000 yen per year of service on the labour market and– for persons with more than 20 years of service – a deduction of 8 million yen plus 700 000 yen per year in excess of 20. The minimum deductible amount is 800 000 yen. Lump sum retirement allowances paid by the social insurance system and qualified pensions are considered as retirement income.

Interest income is generally taxed at a final withholding rate of 20 per cent (15 per cent national, 5 per cent local). Non-resident individuals and foreign corporations are exempt from tax on interest income from government bills and bonds. For people above 65 years, and physically handicapped persons, etc., interest income from *e.g.* postal savings, government bonds and small deposits is exempt (up to a total principal of 3.5 million yen each). Likewise, no income tax is levied on interest from workers' property accumulation, housing and pension savings (up to a total principal of 5.5 million yen).

Dividend income is usually subject to a withholding tax of 20 per cent on distribution. When one-payment dividend from a single corporation (to an individual shareholder) is less than 50 000 yen (less than 100 000 yen, in the case where dividend payments are made once a year), taxpayers can opt to pay the withholding tax as final tax. When one-payment dividends from a single corporation are less than 250 000 yen (less than 500 000 yen, in the case where the payment is made once a year), taxpayers can opt for a final withholding tax at 35 per cent. For one-payment dividends from a single corporation above 250 000 yen (over 500 000 yen if one payment per year) dividend income is subject to comprehensive taxation, and the tax withheld on distribution is credited against personal income tax liability. When taxpayers opt for comprehensive taxation of dividend income, they are entitled to a tax credit of 10 per cent of the dividend (however, if total ordinary income, including dividends, exceeds 10 million yen, a 5 per cent tax credit applies to the dividend income corresponding to the amount in excess of 10 million yen).

Table A1. **Key statutory features of the Japanese tax system, 1999** *(cont.)*

Capital gains for listed stocks are taxed by either of the following options at the taxpayer's choice (for each transaction):

a) Separate tax return: realised gains are taxed at 26 per cent (including 6 per cent local inhabitants tax). Losses can be subtracted only from capital gains for sales of stocks;

b) Separate withholding tax: in the case sales are made through or to security firms, the taxpayer can opt for a withholding tax of 1.05 per cent of the total proceeds as final tax. This option will be abolished as from FY 2001.

Capital gains on non-listed stocks are taxed by option *a)* above. Capital gains on bonds are not taxed.

Fringe benefits, including significant subsidies for employees' dwellings, recreational facilities, transportation, meals, clothing, etc., are not taxed.

Self-employment:

Taxable business income is calculated as gross receipts less "necessary" expenses for the year. Gross receipts are calculated on an accrual basis. Necessary expenses include taxes paid for land and property used for business but not those paid for income tax and inhabitants tax. They also include wages or salaries to family employees (max. 800 000 for spouse and 500 000 for others when certain conditions are met). The self-employed can opt to become so-called "blue return taxpayers". This means that they are committed to meeting certain accounting and book-keeping standards against obtaining some additional tax advantages (*e.g.* unlimited deductibility of wages/salaries for family employees, reserves for bad debt, reserves for retirement allowances, tax credit for incremental R&D expenditure, enhanced depreciation rules, enhanced loss carry-over rules, etc.) as well as a special deduction for filing the blue return.

Table A1. **Key statutory features of the Japanese tax system, 1999** *(cont.)*

2. Memorandum item: Social security contributions

	Coverage	Contribution	Tax treatment of contributions	Benefits	Tax treatment of benefits
A. The pension system (outline of main features)					
A1. Public pensions Tier-1 (flat rate basic benefit) *Kokumin Nenkin*	All residents*	Flat amount of 13 300 yen per month (FY 1999) paid by non-employees	100 per cent deductible	Maximum of 67 017 yen per month (FY 1999 – fixed prices) for people aged 65.	Taxed as public annuity income
Tier-2 (earnings related benefit) *Kosei Nenkin*	Employees and their dependent spouses**	17.35 per cent of standard remuneration (*i.e.* excl. bonuses) and 1 per cent of bonuses to cover both contributions to tier-1 and tier-2. Divided equally between employer and employee. Contributions are capped at monthly standard earnings of 590 000 yen	100 per cent deductible	Monthly benefit = [(0.75 per cent of average monthly standard remuneration)* (number of insured months)] (special rates apply to persons born before 1 April 1946)	Taxed as public annuity income
A2. Occupational pension schemes Tax qualified plan *Tekikaku Taishyoku Nenkin*	Employees	Not fixed – paid by employer. A special 1 per cent tax on fund assets is temporarily abolished	100 per cent deductible for employers. Max. 50 000 per year deductible for employees.	Defined benefit (annuity or lump-sum) on top of public pensions	Taxed as public annuity income or retirement income (after deducting contribution made by a recipient)
Contracted out plans (employees pension plans) *Kosei Nenkin Kikin*	Employees	Not fixed – mainly paid by employer	100 per cent deductible for employers. Max. 50 000 per year deductible for employees	Integrated with public pensions. Annuity or lump sum	Taxed as public annuity income or retirement income

Table A1. **Key statutory features of the Japanese tax system, 1999** *(cont.)*

	Coverage	Contribution	Tax treatment of contributions	Benefits	Tax treatment of benefits
A3. Individual pension savings *Kokumin Nenkin Kikin*	All residents	Not fixed. Paid by individuals	Max. 50 000 yen per year is deductible. For non-employees and their spouses a special type of individual retirement pension account was set up in 1991, which allows a 100 per cent deductible contribution of 68 000 yen per month per person	Depending on individual savings	Taxed as public annuity income

	Employer contribution	Employee contribution
B. Sickness	4.1 per cent of standard remuneration plus 0.5 per cent of bonuses	4.1 per cent of standard remuneration plus 0.3 per cent of bonuses
C. Unemployment	0.75 per cent of total remuneration	0.4 per cent of total remuneration
D. Work injury	0.6-14.4 per cent of total remuneration (depending on industry risk)	None

* The basic scheme – *Kokumin-Nenkin* (KN) – covers all residents between 20 and 60. The standard retirement age is 65 years, but pension may be claimed at any age between 60 and 70 subject to an actuarial reduction before 65 and an increase after 65. The pension benefit depends on the number of years where contributions have been paid.

** The description here concerns private-sector employees and their spouses under the *Kosei-Nenkin-Hoken* (KNH) scheme. It covers 33 million private employees, or 85 per cent of total employees. Government employees, employees in the nationalised industries, private schoolteachers and employees, and employees of agriculture/forestry/fishing organisations are covered under special programmes (mutual aid associations). Dependent spouses are entitled to the flat-rate basic benefit even though they do not contribute to the pension system. Spouses with annual earnings exceeding 1.3 million yen lose their right to be treated as dependent spouses and are obliged to enrol in the basic system.

Table A1. **Key statutory features of the Japanese tax system, 1999** *(cont.)*

3. Corporate taxes

Tax rates	National	Prefectural	Municipal	All-in typical marginal rate
Income tax	30 per cent; 22 per cent reduced rate is applied to a part of income (up to 8 million yen) of small business*			–
Corporate enterprise tax	–	Progressive scale with a top rate of 9.6 per cent of taxable income. The corporate enterprise tax is deductible from the national corporate tax base in the following year. A reduced rate of 5 per cent is levied on a part of income (up to 4 million yen) and a reduced rate of 7.3 per cent is levied on a part of income (from 4 million yen up to 8 million).* Prefectures may raise the tax rate by up to 1.1 times the standard rate	–	–
Inhabitants tax	–	Standard rate: 5 per cent is levied on national income tax (may be raised to a maximum of 6 per cent). + lump sum amount according to size of capital and reserves, max. 0.8 million yen	Standard rate: 12.3 per cent is levied on national income tax (may be raised to a maximum of 14.7 per cent). + lump sum amount according to size of capital, reserves and number of employees, max. 3 million yen.	–

Table A1. **Key statutory features of the Japanese tax system, 1999** *(cont.)*

Tax rates	National	Prefectural	Municipal	All-in typical marginal rate
				40.87 per cent [30 × (1 + 0.05 + 0.123)+ 9.6] [1.096] Low rate of 29.34 per cent (up to 4 million yen) for small business* [22 × (1 + 0.05 + 0.123) +5]/ [1.05]

* For corporations with a capital of less than 100 million yen. The reduced rate also applies to co-operative associations, special medical corporations and profit-making activities of corporations in public interest.

Selected features of the tax base:

Capital gains of corporations are subject to taxation in full as they are realised. These capital gains are taxed at the same rate as operating profits.

Dividends from other corporations are generally excluded from taxable income, but if the recipient corporation owns less than 25 per cent of the domestic corporation distributing the dividends, 20 per cent of net dividend income is included in gross income.

Inventories may be valued at cost or market value, whichever is lower, by using any of eight cost methods of valuation (actual cost, LIFO, FIFO, etc.) or two cost or market – whichever the lower – method.

Standard depreciation is calculated by the straight line method or the declining balance method.

Special tax measures for corporations:

Special depreciation measures are broadly grouped into two categories, *i.e.* increased initial depreciation (which is the more important of the two) and additional depreciation. The increased initial depreciation allows, in addition to the ordinary depreciation, one time special deduction of a portion of the acquisition cost of an asset in the first period in which the asset is used. The additional depreciation allows, in addition to ordinary depreciation, the deduction of a certain percentage of the ordinary depreciation allowance for a specified number of consecutive accounting periods.

Tax free reserves. These are categorised in two groups: one *(Hikiatekin)* which is consistent with generally accepted accounting principles and another *(Junbikin)* which may not be duly justified by these principles, but which serves other specific policy objectives. The *Hikiatekin* reserves include reserves for bad debts, retirement allowances, etc. The Junbikin reserves include, among others, reserves for certain investment losses and for prevention of pollution and disasters. The tax-free reserves provide a tax deferral but not an exemption.

Preferential tax terms (allowances, credits, etc.) are also given to promote policy objectives such as energy conservation, pollution control, promotion of small to medium-sized business and promotion of R&D.

The blue return system also applies to corporations (see under self-employment). For corporations filing blue returns, net operating losses may be offset against the previous years' taxable income or may carry the losses forward for 5 years.

Finally, it should be noted, that *expenditures for entertainment,* which provide some of the fringe benefits for employees and business executives, are generally not deductible from corporate income.

Table A1. **Key statutory features of the Japanese tax system, 1999** *(cont.)*

4. Consumption taxes

Value added tax (VAT)

A tax rate of 5 per cent (of which 1 percentage point is classified as a local tax) is levied on domestic transactions and imported goods. Non-taxable transactions include land transfer or lease, housing lease, interest on loans and insurance premiums, medical care, education, welfare services, etc. Exports are also exempted from taxation.

In contrast with other OECD countries, the Japanese VAT system is not based on invoices but on deduction of taxable purchases as registered in the corporate accounts and proven by business invoices. Taxable purchases include those made from small businesses exempt from VAT. In principle the entire amount of consumption tax paid on purchases related to taxable sales is deductible. In order to limit the deduction (and thus the VAT tax credit) accorded to purchases related to non-taxable sales, the amount deductible must be calculated by the so-called itemised method or the proportional method if non-taxable sales exceed 5 per cent of total sales. The itemised method divides taxable purchases into those related to taxable sales, those related to other sales and those related to both, and then calculates the deduction as the sum of 1) taxes attributable to taxable sales and 2) taxes attributable to both taxed and non-taxed sales multiplied by the proportion of taxable sales to total sales. The proportional method calculates the tax credit by multiplying the amount of tax paid on taxable purchases by the percentage of taxable sales to total sales.

A simplified VAT regime exists for small business, whose taxable sales are below 200 million yen in the two previous years. In this regime, the taxpayer can choose to pay VAT on taxable sales minus an imputed rates of taxable purchases, where the imputed rates of taxable purchases to final taxable sales are:

– class 1 enterprises (wholesalers): 90 per cent
– class 2 enterprises (retailers): 80 per cent
– class 3 enterprises (manufactures, agriculture, etc.): 70 per cent
– class 4 enterprises (restaurants): 60 per cent
– class 5 enterprises (services): 50 per cent

Small enterprises with taxable sales less than 30 million yen in the two previous years are exempted from VAT (and cannot claim VAT credits).

Other consumption taxes	National	Local
	Liquor tax, tobacco tax, gasoline tax, aviation fuel tax, liquefied petroleum gas tax, petroleum tax, motor vehicle tax, custom duties, tonnage duty, local road tax, special tonnage due tax, promotion of power resources development tax	Prefectural tobacco tax, municipal tobacco tax, light oil delivery tax, automobile acquisition tax, special local consumption tax, golf course utilisation tax, bathing tax.

Table A1. **Key statutory features of the Japanese tax system, 1999** *(cont.)*

5. Taxation of property and land

	National tax	Local taxes
Property tax	–	Municipal tax (though prefectures also levy this tax on certain depreciable assets). The value of land, buildings and depreciable assets are taxed at a standard rate of 1.4 per cent or up to a maximum of 2.1 per cent. The tax base is in principle "a fair market price"* which is re-evaluated every 3 years (in the case of land and buildings). Tax exemption thresholds are: – Land: 300 000 yen – Buildings: 200 000 yen – Depreciable assets: 1 500 000 yen
City planning tax	–	Municipal tax. The value of land and buildings within the specified urban or city area are taxed at a maximum rate of 0.3 per cent. The tax base and exemptions are the same as for the property tax
Special landholding tax		Municipal tax, consisting of a tax on holding of land and a tax on acquisition of land. The rates are similar to the property tax and the real property acquisition tax rates, respectively, and these two taxes are credited against the special landholding tax
Business office tax		Municipal tax levied on construction or extension of business floor space (owners of buildings: – 6 000 yen per m) and on business activity – (600 yen per m2 and 0.25 per cent of wage sum)
Land developments tax	–	Municipal tax levied on housing land developers. The tax rate is stipulated by local ordinance and the revenues are earmarked for construction of roads, waterways, parks, etc.

Table A1. **Key statutory features of the Japanese tax system, 1999** *(cont.)*

	National tax	Local taxes
Real property acquisition tax	–	Prefectural tax. A tax rate of 4 per cent (3 per cent for residences) of the appraised value applies to the acquisition of land or houses. The base is subject to substantial deductions, some of which are temporary**
Registration and licence tax and stamp tax on immovable property	The registration and licence tax is levied on the value of property acquired by a rate ranging from 0.1-5.0 per cent depending on whether the registration is for a sale, a gift, a bequest, etc. Stamp tax (200 yen-600 000 yen depending on document) is also payable at the time of making taxable documents (*e.g.* contracts)**	
Inheritance tax	The value of the inheritance (after deductions) in the hand of each heir is taxed at a progressive scale from 10-70 per cent: 10 per cent (less than 8 million yen) 15 per cent (8-16 million yen) 20 per cent (16-30 million yen) 25 per cent (30-50 million yen) 30 per cent (50-100 million yen) 40 per cent (100-200 million yen) 50 per cent (200-400 million yen) 60 per cent (400 million-2 billion yen) 70 per cent (over 2 billion yen) A standard deduction of 50 million yen plus 10 million yen per heir is deducted from the total value of the inheritance. Furthermore, tax credits are given to *e.g.* spouse (160 million yen or inheritance tax corresponding to the legal share of spouse, whichever the higher), minors and handicapped. Agricultural land outside designated cities is exempted so long as it continues to be used for agriculture for at least 20 years. Gifts are likewise taxed progressively on a scale from 10-70 per cent	

Table A1. **Key statutory features of the Japanese tax system, 1999** *(cont.)*

	National tax	Local taxes
Capital gains taxes on transfer of land	Personal taxes Holding period less than 5 years: taxation equals the maximum of *a)* 40 per cent of capital gains or *b)* 110 per cent of difference between the total income tax amount including and excluding capital gains. Holding period more than 5 years: capital gains are taxed at 26 per cent (of which 20 per cent is national tax and 6 per cent is a local tax). Besides a standard deduction of 1 million yen, significantly higher deductions are given for transfer of residential property, land for development projects, land for rationalisation of agricultural land, etc. Moreover, lower rates apply to transfer of land for development of high-quality residential projects (below a certain threshold) and for transfer of residential property held for more than 10 years	Corporate taxes Holding period less than 2 years: capital gains are taxed at normal corporate rate. Holding period 2-5 years: capital gains are taxed at normal corporate rate + 10 percentage points (addition of 10 percentage points temporarily suspended). Holding period more than 5 years: capital gains are taxed at normal corporate rate + 5 percentage points (addition of 5 percentage points temporarily suspended). Special deductions include transfer of land for development projects, land for rationalisation of agricultural land, etc. When transfer of land generates "business income" as defined in the individual tax code, similar additional taxes are imposed on individual business operators (self-employed)

Note: The national land value tax, which was introduced by the land tax reform in 1992, has been suspended as of FY1998.

* The implicit target is a land valuation for property taxation of 70 per cent of the official valuation price. The target for inheritance tax purposes is 80 per cent of the official land valuation (Ishi, 1998). The actual valuation rates for tax purposes have fluctuated between 30 and 40 per cent of the official valuation price.

** The combined effective transaction tax rate on property caused by the (national) registration tax and the (prefectural) real property acquisition tax is 2.6 per cent of the market value (since only part of the applied value is being taxed).

Source: Ernst and Young (1999), *World wide corporate tax guide.*
Hiromitsu Ishi (1998), in *The tax system in industrialized countries* (edited by Ken Messere), Oxford University Press.
Ministry of Finance (1999), *The Japanese tax system.*
Ministry of Finance (1997), *An outline of Japanese taxes.*
Noriyuki Takayama (1996), *Possible effects of ageing on the equilibrium of the public pension system in Japan,* European Economy No. 3.
OECD (1999), *Tax database.*
OECD (1998), *The tax/benefit position of employees.*
OECD (1995), *The labour market and older workers,* Social Policy Studies No. 17.

Table A2. **Personal income taxation in major OECD countries**

1998[1]

	United States	Japan	Germany	France	Italy[6]	United Kingdom	Canada[6]
Taxes raised by central government							
Range of statutory rates (per cent)	15-39.6	10-37	25.9-53.0	10.5-54	18.1-45.5	20-40	17.5-31.3
Number of tax schedules[2]	5	4	3	6	7	3	4
Tax threshold[3] (yen million)	2.8	3.8	4.5	3.6	–	1.2	–
Rates of sub-national taxes (per cent)	0-11.6	5-13	–	–	–	–	0-22.8
Marginal tax rate for top income earners[4] (per cent)	48.1	50	55.9	61.6	50.8	40	54.1
Highest rate starts at (proportion of APW-income[5])	9.7	4.2	2.2	2.2	3.5	1.8	1.8

1. For Japan, 1999-figures are used.
2. Excluding zero band or basic allowance.
3. For an employee with spouse and two children (in Japan, one child is between 16 and 22 years of age, and in the United States one child is below 16). Exchange rates used are: 1 dollar = 130.9 yen; 1 pound = 216.7 yen; 1 deutschmark = 74.4 yen; 1 franc = 22.2 yen (average of daily rates, 1998).
4. Wage income.
5. APW = average production worker in manufacturing.
6. These countries apply tax credits that are not immediately comparable to the allowances shown for the other countries.

Source: Ministry of Finance (1999*a*); OECD Tax Data Base, 1998.

Table A3. **Corporate taxation in major OECD countries**

1998[1]

	Central government basic rate (per cent)	Top marginal rate[2] (per cent)	Dividends wedge[3] (per cent)	Special rates (incl. small profits rate) (per cent)	Loss provision[4] (years)
United States	35.0	39.5	67.7	15.0	20 CF, 2 CB
Japan	**30.0**	**40.9**	**70.5**	**29.3**	**7 CF, 1 CB**
Germany[5]	47.5/30	58.2	49.3	44.31/23.74	U CF, 2 CB
France	33.3	41.7	66.0	19.0	5 CF, 3 CB
Italy	37.0	37.0	44.9		5 CF, no CB
United Kingdom	31.0	31.0	48.3	21.0	U CF, 1 CB
Canada	29.1	46.1	73.6	13.12/22.12	7 CF, 3 CB

1. For Japan, 1999 figures are used.
2. Including local taxation and surcharges.
3. Difference between the pre-tax profit earned by the distributing company and the net dividend received by the shareholder.
4. CF = Carry forward; CB = Carry back; U = Unlimited.
5. These rates apply to retained profits only.

Source: OECD, Tax Data Base, 1998; *The European Tax Handbook,* 1998. Ernst and Young, *The 1999 World Corporate Tax Guide.*

reform perspectives. It consists of thirty regular members and about the same number of supporting members, all of whom are appointed by the Prime Minister. Since these members represent a host of economic and political interests, in particular labour unions and business leaders, the Council has an important function in creating consensus about longer-term tax objectives. Both the Tax Council and LDP's Tax Commission are tightly connected with the Tax Bureau of the Ministry of Finance, and the Tax Council also works closely with the Ministry of Home Affairs since its remit also includes issues related to local government finances.

Main elements of tax reform since 1988[5]

The largest single tax change in the past decade was the introduction of a broadly based value-added tax – initially levied at a rate of 3 per cent – in the tax reform legislated in 1988 and implemented in April 1989. The reform also featured significant rate cuts in both corporate and personal income taxes, and a reduction in the progressivity of the personal tax system by reducing the number of tax brackets from 12 to 5 in the national income tax schedule and from 7 to 3 in the local inhabitants tax schedule. Personal allowances and deductions were also expanded quite substantially, thus narrowing the income tax base. On the other hand, the taxation of capital income was enhanced by new measures to tax dividends and capital gains. The elimination of the tax-free savings system (which was introduced one year earlier) also contributed to a more comprehensive taxation of income. The inheritance tax was reduced by a combination of rate cutting (the marginal rate was lowered from 75 per cent to the current level of 70 per cent) and base narrowing. The 1988 tax reform implied a net tax reduction of around 0.6 percentage point of GDP, which was interpreted by some commentators as the necessary political price for having

Table A4. **Main features of VAT systems in major OECD countries**

1998

	Year VAT introduced	Initial standard rates	Current standard VAT rates[1]	Tax exempt threshold sales for small traders (in dollars)[2]	Departures from standard exemptions[3]		Coverage of lower rates	
					Exemptions other than "standard exemptions"	Taxation of "standard exemptions"	Zero rate[4]	Lower rates
Japan	1989	3	5	269 060	Social welfare services	Letting of commercial buildings, postal services, non-commercial activities of non-profit making organisations, cultural services, supply of buildings	–	–
Germany	1968	10	16	21 352	–	–	–	Books, food, newspapers, certain cultural events, charitable work if not exempt, transport (only applies to passenger transport by ship and to local public passenger transport). Rate = 7 per cent.
France	1964	20	20.6	14 000	–	–	–	Medicine, handicap equipment, books, hotels, entertainment, author's rights, museums, transport, accommodation, agriculture, catering, newspapers and water. Rates = 2.1-5.5 per cent.

Table A4. **Main features of VAT systems in major OECD countries** *(cont.)*

1998

	Year VAT introduced	Initial standard rates	Current standard VAT rates[1]	Tax exempt threshold sales for small traders (in dollar)[2]	Departures from standard exemptions[3]		Coverage of lower rates	
					Exemptions other than "standard exemptions"	Taxation of "standard exemptions"	Zero rate[4]	Lower rates
Italy	1973	12	20	2 821	Municipal passenger transport, burials	Supply and letting of commercial land and buildings (standard rate)	Scrap iron	Food, medicine, medical products/services for the handicapped, telecommunications, accommodation, books, newspapers, magazines, gas, domestic heating oil, urban wastes, depuration facilities, renewable energy, works of art, cultural events and transport. Rates = 4-10 per cent.
United Kingdom	1973	10	17.5	75 441	Burials and cremations, sports competitions, certain luxury hospital care	The freehold sales of new commercial buildings are standard rated for three years from completion date. Furthermore there is an "option to tax" for other supplies of commercial buildings which would ordinarily be exempt from VAT. Gaming machines and certain gambling in licensed clubs	Certain services and goods supplied to charities, children's clothing, food, passenger transport, books, newspapers, domestic sewage and water, prescribed drugs, medicine, certain aids for disabled, new housing, residential and some charity buildings, alterations to listed buildings	Fuel and power for domestic and charity use (5 per cent), certain energy saving materials supplied together with fitting services to recipient of "Passport benefits". Rates = 2.5-5 per cent.

Table A4. **Main features of VAT systems in major OECD countries** *(cont.)*

1998

	Year VAT introduced	Initial standard rates	Current standard VAT rates[1]	Tax exempt threshold sales for small traders (in dollar)[2]	Departures from standard exemptions[3]		Coverage of lower rates	
					Exemptions other than "standard exemptions"	Taxation of "standard exemptions"	Zero rate[4]	Lower rates
Canada	1991	7	7/15	22 760	Child care, legal aid, ferry, road and bridge tolls, standard municipal services	Lotteries and gambling, supply and leasing of commercial land and buildings, domestic postal services	Medicine, basic groceries, exports, certain financial services (usually to non-residents), certain agricultural and fishing products, medical devices, international travel and transportation services, international organisations and officials, precious metals, (sales of 25 cents or less made through mechanical coin-operated devices)	

1. As of 1 January 1998. For Germany, this rate is applied as of 1 April 1998; for the United Kingdom, the standard rate is applied to a reduced value on imports of certain works of art, antiques and collectors items, resulting in an effective rate of 2.5 per cent. For Canada, 15 per cent Harmonised Sales Tax (HST) applies in those provinces that have harmonised their provincial retail sales tax with the federal GST (the 15 per cent HST is composed of a provincial component of 8 per cent and a federal component of 7 per cent).
2. Exchange rates as of January 1997.
3. Standard exemptions are the following: Postal services, transport of sick/injured persons; hospital and medical care; human blood, tissues and organs; dental care; charitable work; education; non-commercial activities of non-profit making organisations; insurance and reinsurance; letting of immovable property; financial services; betting, lotteries and gambling; supply of land and buildings; certain fund-raising events.
4. All countries apply zero rates to exports.

Source: OECD, *Consumption Tax Trends* (1999).

Table A5. **Estimated effect on national tax revenue of tax changes in FY 1998 and 1999**[1]

Trillion yen

	FY 1998	FY 1999
Temporary tax changes		
Personal income tax reduction	–4.2	..
Corporate tax reductions		
Investment in plant and equipment	..	–0.3
Suspension of special corporation tax	..	–0.2
Permanent tax changes		
Personal taxes		
Increase of personal allowances	–0.7	–0.3
Reduction of top rates	..	–0.3
Proportional tax reduction	..	–2.7
Corporate taxes		
Reduction of rates	–13.8[2]	–1.7
Broadening the base	+11.2[2]	..
Housing and land		
Suspension of land value tax	–1.3	..
Tax credit for housing loans	..	–1.2
Others	–0.3	–0.1
Financial markets		
Reduction of securities transactions tax	–1.9	–0.2
Reduction of Bourse tax	–0.2	–0.0
Other	+0.1	–0.2
Total full year effect	–11.1	–7.3
Memorandum item:		
Total fiscal year effect	–11.8	–5.6
In per cent of GDP	–2.4	–1.1

1. Full year effect except for temporary measures (fiscal year effect). "–" indicates a revenue loss, "+" a revenue gain.
2. Assuming 1.75 per cent average annual growth in real GDP over 1998-2003. At 3.5 per cent average growth the numbers would be –14.4 and +12.3, respectively.

Source: Ministry of Finance (1999*a*) and (1998).

the VAT approved by the Diet (see for instance Ishi, 1998). In the early 1990s, the focus shifted towards land taxation – in light of the explosive increases in land prices during the bubble years – and the land tax reform of 1992 increased taxation on land holding and on capital gains from the transfer of land. The treatment of agricultural land for inheritance tax and property tax purposes was also made less favourable.

An adjustment in 1994 further increased personal tax allowances and expanded the personal income tax brackets substantially – basically to accommodate middle class dissatisfaction with perceived high personal tax burdens. Part of the tax cuts were temporary reliefs that were eliminated again in 1997. The 1994 reform also contained an increase in the VAT rate to 5 per cent (implemented in FY 1997) and a broadening of the VAT base with specific reference to the need for preparing for the costs related to the ageing society.

Significant tax changes were introduced in each of the fiscal years 1998 and 1999, with the main line being a continuation of rate reductions and narrowing of the personal income tax base. A variety of temporary tax reductions were also introduced (Table A5). The FY 1998 change contained total tax relief of some 2.4 per cent of GDP[6] of which around ⅓ was temporary income tax reductions. In particular, the temporary increase in personal allowances (from 3.6 million yen to almost 5 million yen for a one-earner family with two children) implied that the share of income tax exempted employees rose from 13 per cent to an estimate of 30 per cent. Other features of the FY 1998 change included a reduction of the top corporate rate from 50 per cent to 46 per cent while broadening the base somewhat by reducing the availability of tax-free reserves and tightening the depreciation rules as well as the valuation of securities. Transaction taxes for financial instruments were also reduced substantially and the land value added tax was suspended. Finally, capital gains taxes on transfer of land for corporations and individuals were lowered.

The FY 1999 tax change was also expansionary, but expected revenue losses were somewhat smaller than the FY 1998 package (totalling some 1.1 per cent of GDP).[7] However, the beneficial economic effects may be more significant in the long run since the package includes a further considerable reduction in corporate rates (from 46 per cent to 41 per cent) as well as a drastic cut in the top personal rate: from 65 per cent to 50 per cent (including local taxes). The FY 1999 measures also contain a considerable extension of the tax credit for housing loans as well as a further reduction of financial transaction taxes (including an exemption of the 20 per cent withholding tax on interest payments for foreign corporations and non-resident individuals). The 1 per cent tax on assets in corporate pension plans is suspended for two years. Personal tax allowances for dependent children are increased on a permanent basis, but with the withdrawal of the temporary allowances in place for FY 1998 the tax-exempt threshold for employees is reduced by almost 25 per cent compared with FY 1998 (for a one-earner family with two children). Finally, the measures include a small reduction in personal capital gains tax on land as well as tax reductions for promoting investment in small and medium-sized enterprises and in information technology.

Notes

1. Table A1 summarises the key statutory features of the tax system.
2. Timber income is also taxed separately – an issue which will not be pursued further here.
3. Japan applies basically a classical system in which dividends are taxed both at the corporate and individual level (although a dividend tax credit alleviates some of the burden of double taxation).
4. The tax credit system for dividends implies that dividends are taxed at a maximum rate of 40 per cent (for incomes below 10 million yen).
5. See Ishi (1993 and 1998) for a thorough overview and discussion of tax changes since 1973. See also Ministry of Finance (1997) for more details on the individual tax reforms.
6. National tax revenues only – see Ministry of Finance (1998).
7. National tax revenues only – see Ministry of Finance (1999*a*).

Annex III

Taxation and private savings

The main economic argument for giving tax privileges to savings is that disincentives to save otherwise embodied in an income tax system (as opposed to an expenditure tax) are thereby reduced. Theoretically and empirically, however, the effects of lower effective taxes on personal savings are ambiguous: the substitution effect will spur savings as returns to savings increase, whereas the income effect could go in either direction, depending on whether or not the household sector is a net creditor. Taxes may also affect private savings indirectly by transferring income between households with different consumption-savings patterns, by substituting public for private savings, and as a result of the interaction between household and company savings. Each of these mechanisms is potentially important to the level of private savings in OECD countries: there has been a significant increase in the overall tax burden, and a major shift in the net tax burden (taxes less transfers) from the older to the younger generations; pension schemes are mainly tax- or contribution-financed and often sheltered from taxation; and company savings tend to be an important part of total private savings (corporate savings averaged 20-30 per cent of total private savings in Japan, the United States, France and Canada during 1980-94).

Empirical evidence on the elasticity of private savings to changes in after-tax returns generally gives mixed and not very robust results, at least when applied to single countries. Bosworth (1993) finds positive and significant elasticities for only two out of 13 OECD countries. Masson *et al.* (1995) find that, based on a panel (cross country) estimate of 21 OECD countries, an increase in the real after-tax interest rate of between 4 and 6 percentage points would raise private savings by 1 percentage point of GDP. A recent study by Tanzi and Zee (1998) incorporates panel data for most industrial countries over several decades and finds a significant negative effect of tax levels on household savings, with income taxes appearing to have a far stronger effect than consumption taxes. Tachibanaki (1997) surveys a range of studies for Japan and concludes that although estimation procedures and data bases differ considerably among these studies, they typically find very small or statistically insignificant effects of changes in the after-tax interest rate on the level of private savings.[1] Nearly all the studies thus support the classic study by Komiya (1966), which concluded that there was no evidence of a strong effect on private savings levels from the preferential tax treatment of interest and dividend income. Several studies have been concerned with the effect of the 1988 tax reform which abolished the general tax exemption of savings and introduced a 20 per cent withholding tax on interest payments. Hayashi *et al.* (1989) suggested that the abolition of tax-free savings accounts would cause a drop in the steady state savings rate by only a few percentage points, and that introducing tax deductibility of mortgage interest payments would barely affect the aggregate savings rate.

One particular area of interest is the tax privileges provided to pension savings in almost all OECD countries (Box A1).[2] From an economic perspective the question arises as

Box A1. Tax treatment of private pensions

Three main transactions constitute most private pension schemes:

– contributions to the scheme, from employers or employees;
– income derived from the investment of contributions;
– payment of retirement benefits from the accumulated fund.

There are examples within the OECD of regimes that tax pensions at almost every combination of these points. The table below illustrates four possible tax regimes in a very simplistic manner, assuming a single income tax rate of 25 per cent and a rate of return on investment of 10 per cent, and considering a single contribution derived from earned income of 100, five years before retirement.

Alternative tax regimes

	A (EET)	B (TEE)	C (TTE)	D (ETT)
Contributions	100	100	100	100
Tax	–	25	25	–
Fund income	100	75	75	100
Net income over 5 years	61.05	45.79	32.67	43.56
Fund at retirement	161.05	120.79	107.67	143.56
Tax on withdrawal	40.26	–	–	35.89
Benefit withdrawn	120.79	120.79	107.67	107.67

Note: Inflation is assumed to be zero.

A. *Tax-free contributions and fund income, taxed benefits (EET).* This regime is the most commonplace way of taxing pension savings in OECD countries. Pension contributions are deductible from taxable income, allowing the whole of the 100 of earnings into the pension fund. The investment income of the fund is not axed, but benefits are taxed in full on withdrawal. This type of tax treatment can be referred to as EET, for exempt, exempt, taxed. It confers a post-tax rate of return on saving equal to the pre-tax rate of return. An individual earning 100 can choose either to spend now, paying 25 of tax and consuming goods worth 75, or to save now and consume goods worth 120.79 in five years [120.79 is simply 75 multiplied by $(1.1)^5$].

B. *Taxed contributions, tax-free fund income and benefits (TEE).* This regime does not allow deduction of contributions, thus reducing the initial size of the fund from 100 to 75. Investment income is free of tax, as in regime A. No tax is due on withdrawal of retirement benefits. This type of tax treatment can be referred to as TEE, for taxed, exempt, exempt. Like regime A, it preserves the equality of pre- and post-tax rates of return.

(continued on next page)

(continued)

C. *Taxed contributions taxed fund income, tax-free benefits (TTE).* This regime is basically that applied to interest-bearing short-term saving in most OECD countries. Its acronym is TTE, for taxed, taxed, exempt: contributions are not tax-deductible, investment income is taxed in full, and there is no tax on withdrawal of benefits. Unlike regimes A and B, this tax treatment brings the post-tax rate of return below the pre-tax rate of return. Here the post-tax rate of return is 7.5 per cent [$107.67 = 75 \times (1.075)^5$].

D. *Tax-free contributions, taxed fund income, taxed benefits (ETT).* This type of regime produces the same outcome as regime C, and therefore the same post-tax rate of return. Its acronym is ETT, for exempt, taxed, taxed: taxation of benefits and exemption of contributions are substituted for taxation of contributions and exemption of benefits.

Regimes of type A and B are examples of an expenditure tax approach, while regimes C and D correspond to a comprehensive income tax approach. Other combinations of taxing and relieving at each of the three points are possible, and indeed exist. In the OECD countries there are no regimes less favourable than C, but many more favourable than A or B.

Source: Adapted from Dilnot (1992).

to why incentives are required to encourage individuals to save more for retirement than they would otherwise do. The main argument is based on moral hazard: given the existence of public pension schemes, people may choose not to save "enough" for retirement so as to be able to receive social assistance. Indeed, if governments can encourage more people to save for their retirement and also encourage those already saving to save more, expenditure on means-tested benefits for the retired will fall. However, the resulting effect on government budgets depends on the balance between the future reduction in government expenditure and the revenue loss incurred by giving tax subsidies to savings by those who will not receive means-tested benefits anyway. There is considerable debate among economists about whether tax-favoured pension schemes increase private net savings or largely encourage households to switch the form in which they save. A massive amount of empirical research has been carried out on the extent to which tax-sheltered retirement savings schemes (like the Individual Retirement Accounts (IRAs) and the 401(k)s in the United States) have raised private and national savings.[3] However, there is no consensus as to whether these tax-sheltered pension savings schemes contribute to higher private savings and even less so with respect to national savings.

Besides personal savings, private-sector savings also include corporate savings. In principle, total private savings may be unresponsive to policies that induce corporations to save less, for instance a reduction in the dividends tax or an increase in the capital gains tax. Since the households own corporations, sophisticated shareholders should be able to "see through the corporate veil" and realise that corporations are saving less on their behalf, and hence increase personal savings by an offsetting amount in order to

re-establish his or her optimal life-cycle allocation between current and future consumption. In practice, however, the offset is not complete, since households may be liquidity constrained, short-sighted, etc. Poterba (1987) found for the United States that household savings would decline by 75 cents if the corporate savings increased by $1, *i.e.* a 75 per cent offset. Bosworth (1993) found an offset coefficient close to unity for the United States, Germany, France, Canada, Belgium and Finland, while an offset coefficient of one was rejected for Japan, Sweden and the United Kingdom. Leibfritz *et al.* (1997) estimated an average offset coefficient of 60 per cent for a panel of five large OECD countries, including Japan.

Notes

1. In measuring savings behaviour there is a peculiar data problem for Japan, namely the so-called "extended household". Elderly people living in their children's households are usually counted as dependants of the younger household, and their savings are not reported in savings data, whereas the savings of independent elderly people are reported.
2. See also Table 35 of the main text.
3. Leibfritz *et al.* (1997) and Bernheim (1999) survey the literature.

Annex IV

Chronology of main economic events

1998

November

Fuji Bank and Dai-ichi Kangyo Bank announced a tie-up in the field of trust banking by setting up a new trust bank to take over the operations of Yasuda Trust Bank.

The Bank of Japan (BOJ) decided to support corporate cash flows more aggressively through CP operations, an emergency loan programme and the introduction of a new operation using corporate bonds as collateral.

The government decided a fiscal stimulus package, including measures amounting to more than 17 trillion yen on a project cost basis. This was implemented in December through a third FY 1998 supplementary budget, which increased expenditures by 5.7 trillion yen in total.

The Decentralisation Committee compiled its fifth report, recommending, among other things, reduced involvement of central government in implementing public works by local governments.

December

A package of laws for the "Big Bang" reform of the financial system takes effect, thereby allowing banks, life and non-life insurance companies to sell trust funds, setting up safety nets for the securities and life insurance industries, and eliminating the obligation to transact on the Security Exchanges.

The Diet decided to freeze the Fiscal Structural Reform Law for an unspecified period.

The government decided to place the Nippon Credit Bank under temporary nationalisation.

The Financial Reconstruction Commission (FRC) was established. It will take over oversight of the financial industry until the end of FY 2000.

Prime Minister Obuchi announced the "New Miyazawa Initiative", aiming to ease the financial difficulties of Asian crisis-afflicted countries by providing funds amounting to US$30 billion in total.

The government agreed to the tariffication of rice imports as from April 1999.

The government decided the outline of tax reform in FY 1999, including a permanent income tax cut, corporate income tax cuts and housing tax credits.

The Ministry of Finance (MOF) announced measures to promote the yen's internationalisation, including beginning to sell Finance Bills in the market.

The government decided the draft initial FY 1999 budget and the initial Fiscal Investment and Loan Programme (FILP).

The government approved its official economic forecast of 0.5 per cent growth in FY 1999.

The Fair Trade Commission released new merger guidelines under the anti-monopoly law.

Finance Minister Miyazawa hinted at a possible termination of the Trust Fund Bureau's purchase of government bonds, causing a sharp rise in long-term interest rates.

1999

January

Prime Minister Obuchi formed a new coalition cabinet of the Liberal Democratic and Liberal Parties.

The FRC published its basic operating policies.

Mitsui Trust and Chuo Trust Banks announced their merger as of April 2000.

The Reform Promotion Headquarters of Central Ministries and Agencies approved an outline of the government reform plan scheduled for January 2001.

The government approved the Doubling Plan for Living Space and the Industry Revitalisation Plan.

The Ministry of Home Affairs reported the FY 1999 Local Public Finance Programme to the cabinet with estimated total outlays of 88.5 trillion yen.

The BOJ released its streamlining plan, which consists of selling and scaling down the houses for general managers of the Bank's branches and disposing of all the Bank-owned recreation facilities for its employees.

February

The government decided to merge six government financial institutions into three.

The BOJ lowered its target call rate from 0.25 to 0.15 per cent, and ultimately as low as possible.

The Ministry of Health and Welfare released its outline of the pensions system reform.

The Economic Strategy Council submitted its final report to Prime Minister Obuchi. It included a scenario of economic recovery by 2001 and restoring the health of public finances thereafter.

March

The FRC decided to inject about 7.5 trillion yen in new capital into 15 major banks.

The Nissan Motor Company announced a tie-up with the French company Renault.

The Diet passed the initial FY 1999 budget and FILP. It also enacted the FY 1999 tax cuts.

The government decided a draft Decentralisation Promotion Law, including abolition of the system of local governors administrating on behalf of the central government.

The government revised the three-year Deregulation Promotion Plan (FY 1998 to FY 2000), incorporating about 250 new items.

April

The revised Equal Job Opportunities Law took effect, prohibiting unequal treatment of female workers.

The Resolution and Collection Bank was merged with the Housing Loan Administration Corporation to form the Resolution and Collection Corporation.

The Financial Supervisory Agency (FSA) released its final report of the working group on the new financial inspection manual.

The Transportation Policy Council recommended the abolition of demand-supply regulation of the taxi business in FY 2001.

The FRC decided to place the operation and assets of the Kokumin Bank, a regional tier-II bank, under financial reorganisation administrators.

A law passed the Diet, allowing non-banks to issue corporate bonds.

The Diet enacted the law to reorganise government financial institutions.

The Council on Foreign Exchange and Other Transactions submitted its report to the Finance Minister, recommending further measures to enhance the yen's international standing.

The Economic Planning Agency released a follow-up report of the November 1998 fiscal package.

The government approved 17 draft laws to reorganise the central 22 ministries and agencies into 12.

May

The Law Concerning Access to Information Held by Administrative Organs passed the Diet, imposing obligations on the administrative organs to disclose administrative documents, in principle.

Sakura Bank, Mitsui Marine and Fire, Mitsui Trust Bank and Mitsui Life announced a business tie-up in the field of defined contribution pensions.

Osaka Bank and Kinki Bank announced their intention to merge as of April 2000.

The Kofuku Bank, a regional tier-II bank in Osaka, applied to the FSA for voluntary closure of business after the agency applied the prompt corrective action clause for the first time.

June

The Tokyo Court formally declared Yamaichi Securities to be bankrupt.

Toho Life Insurance announced its insolvency with a capital shortfall of 0.2 trillion yen and informed the FSA of its voluntary closure.

A British company, Cable & Wireless, competed successfully with NTT to purchase most of the shares of a communication service company, IDC.

The government announced measures to create more than 700 000 jobs and strengthen industrial competitiveness.

The FRC released a basic guideline for capital injections into regional banks, aiming at a capital adequacy ratio of at least 8 per cent.

The FRC decided to place the operation and assets of the Tokyo Sowa Bank, a regional tier-II bank in Tokyo, under financial reorganisation administrators.

A Japanese software company, Softbank, and NASDAQ jointly announced that a new stock market for SMEs, NASDAQ Japan, would be established in 2000.

The FSA decided to apply the prompt corrective action clause to the Namihaya Bank, a regional tier-II bank in Osaka.

Two laws passed the Diet, liberalising the personnel dispatching and job placement businesses.

July

NTT was reorganised into a holding company, east and west regional monopolies and a long-distance communication company.

The Diet enacted the laws for reorganising central ministries and agencies and laws for promoting decentralisation.

The Economic Council submitted to Prime Minister Obuchi a new economic plan describing a scenario up to around year 2010.

A draft law for reforming the pensions system was approved by the government, including measures to raise the share of tax financing of basic pension benefits from a third to a half by 2004.

The Diet passed a law to introduce deputy-Ministers, thereby limiting bureaucrats' participation in Diet debate.

The FRC sanctioned the Credit Suisse Group for a variety of reasons including obstruction of the FSA's inspections.

The government decided its guideline for the initial FY 2000 budget.

August

The Diet passed the Special Law for Industry Revitalisation.

The FRC decided to place the operation and assets of the Namihaya Bank under financial administration.

The government decided on a revised medium-term policy regarding official development assistance, focusing on aid to Asian countries' structural reform.

Softbank, Tokyo Power and Microsoft announced that they would jointly establish a new communication service company to provide internet access at lower rates than NTT.

Industrial Bank of Japan, Dai-ichi Kangyo and Fuji Banks announced that they would jointly set up a holding company in the autumn of 2000 and reorganise their businesses as from spring of 2002.

September

Mr. Obuchi was re-elected as president of LDP.

The BOJ policy board decided to maintain its current policy stance, disappointing market participants and causing further appreciation of the yen.

The FRC decided to sell the Long Term Credit Bank, still under public management, to an international group of financial institutions led by Ripplewood Holdings.

October

Regulations on commissions on stock transactions were fully lifted.

Mr. Obuchi formed his second cabinet based on a new coalition among LDP, Liberal Party and Komei party.

Tokai Bank and Asahi Bank announced their intention to speed up their merger and to set up a holding company in October 2001.

The BOJ policy board decided to use a wider range of money market operations, such as outright purchases of short-term government securities, so as to assure further permeation of the effects of monetary easing.

Sumitomo Bank and Sakura Bank announced their intention to merge by April 2002.

The Minister of International Trade and Industry announced his intention to ask WTO to examine the United State's anti-dumping actions against Japanese steel exports.

The Nissan Motor Company announced its restructuring plan to close five factories and cut 21 000 workers, thereby reducing its production capacity by 30 per cent.

Three non-life insurance companies, Mitsui Marine, Nippon Fire and Marine, and Koa Fire and Marine, announced a future merger.

BASIC STATISTICS:

INTERNATIONAL COMPARISONS

BASIC STATISTICS: INTERNATIONAL COMPARISONS

	Units	Reference period[1]	Australia	Austria	Belgium	Canada	Czech Republic	Denmark	Finland	France	Germany	Greece
Population												
Total	Thousands	1996	18 289	8 060	10 157	29 964	10 316	5 262	5 125	58 380	81 877	10 465
Inhabitants per sq. km	Number	1996	2	96	333	3	131	122	15	106	229	79
Net average annual increase over previous 10 years	%	1996	1.3	0.6	0.3	1.3	0	0.3	0.4	0.5	3	0.5
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	8 344	3 737 (94)	3 675 (95)	13 676	4 918	2 593	2 087	21 951	35 360	3 824 (95)
of which: Agriculture	% of TCE	1996	5.1	7.2 (94)	2.5 (95)	4.1	6.3	4	7.1	4.6	3.3	20.4 (95)
Industry	% of TCE	1996	22.5	33.2 (94)	26.7 (95)	22.8	42	27	27.6	25.9	37.5	23.2 (95)
Services	% of TCE	1996	72.4	59.6 (94)	71.4 (95)	73.1	51.7	69	65.3	69.5	59.1	56.4 (95)
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	398.9	228.7	268.2	579.2	56.2	174.9	125.1	1 536.6	2 353.5	91.2 (95)
Per capita	US$	1996	21 812	28 384	26 409	19 330	5 445	33 230	24 420	26 323	28 738	8 722 (95)
At current prices using current PPPs[3]	Bill. US$	1996	372.7	172.4	222	645.1	..	118	96.7	1 198.6	1 736.1	133.5
Per capita	US$	1996	20 376	21 395	21 856	21 529	..	22 418	18 871	20 533	21 200	12 743
Average annual volume growth over previous 5 years	%	1996	3.9	1.6	1.2	2.2	2	2.2	1.6	1.2	1.4	1.3 (95)
Gross fixed capital formation (GFCF)	% of GDP	1996	20.3	23.8	17.3	17.7	33	16.7	16.1	17.4	20.6	17 (95)
of which: Machinery and equipment	% of GDP	1996	10.2 (95)	8.8 (95)	7.5 (95)	6.6	..	7.9 (95)	6.4 (95)	7.8	7.6	7.7 (95)
Residential construction	% of GDP	1996	4.6 (95)	5.9 (95)	4.6 (95)	5.4	..	3.2 (95)	3.5 (95)	4.4	7.3	3.3 (95)
Average annual volume growth over previous 5 years	%	1996	5.6	2.1	0.3	2.2	9.4	2	–4.1	–1.5	0.2	0.5 (95)
Gross saving ratio[4]	% of GDP	1996	18	21.9	22.2	17.8	..	17.6	19.6	18.7	20	16 (95)
General government												
Current expenditure on goods and services	% of GDP	1996	17	19.8	14.5	18.7	21.5	25.2	21.9	19.4	19.8	20.8 (95)
Current disbursements[5]	% of GDP	1995	35.6	48.6	52.2	45.8	..	59.6	55.9	50.9	46.6	52.1
Current receipts	% of GDP	1995	34.9	47.4	49.9	42.7	..	58.1	52.8	46.9	45.9	45
Net official development assistance	% of GNP	1995	0.36	0.33	0.38	0.38	..	0.96	0.32	0.55	0.31	0.13
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	12 596	12 152	13 793	12 959	..	12 027	10 282	12 506	12 244	9 473
Passenger cars, per 1 000 inhabitants	Number	1994	460	433	416	466	282	312	368	430	488	199
Telephones, per 1 000 inhabitants	Number	1994	496	466	449	576	209	604	551	547	483[8]	478
Television sets, per 1 000 inhabitants	Number	1993	489	479	453	618	476	538	504	412	559	202
Doctors, per 1 000 inhabitants	Number	1995	2.2 (91)	2.7	3.7 (94)	2.2	2.9	2.9 (94)	2.8	2.9	3.4	3.9 (94)
Infant mortality per 1 000 live births	Number	1995	5.7	5.4	7.6 (94)	6.3 (94)	7.7	5.5	4	5.8 (94)	5.3	8.1
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	1.7	5.2	2.7	2.4	..	3.2	3.8	2.6	4.2	11.8
Consumer prices	%	1996	2.4	2.9	2.2	1.4	11.9	1.9	1.5	2	3.1	11.6
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	60 288	57 870	170 223[7]	202 320	21 910	51 030	40 576	288 450	521 263	11 501
As % of GDP	%	1996	15.1	25.3	63.5	34.9	39	29.2	32.4	18.8	22.1	12.9 (95)
Average annual increase over previous 5 years	%	1996	7.5	7.1	7.6	9.7	..	6.2	12.1	6.3	5.4	5.8
Imports of goods, cif*	Mill. US$	1996	61 374	67 376	160 917[7]	170 931	27 721	44 987	30 911	271 348	455 741	27 402
As % of GDP	%	1996	15.4	29.5	60	29.5	49.3	25.7	24.7	17.7	19.4	30.4 (95)
Average annual increase over previous 5 years	%	1996	9.7	5.9	5.9	7.7	..	5.6	7.3	3.9	3.3	6.6
Total official reserves[6]	Mill. SDRs	1996	10 107	15 901	11 789[7]	14 202	8 590	9 834	4 810	18 635	57 844	12 171
As ratio of average monthly imports of goods	Ratio	1996	2	2.8	0.9	1	..	2.6	1.9	0.8	1.5	5.3

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Hungary	Iceland	Ireland	Italy	Japan	Korea	Luxembourg	Mexico	Netherlands	New Zealand
Population												
Total	Thousands	1996	10 193	270	3 621	57 473	125 864	45 545	418	96 582	15 494	3 640
Inhabitants per sq. km	Number	1996	77	3	52	191	333	458	161	48	380	14
Net average annual increase over previous 10 years	%	1996	–0.3	1.1	0.2	0	0.4	1	1.3	2	0.6	1.1
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	3 605	142	1 307	20 036	64 860	20 764	212 (95)	32 385 (95)	6 983	1 688
of which: Agriculture	% of TCE	1996	8.4	9.2	10.7	7	5.5	11.6	2.8 (95)	23.5 (95)	3.9	9.5
Industry	% of TCE	1996	33	23.9	27.2	32.1	33.3	32.5	30.7 (90)	21.7 (95)	22.4	24.6
Services	% of TCE	1996	58.6	66.2	62.3	60.9	61.2	55.9	66.1 (90)	54.8 (95)	73.8	65.9
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	43.7 (95)	7.3	70.7	1 214.2	4 595.2	484.8	17	329.4	396	65.9
Per capita	US$	1996	4 273 (95)	27 076	19 525	21 127	36 509	10 644	40 791	3 411	25 511	18 093
At current prices using current PPPs[3]	Bill. US$	1996	..	6.3	68.8	1 148	2 924.5	618.5	13.5	751.1	324.5	63.6
Per capita	US$	1996	..	23 242	18 988	19 974	23 235	13 580	32 416	7 776	20 905	17 473
Average annual volume growth over previous 5 years	%	1996	–2.4 (95)	1.5	7.1	1	1.5	7.1	4.8	1.7	2.3	3.7
Gross fixed capital formation (GFCF)	% of GDP	1996	19.3 (95)	17.5	17.2	17	29.7	36.8	20.8	18	19.7	20.9
of which: Machinery and equipment	% of GDP	1996	..	6.7	5.5 (95)	8.8	10.1 (95)	13	..	8.8	9.4	10
Residential construction	% of GDP	1996	. .	3.9	4.9 (95)	4.5	5.3 (95)	7.6	..	4.7	5	5.6
Average annual volume growth over previous 5 years	%	1996	–0.9 (95)	–1.4	6	–1.4	1.3	6.9	0.2	–0.7	2.2	9.6
Gross saving ratio[4]	% of GDP	1996	..	15.6	21.7	20.5	31.4	34.2	37.5	22.7	25.7	16
General government												
Current expenditure on goods and services	% of GDP	1996	24.9 (95)	20.8	14.1	16.4	9.7	10.6	13.6	9.7[10]	14	14.4
Current disbursements[5]	% of GDP	1995	..	35.1	39.2 (94)	49.5	28.5	15.1	..	..	51.8	..
Current receipts	% of GDP	1995	..	36	39.3 (94)	44.5	32	25.1	..	..	50	..
Net official development assistance	% of GNP	1995	..	..	0.29	0.15	0.28	0.03	0.36	..	0.81	0.23
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	..	14 244	10 020	12 224	13 912	7 354	17 811	5 045	12 477	10 895
Passenger cars, per 1 000 inhabitants	Number	1994	212	434	264	517	342	115	544	91	383	457
Telephones, per 1 000 inhabitants	Number	1994	170	557	350	429	480	397	564	93	509	470
Television sets, per 1 000 inhabitants	Number	1993	427	335	301	429	618	215	261	150	491	
Doctors, per 1 000 inhabitants	Number	1995	3.4	3.9 (94)	3.4	3.0 (94)	1.7	1.6 (92)	1.8 (94)	1.1	2.2 (93)	1.6
Infant mortality per 1 000 live births	Number	1995	11	6.1	6.3	6.6 (94)	4.3	9	5.3 (94)	17 (94)	5.5	7.2 (94)
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	..	..	3.7	3.5	1.8	..	..	–1.6	2.4	1.5
Consumer prices	%	1996	23.2	2.6	2.2	4.5	0.7	5.3	2.4	19.7	2.5	2
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	15 674	1 891	48 416	250 842	411 067	129 715	..	96 000	203 953	14 316
As % of GDP	%	1996	35.9	26	68.5	20.7	8.9	26.8	..	29.1	51.5	21.7
Average annual increase over previous 5 years	%	1996	8.9	4	14.8	8.2	5.5	12.5	..	17.6	8.9	8.2
Imports of goods, cif*	Mill. US$	1996	18 105	2 032	35 763	206 904	349 149	150 340	..	89 469	184 389	14 682
As % of GDP	%	1996	41.4	27.9	50.6	17	7.6	31	..	27.2	46.6	22.3
Average annual increase over previous 5 years	%	1996	9.6	3.4	11.5	2.5	8	13.9	..	12.4	7.8	11.8
Total official reserves[6]	Mill. SDRs	1996	6 812	316	5 706	31 954	150 663	23 670	..	13 514	18 615	4 140
As ratio of average monthly imports of goods	Ratio	1996	..	1.9	1.9	1.9	5.2	..	..	1.8	1.2	3.4

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Norway	Poland	Portugal	Spain	Sweden	Switzerland	Turkey	United Kingdom	United States
Population											
Total	Thousands	1996	4 370	38 618	9 935	39 270	8 901	7 085	62 695	58 782	265 557
Inhabitants per sq. km	Number	1996	13	123	108	78	20	172	80	240	28
Net average annual increase over previous 10 years	%	1996	0.5	0.3	–0.1	0.2	0.6	0.8	2	0.3	1
Employment											
Total civilian employment (TCE)[2]	Thousands	1996	2 110	14 977	4 475	12 394	3 963	3 803	20 895	26 088	126 708
of which: Agriculture	% of TCE	1996	5.2	22.1	12.2	8.7	2.9	4.5	44.9	2	2.8
Industry	% of TCE	1996	23.4 (95)	31.7	31.4	29.7	26.1	27.7	22	27.4	23.8
Services	% of TCE	1996	71.5 (95)	46.2	56.4	61.6	71	67.4	33.1	71	73.3
Gross domestic product (GDP)											
At current prices and current exchange rates	Bill. US$	1996	157.8	117.9 (95)	103.6	584.9	251.7	294.3	181.5	1 153.4	7 388.1
Per capita	US$	1996	36 020	3 057 (95)	10 425	14 894	28 283	41 411	2 894	19 621	27 821
At current prices using current PPPs[3]	Bill. US$	1996	106.7	..	130.1	587.2	171.4	180.6	383.3	1 095.5	7 388.1
Per capita	US$	1996	24 364	..	13 100	14 954	19 258	25 402	6 114	18 636	27 821
Average annual volume growth over previous 5 years	%	1996	4.1	2.2 (95)	1.5	1.3	1	0.1	4.4	2.2	2.8
Gross fixed capital formation (GFCF)	% of GDP	1996	20.5	17.1 (95)	24.1	20.1	14.8	20.2	25	15.5	17.6
of which: Machinery and equipment	% of GDP	1996	8.4	..	11.7 (93)	6.1 (95)	7.9	9.3	11.9	7.6	8.3 (95)
Residential construction	% of GDP	1996	2.6 (94)	..	5.2 (93)	4.3 (95)	1.9	11[9]	8.4 (95)	3	4.1 (95)
Average annual volume growth over previous 5 years	%	1996	2.8	5.4 (95)	2.2	–1	–2.6	–0.8	6.9	1.3	6.9
Gross saving ratio[4]	% of GDP	1996	29.9	..	21.6	20.7	16	27.1	22.3	14.6	16.6
General government											
Current expenditure on goods and services	% of GDP	1996	20.5	16.9 (95)	18.5	16.3	26.2	14.3	11.6	21.1	15.6
Current disbursements[5]	% of GDP	1995	45.8	..	42.5 (93)	41.2	63.8	47.7	..	42.3 (94)	34.3
Current receipts	% of GDP	1995	50.9	..	39.8 (93)	37.9	57.5	53.8	..	37.2 (94)	32.1
Net official development assistance	% of GNP	1995	0.87	..	0.27	0.24	0.77	0.34	0.07	0.28	0.1
Indicators of living standards											
Private consumption per capita using current PPPs[3]	US$	1996	11 593	..	8 522	9 339	10 096	15 632	4 130	11 865	18 908
Passenger cars, per 1 000 inhabitants	Number	1994	381	186	357	351	406 (93)	451	47	372	565
Telephones, per 1 000 inhabitants	Number	1994	554	131	350	371	683	597	201	489	602
Television sets, per 1 000 inhabitants	Number	1993	427	298	190	400	470	400	176	435	816
Doctors, per 1 000 inhabitants	Number	1995	2.8	2.3	3	4.1 (93)	3.1	3.1 (94)	1.2	1.6 (94)	2.6 (94)
Infant mortality per 1 000 live births	Number	1995	4	13.6	7.4	6 (94)	4	5	46.8 (94)	6.2 (94)	8 (94)
Wages and prices (average annual increase over previous 5 years)											
Wages (earnings or rates according to availability)	%	1996	3.2	..	..	5.8	4.8	..	..	4.9	2.7
Consumer prices	%	1996	1.9	..	5.6	4.7	2.7	2.2	81.6	2.7	2.9
Foreign trade											
Exports of goods, fob*	Mill. US$	1996	49 576	24 417	24 614	102 067	84 836	79 581	23 301	259 941	625 075
As % of GDP	%	1996	31.4	20.7	23.8	17.5	33.7	27	12.8	22.5	8.5
Average annual increase over previous 5 years	%	1996	7.8	..	8.6	11.2	9	5.3	11.1	7	8.2
Imports of goods, cif*	Mill. US$	1996	35 575	37 185	35 192	121 838	66 825	78 052	43 094	287 033	795 289
As % of GDP	%	1996	22.5	31.5	34	20.8	26.5	26.5	23.7	24.9	10.8
Average annual increase over previous 5 years	%	1996	6.9	..	6.1	5.5	6	3.2	15.1	6.5	10.3
Total official reserves[6]	Mill. SDRs	1996	18 441	12 409	11 070	40 284	13 288	26 727	11 430	27 745	44 536
As ratio of average monthly imports of goods	Ratio	1996	6.2	..	3.8	4	2.4	4.1	3.2	1.2	0.7

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

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